



# **Sentinel-5 precursor/TROPOMI**

## **Level 2 Product User Manual**

### **Sulphur Dioxide SO<sub>2</sub>**



**TROPOMI**



**sentinel-5p**

document number : S5P-L2-DLR-PUM-400E

authors : Fabian Romahn, Mattia Pedernana, Diego Loyola, Arnoud Apituley, Maarten Sneep,  
J. Pepijn Veefkind

CI identification : CI-400E-PUM

issue : 02.07.00

date : 2024-10-04

status : released

## **Document approval record**

- This document was prepared by Fabian Romahn, Mattia Pedernana, Diego Loyola, Arnoud Apituley, Maarten Sneepe, J. Pepijn Veeffkind.

# 1 Introduction

## 1.1 Identification

This document, identified as S5P-L2-DLR-PUM-400E, describes the technical characteristics of the S5p/TROPOMI Level 2 products that are needed for efficient and correct use of the data contained. This product user manual is specific for Sulphur Dioxide SO<sub>2</sub>.

## 1.2 Purpose and objective

The Sentinel-5 Precursor (S5p) mission is a low Earth orbit polar satellite system to provide information and services on air quality, climate and the ozone layer. The S5p mission is part of the Global Monitoring of the Environment and Security (GMES/COPERNICUS) space component programme. The S5p mission consists of a satellite bus, the payload consisting of the TROPOspheric Monitoring Instrument (TROPOMI), and a ground system. A journal paper describing the mission and its objectives can be found in [RD1], while a comprehensive description of the mission can be found in [RD2]. Furthermore, various websites are maintained with S5p/TROPOMI information, e.g. [ER1, ER2].

From the data collected by the TROPOMI instrument, a number of geophysical (L2) products are derived. The algorithms for the raw data treatment (L0 – L1b) and the actual L2 data processing are each described in an algorithm theoretical basis document (ATBD). This Product User Manual (PUM) describes the technical characteristics of the S5p/TROPOMI Level 2 geophysical data products that are needed for efficient and correct use of the data contained.

In the PUM, the common structure of the datafiles and metadata used in all the delivered products as well as a specific section related to the Sulphur Dioxide SO<sub>2</sub> product are described.

## 1.3 Document overview

We start with a summary of the S5p L2 products and information needed to obtain and inspect data, as well as how to obtain product support. The Sulphur Dioxide SO<sub>2</sub> data product is described next, with examples, and information about the use of the data. Format, L2 structure and metadata are addressed in the next chapter, followed by the detailed description of the Sulphur Dioxide SO<sub>2</sub> data. We then continue with a discussion of units and quality assurance parameters. The final chapter contains information about generic metadata and the Appendix lists measurement flags, processing quality flags, and surface classifications.

## 2 Applicable and reference documents

### 2.1 Applicable documents

[AD1] Tailoring of the Earth Observation File Format Standard for the Sentinel 5 precursor Ground Segment.  
**source:** ESA/ESTEC; **ref:** S5P-TN-ESA-GS-106; **issue:** 2.2; **date:** 2015-02-20.

### 2.2 Standard documents

There are no standard documents

### 2.3 Reference documents

[RD1] J. P. Veefkind, I. Aben, K. McMullan *et al.*; TROPOMI on the ESA Sentinel-5 Precursor: A GMES mission for global observations of the atmospheric composition for climate, air quality and ozone layer applications. *Remote Sens. Environ.*; **120** (2012), 70; 10.1016/j.rse.2011.09.027.

[RD2] Input/output data specification for the TROPOMI L01b data processor.  
**source:** KNMI; **ref:** S5P-KNMI-L01B-0012-SD; **issue:** 11.0.0; **date:** 2022-03-31.

[RD3] S5P/TROPOMI ATBD Cloud Products.  
**source:** DLR; **ref:** S5P-DLR-L2-ATBD-400I; **issue:** 2.3; **date:** 2021-06-25.

[RD4] Sentinel-5 precursor/TROPOMI Level 2 Product User Manual Cloud.  
**source:** DLR; **ref:** S5P-L2-DLR-PUM-400I; **issue:** 2.6.1; **date:** 2023-11-10.

[RD5] S5P-NPP Cloud Processor ATBD.  
**source:** RAL Space; **ref:** S5P-NPPC-RAL-ATBD-0001; **issue:** 1.0.0; **date:** 2016-02-12.

[RD6] S5P/TROPOMI HCHO ATBD.  
**source:** BIRA; **ref:** S5P-BIRA-L2-400F-ATBD; **issue:** 2.4.1; **date:** 2022-06-22.

[RD7] Sentinel-5 precursor/TROPOMI Level 2 Product User Manual HCHO.  
**source:** DLR; **ref:** S5P-L2-DLR-PUM-400F; **issue:** 2.4.0; **date:** 2022-07-11.

[RD8] S5P/TROPOMI SO<sub>2</sub> ATBD.  
**source:** BIRA; **ref:** S5P-BIRA-L2-400E-ATBD; **issue:** 2.6.0; **date:** 2024-10-08.

[RD9] S5P/TROPOMI Total ozone ATBD.  
**source:** DLR/BIRA; **ref:** S5P-L2-DLR-ATBD-400A; **issue:** 2.4.0; **date:** 2022-07-11.

[RD10] Sentinel-5 precursor/TROPOMI Level 2 Product User Manual Total Ozone Column.  
**source:** DLR; **ref:** S5P-L2-DLR-PUM-400A; **issue:** 2.4.0; **date:** 2022-07-11.

[RD11] TROPOMI ATBD of tropospheric ozone data products.  
**source:** DLR/IUP; **ref:** S5P-DLR-IUP-L2-400C; **issue:** 2.3.0; **date:** 2021-06-04.

[RD12] Sentinel-5 precursor/TROPOMI Level 2 Product User Manual Ozone Tropospheric Column.  
**source:** DLR; **ref:** S5P-L2-DLR-PUM-400C; **issue:** 2.3.0; **date:** 2021-06-04.

[RD13] TROPOMI ATBD of the Aerosol Layer Height product.  
**source:** KNMI; **ref:** S5P-KNMI-L2-0006-RP; **issue:** 2.6.0; **date:** 2023-10-20.

[RD14] Sentinel-5 precursor/TROPOMI Level 2 Product User Manual Aerosol Layer Height.  
**source:** KNMI; **ref:** S5P-KNMI-L2-0022-MA; **issue:** 2.7.0; **date:** 2024-04-04.

[RD15] TROPOMI ATBD of the UV aerosol index.  
**source:** KNMI; **ref:** S5P-KNMI-L2-0008-RP; **issue:** 2.1.0; **date:** 2022-07-22.

[RD16] Sentinel-5 precursor/TROPOMI Level 2 Product User Manual Aerosol Index.  
**source:** KNMI; **ref:** S5P-KNMI-L2-0026-MA; **issue:** 2.7.0; **date:** 2024-04-04.

- [RD17] TROPOMI ATBD Ozone profile and tropospheric profile.  
**source:** KNMI; **ref:** S5P-KNMI-L2-0004-RP; **issue:** 1.0.0; **date:** 2021-10-22.
- [RD18] Sentinel-5 precursor/TROPOMI Level 2 Product User Manual Ozone Profile and Tropospheric Ozone Profile.  
**source:** KNMI; **ref:** S5P-KNMI-L2-0020-MA; **issue:** 2.7.0; **date:** 2024-04-04.
- [RD19] TROPOMI ATBD of the total and tropospheric NO<sub>2</sub> data products.  
**source:** KNMI; **ref:** S5P-KNMI-L2-0005-RP; **issue:** 2.7.0; **date:** 2024-07-18.
- [RD20] Sentinel-5 precursor/TROPOMI Level 2 Product User Manual Nitrogen Dioxide.  
**source:** KNMI; **ref:** S5P-KNMI-L2-0021-MA; **issue:** 4.3.0; **date:** 2024-04-04.
- [RD21] Algorithm Theoretical Baseline Document for Sentinel-5 Precursor: Carbon Monoxide Total Column Retrieval.  
**source:** SRON; **ref:** SRON-S5P-LEV2-RP-002; **issue:** 2.5.0; **date:** 2024-06-11.
- [RD22] Sentinel-5 precursor/TROPOMI Level 2 Product User Manual Carbon Monoxide Column.  
**source:** SRON/KNMI; **ref:** SRON-S5P-LEV2-MA-002; **issue:** 2.7.0; **date:** 2024-04-04.
- [RD23] Algorithm Theoretical Baseline Document for Sentinel-5 Precursor methane retrieval.  
**source:** SRON; **ref:** SRON-S5P-LEV2-RP-001; **issue:** 2.5.0; **date:** 2024-06-11.
- [RD24] Sentinel-5 precursor/TROPOMI Level 2 Product User Manual Methane.  
**source:** SRON/KNMI; **ref:** SRON-S5P-LEV2-MA-001; **issue:** 2.7.0; **date:** 2024-04-04.
- [RD25] Tailoring of the Earth Observation File Format Standard for the Sentinel 5 precursor Ground Segment.  
**source:** ESA/ESTEC; **ref:** S5P-TN-ESA-GS-106; **issue:** 2.2; **date:** 2015-02-20.
- [RD26] Earth Observation – Ground segment file format standard.  
**source:** ESA/ESTEC; **ref:** PE-TN-ESA-GS-0001; **issue:** 2.0; **date:** 2012-05-03.
- [RD27] Geographic information – Metadata.  
**source:** ISO; **ref:** ISO 19115:2003(E); **issue:** 1; **date:** 2003-05-01.
- [RD28] Geographic information – Metadata – Part 2: Extensions for imagery and gridded data.  
**source:** ISO; **ref:** ISO 19115-2:2009(E); **issue:** 1; **date:** 2009-02-12.
- [RD29] Geographic information – Data quality.  
**source:** ISO; **ref:** ISO 19157; **issue:** 1; **date:** 2013-10-10.
- [RD30] Earth Observation Metadata profile of Observations & Measurements.  
**source:** Open Geospatial Consortium; **ref:** OGC 10-157r3; **issue:** 1.0; **date:** 2012-06-12.
- [RD31] Data Standards Requirements for CCI Data Producers.  
**source:** ESA; **ref:** CCI-PRGM-EOPS-TN-13-0009; **issue:** 1.1; **date:** 2013-05-24.
- [RD32] Metadata specification for the TROPOMI L1b products.  
**source:** KNMI; **ref:** S5P-KNMI-L01B-0014-SD; **issue:** 2.0.0; **date:** 2014-12-09.
- [RD33] Sentinel-4 UVN Phase B2, C/D and support to phase E1 – Level 0 to Level 1b data processing software Input/Output Data Specification (IODS): Level 1b output products and metadata contents and format.  
**source:** ESA/ESTEC; **ref:** S4.ESA.UVN.TN.1206; **issue:** 1.0; **date:** 2011-06-23.
- [RD34] Algorithm theoretical basis document for the TROPOMI L01b data processor.  
**source:** KNMI; **ref:** S5P-KNMI-L01B-0009-SD; **issue:** 10.0.0; **date:** 2022-03-31.
- [RD35] Data elements and interchange formats – Information interchange – Representation of dates and times.  
**source:** ISO; **ref:** ISO 8601:2004(E); **issue:** 3; **date:** 2004-12-01.
- [RD36] John Caron; Annotated Schema for NcML (2011). URL <http://www.unidata.ucar.edu/software/netcdf/ncml/v2.2/AnnotatedSchema4.html>.

- [RD37] INSPIRE Metadata Regulation, Commission Regulation (EC), No1205/2008.  
**source:** EC; **ref:** Commission Regulation (EC) No 1205/2008; **date:** 2008-12-03.
- [RD38] INSPIRE Metadata Implementing Rules: Technical Guidelines based on EN ISO 19115 and EN ISO 19119.  
**source:** EC JRC; **ref:** MD\_IR\_and\_ISO\_v1\_2\_20100616; **issue:** 1.2; **date:** 2010-06-16.
- [RD39] Geographic Information – Observations and Measurements.  
**source:** ISO; **ref:** ISO 19156:2011(E); **date:** 2011-12-20.
- [RD40] Geographic information – Metadata – XML schema implementation.  
**source:** ISO; **ref:** ISO 19139:2007(E); **issue:** 1; **date:** 2010-12-13.
- [RD41] Observations and Measurements - XML Implementation.  
**source:** OGC; **ref:** OGC 10-025r1; **issue:** 2.0; **date:** 2011-03-22.
- [RD42] Observations and Measurements – XML Implementation..  
**source:** Open Geospatial Consortium; **ref:** OGC 10-025r1; **issue:** 2.0; **date:** 2011-03-22.
- [RD43] Sentinel 5 precursor/TROPOMI KNMI and SRON level 2 Input Output Data Definition.  
**source:** KNMI; **ref:** S5P-KNMI-L2-0009-SD; **issue:** 16.0.0; **date:** 2022-02-25.
- [RD44] Sentinel-5 Precursor Level 2 UPAS Processor Input/Output Definition Document.  
**source:** DLR-IMF; **ref:** S5P-L2-DLR-IODD-3002; **issue:** 3.7.6; **date:** 2024-10-04.
- [RD45] S5P-NPP Cloud Processor IODD.  
**source:** RAL; **ref:** S5P-NPPC-RAL-IODD-0001; **issue:** 0.10.0; **date:** 2014-05-28.
- [RD46] M.L. Carroll, J.R. Townshend, C.M. DiMiceli *et al.*; A new global raster water mask at 250 m resolution. *International Journal of Digital Earth*; **2** (2009) (4), 291; 10.1080/17538940902951401.

## 2.4 Electronic references

- [ER1] Tropomi official website. URL <http://www.tropomi.eu>.
- [ER2] S5P official website. URL <https://sentinel.esa.int/web/sentinel/missions/sentinel-5p>.
- [ER3] Robert B. Schmunk; Panoply netCDF, HDF and GRIB Data Viewer. URL <http://www.giss.nasa.gov/tools/panoply/>.
- [ER4] Infrastructure for Spatial Information in the European Community (INSPIRE) Directive 2007/2/EC. URL <http://inspire.jrc.ec.europa.eu/>.
- [ER5] Brian Eaton, Jonathan Gregory, Bob Drach *et al.*; *NetCDF Climate and Forecast (CF) Metadata Conventions*. Lawrence Livermore National Laboratory (2014). Version 1.7 draft; URL <http://cfconventions.org>.
- [ER6] ESIP; *Attribute Conventions for Dataset Discovery (ACDD)*. 1st edition (2013). URL [http://wiki.esipfed.org/index.php/Attribute\\_Convention\\_for\\_Data\\_Discovery\\_\(ACDD\)](http://wiki.esipfed.org/index.php/Attribute_Convention_for_Data_Discovery_(ACDD)).
- [ER7] NetCDF Users Guide (2011). URL <http://www.unidata.ucar.edu/software/netcdf/docs/netcdf.html>.
- [ER8] Unidata; *NetCDF library and documentation*. URL <http://www.unidata.ucar.edu/software/netcdf/>.
- [ER9] UDUNITS 2 Manual (2011). URL <http://www.unidata.ucar.edu/software/udunits/>.
- [ER10] Cooperative Ocean/Atmosphere Research Data Service; Conventions for the standardization of NetCDF files (1995). URL [http://ferret.wrc.noaa.gov/noaa\\_coop/coop\\_cdf\\_profile.html](http://ferret.wrc.noaa.gov/noaa_coop/coop_cdf_profile.html).
- [ER11] USGS; Global Land Cover Characteristics Data Base Version 2.0 (2012). Website last visited on March 6, 2017; URL [https://lta.cr.usgs.gov/glcc/globdoc2\\_0](https://lta.cr.usgs.gov/glcc/globdoc2_0).

162 [ER12] The ECS SDP Toolkit (2012). DEM and land-sea mask data itself is available from [ftp://](ftp://edhs1.gsfc.nasa.gov/edhs/sdptk/DEMdata)  
163 [edhs1.gsfc.nasa.gov/edhs/sdptk/DEMdata](ftp://edhs1.gsfc.nasa.gov/edhs/sdptk/DEMdata); URL [http://newsroom.gsfc.nasa.gov/](http://newsroom.gsfc.nasa.gov/sdptoolkit/TKDownload.html)  
164 [sdptoolkit/TKDownload.html](http://newsroom.gsfc.nasa.gov/sdptoolkit/TKDownload.html).

### 3 Terms, definitions and abbreviated terms

Terms, definitions, and abbreviated terms that are specific for this document can be found below.

#### 3.1 Terms and definitions

|      |                                      |
|------|--------------------------------------|
| ATBD | Algorithm Theoretical Basis Document |
| TBA  | To be Added                          |
| TBC  | To be Confirmed                      |
| TBD  | To be Defined                        |

#### 3.2 Acronyms and Abbreviations

|        |                                                          |
|--------|----------------------------------------------------------|
| ATBD   | Algorithm Theoretical Basis Document                     |
| DLR    | Deutsches Zentrum für Luft- und Raumfahrt                |
| ESA    | European Space Agency                                    |
| KNMI   | Koninklijk Nederlands Meteorologisch Instituut           |
| IODD   | Input Output Data Definition                             |
| OCRA   | Optical Cloud Recognition Algorithm                      |
| PUM    | Product User Manual                                      |
| ROCINN | Retrieval of Cloud Information using Neural Networks     |
| QA     | Quality Assurance                                        |
| UPAS   | Universal Processor for UV/VIS Atmospheric Spectrometers |

### 4 Overview of the Sentinel 5 precursor/TROPOMI Level 2 Products

The Sentinel 5 Precursor mission aims at providing information and services on air quality and climate in the timeframe 2017–2023. The S5p mission is part of the Global Monitoring of the European Programme for the establishment of a European capacity for Earth Observation (COPERNICUS). TROPOMI makes daily global observations of key atmospheric constituents, including ozone, nitrogen dioxide, sulfur dioxide, carbon monoxide, methane, formaldehyde as well as cloud and aerosol properties. The list of standard S5p/TROPOMI L2 products is given in table 1. Other products, such as UV index, are under development and will made available at a later date.

**Table 1:** Standard S5P L2 products with name, identifier, and responsible institutes.

| Product                             | ATBD   | PUM           | Identifier  | Institution |
|-------------------------------------|--------|---------------|-------------|-------------|
| Cloud                               | [RD3]  | [RD4]         | L2__CLOUD__ | DLR         |
| NPP-VIIRS Clouds                    | [RD5]  | [RD5]         | L2__NP_BDx  | RAL         |
| HCHO                                | [RD6]  | [RD7]         | L2__HCHO__  | BIRA/DLR    |
| SO <sub>2</sub>                     | [RD8]  | This document | L2__SO2__   | BIRA/DLR    |
| O <sub>3</sub> Total Column         | [RD9]  | [RD10]        | L2__O3__    | BIRA/DLR    |
| O <sub>3</sub> Tropospheric Column  | [RD11] | [RD12]        | L2__O3_TCL  | IUP/DLR     |
| Aerosol layer height                | [RD13] | [RD14]        | L2__AER_LH  | KNMI        |
| Ultra violet aerosol index          | [RD15] | [RD16]        | L2__AER_AI  | KNMI        |
| O <sub>3</sub> Full Profile         | [RD17] | [RD18]        | L2__O3__PR  | KNMI        |
| O <sub>3</sub> Tropospheric Profile | [RD17] | [RD18]        | L2__O3_TPR  | KNMI        |
| NO <sub>2</sub>                     | [RD19] | [RD20]        | L2__NO2__   | KNMI        |
| CO                                  | [RD21] | [RD22]        | L2__CO__    | SRON/KNMI   |
| CH <sub>4</sub>                     | [RD23] | [RD24]        | L2__CH4__   | SRON/KNMI   |

## 4.1 File name convention

The table specifies an identifier that is a substring of real name. The complete filename conventions for all the S5p products can be found in [RD25, chapter 4]. Note that intermediate L2 products beside those listed in table 1 may exist within the PDGS framework. For each of the products listed in the table, a PUM is available. Note that product documentation, e.g. ATBDs and PUMs, will be updated with new releases of processors. User documentation is distributed through the tropomi website [ER1]. Information about S5p mission can be found at the official ESA website for the Sentinel 5 precursor mission [ER2].

In the current PUM the Sulphur Dioxide SO<sub>2</sub> product is described and an example of the full real name is as following:

S5P\_NRTI\_L2\_SO2\_20140101T000000\_20140102T000000\_00099\_01\_000200\_20141010T173511.nc

The components of this file name are given in table 2

**Table 2:** Components of an S5P product file name. Components are separated by underscores, except for the file extension at the end, which is separated by a period. Character indices start counting at 0, the end-index is a Python style index, it lists the first character not in the block.

| Start | End | Length | Meaning                                                                                                                            |
|-------|-----|--------|------------------------------------------------------------------------------------------------------------------------------------|
| 0     | 3   | 3      | Mission name, always “S5P”                                                                                                         |
| 4     | 8   | 4      | Processing stream, one of “NRTI” (near real-time), “OFFL” (offline) or “RPRO” (reprocessing)                                       |
| 9     | 19  | 10     | Product identifier, as listed in table 1                                                                                           |
| 20    | 35  | 15     | Start of granule in UTC as “YYYYMMDDTHHMMSS”. The “T” is a fixed character.                                                        |
| 36    | 51  | 15     | End of the granule in UTC as “YYYYMMDDTHHMMSS”. The “T” is a fixed character.                                                      |
| 52    | 57  | 5      | Orbit number                                                                                                                       |
| 58    | 60  | 2      | Collection number                                                                                                                  |
| 61    | 67  | 6      | Processor version number as “MMmmpp”, with “MM” the major version number, “mm” the minor version number, and “pp” the patch level. |
| 68    | 83  | 15     | The time of processing for this granule in UTC as “YYYYMMDDTHHMMSS”. The “T” is a fixed character.                                 |
| 84    | 86  | 2      | The file name extension. All Sentinel 5 precursor files are netCDF-4 files and use the extension “.nc”                             |

## 5 Data Distribution and Product Support

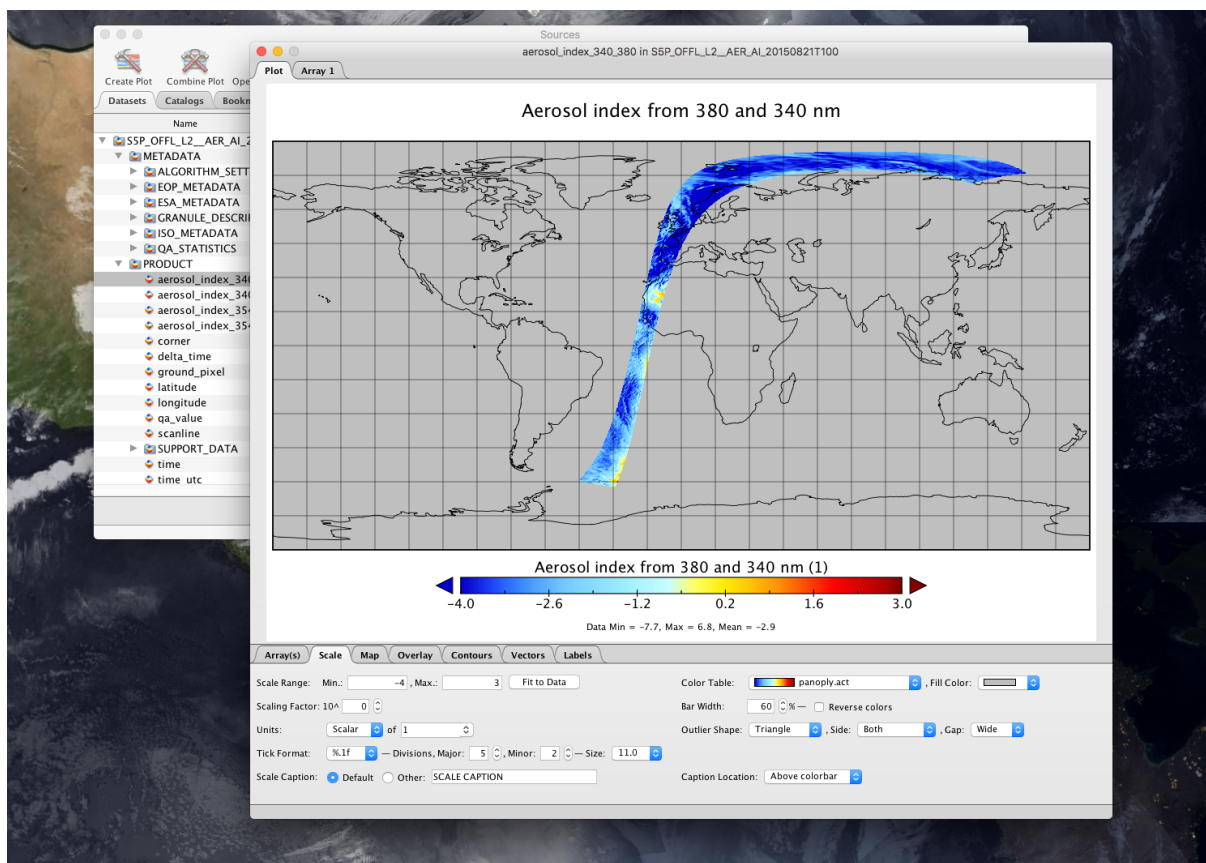
The TROPOMI Sulphur Dioxide SO<sub>2</sub> product data are available from the Copernicus Open Data Hub <https://scihub.copernicus.eu>.

The access and use of any Copernicus Sentinel data available through the Sentinel Data Hub is governed by the Legal Notice on the use of Copernicus Sentinel Data and Service Information and is given here: [https://sentinels.copernicus.eu/documents/247904/690755/Sentinel\\_Data\\_Legal\\_Notice](https://sentinels.copernicus.eu/documents/247904/690755/Sentinel_Data_Legal_Notice).

### 5.1 Information to supply with a support request

We have been very careful in the preparation of the processors, the processing system, the data distribution system and all other components that generate the level 2 products for the Sentinel 5 precursor mission. You may encounter problems when reading the level 2 files despite our care, or you may not understand what we have written in the product user manual or the ATBD. You can contact us through the earth observation help desk operated by ESA at EOSupport@copernicus.esa.int. Please clearly indicate that you are requesting support for Sentinel 5 precursor (S5p) / TROPOMI mission.

If you are requesting technical support it is helpful to provide us with details of the file you are trying to read. The easiest way to do this is to provide a “dump” of the header of the file. This can be generated using the “ncdump” tool provided with the netCDF-4 library. Only the header is required, so “ncdump -h FILE.nc > FILE.cdl” will provide us with all metadata in the file and help us pinpoint how the file was produced. Here you replace FILE.nc with the actual file name on the command line.



**Figure 1: Panoply**

If generating the header fails, please provide us with the exact original file name of the granule you are trying to read, the exact error message you get and the exact version of the software you are using, including the versions of netCDF-4 and HDF-5. Providing us with a checksum to verify file integrity can also speed up our response.

## 6 General Reader and Visualisation Tools

For reading and visualising you may find Panoply [ER3] a useful tool. Panoply is a cross-platform application that plots geo-gridded and other arrays from netCDF, HDF, GRIB, and other datasets, including the Sentinel 5 precursor Level 2 datafiles. With Panoply 4 you can:

- Slice and plot geo-gridded latitude-longitude, latitude-vertical, longitude-vertical, or time-latitude arrays from larger multidimensional variables.
- Slice and plot "generic" 2D arrays from larger multidimensional variables.
- Slice 1D arrays from larger multidimensional variables and create line plots.
- Combine two geo-gridded arrays in one plot by differencing, summing or averaging.
- Plot lon-lat data on a global or regional map using any of over 100 map projections or make a zonal average line plot.
- Overlay continent outlines or masks on lon-lat map plots.
- Use any of numerous color tables for the scale colorbar, or apply your own custom ACT, CPT, or RGB color table.
- Save plots to disk GIF, JPEG, PNG or TIFF bitmap images or as PDF or PostScript graphics files.
- Export lon-lat map plots in KMZ format.
- Export animations as AVI or MOV video or as a collection of individual frame images.

## 7 S5p/TROPOMI L2 Sulphur Dioxide SO<sub>2</sub> Product Description

Sulphur dioxide (SO<sub>2</sub>) enters the Earth's atmosphere through both natural and anthropogenic processes. It plays a role in chemistry on a local and global scale and its impact ranges from short term pollution to effects on climate. Only about 30% of the emitted SO<sub>2</sub> comes from natural sources; the majority is of anthropogenic origin. SO<sub>2</sub> emissions affect human and animal health and air quality. SO<sub>2</sub> has an effect on climate through radiative forcing, via the formation of sulphate aerosols. Volcanic SO<sub>2</sub> emissions may also pose a threat to aviation, along with volcanic ash. The S5p sensor TROPOMI samples the Earth's surface with a revisit time of one day and with an unprecedented spatial resolution of 7.0x3.5 km<sup>2</sup>, respectively 5.5 × 3.5 km<sup>2</sup> (since 6th of August 2019). This allows the resolution of fine details and S5p to arguably be a valuable tool to better study anthropogenic SO<sub>2</sub> emissions but also volcanic emissions, from degassing to eruptive processes.

The retrieval of SO<sub>2</sub> vertical columns is performed in near-real time (i.e. typically 3 hours after measurement) based on a combination of DOAS and the COvariance-Based Retrieval Algorithm (COBRA), involving two main steps:

First, the effective slant column amount  $N_s$  (corresponding to the integrated SO<sub>2</sub> concentration along the mean atmospheric optical path) is derived using a measurement error covariance matrix to fully represent the SO<sub>2</sub>-free radiance variability, so that the SO<sub>2</sub> slant column density is the only retrieved parameter of the algorithm. The strength of the technique is that the covariance matrix is calculated dynamically based on a set of TROPOMI radiance measurements directly representative of the observation to be analysed. This allows to capture the instrument and atmospheric variability of the spectra in an optimal way. If the SO<sub>2</sub> columns are high enough to trigger alternative fit windows, a least-squares fit of the measured Earth reflectance spectrum to laboratory absorption cross-sections is performed (i.e. the traditional DOAS method).

Second, slant columns are converted into vertical columns by means of air mass factors (AMF) obtained from suitable radiative transfer calculations, accounting for the presence of clouds and aerosols, surface properties and best-guess SO<sub>2</sub> vertical profiles.

In addition to the retrieval of the SO<sub>2</sub> vertical column also the SO<sub>2</sub> layer height is retrieved using a coupled PCA-NeuralNetwork approach, in which a pre-trained NN is applied to the measured spectra. This LH retrieval is triggered for pixels with increased SO<sub>2</sub> vertical column of >15 DU. With the retrieved LH, the slant columns are converted into vertical columns by AMFs for this layer height. In this way the user gets the correct SO<sub>2</sub> vertical column.

Refer to the specific ATBD [RD8] documentation for further information about the L2 Sulphur Dioxide SO<sub>2</sub>.

### 7.1 Data Product Examples

Quicklooks are reported in this section as a data product examples of the Sulphur Dioxide SO<sub>2</sub> product. The Sulphur Dioxide SO<sub>2</sub> total vertical column is shown in Figure 2. A zoom-in of the Ambae and Kadovar volcanos is reported in Figure 3 and 4, respectively. Further quicklooks may be found here: <https://atmos.eoc.dlr.de/tropomi>

### 7.2 Product Geophysical Validation

The Mission Performance Centre (MPC) validation reports are regularly updated and can be found here: <http://mpc-vdaf.tropomi.eu>.

### 7.3 History of product changes

This manual describes the L2 Sulphur Dioxide SO<sub>2</sub> product. A brief description of data product changes is given here. Detailed description of the changes can be found in appropriate versions of the ATBD.

**Table 3:** History of product changes of Sulphur Dioxide SO<sub>2</sub>

| Version | Description                                                     |
|---------|-----------------------------------------------------------------|
| 2.7.0   | Added <i>COBRA</i> related variables                            |
| 2.5.0   | Added <i>SO2_LAYER_HEIGHT</i> group with it's related variables |
| 2.3.1   | Added metadata related to fallback background correction        |
| 2.1     | Added surface temperature, DOAS polynomial coefficients         |

History of product changes of Sulphur Dioxide SO<sub>2</sub> (continued).

| Name | Description                                                                                              |
|------|----------------------------------------------------------------------------------------------------------|
| 2.1  | Added additional fields with background data information                                                 |
| 2.0  | Added tropopause layer index information from TM5 (tm5_tropopause_layer_index)                           |
| 2.0  | Additional reference region to generate Earthshine reference spectrum                                    |
| 2.0  | Added 10m Wind information from ECMWF (eastward_wind, northward_wind)                                    |
| 2.0  | Updated QA value calculation                                                                             |
| 2.0  | Added QA value for box SO <sub>2</sub> profiles (qa_value_box_profile)                                   |
| 1.1  | Flagging of enhanced SO <sub>2</sub> levels has been implemented ( <i>sulfurdioxide_detection_flag</i> ) |
| 1.1  | New kernels variables added due to updated prototype algorithms                                          |
| 1.1  | Added <i>BACKGROUND_CORRECTION</i> group including the information of the <i>AUX_BGSO2</i> file          |
| 0.12 | Official version for E2 delivery                                                                         |

## 7.4 Using the S5p/TROPOMI L2 Sulphur Dioxide SO<sub>2</sub>

The Product Readme File (PRF, available here: <http://www.tropomi.eu/document/product-readme-file-sulphur-dioxide>) for SO<sub>2</sub> describes the current processing baseline, product and quality limitations, and product availability status. More information on this data product is available from the Sentinel product webpage (<https://sentinels.copernicus.eu/web/sentinel/technical-guides/sentinel-5p/products-algorithms>) and from the TROPOMI product webpage (<http://www.tropomi.eu/data-products>).

# SO<sub>2</sub>

Sulphur Dioxide total column  
Phase E1 (Commissioning phase)

## Acquisition Time

29-March-2018 00:38:27  
30-March-2018 01:17:54

## Plot Range

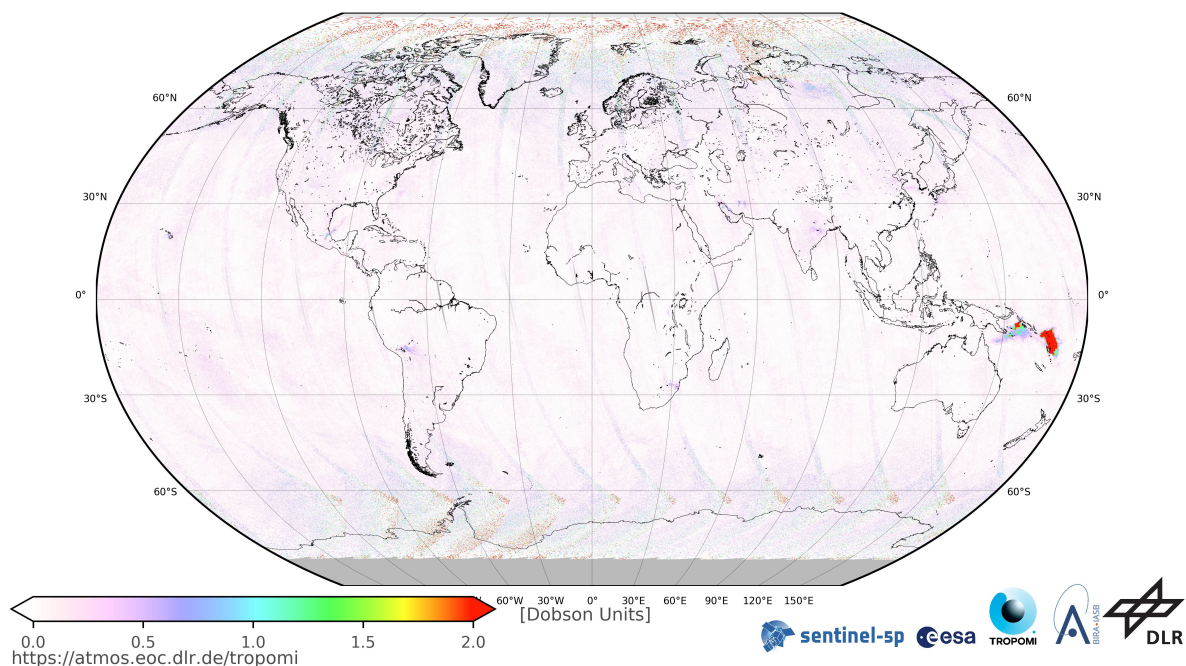
Min: -80.4202 - Max: 84.72098

## Sensor

TROPOMI  
S5P

## Algorithm

UPAS-SO2-DOAS\_LUT-5.0.9  
UPAS2 02.30.73



**Figure 2:** A full day plot of the Sulphur Dioxide SO<sub>2</sub> product acquired on 29th March 2018. Further quicklooks may be found here: <https://atmos.eoc.dlr.de/tropomi>

# SO<sub>2</sub>

Sulphur Dioxide total column  
Phase E1 (Commissioning phase)

## Acquisition Time

29-March-2018 00:38:27  
30-March-2018 01:17:54

## Plot Range

Min: -80.4202 - Max: 84.72098

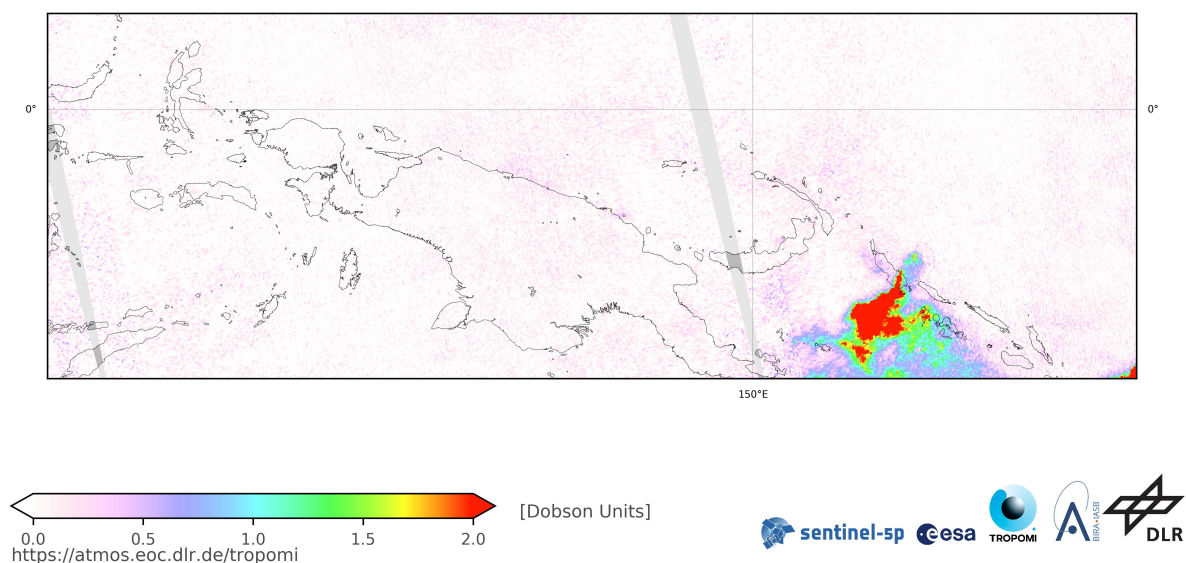
## Sensor

TROPOMI  
S5P

## Algorithm

UPAS-SO2-DOAS\_LUT-5.0.9  
UPAS2 02.30.73

## Kadovar



**Figure 3:** The eruption of Kadovar Volcano acquired on 29th March 2018. Further quicklooks may be found here: <https://atmos.eoc.dlr.de/tropomi>

# SO<sub>2</sub>

Sulphur Dioxide total column  
Phase E1 (Commissioning phase)

## Acquisition Time

29-March-2018 00:38:27  
30-March-2018 01:17:54

## Plot Range

Min: -80.4202 - Max: 84.72098

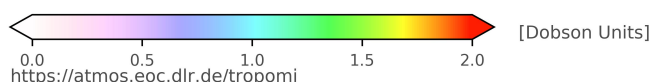
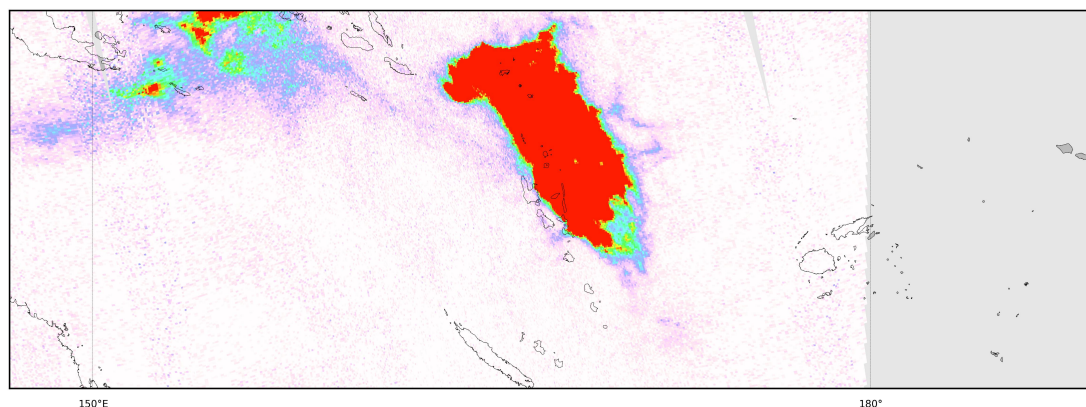
## Sensor

TROPOMI  
S5P

## Algorithm

UPAS-SO2-DOAS\_LUT-5.0.9  
UPAS2 02.30.73

## Ambae



**Figure 4:** The eruption of Ambae Volcano acquired on 29th March 2018. Further quicklooks may be found here: <https://atmos.eoc.dlr.de/tropomi>

## 8 General structure of S5P/TROPOMI Level 2 files

This section gives an overview of the basic structure of all Sentinel 5 precursor level 2 files. In subsections 8.2–8.3 and sections 9–11 some details are provided on the background of the structure of the level 2 files of Sentinel 5 precursor. A complete description of the variables in the Sulphur Dioxide SO<sub>2</sub> files is given in section 12. Figure 5 gives a graphical representation of the generic structure of a TROPOMI Level 2 file. The outermost layer is the file itself. Within the file different groups are used to organise the data and make it easier to find what you are looking for. Within the file there are two groups: “PRODUCT” and “METADATA”. Both of these groups contain sub-groups. The purpose of each group are discussed below.

**PRODUCT** The variables in this group will answer the questions *what*, *when*, *where* and *how well*. This group stores the main data fields of the product, including the precision of the main parameters, latitude, longitude and variable to determine the observation time and the dimensions needed for the data (a time reference dimension (time), the number of measurements in the granule (scanline), the number of spectra in a measurement (ground\_pixel) and depending on the product also a pressure-level dimension, or state-vector dimensions). The “qa\_value” parameter summarizes the processing flags into a continuous value, giving a quality percentage: 100 % is the most optimal value, 0 % is a processing failure, in between lies a continuum of values<sup>1</sup>.

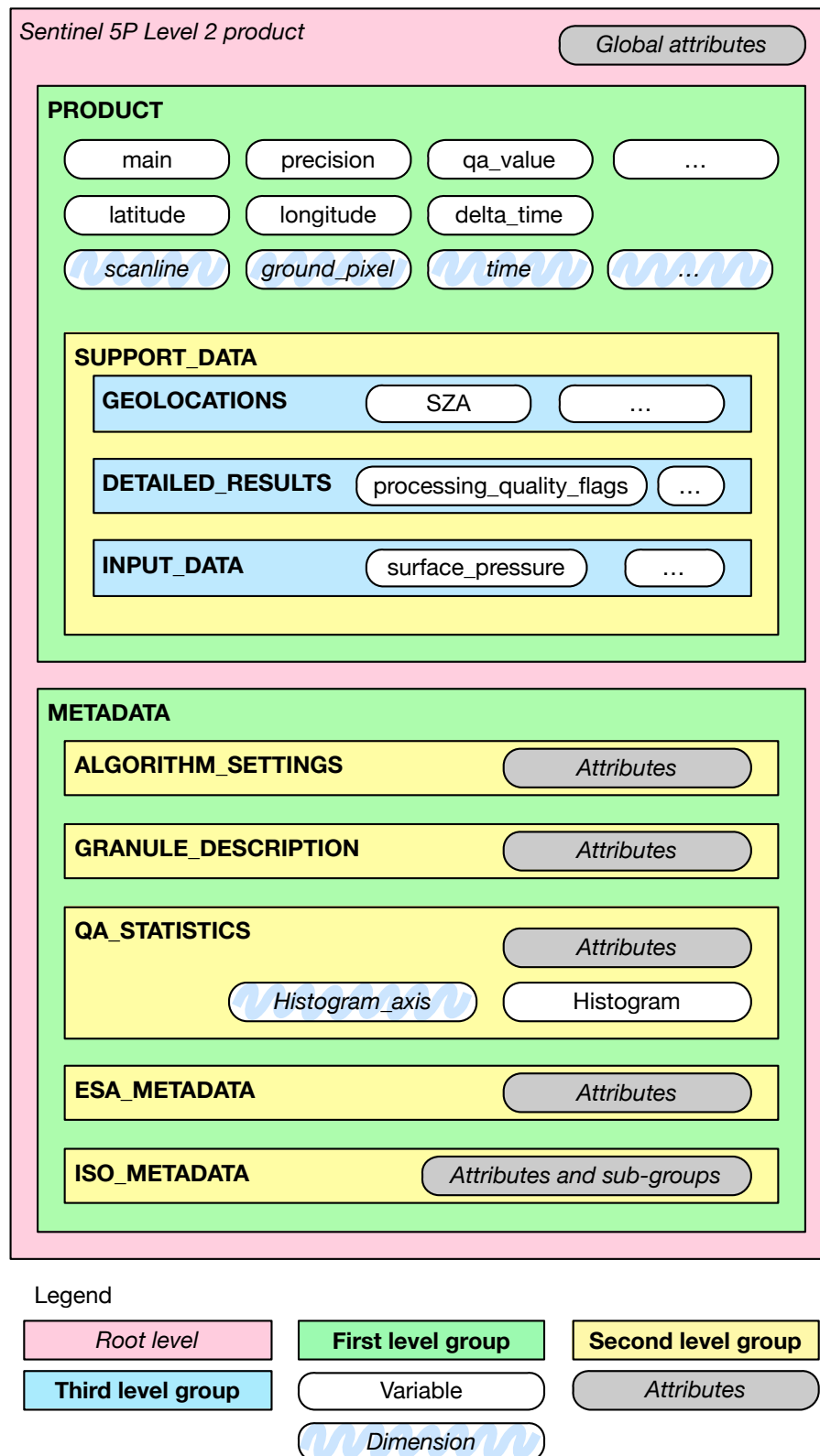
In the ‘PRODUCT’ group a sub-group ‘SUPPORT\_DATA’ can be found:

**SUPPORT\_DATA** Additional data that is not directly needed for using and understanding the main data product is stored in sub-groups of this group.

The data in this group is further split up into the following sub groups:

**GEOLOCATIONS** Additional geolocation and geometry related fields, including the pixel boundaries (pixel corners), viewing- and solar zenith angles, azimuth angles, and spacecraft location.

<sup>1</sup> More detailed processing flags indicating precisely why the 100 % value isn't reached, are available elsewhere in the product.



**Figure 5:** Graphical description of the generic structure of a Level 2 file. The elements labeled as a dimension are coordinate variables. See section 8 for a full description.

**DETAILED\_RESULTS** Additional output, including state-vector elements that are not the main parameter(s), output describing the quality of the retrieval result, such as a  $\chi^2$  value, and detailed processing flags. The Cloud L2 product additionally contains the **COREGISTRATION** sub-group which contains variables with information about the coregistration between the UV/VIS and NIR bands.

**INPUT\_DATA** Additional input data, such as meteorological input data, surface albedo values, surface altitude and other data that was used to derive the output. Note that input profile information is not stored here, but is available for download from elsewhere.

The Sulphur dioxide (SO<sub>2</sub>) L2 product additionally contains the **SO2\_LAYER\_HEIGHT** sub-group:

**SO2\_LAYER\_HEIGHT** This groups contains the data fields related to the SO<sub>2</sub> layer height retrieval. It also contains a “qa\_value\_layer\_height” parameter, that summarizes the layer height processing flags into a continuous value.

**METADATA** This is a group to collect metadata items, such as the items that appear in the header file [RD26, section 7] and items required by INSPIRE [ER4], ISO 19115 [RD27], ISO 19115-2 [RD28], ISO 19157 [RD29] and OGC 10-157r3 [RD30]. These metadata standards are all meant to facilitate dataset discovery.

The metadata will be stored as attributes, while grouping attributes that belong to a specific standard will be done by using sub-groups in the Metadata group. Some attributes are required to be attached to the global level by convention, such as the CF metadata conventions [ER5], the Attribute Convention for Dataset Discovery [ER6], the NetCDF-4 user guide [ER7] and the ESA CCI project [RD31]. For interoperability reasons the conventions are followed, and the specified global attributes are added to the output files at the root-level.

**ALGORITHM\_SETTINGS** An attribute is added to this group for each key in the configuration file. The exact contents differ for each processor.

**GRANULE\_DESCRIPTION** Parameters describing the granule, such as an outline of the geolocations covered in the granule, the time coverage, and processing facility.

**QA\_STATISTICS** Quality assurance statistics. This group contains two types of data:

1. The total number of pixel matching a certain criterion: number of input pixels, number of pixels successfully processed and the number of pixels that failed for specific reasons. Also part of the pixel counting are the number of warnings that were raised, including those for the south Atlantic anomaly, sun glint and solar eclipse. This is collectively known as ‘event counting’.
2. Histogram(s) of the main parameter(s) in the file. Histograms are additive and allow for easy monitoring of changes over time. This can be a valuable addition for quality monitoring of the science data.

**ESA\_METADATA** The metadata items that are required in the ESA header.

**ISO\_METADATA** The ISO metadata items, organized in subgroups.

**EOP\_METADATA** The EOP metadata items, organized in subgroups.

The work of Level 1B on metadata as described in the metadata specification for TROPOMI L01b data processor [RD32] is used as the basis for the level 2 metadata, in particular for the items in the ‘ISO\_METADATA’ and ‘EOP\_METADATA’ subgroups. The listed metadata standards give a data model and an implementation guideline for producing an XML file with the metadata – as a side-file to the data-file itself. The Level 1B IODS [RD2] describes a method to store the metadata in the NetCDF-4 file, and produce XML side-files as needed. A detailed discussion on metadata as it applies to Level 2 can be found in section 11.

Details of the specific format of the level 2 product file for the Sulphur Dioxide SO<sub>2</sub> product is given in section 12. Here all variables are described in detail. A dump output of the final structure proposed in Figure 5 shall have a hierarchy as follows:

```
/root/PRODUCT
/root/PRODUCT/SUPPORT_DATA
/root/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS
/root/PRODUCT/SUPPORT_DATA/GEOLOCATION
/root/PRODUCT/SUPPORT_DATA/INPUT_DATA
/root/PRODUCT/SO2_LAYER_HEIGHT
/root/METADATA
```

351 /root/METADATA/ALGORITHM\_SETTINGS  
352 /root/METADATA/ESA\_METADATA  
353 /root/METADATA/GRANULE\_DESCRIPTION  
354 /root/METADATA/ISO\_METADATA  
355 /root/METADATA/QA\_STATISTICS

356 Where `root` is the file itself or the outer level.

357 The geo-coordinates of the pixel corners are shown in Figure 7. Note that this choice follows the CF metadata  
358 standard [ER5, section 7.1].  
359

## 360 8.1 S5p/TROPOMI L2 File Format

361 The file format used for all the L2 product is **netCDF-4** [ER8]. This file format is very versatile and flexible and  
362 will be used for other Sentinel missions, e.g. S4 mission [RD33], as well as other ESA and NASA missions.  
363 The netCDF-4 library is built on top of NetCDF-3 and HDF-5 libraries and it allows a grouping mechanism  
364 as well as a wide collection of datatypes and other features tailored from the HDF-5 library. This permits the  
365 user to use either the netCDF-4 or HDF-5 APIs in order to read the data. Those APIs are written in many  
366 data-analysis packages such as IDL, NCO, Matlab, R, and Mathematica or in general programming languages  
including Python, Ruby, C, C++, Java and Fortran 90.

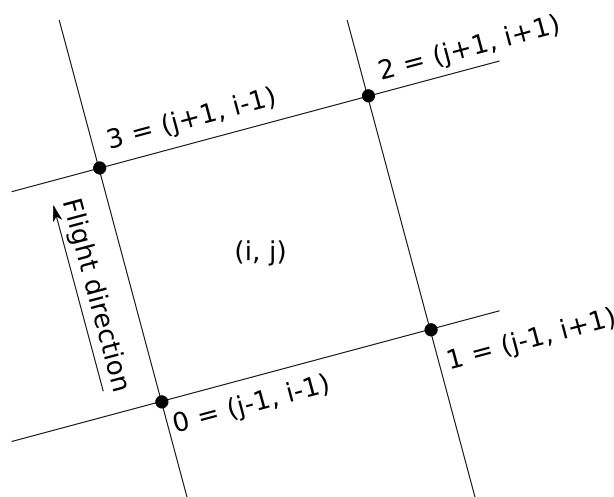


Figure 6: Pixel corner coordinates following [ER5, section 7.1].

## 368 8.2 Dimensions and dimension ordering

369 All variables in a NetCDF-4 file use named and shared dimensions. This explicitly connects variables to  
370 dimensions, and to each other. A few of the dimension names were already shown in figure 5.

371 **time** A time dimension. The length of this dimension is 1, at least for S5P. The reason this dimension is used  
372 are compatibility with Level 1B, and forward compatibility with Sentinel 4 and Level 3 output. Details are  
373 provided in sections 8.4.

374 **scanline** The dimension that indicates the flight direction.

375 **ground\_pixel** The dimension perpendicular to the flight direction.

376 Other dimensions can be added as needed, but these names shall be the default for these roles.

377 The climate and forecast metadata conventions recommend a specific order for dimensions in a vari-  
378 able [ER5, section 2.4]. Spatiotemporal dimensions should appear in the relative order: “date or time” (*T*),  
379 “height or depth” (*Z*), “latitude” (*Y*), and “longitude” (*X*). Note that the ordering of the dimensions in CDL, our  
380 documentation and C/C++ is row-major: the last dimension is stored contiguously in memory<sup>2</sup>.

<sup>2</sup> Fortran uses column-major order, effectively reversing the dimensions in the code compared to the documentation.

Using straight latitude and longitude is fine with model parameters, but the S5P/TROPOMI Level 1B/Level 2 observation grid is not a regular grid. Because of the polar orbit, the across track dimension ('ground\_pixel') corresponds most closely with the longitude, and therefore is associated with the  $X$ -dimension, while the along track dimensions ('scanline') corresponds most directly with latitude, and is therefore labelled as the  $Y$ -dimension.

However, in the CF conventions goes on to recommend that additional dimensions are added before the  $(T, Z, Y, X)$  axes, that is to have contiguous  $(T, Z, Y, X)$  hyperslabs, and spread out the data in other dimensions. We do not follow this recommendation. Instead we recommend to keep units that are likely to be accessed as a unit together in memory, but following the recommended order for  $(T, Y, X)$ . Note that we do not follow the CF conventions for profiles as they are more likely accessed as complete profiles rather than horizontal slices. A few examples will help:

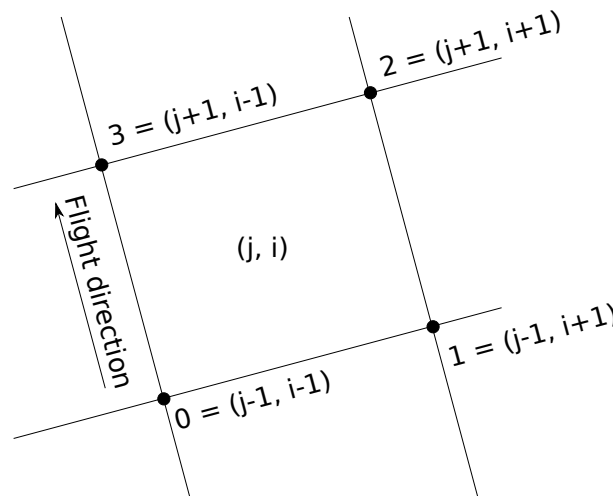
**Tropospheric NO<sub>2</sub> column** This variable contains a single value per ground pixel, and the dimensions are (time, scanline, ground\_pixel).

The `state_vector_length` variable that accompanies the `state_vector_length` dimension is a string array, giving the names of the state vector elements.

### 8.3 Geolocation, pixel corners and angles

The latitude, longitude, pixel corner coordinates and related angles and satellite position in the level 2 files are copied from the level 1B input data [RD34, chapters 26 and 27]. Details about the definitions can be found there. Note that the latitude and longitude have not been corrected for the local surface altitude, but are instead given at the intersection of the line of sight with the WGS84 ellipsoid.

The geo-coordinates of the pixel corners are shown in Figure 7. Note that this choice follows the CF metadata standard [ER5, section 7.1].



**Figure 7:** Pixel corner coordinates. The sequence  $\{0, 1, 2, 3\}$  refers to the elements in the `corner` dimension.

The azimuth angles, i.e. the solar azimuth angle  $\varphi_0$  and the viewing azimuth angle  $\varphi$  give the angle of the sun and the instrument respectively at the intersection of the line of sight with the WGS84 ellipsoid. Both angles are given as degrees east relative to the local north. This definition is identical to the definition of the azimuth angles in both the OMI and GOME-2 instruments, but requires some care when comparing to a radiative transfer model. A radiative transfer model will typically use  $\varphi - \varphi_0$  which differs by  $180^\circ$  as it follows the path of the light.

### 8.4 Time information

Time information is stored in two steps. We have the time dimension, which indicates the reference time. This reference time is defined to be UTC midnight before the start of the orbit, which itself is defined by spacecraft midnight. The `time` variable contains the reference time in seconds since 2010-01-01, UTC midnight.

Alternative representations of the reference time are listed in table 4. The offset of individual measurements within the granule is given in milliseconds with respect to this reference time in the variable `delta_time`.

The reason for this double reference is to more closely follow the CF conventions. Because the flight direction relates the latitude and the time within the orbit, we have  $Y$  and  $T$  dimensions that are closely related. By separating these into a `time` dimension of length 1 and a `scanline` dimension, we obtain independent  $Y$  and  $T$  dimensions. The actual observation time of an individual observation must be reconstructed from an offset and a time-delta.

As a service to the users, the time is also stored in the ‘`time_utc`’ variable. This variable is a string array, with each observation time stored as an ISO date string [RD35].

**Table 4:** Reference times available in a S5P L2 file. Types: (A) global attribute, (D) dimensional variable, (V) variable. All reference times ignore leap seconds.

| Name                                           | Type | Description                                                                                                                                                  |
|------------------------------------------------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>time_reference</code>                    | (A)  | ISO date/time string [RD35]                                                                                                                                  |
| <code>time_reference_days_since_1950</code>    | (A)  | The number of days since January first, 1950, UTC midnight, as used in several weather and climate models (ECMWF, TM5).                                      |
| <code>time_reference_julian_day</code>         | (A)  | The Julian date of the reference time as used in astronomy. This is the reference time system as used in IDL.                                                |
| <code>time_reference_seconds_since_1970</code> | (A)  | The number of seconds since January first, 1970, UTC midnight. This is also known as the unix epoch. Time functions on many systems will accept this number. |
| <code>time</code>                              | (D)  | This variable contains the number of seconds since 2010-01-01, UTC midnight.                                                                                 |
| <code>time_utc</code>                          | (V)  | Array of ISO date/time strings [RD35], one for each observation, i.e. one for each element in the scanline dimension                                         |

## 8.5 Vertical coordinates

Different ATBD authors have specified different vertical grids for the retrieval, which means that the various Level 2 products are not consistent in this respect. There are several options, depending on the choice made by the authors of the retrieval algorithm. Some authors choose to use a vertical grid on a fixed height scale<sup>3</sup>, others use a grid that is defined in pressure relative to the surface pressure, similar to the ECMWF vertical grid.

The ECMWF vertical grid is a “atmosphere hybrid sigma pressure coordinate” in CF conventions terminology [ER5, appendix D].

$$p(n, k, j, i) = a_p(k) + b(k)p_s(n, j, i) \quad (1)$$

where  $p(n, k, j, i)$  is the pressure at gridpoint  $(n, k, j, i)$  on the  $(T, Z, Y, X)$  axes;  $a_p(k)$  and  $b(k)$  the components of the hybrid coordinate at level  $k$  and  $p_s(n, j, i)$  the surface pressure at coordinate  $(n, j, i)$ . As a consequence the surface pressure must be added to the output file, otherwise the pressure levels on which the profiles are reported cannot be reconstructed. In addition the  $a_p(k)$  and  $b(k)$  coefficients must be added to the output as separate variables.

For the fixed height grid there is no reduced pressure grid available, and similarly calculating a height from the pressure profile requires some assumptions. In some cases the full four-dimensional pressure grid will be given.

## 9 Units

The `units` attribute originates from the NetCDF-4 users guide [ER7]. This means that the use of this attribute is integral to the use of NetCDF-4 itself, and that the use of the `units` attribute in the NetCDF-4 users guide is a hard requirement. The NetCDF-4 users guide [ER7] strongly suggests to use the UDUnits [ER9] package to handle units. The CF metadata conventions reinforce this requirement [ER5, sections 1.3 and 3.1].

<sup>3</sup> This is ‘height’ as defined by the CF conventions: distance above the surface; ‘altitude’ is the distance above the geoid or approximate sea level.

Making the UDUUnits package [ER9] a requirement, and thereby forcing all units to be compliant with formal SI units<sup>4</sup> is a good thing for consistency and will help avoid confusion in the long run. In the short term it will require adjustments within the earth observation community, as many of the units that the user community is accustomed to are not SI, and are therefore not available within the UDUUnits package. The MAG has decided that Sentinel 5 precursor will represent all level 2 output in SI units. In particular, all column amounts will be given in mol m<sup>-2</sup>.

To make it easier for end-users to adjust to these ‘new’ units, conversion factors are attached to the appropriate variables.

**multiplication\_factor\_to\_convert\_to\_molecules\_perkm2** Multiply the contents of the variable with this scale factor ( $6.02214 \times 10^{+19}$ ) to obtain columns in molecules cm<sup>-2</sup>

**multiplication\_factor\_to\_convert\_to\_DU** Multiply the contents of the variable with this scale factor (2241.15) to obtain columns in DU.

**multiplication\_factor\_to\_convert\_to\_photons\_persecond\_perkm2\_persr** Multiply the contents of the variable with this scale factor ( $6.02214 \times 10^{+19}$ ) to obtain a radiance in photons s<sup>-1</sup> nm<sup>-1</sup> cm<sup>-2</sup> sr<sup>-1</sup>.

## 10 Quality Assurance parameters

The Level 2 output will include automated quality assurance parameters. These include ‘event counters’ for each of the flags defined in the processing quality flags, see tables 12 and 13. These processing quality flags are made uniform across all products, and include flags that may not be applicable to a particular algorithm. We still count all flags, so this list is the same for all products, a list is provided in table 5.

In addition to these ‘event counters’, we also store a histogram of the main parameters. Storing a histogram of retrieved values is easy during processing, and allows for continuous statistical quality monitoring of the retrieval. It also makes it easy to collect histograms of S5P/TROPOMI data for longer periods. The bins for the histogram depend on the parameter in the Level 2 product, and are defined in the configuration file.

In addition to the histogram an approximation of a probability density function can be created:

$$f_{\text{pdf}}(x_j) = \frac{1}{N} \sum_{i=0}^N \frac{\cos(\delta_{\text{geo},i})}{\sigma_i \sqrt{2\pi}} \exp \left[ -\frac{(x_j - x_i)^2}{2\sigma_i^2} \right] \quad (2)$$

This is a discrete approximation of a continuous probability density function, for discrete values  $x_j$  for all successful retrievals  $i = 1, \dots, N$ . The value of  $\cos(\delta_{\text{geo},i})$  is used to make the result less sensitive to the relative oversampling of S5P at high latitude.

The mission performance center for Sentinel 5 precursor maintains a record of quality control/quality assurance parameters for monitoring purposes.

<sup>4</sup> And some deeply entrenched non-SI units such as DU.

**Table 5:** Common quality assurance parameters. The actual integer values of incident occurrences are stored. Using percentages stored as integers will hide potential issues, especially given the total number of pixels in a S5P/TROPOMI granule.

| Name                                                       | Description                                                                                                                                                                                                            |
|------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>number_of_groundpixels</code>                        | Number of ground pixels in the file.                                                                                                                                                                                   |
| <code>number_of_processed_pixels</code>                    | Number of ground pixels where a retrieval was attempted. This is the <code>number_of_groundpixels</code> minus the pixels that were rejected on trivial grounds, such as the solar zenith angle.                       |
| <code>number_of_successfully_processed_pixels</code>       | Number of ground pixels where a retrieval was successful.                                                                                                                                                              |
| <code>number_of_rejected_pixels_not_enough_spectrum</code> | Number of ground pixels where a retrieval was not attempted because too many spectral pixels were flagged as bad.                                                                                                      |
| <code>number_of_failed_retrievals</code>                   | Number of pixels that were attempted but failed.                                                                                                                                                                       |
| <code>number_of_ground_pixels_with_warnings</code>         | Number of pixels with one or more warnings.                                                                                                                                                                            |
| <code>number_of_missing_scanlines</code>                   | Number of scanlines that are missing from the input, presumably transmission errors.                                                                                                                                   |
| <code>number_of_radiance_missing_occurrences</code>        | Number of ground pixels where “the number of spectral pixels in the radiance due to flagging is too small to perform the fitting” occurred.                                                                            |
| <code>number_of_irradiance_missing_occurrences</code>      | Number of ground pixels where “the number of spectral pixels in the irradiance due to flagging is too small to perform the fitting” occurred.                                                                          |
| <code>number_of_input_spectrum_missing_occurrences</code>  | Number of ground pixels where “the reflectance spectrum does not contain enough points to perform the retrieval. This is different from (ir)radiance_ missing in that the missing points may not be aligned” occurred. |
| <code>number_of_reflectance_range_error_occurrences</code> | Number of ground pixels where “any of the reflectances is out of bounds ( $R < 0$ or $R > R_{\max}$ )” occurred.                                                                                                       |
| <code>number_of_ler_range_error_occurrences</code>         | Number of ground pixels where “lambert-equivalent reflectivity out of range error” occurred.                                                                                                                           |
| <code>number_of_snr_range_error_occurrences</code>         | Number of ground pixels where “too low signal to noise to perform retrieval” occurred.                                                                                                                                 |
| <code>number_of_sza_range_error_occurrences</code>         | Number of ground pixels where “solar zenith angle out of range, maximum value from configuration” occurred.                                                                                                            |
| <code>number_of_vza_range_error_occurrences</code>         | Number of ground pixels where “viewing zenith angle out of range, maximum value from configuration” occurred.                                                                                                          |
| <code>number_of_lut_range_error_occurrences</code>         | Number of ground pixels where “extrapolation in lookup table (airmass factor, cloud radiances)” occurred.                                                                                                              |

**Table 5:** Common quality assurance parameters. (continued).

| Name                                                       | Description                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| number_of_ozone_range_error_occurrences                    | Number of ground pixels where “ozone column significantly out of range of profile climatology” occurred.                                                                                                                                                                                                                                                                                                    |
| number_of_wavelength_offset_error_occurrences              | Number of ground pixels where “wavelength offset exceeds maximum from configuration” occurred.                                                                                                                                                                                                                                                                                                              |
| number_of_initialization_error_occurrences                 | Number of ground pixels where “an error occurred during the processing of the pixel, no output was generated. The following errors raise this flag: Mismatch between irradiance and radiance wavelengths; The on-ground distance between band 1 and band 2 ground pixels exceeds a threshold set in the configuration. Derived a-priori information does not validate, no processing is possible” occurred. |
| number_of_memory_error_occurrences                         | Number of ground pixels where “memory allocation or deallocation error” occurred.                                                                                                                                                                                                                                                                                                                           |
| number_of_assertion_error_occurrences                      | Number of ground pixels where “error in algorithm detected during assertion” occurred.                                                                                                                                                                                                                                                                                                                      |
| number_of_io_error_occurrences                             | Number of ground pixels where “error detected during transfer of data between algorithm and framework” occurred.                                                                                                                                                                                                                                                                                            |
| number_of_numerical_error_occurrences                      | Number of ground pixels where “general fatal numerical error occurred during inversion” occurred.                                                                                                                                                                                                                                                                                                           |
| number_of_lut_error_occurrences                            | Number of ground pixels where “error in accessing the lookup table” occurred.                                                                                                                                                                                                                                                                                                                               |
| number_of_ISRF_error_occurrences                           | Number of ground pixels where “error detected in the input instrument spectral response function input data” occurred.                                                                                                                                                                                                                                                                                      |
| number_of_convergence_error_occurrences                    | Number of ground pixels where “the main algorithm did not converge” occurred.                                                                                                                                                                                                                                                                                                                               |
| number_of_cloud_filter_convergence_error_occurrences       | Number of ground pixels where “the cloud filter did not converge” occurred.                                                                                                                                                                                                                                                                                                                                 |
| number_of_max_iteration_convergence_error_occurrences      | Number of ground pixels where “no convergence because retrieval exceeds maximum number of iterations. Maximum value from configuration” occurred.                                                                                                                                                                                                                                                           |
| number_of_aot_lower_boundary_convergence_error_occurrences | Number of ground pixels where “no convergence because the aerosol optical thickness crosses lower boundary twice in succession” occurred.                                                                                                                                                                                                                                                                   |
| number_of_other_boundary_convergence_error_occurrences     | Number of ground pixels where “no convergence because a state vector element crosses boundary twice in succession. Note that a separate failure flag is defined for non-convergence due to crossing of lower AOT boundary” occurred.                                                                                                                                                                        |

**Table 5:** Common quality assurance parameters. (continued).

| Name                                                            | Description                                                                                                                                               |
|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| number_of_geolocation_error_occurrences                         | Number of ground pixels where “geolocation out of range” occurred.                                                                                        |
| number_of_ch4_noscat_zero_error_occurrences                     | Number of ground pixels where “the CH <sub>4</sub> column retrieved by the non-scattering CO algorithm from the weak band or strong band is 0” occurred.  |
| number_of_h2o_noscat_zero_error_occurrences                     | Number of ground pixels where “the H <sub>2</sub> O column retrieved by the non-scattering CO algorithm from the weak band or strong band is 0” occurred. |
| number_of_max_optical_thickness_error_occurrences               | Number of ground pixels where “maximum optical thickness exceeded during iterations” occurred.                                                            |
| number_of_aerosol_boundary_error_occurrences                    | Number of ground pixels where “boundary hit of aerosol parameters at last iteration” occurred.                                                            |
| number_of_boundary_hit_error_occurrences                        | Number of ground pixels where “fatal boundary hit during iterations” occurred.                                                                            |
| number_of_chi2_error_occurrences                                | Number of ground pixels where “ $\chi^2$ is not-a-number or larger than 10 <sup>10</sup> ” occurred.                                                      |
| number_of_svd_error_occurrences                                 | Number of ground pixels where “singular value decomposition failure” occurred.                                                                            |
| number_of_dfs_error_occurrences                                 | Number of ground pixels where “degree of freedom is not-a-number” occurred.                                                                               |
| number_of_radiative_transfer_error_occurrences                  | Number of ground pixels where “errors occurred during the radiative transfer computations, no processing possible” occurred.                              |
| number_of_optimal_estimation_error_occurrences                  | Number of ground pixels where “errors occurred during the optimal estimation, processing has been terminated” occurred.                                   |
| number_of_profile_error_occurrences                             | Number of ground pixels where “flag that indicates if there were any errors during the computation of the ozone profile” occurred.                        |
| number_of_cloud_error_occurrences                               | Number of ground pixels where “no cloud data” occurred.                                                                                                   |
| number_of_model_error_occurrences                               | Number of ground pixels where “forward model failure” occurred.                                                                                           |
| number_of_number_of_input_data_points_too_low_error_occurrences | Number of ground pixels where “not enough input ozone columns to calculate a tropospheric column” occurred.                                               |
| number_of_cloud_pressure_spread_too_low_error_occurrences       | Number of ground pixels where “cloud pressure variability too low to estimate a tropospheric column” occurred.                                            |
| number_of_cloud_too_low_level_error_occurrences                 | Number of ground pixels where “clouds are too low in the atmosphere to assume sufficient shielding” occurred.                                             |
| number_of_generic_range_error_occurrences                       | Number of ground pixels where “generic range error” occurred.                                                                                             |
| number_of_generic_exception_occurrences                         | Number of ground pixels where “catch all generic error” occurred.                                                                                         |

**Table 5:** Common quality assurance parameters. (continued).

| Name                                                 | Description                                                                                                                           |
|------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| number_of_input_spectrum_alignment_error_occurrences | Number of ground pixels where “input radiance and irradiance spectra are not aligned correctly” occurred.                             |
| number_of_abort_error_occurrences                    | Number of ground pixels where “not processed because processor aborted prematurely (time out or user abort)” occurred.                |
| number_of_wrong_input_type_error_occurrences         | Number of ground pixels where “wrong input type error, mismatch between expectation and received data” occurred.                      |
| number_of_wavelength_calibration_error_occurrences   | Number of ground pixels where “an error occurred in the wavelength calibration of this pixel” occurred.                               |
| number_of_coregistration_error_occurrences           | Number of ground pixels where “no colocated pixels found in a supporting band” occurred.                                              |
| number_of_slant_column_density_error_occurrences     | Number of ground pixels where “slant column fit returned error, no values can be computed” occurred.                                  |
| number_of_airmass_factor_error_occurrences           | Number of ground pixels where “airmass factor could not be computed” occurred.                                                        |
| number_of_vertical_column_density_error_occurrences  | Number of ground pixels where “vertical column density could not be computed” occurred.                                               |
| number_of_signal_to_noise_ratio_error_occurrences    | Number of ground pixels where “the signal to noise ratio for this spectrum is too low for processing” occurred.                       |
| number_of_configuration_error_occurrences            | Number of ground pixels where “error while parsing the configuration” occurred.                                                       |
| number_of_key_error_occurrences                      | Number of ground pixels where “key does not exist” occurred.                                                                          |
| number_of_saturation_error_occurrences               | Number of ground pixels where “saturation in input spectrum” occurred.                                                                |
| number_of_max_num_outlier_exceeded_error_occurrences | Number of ground pixels where “the number of outliers detected in the DOAS fit exceeds a maximum set for healthy spectra. ” occurred. |
| number_of_solar_eclipse_filter_occurrences           | Number of ground pixels where “solar eclipse” occurred.                                                                               |
| number_of_cloud_filter_occurrences                   | Number of ground pixels where “the cloud filter triggered causing the pixel to be skipped” occurred.                                  |
| number_of_altitude_consistency_filter_occurrences    | Number of ground pixels where “too large difference between ECMWF altitude and DEM altitude value” occurred.                          |
| number_of_altitude_roughness_filter_occurrences      | Number of ground pixels where “too large standard deviation of altitude in DEM” occurred.                                             |

**Table 5:** Common quality assurance parameters. (continued).

| Name                                                  | Description                                                                                                                                                                                                                                          |
|-------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| number_of_sun_glint_filter_occurrences                | Number of ground pixels where “for pixels over water, viewing direction inside sun glint region. Definition of sun glint angle and threshold value from ATBD” occurred.                                                                              |
| number_of_mixed_surface_type_filter_occurrences       | Number of ground pixels where “pixel contains land and water areas (e.g. coastal pixel)” occurred.                                                                                                                                                   |
| number_of_snow_ice_filter_occurrences                 | Number of ground pixels where “pixel contains snow/ice: Snow/ice flag according to dynamic input OR climatological surface albedo at VIS wavelength is larger than 0.5” occurred.                                                                    |
| number_of_aai_filter_occurrences                      | Number of ground pixels where “AAI smaller than 2.0” occurred.                                                                                                                                                                                       |
| number_of_cloud_fraction_fresco_filter_occurrences    | Number of ground pixels where “pixel contains clouds: The FRESCO effective cloud fraction is larger than threshold. Threshold value from ATBD” occurred.                                                                                             |
| number_of_aai_scene_albedo_filter_occurrences         | Number of ground pixels where “pixel contains clouds: The difference between scene albedo at 380 nm from AAI calculation and the climatological surface albedo exceeds threshold. Threshold value from ATBD. This test filters out clouds” occurred. |
| number_of_small_pixel_radiance_std_filter_occurrences | Number of ground pixels where “pixel contains clouds: Standard deviation of radiances in small-pixel column exceeds threshold. Threshold value from ATBD” occurred.                                                                                  |
| number_of_cloud_fraction_viirs_filter_occurrences     | Number of ground pixels where “pixel contains clouds: The cloud fraction from VIIRS / NPP exceeds threshold. Threshold value from ATBD” occurred.                                                                                                    |
| number_of_cirrus_reflectance_viirs_filter_occurrences | Number of ground pixels where “pixel contains clouds: Cirrus reflectance from VIIRS / NPP exceeds threshold. Threshold value from ATBD” occurred.                                                                                                    |
| number_of_cf_viirs_swir_ifov_filter_occurrences       | Number of ground pixels where “fraction of cloudy VIIRS pixels within S5P SWIR ground pixel exceeds a priori threshold from configuration” occurred.                                                                                                 |
| number_of_cf_viirs_swir_ofova_filter_occurrences      | Number of ground pixels where “fraction of cloudy VIIRS pixels within S5P SWIR OFOVa exceeds a priori threshold from configuration” occurred.                                                                                                        |
| number_of_cf_viirs_swir_ofovb_filter_occurrences      | Number of ground pixels where “fraction of cloudy VIIRS pixels within S5P SWIR OFOVb exceeds a priori threshold from configuration” occurred.                                                                                                        |
| number_of_cf_viirs_swir_ofovc_filter_occurrences      | Number of ground pixels where “fraction of cloudy VIIRS pixels within S5P SWIR OFOVc exceeds a priori threshold from configuration” occurred.                                                                                                        |
| number_of_cf_viirs_nir_ifov_filter_occurrences        | Number of ground pixels where “fraction of cloudy VIIRS pixels within S5P NIR ground pixel exceeds a priori threshold from configuration” occurred.                                                                                                  |

**Table 5:** Common quality assurance parameters. (continued).

| Name                                                 | Description                                                                                                                                                                                                                                    |
|------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| number_of_cf_viirs_nir_ofova_filter_occurrences      | Number of ground pixels where “fraction of cloudy VIIRS pixels within S5P NIR OFOVa exceeds a priori threshold from configuration” occurred.                                                                                                   |
| number_of_cf_viirs_nir_ofovb_filter_occurrences      | Number of ground pixels where “fraction of cloudy VIIRS pixels within S5P NIR OFOVb exceeds a priori threshold from configuration” occurred.                                                                                                   |
| number_of_cf_viirs_nir_ofovc_filter_occurrences      | Number of ground pixels where “fraction of cloudy VIIRS pixels within S5P NIR OFOVc exceeds a priori threshold from configuration” occurred.                                                                                                   |
| number_of_refl_cirrus_viirs_swir_filter_occurrences  | Number of ground pixels where “average VIIRS cirrus reflectance within SWIR ground pixel exceeds a priori threshold from configuration” occurred.                                                                                              |
| number_of_refl_cirrus_viirs_nir_filter_occurrences   | Number of ground pixels where “average VIIRS cirrus reflectance within NIR ground pixel exceeds a priori threshold from configuration” occurred.                                                                                               |
| number_of_diff_refl_cirrus_viirs_filter_occurrences  | Number of ground pixels where “difference in VIIRS average cirrus reflectance between SWIR and NIR ground pixel exceeds a priori threshold from configuration” occurred.                                                                       |
| number_of_ch4_noscat_ratio_filter_occurrences        | Number of ground pixels where “the ratio between [CH <sub>4</sub> ] <sub>weak</sub> and [CH <sub>4</sub> ] <sub>strong</sub> is below or exceeds a priori thresholds from configuration” occurred.                                             |
| number_of_ch4_noscat_ratio_std_filter_occurrences    | Number of ground pixels where “the standard deviation of [CH <sub>4</sub> ] <sub>weak</sub> /[CH <sub>4</sub> ] <sub>strong</sub> within the SWIR pixel and the 8 neighbouring pixels exceeds a priori threshold from configuration” occurred. |
| number_of_h2o_noscat_ratio_filter_occurrences        | Number of ground pixels where “the ratio between [H <sub>2</sub> O] <sub>weak</sub> and [H <sub>2</sub> O] <sub>strong</sub> is below or exceeds a priori thresholds from configuration” occurred.                                             |
| number_of_h2o_noscat_ratio_std_filter_occurrences    | Number of ground pixels where “the standard deviation of [H <sub>2</sub> O] <sub>weak</sub> /[H <sub>2</sub> O] <sub>strong</sub> within the SWIR pixel and the 8 neighbouring pixels exceeds a priori threshold from configuration” occurred. |
| number_of_diff_psurf_fresco_ecmwf_filter_occurrences | Number of ground pixels where “difference between the FRESCO apparent surface pressure and the ECMWF surface pressure exceeds a priori threshold from configuration” occurred.                                                                 |
| number_of_psurf_fresco_stdv_filter_occurrences       | Number of ground pixels where “the standard deviation of the FRESCO apparent surface pressure in the NIR pixel and the 8 surrounding pixels exceeds a priori threshold from configuration” occurred.                                           |
| number_of_ocean_filter_occurrences                   | Number of ground pixels where “the ground pixel is over ocean (and ocean glint retrievals are not switched on)” occurred.                                                                                                                      |

**Table 5:** Common quality assurance parameters. (continued).

| Name                                                 | Description                                                                                                                                                                                                                                                                                |
|------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| number_of_time_range_filter_occurrences              | Number of ground pixels where “time is out of the range that is to be processed” occurred.                                                                                                                                                                                                 |
| number_of_pixel_or_scanline_index_filter_occurrences | Number of ground pixels where “not processed because pixel index does not match general selection criteria” occurred.                                                                                                                                                                      |
| number_of_geographic_region_filter_occurrences       | Number of ground pixels where “pixel falls outside the specified regions of interest” occurred.                                                                                                                                                                                            |
| number_of_input_spectrum_warning_occurrences         | Number of ground pixels where “number of good pixels in radiance, irradiance or calculated reflectance below threshold from configuration” occurred.                                                                                                                                       |
| number_of_wavelength_calibration_warning_occurrences | Number of ground pixels where “offset from wavelength fit is larger than limit set in configuration” occurred.                                                                                                                                                                             |
| number_of_extrapolation_warning_occurrences          | Number of ground pixels where “pressure or temperature outside cross section LUT range, other lookup table extrapolation” occurred.                                                                                                                                                        |
| number_of_sun_glint_warning_occurrences              | Number of ground pixels where “sun glint possibility warning” occurred.                                                                                                                                                                                                                    |
| number_of_south_atlantic_anomaly_warning_occurrences | Number of ground pixels where “TROPOMI is inside the south Atlantic anomaly while taking these measurements” occurred.                                                                                                                                                                     |
| number_of_sun_glint_correction_occurrences           | Number of ground pixels where “A sun glint correction has been applied” occurred.                                                                                                                                                                                                          |
| number_of_snow_ice_warning_occurrences               | Number of ground pixels where “snow/ice flag is set, i.e. using scene data from the cloud support product” occurred.                                                                                                                                                                       |
| number_of_cloud_warning_occurrences                  | Number of ground pixels where “cloud filter based on FRESCO apparent surface pressure (VIIRS not available), cloud fraction above threshold or cloud pressure adjusted to force cloud above surface. In case of Cloud product this flag indicates the possibility of ice-clouds” occurred. |
| number_of_AAI_warning_occurrences                    | Number of ground pixels where “possible aerosol contamination as either indicated by the AAI (O <sub>3</sub> profile) or other criteria (Cloud)” occurred.                                                                                                                                 |
| number_of_pixel_level_input_data_missing_occurrences | Number of ground pixels where “dynamic auxiliary input data (e.g.. cloud) is missing for this ground pixel. A fallback option is used” occurred.                                                                                                                                           |

**Table 5:** Common quality assurance parameters. (continued).

| Name                                                      | Description                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| number_of_data_range_warning_occurrences                  | Number of ground pixels where “carbon monoxide column tends to negative values; Water column tends to negative values; Heavy water (HDO) column tends to negative values; others. In case of the O <sub>3</sub> product this flag indicates VCD or effective albedo values outside a valid range. In case of the SO <sub>2</sub> or the HCHO product this flag indicates AMF values outside a valid range” occurred. |
| number_of_low_cloud_fraction_warning_occurrences          | Number of ground pixels where “low cloud fraction, therefore no cloud pressure retrieved” occurred.                                                                                                                                                                                                                                                                                                                  |
| number_of_altitude_consistency_warning_occurrences        | Number of ground pixels where “difference between ECMWF surface elevation and high-resolution surface elevation exceeds threshold from configuration” occurred.                                                                                                                                                                                                                                                      |
| number_of_signal_to_noise_ratio_warning_occurrences       | Number of ground pixels where “signal to noise ratio in SWIR and/or NIR band below threshold from configuration. For the O <sub>3</sub> and HCHO products this flag indicates an RMS above a certain threshold” occurred.                                                                                                                                                                                            |
| number_of_deconvolution_warning_occurrences               | Number of ground pixels where “failed deconvolution irradiance spectrum (not pixel-specific, but row-specific)” occurred.                                                                                                                                                                                                                                                                                            |
| number_of_so2_volcanic_origin_likely_warning_occurrences  | Number of ground pixels where “warning for SO <sub>2</sub> BL product, UTLS products: volcanic origin except for heavily polluted sites” occurred.                                                                                                                                                                                                                                                                   |
| number_of_so2_volcanic_origin_certain_warning_occurrences | Number of ground pixels where “warning for SO <sub>2</sub> BL product, UTLS products: volcanic origin certain” occurred.                                                                                                                                                                                                                                                                                             |
| number_of_interpolation_warning_occurrences               | Number of ground pixels where “warning for interpolation on partially missing data. In this case the valid available data is used, potentially leading to a bias” occurred.                                                                                                                                                                                                                                          |
| number_of_saturation_warning_occurrences                  | Number of ground pixels where “saturation occurred spectrum, possibly causing biases in the retrieval” occurred.                                                                                                                                                                                                                                                                                                     |
| number_of_high_sza_warning_occurrences                    | Number of ground pixels where “warning for high solar zenith angle. In this case, the processing can be performed with less final quality” occurred.                                                                                                                                                                                                                                                                 |
| number_of_cloud_retrieval_warning_occurrences             | Number of ground pixels where “warning occurring when the retrieval diagnostic indicates a degraded quality of the cloud retrieval” occurred.                                                                                                                                                                                                                                                                        |
| number_of_cloud_inhomogeneity_warning_occurrences         | Number of ground pixels where “the cloud coregistration inhomogeneity parameter is above a given threshold” occurred.                                                                                                                                                                                                                                                                                                |

## 11 Generic metadata and attributes

Metadata gives information about the satellite, algorithms, configuration as well as other parameters useful for the interpretation of the processed data and tracing the production process of the level 2 files. The Sentinel 5 precursor product files, both for level 1B and level 2 contain a rich amount of metadata, both at the variable level and at the granule level. The full description of the metadata in the files for the Sulphur Dioxide SO<sub>2</sub> product is given in the file format description, in section 12.2. Here we provide some background on what can be found in which location. The abbreviations listed in table 6 are used in the following part of this document to better identify the nature of the attributes.

**Table 6:** The abbreviations used in metadata descriptions to indicate the origin of a specific attribute, and the abbreviations used to indicate the type of an attribute.

| Abbreviation | Description                                                                                                        |
|--------------|--------------------------------------------------------------------------------------------------------------------|
| NUG          | netCDF-4 Users Guide [ER7]                                                                                         |
| CF           | Climate and Forecast metadata conventions [ER5], which includes the COARDS [ER10] conventions                      |
| ISO          | ISO standards 19115, 19115-2 and 19157 [RD27, RD28, RD29]                                                          |
| Inspire      | Inspire directive [ER4]                                                                                            |
| ACDD         | ESIP-ACDD Attribute convention for dataset discovery [ER6]                                                         |
| CCI          | Attributes requested by the ESA climate change initiative project. These largely overlap with the ACDD attributes. |
| ESA          | Fixed ESA Header [RD26]                                                                                            |
| S5P          | Internal use – mostly for retrieval settings, possibly as an extension to ISO 19115 [RD27]                         |
| S            | Attribute is a string attribute                                                                                    |
| P            | Attribute has the data-type of the variable with which it is associated ('parent' data type).                      |
| I            | Attribute is an integer value                                                                                      |
| F            | Attribute is a floating point value (either 32-bit or 64-bit).                                                     |
| T            | Attribute is a CCSDS-ASCII time representation ("UTC=" + ISO 8601 [RD35])                                          |

We follow several metadata conventions in the S5P level 2 files, as can be seen in table 6. These include ISO 19115-2 [RD28], OGC 10.157r3 [RD30], the ESA earth observation header [RD26] and the Climate and Forecast metadata conventions [ER5]. Following ISO 19115-2 also ensures compliance with the Inspire directive, with the provision that a few items that are optional in the ISO standard are required by Inspire. These metadata standards prescribe the generation of XML files as side-files to the main product file. These metadata standards are mostly intended for data discovery and data dissemination. This means that the metadata must be ingested by a server so that it can be stored in a database. This database will end users help to find the data they need. Ingestion of this metadata is facilitated by storing the metadata in a predefined XML format. While it is possible to store the required XML directly in a NetCDF variable or attribute, it is hard to use these directly to extract metadata. Using attributes for the individual metadata fields makes it far easier for users to read the metadata from their programs, as the interface becomes uniform: just netCDF-4.

Then the question becomes how to store the metadata for the ISO 19115-2, OGC 10.157r3 and the ESA earth observation header in the NetCDF datafile, in a way that facilitates automated creation of the XML side files for ingestion into the database for dissemination and discovery. Fortunately this problem has already been solved by the S5P L1B team, and a description can be found in the L1B input/output data specification and the metadata specification [RD2, RD32]. The short version is that the attributes in the data file can be exported as NcML [RD36], which can be translated into the desired output using an XSLT transformation. Support attributes are added to the data file to facilitate this. Creating such a transformation script has been declared out of scope for the level 1B and level 2 processor CFI providers.

### 11.1 The Climate and Forecast conventions

The CF metadata conventions [ER5] provide guidelines for attributes for variables so that the link between data and its geolocation and time of observation can be made automatically. Applying the CF-metadata conventions to the output products already limits the number of choices we will have to make. Units and other attributes are

already defined and some structure is provided by the CF-conventions, for instance in linking data fields with geolocation.

## 11.2 NetCDF User Guide Conventions

A full description of the conventions might be found in the NetCDF user manual [ER7]]. In general, names starting with underscore character are always reserved for use by the NetCDF library. NUG conventions are a subset of the CF-conventions.

## 11.3 Global attributes

Global attributes that are present at the `root` level of a S5p L2 product as described in section 12. These are mostly `string` attributes.

## 11.4 ESA earth observation header

The ESA earth observations file format guidelines and tailoring for S5P [RD26, RD25] specify the creation of a header file with a basic description of the contents of an output file. This header file consists of a fixed part and a customizable variable part. The variable part contains the lineage of the product is repeated, see section 12.2.3.91 for a description the the attributes contained in this part of the header. The fixed header is described in tables 7–9.

**Table 7:** Metadata in the fixed header required by the ESA earth observation file format standard. The data types refer to the short list in table 6.

| Name             | Data type          | Definition                                                                                                                                                                          |
|------------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| File_Name        | S                  | File name of the product without extension.                                                                                                                                         |
| File_Description | S                  | Description of the file type.                                                                                                                                                       |
| Notes            | S                  | Any type of notes/comments (multi-lines).                                                                                                                                           |
| Mission          | S                  | Description of the mission (Fixed to “S5P”)                                                                                                                                         |
| File_Class       | S                  | Description of the file class. It is redundant with the File Class element embedded in the File Name.(e.g., “NRTI”)                                                                 |
| File_Type        | S                  | Description of the file type, for the current product it is set to “L2__ - Sulphur Dioxide SO <sub>2</sub> ”. It is redundant with the File Type element embedded in the File Name. |
| Validity_Period  | Group, see table 8 | Time coverage of the data.                                                                                                                                                          |
| File_Version     | I                  | It is redundant with the File Version element embedded in the File Name.                                                                                                            |
| Source           | Group, see table 9 | Information about the ground segment facility where the product was generated.                                                                                                      |

**Table 8:** Fields in the Validity\_Period group. The data types refer to the short list in table 6.

| Name           | Data type | Definition                                                                                                                                           |
|----------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Validity_Start | T         | This is the UTC Validity Start Time, the same as the Validity Start Time in the File Name and the <code>time_coverage_start</code> global attribute. |
| Validity_Stop  | T         | This is the UTC Validity Stop Time, the same as the Validity Stop Time in the File Name and the <code>time_coverage_end</code> global attribute.     |

**Table 9:** Fields in the source group. The data types refer to the short list in table 6.

| Name            | Data type | Definition                                                                                                        |
|-----------------|-----------|-------------------------------------------------------------------------------------------------------------------|
| System          | S         | Name of the Ground Segment element creating the file.                                                             |
| Creator         | S         | Name of the facility or tool, within the Ground Segment element, creating the file.                               |
| Creator_Version | S         | Version of the tool.                                                                                              |
| Creation_Date   | T         | This is the UTC Creation Date. This field also appears in the file name and in the date_created global attribute. |

## 11.5 Inspire directive

INSPIRE is based on the infrastructures for spatial information established and operated by the 27 Member States of the European Union. The INSPIRE directive came into force on 15 May 2007 and will be developed in several stages until a complete release with due date set in 2019. The INSPIRE directive aims to create a European Union (EU) spatial data infrastructure. This will enable the sharing of environmental spatial information among public sector organizations and better facilitate public access to spatial information across Europe. The European Commission issued a Metadata Regulation [RD37] which aims at setting the requirements for the creation and maintenance of metadata for spatial data sets, spatial data set series and spatial data services corresponding to the themes listed in the annexes of the regulation.

Since many different standard are involved, collisions may occur. The INSPIRE Metadata Implementing Rules [RD38] define how the Regulation can be implemented using ISO 19115. As also reported in [RD32], the conclusion of the study pointed out the following:

1. The conformance of an ISO 19115 metadata set to the ISO 19115 Core does not guarantee the conformance to INSPIRE.
2. The use of these guidelines to create INSPIRE metadata ensures that the metadata is not in conflict with ISO 19115. However, full conformance to ISO 19115 implies the provision of additional metadata elements which are not required by INSPIRE.

## 11.6 ISO and OGC standards

Two ISOs standards useful for the description of collection of Earth Observation products and to the description of individual EO products are ISO 19115-2 [RD28] and ISO 19156 [RD39], respectively. However, these two ISOs do not provide any encoding syntax but they are merely conceptual models. On the other hand, standards that provide encoding and XML schema for describing, validating and exchanging metadata about geographic datasets and for observations and measurements are:

1. ISO 19139 [RD40]
2. OGC 10-025C [RD41]
3. OGC 10-157 [RD30]

Full description of all above mentioned standard is not part of this document. The S5p L01B development team have addressed and analyzed the complex structure of the application of all those ISOs and OGC standard in the S5P L01B metadata specification [RD32].

## 11.7 Attributes

In Table 11 a list of attributes that can be appended to variables in S5p products. Not all of these attributes will be used on all variables, but for each variables an appropriate selection is made. The different types with their respective abbreviations are shown in Table 6. The NetCDF attribute `_FillValue` which represents missing or undefined data can assume the default values listed in Table 10.

**Table 10:** netCDF-4 type definitions and fill values. In order to avoid rounding errors, it is recommended to use the hexadecimal notation when specifying fill values for float and double types. Note that these are the netCDF-4 default fill values, there should be no need to specify these values explicitly. In some cases the fill value for float or double variables may fall within the valid range of a variable. For those cases an explicit fill value must be set, the value  $-9.9692099683868690 \times 10^{36}$  (hex:  $-0 \times 1.\text{ep}+122$ ) is recommended for these cases.

| Type   | Description             | Fill value                                                             |
|--------|-------------------------|------------------------------------------------------------------------|
| byte   | 8-bit signed integer    | −127                                                                   |
| ubyte  | 8-bit unsigned integer  | 255                                                                    |
| short  | 16-bit signed integer   | −32767                                                                 |
| ushort | 16-bit unsigned integer | 65535                                                                  |
| int    | 32-bit signed integer   | −2147483647                                                            |
| uint   | 32-bit unsigned integer | 4294967295                                                             |
| float  | 32-bit floating point   | $9.9692099683868690 \times 10^{36}$ (hex: $0 \times 1.\text{ep}+122$ ) |
| double | 64-bit floating point   | $9.9692099683868690 \times 10^{36}$ (hex: $0 \times 1.\text{ep}+122$ ) |

**Table 11:** Attributes for variables used in S5p netCDF-4 files. The data types refer to the short list in table 6.

| Name                | Type | Std. | Description                                                                                                                                               |
|---------------------|------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| ancillary_variables | S    | CF   | Identifies a variable that contains closely associated data, e.g. the measurement uncertainties of instrument data.                                       |
| bounds              | S    | CF   | Connects a boundary variable to a coordinate variable.                                                                                                    |
| cell_measures       | S    | CF   | Identifies variables that contain cell areas or volumes. This can be used to connect approximate ground pixel coverage in km <sup>2</sup> to data-fields. |
| comment             | S    | CF   | Miscellaneous information about the data or methods used to produce it.                                                                                   |
| coordinates         | S    | CF   | Identifies auxiliary coordinate variables, providing a connection between data and geolocation, time.                                                     |
| _FillValue          | P    | NUG  | Value to represent missing or undefined data. Recommended (default) values are given in table 10.                                                         |
| flag_masks          | P    | CF   | Provides a list of bit fields expressing Boolean or enumerated flags.                                                                                     |
| flag_meanings       | S    | CF   | Use in conjunction with flag_values to provide descriptive words or phrases for each flag value.                                                          |
| flag_values         | P    | CF   | Provides a list of the flag values. Use in conjunction with flag_meanings.                                                                                |
| formula             | S    | CF   | Formula to calculate the values for an adaptive grid, for instance for a dimensionless vertical coordinate. Example: “hyam hybm (mlev=hyam+hybm*aps)”.    |
| formula_terms       | S    | CF   | Identifies variables that correspond to the terms in a formula, for instance for a dimensionless vertical coordinate. Example: “ap: hyam b: hybm ps: aps” |
| institution         | S    | CF   | Specifies where the original data was produced.                                                                                                           |
| long_name           | S    | CF   | A descriptive name that indicates a variable's content. This name is not standardized.                                                                    |
| positive            | S    | CF   | Direction of increasing vertical coordinate value ('up' for $z$ in m or 'down' for $p$ in hPa).                                                           |
| references          | S    | CF   | References that describe the data or methods used to produce it.                                                                                          |
| source              | S    | CF   | Method of production of the original data.                                                                                                                |

**Table 11:** Attributes for variables used in S5p netCDF-4 files (continued).

| Name                      | Type | Std. | Description                                                                                                                                                                                                                              |
|---------------------------|------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| standard_error_multiplier | F    | CF   | If a data variable with a standard_name modifier of standard_error has this attribute, it indicates that the values are the stated multiple of one standard error. The only allowed value for S5p files is 1, used only to disambiguate. |
| standard_name             | S    | CF   | A standard name that references a description of a variable's content in the standard name table.                                                                                                                                        |
| units                     | S    | CF   | Units of a variable's content. See section 9 for a detailed discussion.                                                                                                                                                                  |
| valid_max                 | P    | NUG  | Largest valid value of a variable.                                                                                                                                                                                                       |
| valid_min                 | P    | NUG  | Smallest valid value of a variable.                                                                                                                                                                                                      |
| valid_range               | P[2] | NUG  | Smallest and largest valid values of a variable. This attribute should not be combined with either valid_min or valid_max                                                                                                                |

## 12 Description of the SO<sub>2</sub> product

Description of the main output file for the Sulphur dioxide SO<sub>2</sub> product from the TROPOMI instrument on the Sentinel 5-precursor mission.

These are the file-level attributes.

These are the file-level attributes, DLR-L2 specific.

If the ECMWF dynamic auxiliary data is not available a fallback solution will be used. In this case the Level 2 output file will be flagged using the “Status\_MET\_2D” global attribute.

If the NISE dynamic auxiliary data is not available a fallback solution will be used. In this case the Level 2 output file will be flagged using the “Status\_NISE\_” global attribute.

If the TM5 dynamic auxiliary data is not available a fallback solution will be used. In this case the Level 2 output file will be flagged using the “Status\_CTMFCT\_CTMANA” global attribute.

In case of unavailability of background correction auxiliary data, the correction will be not applied and the L2 output file will be flag accordingly.

In case of unavailability of Aerosol Index auxiliary data, fallback solution will be used and the L2 output file will be flagged accordingly.

If the L2 Cloud auxiliary data is not available or not valid it will be automatically internally computed. In this case the Level 2 output file will be flagged using the “Status\_L2\_CLOUD\_” global attribute as “Internal”.

The information tracks if the earthshine spectrum was used instead of the solar spectrum. In this case the Level 2 output file will be flagged using the “Status\_reference\_spectrum” global attribute.

### Global attributes in SO2\_

Group attributes attached to SO2\_

| Name                                                                                                                                                                                                                                                                                                                                                                                                                                       | Value                                                                     | Type      |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|-----------|
| <b>Conventions</b>                                                                                                                                                                                                                                                                                                                                                                                                                         | ‘CF-1.7’ (static)                                                         | NC_STRING |
| Name of the conventions followed by the dataset. Note that while we try to follow the climate and forecast metadata conventions, there are some features – notably the use of groups to hierarchically organize the data – that are not part of version 1.6 of the CF metadata conventions. In those cases we try to follow the spirit of the conventions. This attribute originates from the NUG standard.                                |                                                                           |           |
| <b>institution</b>                                                                                                                                                                                                                                                                                                                                                                                                                         | ‘%(institute)s’ (dynamic)                                                 | NC_STRING |
| The institute where the original data was produced. The actual processing center is given in the <code>ProcessingCenter</code> attribute, here we would like to indicate the responsible parties. The value is a combination from BIRA, DLR, ESA, FMI, IUP, KNMI, MPIC, SRON, ... The actual value is a combination of the ATBD institute and the institute that developed the processor. This attribute originates from the NUG standard. |                                                                           |           |
| <b>source</b>                                                                                                                                                                                                                                                                                                                                                                                                                              | ‘Sentinel 5 precursor, TROPOMI, space-borne remote sensing, L2’ (dynamic) | NC_STRING |
| Method of production of the original data. Value includes instrument, generic description of retrieval, product level, and adds a short product name and processor version. This attribute originates from the CF standard.                                                                                                                                                                                                                |                                                                           |           |
| <b>history</b>                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                           | NC_STRING |
| Provides an audit trail for modifications to the original data. Well-behaved generic netCDF filters will automatically append their name and the parameters with which they were invoked to the global history attribute of an input netCDF file. Each line shall begin with a timestamp indicating the date and time of day that the program was executed. This attribute originates from the NUG, CF standards.                          |                                                                           |           |
| <b>summary</b>                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                           | NC_STRING |
| Miscellaneous information about the data or methods used to produce it.                                                                                                                                                                                                                                                                                                                                                                    |                                                                           |           |
| If processing in a degraded mode occurred, then a note should be placed in this attribute. A degraded processing mode can occur for several reasons, for instance the use of static backup data for nominally dynamic input or an irradiance product that is older than a few days. A machine-parseable description is available in the “processing_status” attribute. This attribute originates from the CF standard.                     |                                                                           |           |
| <b>tracking_id</b>                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                           | NC_STRING |

This unique tracking ID is proposed by the Climate Change Initiative – European Space Agency project. This ID is a UUID and allows files to be referenced, and linked up to processing description, input data, documentation, etc. The CCI-ESA project uses version 4 UUIDs (random number based) for consistency with CMIP5. This attribute originates from the CCI standard.

|                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                            |           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>id</b>                                                                                                                                                                                                                                                                                                                                                                                                                    | '%(logical_filename)s' (dynamic)                                                                                                                                           | NC_STRING |
| The “id” and “naming_authority” attributes are intended to provide a globally unique identification for each dataset. The “id” value should attempt to uniquely identify the dataset. The naming authority allows a further refinement of the “id”. The combination of the two should be globally unique for all time. We use the logical file name for the “id” attribute. This attribute originates from the CCI standard. |                                                                                                                                                                            |           |
| <b>time_reference</b>                                                                                                                                                                                                                                                                                                                                                                                                        | 'YYYY-MM-DDT00:00:00Z' (dynamic)                                                                                                                                           | NC_STRING |
| UTC time reference as an ISO 8601 [RD35] string. This corresponds to the UTC value in the <code>time</code> dimensional variable. By definition it indicates UTC midnight before the start of the granule.                                                                                                                                                                                                                   |                                                                                                                                                                            |           |
| <b>time_reference_days_since_1950</b>                                                                                                                                                                                                                                                                                                                                                                                        | 0 (dynamic)                                                                                                                                                                | NC_INT    |
| The reference time expressed as the number of days since 1950-01-01. This is the reference time unit used by both TM5 and ECMWF.                                                                                                                                                                                                                                                                                             |                                                                                                                                                                            |           |
| <b>time_reference_julian_day</b>                                                                                                                                                                                                                                                                                                                                                                                             | 0.0 (dynamic)                                                                                                                                                              | NC_DOUBLE |
| The reference time expressed as a Julian day number.                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                            |           |
| <b>time_reference_seconds_since_1970</b>                                                                                                                                                                                                                                                                                                                                                                                     | 0 (dynamic)                                                                                                                                                                | NC_INT64  |
| The reference time expressed as the number of seconds since 1970-01-01 00:00:00 UTC. This is the reference time unit used by Unix systems.                                                                                                                                                                                                                                                                                   |                                                                                                                                                                            |           |
| <b>time_coverage_start</b>                                                                                                                                                                                                                                                                                                                                                                                                   | 'YYYY-MM-DDTHH:MM:SS.mmmmmmmZ' (dynamic)                                                                                                                                   | NC_STRING |
| Start of the data granule in UTC as an ISO 8601 [RD35] string. See the discussion of the <code>delta_time</code> variable on page 41 for details.                                                                                                                                                                                                                                                                            |                                                                                                                                                                            |           |
| <b>time_coverage_end</b>                                                                                                                                                                                                                                                                                                                                                                                                     | 'YYYY-MM-DDTHH:MM:SS.mmmmmmmZ' (dynamic)                                                                                                                                   | NC_STRING |
| End of the data granule in UTC as an ISO 8601 [RD35] string. See the discussion of the <code>delta_time</code> variable on page 41 for details.                                                                                                                                                                                                                                                                              |                                                                                                                                                                            |           |
| <b>time_coverage_duration</b>                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                            | NC_STRING |
| Duration of the data granule as an ISO 8601 [RD35] duration string (“PT%(duration_seconds)s”). This attribute originates from the CCI standard.                                                                                                                                                                                                                                                                              |                                                                                                                                                                            |           |
| <b>time_coverage_resolution</b>                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                            | NC_STRING |
| Interval between measurements in the data granule as an ISO 8601 [RD35] duration string (“PT%(interval_seconds)fS”). For most products this is 840 or 1080 ms in nominal operation, except for “L2__O3__PR”, which uses 3240 ms due to coaddition. This attribute originates from the CCI standard.                                                                                                                          |                                                                                                                                                                            |           |
| <b>orbit</b>                                                                                                                                                                                                                                                                                                                                                                                                                 | 0 (dynamic)                                                                                                                                                                | NC_INT    |
| The absolute orbit number, starting at 1 – first ascending node crossing after spacecraft separation. For pre-launch testing this value should be set to “–1”.                                                                                                                                                                                                                                                               |                                                                                                                                                                            |           |
| <b>references</b>                                                                                                                                                                                                                                                                                                                                                                                                            | '%(references)s' (static)                                                                                                                                                  | NC_STRING |
| References that describe the data or methods used to produce it. This attribute originates from the CF standard.                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                            |           |
| <b>processor_version</b>                                                                                                                                                                                                                                                                                                                                                                                                     | '%(version)s' (dynamic)                                                                                                                                                    | NC_STRING |
| The version of the data processor, as string of the form “major.minor.patch”.                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                            |           |
| <b>keywords_vocabulary</b>                                                                                                                                                                                                                                                                                                                                                                                                   | 'AGU index terms, <a href="http://publications.agu.org/author-resource-center/index-terms/">http://publications.agu.org/author-resource-center/index-terms/</a> ' (static) | NC_STRING |
| The guidelines followed for the keywords attribute. We use the index terms published by the AGU.                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                            |           |
| <b>keywords</b>                                                                                                                                                                                                                                                                                                                                                                                                              | '%(keywords_agu)s' (dynamic)                                                                                                                                               | NC_STRING |
| Keywords from the “keywords_vocabulary” describing the contents of the file. To be provided by the ATBD authors.                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                            |           |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                               |           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>standard_name_vocabulary</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 'NetCDF Climate and Forecast Metadata Conventions Standard Name Table (v29, 08 July 2015), <a href="http://cfconventions.org/standard-names.html">http://cfconventions.org/standard-names.html</a> ' (static) | NC_STRING |
| The table followed for the standard_name attributes.                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                               |           |
| <b>naming_authority</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | '%(naming_authority)s' (dynamic)                                                                                                                                                                              | NC_STRING |
| Specify who is giving out the id attribute. This attribute originates from the CCI standard.                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                               |           |
| <b>cdm_data_type</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 'Swath' (static)                                                                                                                                                                                              | NC_STRING |
| The THREDDS data type appropriate for this dataset, fixed to "Swath" for S5P level 2 products. This attribute originates from the CCI standard.                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                               |           |
| <b>date_created</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 'YYYY-mm-ddTHH:MM:SS.ffffffZ' (dynamic)                                                                                                                                                                       | NC_STRING |
| The date on which this file was created. This attribute originates from the CCI standard.                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                               |           |
| <b>creator_name</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | '%(credit)s' (dynamic)                                                                                                                                                                                        | NC_STRING |
| The name of the creator, equal to the value of the "gmd:credit" attribute. For S5P this attribute is set to "The Sentinel 5 Precursor TROPOMI Level 2 products are developed with funding from the European Space Agency (ESA), the Netherlands Space Office (NSO), the Belgian Science Policy Office, the German Aerospace Center (DLR) and the Bayerisches Staatsministerium für Wirtschaft und Medien, Energie und Technologie (StMWi)." This attribute originates from the CCI standard. |                                                                                                                                                                                                               |           |
| <b>creator_url</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | '%(creator_url)s' (dynamic)                                                                                                                                                                                   | NC_STRING |
| Hyperlink to a location where more information on the product can be found. Set to <a href="http://www.tropomi.eu/">http://www.tropomi.eu/</a> . This attribute originates from the CCI standard.                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                               |           |
| <b>creator_email</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 'EOSupport@Copernicus.esa.int' (dynamic)                                                                                                                                                                      | NC_STRING |
| Point of contact for more information and support for this product. Set to "mailto:EOSupport@Copernicus.esa.int". This attribute originates from the CCI standard.                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                               |           |
| <b>project</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 'Sentinel 5 precursor/TROPOMI' (dynamic)                                                                                                                                                                      | NC_STRING |
| The name of the scientific project that created the data. This attribute originates from the CCI standard.                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                               |           |
| <b>geospatial_lat_min</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                               | NC_FLOAT  |
| Lowest latitude present in the file in decimal degrees. This attribute originates from the CCI standard.                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                               |           |
| <b>geospatial_lat_max</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                               | NC_FLOAT  |
| Highest latitude present in the file in decimal degrees. This attribute originates from the CCI standard.                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                               |           |
| <b>geospatial_lon_min</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                               | NC_FLOAT  |
| Lowest longitude present in the file in decimal degrees. This attribute originates from the CCI standard.                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                               |           |
| <b>geospatial_lon_max</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                               | NC_FLOAT  |
| Highest longitude present in the file in decimal degrees. This attribute originates from the CCI standard.                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                               |           |
| <b>license</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 'No conditions apply' (static)                                                                                                                                                                                | NC_STRING |
| describe the restrictions to data access and distribution. For S5P "No conditions apply". This attribute originates from the CCI standard.                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                               |           |
| <b>platform</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 'S5P' (static)                                                                                                                                                                                                | NC_STRING |
| Name of the satellite, set to "S5P". This attribute originates from the CCI standard.                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                               |           |
| <b>sensor</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 'TROPOMI' (static)                                                                                                                                                                                            | NC_STRING |
| Name of the sensor, set to "TROPOMI". This attribute originates from the CCI standard.                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                               |           |
| <b>spatial_resolution</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                               | NC_STRING |
| Spatial resolution at nadir. For most products this is either "7.2 × 3.6 km <sup>2</sup> " or "5.6 × 3.6 km <sup>2</sup> " (for the new pixel size since 6th of August 2019), except for "L2__O3__PR", which uses "28 × 21 km <sup>2</sup> " and "L2__CO__" and "L2__CH4__", which both use "7 × 7 km <sup>2</sup> ". This attribute originates from the CCI standard.                                                                                                                       |                                                                                                                                                                                                               |           |
| <b>cpp_compiler_version</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                               | NC_STRING |
| The version of the compiler used for the C++ code. The value of this attribute is set via the Makefile.                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                               |           |
| <b>cpp_compiler_flags</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                               | NC_STRING |
| The compiler flags passed to the C++ compiler. The value of this attribute is set via the Makefile.                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                               |           |
| <b>f90_compiler_version</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                               | NC_STRING |

The version of the compiler version used for the Fortran code. The value of this attribute is set via the Makefile. Note that not all processors make use of Fortran code.

|                           |           |
|---------------------------|-----------|
| <b>f90_compiler_flags</b> | NC_STRING |
|---------------------------|-----------|

The compiler flags passed to the Fortran compiler. The value of this attribute is set via the Makefile. Note that not all processors make use of Fortran code.

|                         |           |
|-------------------------|-----------|
| <b>exe_linker_flags</b> | NC_STRING |
|-------------------------|-----------|

The flags will be used by the linker when creating an executable. The value of this attribute is set via the Makefile.

|                   |           |
|-------------------|-----------|
| <b>build_date</b> | NC_STRING |
|-------------------|-----------|

The date on which the processor was built.

|                                    |                                                    |           |
|------------------------------------|----------------------------------------------------|-----------|
| <b>revision_control_identifier</b> | '%(revision_control_source_identifier)s' (dynamic) | NC_STRING |
|------------------------------------|----------------------------------------------------|-----------|

Revision control system identifier for the source used to build this processor.

|                                   |        |
|-----------------------------------|--------|
| <b>geolocation_grid_from_band</b> | NC_INT |
|-----------------------------------|--------|

The band from which the geolocation was taken, useful for colocating the level 2 output with other products.

|                               |                             |           |
|-------------------------------|-----------------------------|-----------|
| <b>identifier_product_doi</b> | '%(product_doi)s' (dynamic) | NC_STRING |
|-------------------------------|-----------------------------|-----------|

This is the DOI ("Digital Object Identifier") of the current product. It allows to easily find download and background information, even if that location is moved after the file has been created.

|                                         |                               |           |
|-----------------------------------------|-------------------------------|-----------|
| <b>identifier_product_doi_authority</b> | 'http://dx.doi.org/' (static) | NC_STRING |
|-----------------------------------------|-------------------------------|-----------|

This attribute defines the authoritative service for use with DOI values in resolving to the URL location.

|                          |                                   |           |
|--------------------------|-----------------------------------|-----------|
| <b>algorithm_version</b> | '%(algorithm_version)s' (dynamic) | NC_STRING |
|--------------------------|-----------------------------------|-----------|

The algorithm version, separate from the processor (framework) version, to accomodate different release schedules for different products.

|                        |                   |           |
|------------------------|-------------------|-----------|
| <b>product_version</b> | '0.0.0' (dynamic) | NC_STRING |
|------------------------|-------------------|-----------|

The product version, separate from the processor (framework) and algorithm version.

|                          |                     |           |
|--------------------------|---------------------|-----------|
| <b>processing_status</b> | 'Nominal' (dynamic) | NC_STRING |
|--------------------------|---------------------|-----------|

Description the processing status of the granule on a global level, mainly based on the availability of auxiliary input data.

Possible values: Nominal, Degraded

|                   |           |
|-------------------|-----------|
| <b>cloud_mode</b> | NC_STRING |
|-------------------|-----------|

The attribute aims at identifying the source of the cloud parameter, either "cal" or "crb".

Possible values: crb, cal

|              |                                                          |           |
|--------------|----------------------------------------------------------|-----------|
| <b>title</b> | 'TROPOMI/S5P Sulphur Dioxide SO <sub>2</sub> ' (dynamic) | NC_STRING |
|--------------|----------------------------------------------------------|-----------|

This is a short description of the product. In near-realtime processing the granule may be shorter than 1 orbit. In that case the title will become dynamic. This attribute originates from the NUG standard.

|                   |                                             |           |
|-------------------|---------------------------------------------|-----------|
| <b>references</b> | 'https://atmos.eoc.dlr.de/tropomi' (static) | NC_STRING |
|-------------------|---------------------------------------------|-----------|

References that describe the data or methods used to produce it. A URI to the ATBD seems to be an appropriate starting point. This attribute originates from the CF standard.

|                      |           |
|----------------------|-----------|
| <b>Status_MET_2D</b> | NC_STRING |
|----------------------|-----------|

The status of ECMWF input, either "Nominal" or "Fallback". Note that the "MET\_2D" auxiliary input is used as an anchor point for *all* meteorological data (where applicable).

Possible values: Nominal, Fallback

|                      |           |
|----------------------|-----------|
| <b>Status_NISE__</b> | NC_STRING |
|----------------------|-----------|

The status of NISE input, either "Nominal" or "Fallback".

Possible values: Nominal, Fallback

|                             |           |
|-----------------------------|-----------|
| <b>Status_CTMFCT_CTMANA</b> | NC_STRING |
|-----------------------------|-----------|

The status of TM5 Temperature profiles, NO<sub>2</sub>, SO<sub>2</sub> and HCHO profiles, either "Nominal" or "Fallback". Possible types of TM5 data are "AUX\_CTMFCT" for forecast or "AUX\_CTMANA" for analysis.

Possible values: Nominal, Fallback

|                                                                                                                                                                                                                                                        |           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>Status_BG</b>                                                                                                                                                                                                                                       | NC_STRING |
| The status of background correction input, either “Nominal” or “Fallback”.<br>Possible values: Nominal, Fallback                                                                                                                                       |           |
| <b>Status_AER_AI</b>                                                                                                                                                                                                                                   | NC_STRING |
| The status of Aerosol Index input, either “Nominal”, “Fallback” or “Unneeded”.<br>Possible values: Nominal, Fallback, Unneeded                                                                                                                         |           |
| <b>Status_L2_CLOUD</b>                                                                                                                                                                                                                                 | NC_STRING |
| The status of L2 cloud input, either “External” or “Internal”.<br>Possible values: External, Internal                                                                                                                                                  |           |
| <b>Status_reference_spectrum</b> ‘solar earth’ (dynamic)                                                                                                                                                                                               | NC_STRING |
| The status of reference spectrum, either “earth” or “solar”. Note that the earthshine spectrum is calculated from the auxiliary BG processor and it is read in the L2 processor from this intermediate file as input.<br>Possible values: earth, solar |           |
| <b>Status_L2_O3</b>                                                                                                                                                                                                                                    | NC_STRING |
| The status of L2 Ozone input, either “External” or “Internal”.<br>Possible values: External, Internal                                                                                                                                                  |           |

## 12.1 Group “PRODUCT” in “SO2\_\_\_”

This is the main group containing the SO<sub>2</sub> product. At this level the dimensions are defined, the actual data can be found one level deeper.

The dimensions that are common to all products. These are all located in the “PRODUCT” group, and can be accessed from that group and all sub-groups of the “PRODUCT” group, that is everywhere except the “METADATA” group.

The corner dimension is common to certain products. These are all located in the “PRODUCT” group, and can be accessed from that group and all sub-groups of the “PRODUCT” group, that is everywhere except the “METADATA” group.

All dimensions have an associated variable. These variables give a meaning to the dimension, spanning the axis of other variables.

All dimensions have an associated variable. Corner dimension is included in a separated file.

The latitude and longitude. Used in all products, placed in the “PRODUCT” group.

### Dimensions in SO2\_\_\_/PRODUCT

**scanline** The number of measurements along the swath, in the flight-direction.

**size** Unlimited.

**ground\_pixel** The number of ground pixels across track. This depends on the product and will follow the dimension found in the main input Level 1B product.

**size** -1 (dynamic)

**source** L1B.

**time** The time dimension. See the discussion of the associated dimensional variable on page 39 for details.

**size** 1 (fixed)

**corner** The number of corners for a pixel.

**size** 4 (fixed)

**layer** The number of layers in the SO<sub>2</sub> profile data and averaging kernels.

**size** 1 (dynamic)

**source** Processor.

### Variables in SO2\_\_\_/PRODUCT

#### **scanline** in SO2\_\_\_/PRODUCT

**Description:** The coordinate variable `scanline` refers to the along-track dimension of the measurement. The scanlines are time-ordered, meaning that “earlier” measurements have a lower index than “later” measurements. This variable merely contains an index to ensure that when indicating a pixel in a file the same index is used. This avoids the off-by-one confusion that frequently occurred in OMI discussions.

**Dimensions:** `scanline` (coordinate variable).

**Type:** NC\_INT.

**Source:** Processor.

| Attributes: | Name                                                                                 | Value                                                                                  | Type      |
|-------------|--------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|-----------|
|             | <b>units</b>                                                                         | ‘1’ (static)                                                                           | NC_STRING |
|             | Dimensionless, no physical quantity. This attribute originates from the CF standard. |                                                                                        |           |
|             | <b>axis</b>                                                                          | ‘Y’ (static)                                                                           | NC_STRING |
|             | <b>long_name</b>                                                                     | ‘along-track dimension index’ (static)                                                 | NC_STRING |
|             | <b>comment</b>                                                                       | ‘This coordinate variable defines the indices along track; index starts at 0’ (static) | NC_STRING |

#### **ground\_pixel** in SO2\_\_\_/PRODUCT

**Description:** The coordinate variable `ground_pixel` refers to the across-track dimension of the measurement. The `ground_pixel` ordering is from left to right with respect to the flight direction. For the Sentinel 5 precursor orbit this corresponds to west to east during the ascending part of the orbit, i.e. a higher index corresponds to a higher longitude. This variable merely contains an index to ensure that when indicating a pixel in a file the same index is used. This avoids the off-by-one confusion that frequently occurred in OMI discussions.

**Dimensions:** `ground_pixel` (coordinate variable).

**Type:** NC\_INT.

**Source:** Processor.

| Attributes: | Name                                                                                 | Value                                                                                                      | Type      |
|-------------|--------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|-----------|
|             | <b>units</b>                                                                         | ‘1’ (static)                                                                                               | NC_STRING |
|             | Dimensionless, no physical quantity. This attribute originates from the CF standard. |                                                                                                            |           |
|             | <b>axis</b>                                                                          | ‘X’ (static)                                                                                               | NC_STRING |
|             | <b>long_name</b>                                                                     | ‘across-track dimension index’ (static)                                                                    | NC_STRING |
|             | <b>comment</b>                                                                       | ‘This coordinate variable defines the indices across track, from west to east; index starts at 0’ (static) | NC_STRING |

#### **time** in SO2\_\_\_/PRODUCT

**Description:** The variable `time(time)` is the reference time of the measurements. The reference time is set to YYYY-MM-DDT00:00:00 UTC, midnight UTC before spacecraft midnight, the formal start of the current orbit. The `delta_time(scanline)` variable indicates the time difference of the observations with the reference time. Thus combining the information of `time(time)` and `delta_time(scanline)` yields the measurement time for each scanline as UTC time. The reference `time(time)` corresponds to the global attribute `time_reference` which is specified as a UTC time specified as an ISO 8601 [RD35] date.

**Dimensions:** `time` (coordinate variable).

**Type:** NC\_INT.

**Source:** Processor.

| Attributes: | Name                 | Value                                          | Type      |
|-------------|----------------------|------------------------------------------------|-----------|
|             | <b>units</b>         | ‘seconds since 2010-01-01 00:00:00’ (dynamic)  | NC_STRING |
|             | <b>standard_name</b> | ‘time’ (static)                                | NC_STRING |
|             | <b>axis</b>          | ‘T’ (static)                                   | NC_STRING |
|             | <b>long_name</b>     | ‘reference time for the measurements’ (static) | NC_STRING |

|                |                                                                                                     |  |           |
|----------------|-----------------------------------------------------------------------------------------------------|--|-----------|
| <b>comment</b> | 'The time in this variable corresponds to the time in the time_reference global attribute' (static) |  | NC_STRING |
|----------------|-----------------------------------------------------------------------------------------------------|--|-----------|

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                      |             |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| <b>corner</b> in SO2___/PRODUCT |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                      |             |
| Description:                    | An index for the pixel corners. We follow the CF-Metadata conventions [ER5, section 7.1]. The full coordinate system is right-handed, and the order of the pixel corners is counter-clockwise, starting in the "lower-left" corner (i.e. the smallest value in both latitude and longitude on the ascending part of the orbit, or equivalently for TROPOMI the lowest value for both the ground_pixel and scanline indices). See figure 7 on page 18 for a graphical depiction of the corners. |                                                                                                                                                                                                      |             |
| Dimensions:                     | corner (coordinate variable).                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                      |             |
| Type:                           | NC_INT.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                      |             |
| Source:                         | Processor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                      |             |
| Attributes:                     | <i>Name</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <i>Value</i>                                                                                                                                                                                         | <i>Type</i> |
|                                 | <b>units</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | '1' (static)                                                                                                                                                                                         | NC_STRING   |
|                                 | Dimensionless, no physical quantity. This attribute originates from the CF standard.                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                      |             |
|                                 | <b>long_name</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 'pixel corner index' (static)                                                                                                                                                                        | NC_STRING   |
|                                 | <b>comment</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 'This coordinate variable defines the indices for the pixel corners; index starts a 0 (counter-clockwise, starting from south-western corner of the pixel in ascending part of the orbit).' (static) | NC_STRING   |

|                                   |                                                                                                                                                                                                     |                                                               |             |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|-------------|
| <b>latitude</b> in SO2___/PRODUCT |                                                                                                                                                                                                     |                                                               |             |
| Description:                      | The latitude of the pixel centers of the ground pixels in the data. Latitude, longitude coordinates for the ground pixel center and the ground pixel corners are calculated at the WGS84 ellipsoid. |                                                               |             |
| Dimensions:                       | time, scanline, ground_pixel.                                                                                                                                                                       |                                                               |             |
| Type:                             | NC_FLOAT.                                                                                                                                                                                           |                                                               |             |
| Source:                           | Processor.                                                                                                                                                                                          |                                                               |             |
| Attributes:                       | <i>Name</i>                                                                                                                                                                                         | <i>Value</i>                                                  | <i>Type</i> |
|                                   | <b>long_name</b>                                                                                                                                                                                    | 'pixel center latitude' (static)                              | NC_STRING   |
|                                   | <b>units</b>                                                                                                                                                                                        | 'degrees_north' (static)                                      | NC_STRING   |
|                                   | <b>standard_name</b>                                                                                                                                                                                | 'latitude' (static)                                           | NC_STRING   |
|                                   | <b>valid_min</b>                                                                                                                                                                                    | -90.0 (static)                                                | NC_FLOAT    |
|                                   | <b>valid_max</b>                                                                                                                                                                                    | 90.0 (static)                                                 | NC_FLOAT    |
|                                   | <b>bounds</b>                                                                                                                                                                                       | '/PRODUCT/SUPPORT_DATA/GEOLOCATIONS/latitude_bounds' (static) | NC_STRING   |
|                                   | A link to the boundary coordinates, i.e. the pixel corners. Note that the use of group-names in this attribute is an extension of the climate and forecasting metadata conventions.                 |                                                               |             |

|                                    |                                                                                                                                                                                                      |                                   |             |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|-------------|
| <b>longitude</b> in SO2___/PRODUCT |                                                                                                                                                                                                      |                                   |             |
| Description:                       | The longitude of the pixel centers of the ground pixels in the data. Latitude, longitude coordinates for the ground pixel center and the ground pixel corners are calculated at the WGS84 ellipsoid. |                                   |             |
| Dimensions:                        | time, scanline, ground_pixel.                                                                                                                                                                        |                                   |             |
| Type:                              | NC_FLOAT.                                                                                                                                                                                            |                                   |             |
| Source:                            | Processor.                                                                                                                                                                                           |                                   |             |
| Attributes:                        | <i>Name</i>                                                                                                                                                                                          | <i>Value</i>                      | <i>Type</i> |
|                                    | <b>long_name</b>                                                                                                                                                                                     | 'pixel center longitude' (static) | NC_STRING   |
|                                    | <b>units</b>                                                                                                                                                                                         | 'degrees_east' (static)           | NC_STRING   |
|                                    | <b>standard_name</b>                                                                                                                                                                                 | 'longitude' (static)              | NC_STRING   |
|                                    | <b>valid_min</b>                                                                                                                                                                                     | -180.0 (static)                   | NC_FLOAT    |
|                                    | <b>valid_max</b>                                                                                                                                                                                     | 180.0 (static)                    | NC_FLOAT    |

|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                               |             |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
|              | <b>bounds</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | '/PRODUCT/SUPPORT_DATA/GEOLOCATIONS/<br>longitude_bounds' (static)<br><br>A link to the boundary coordinates, i.e. the pixel corners. Note that the use of group-names in this attribute is an extension of the climate and forecasting metadata conventions. | NC_STRING   |
|              | <b>delta_time</b> in SO2___/PRODUCT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                               |             |
| Description: | The <code>delta_time(scanline)</code> variable indicates the time difference with the reference time <code>time(time)</code> (see page 39). Thus combining the information of <code>time(time)</code> and <code>delta_time(scanline)</code> yields the start of the measurement time for each scanline as TAI2010 time. Combining the information in the global attribute <code>time_reference</code> with <code>delta_time(scanline)</code> yields the start of the measurement time in UTC time. The UTC time derived for the first scanline corresponds to the global attribute <code>time_coverage_start</code> . However, the UTC time derived for the last scanline does not correspond to global attribute <code>time_coverage_end</code> . One scanline measurement is the result of adding independent measurements during one coaddition period. The scanline measurement is given the measurement time of the first sample in this co-addition. It is the measurement time of the last sample in the coaddition period of the last scanline that corresponds to <code>time_coverage_end</code> .<br><br>This variable gives the time offset in ms accuracy. |                                                                                                                                                                                                                                                               |             |
| Dimensions:  | time, scanline, ground_pixel.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                               |             |
| Type:        | NC_INT.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                               |             |
| Source:      | Processor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                               |             |
| Attributes:  | <i>Name</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <i>Value</i>                                                                                                                                                                                                                                                  | <i>Type</i> |
|              | <b>long_name</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 'offset from reference start time of measurement' (static)                                                                                                                                                                                                    | NC_STRING   |
|              | <b>units</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 'milliseconds since %(ref_time)s' (static)                                                                                                                                                                                                                    | NC_STRING   |
|              | <b>time_utc</b> in SO2___/PRODUCT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                               |             |
| Description: | The time of observation expressed as ISO 8601 [RD35] date-time string.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                               |             |
| Dimensions:  | time, scanline.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                               |             |
| Type:        | NC_STRING.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                               |             |
| Source:      | Processor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                               |             |
| Attributes:  | <i>Name</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <i>Value</i>                                                                                                                                                                                                                                                  | <i>Type</i> |
|              | <b>long_name</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 'Time of observation as ISO 8601 date-time string' (static)                                                                                                                                                                                                   | NC_STRING   |
|              | <b>qa_value</b> in SO2___/PRODUCT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                               |             |
| Description: | A continuous quality descriptor, varying between 0 (no data) and 1 (full quality data). The value will change based on observation conditions and retrieval flags. Detailed quality flags are provided in the <code>processing_quality_flags</code> elsewhere in the product.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                               |             |
| Dimensions:  | time, scanline, ground_pixel.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                               |             |
| Type:        | NC_UBYTE.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                               |             |
| Source:      | Processor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                               |             |
| Attributes:  | <i>Name</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <i>Value</i>                                                                                                                                                                                                                                                  | <i>Type</i> |
|              | <b>units</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | '1' (static)                                                                                                                                                                                                                                                  | NC_STRING   |
|              | <b>scale_factor</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 0.01 (static)                                                                                                                                                                                                                                                 | NC_FLOAT    |
|              | <b>add_offset</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 0 (static)                                                                                                                                                                                                                                                    | NC_FLOAT    |
|              | <b>valid_min</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 0 (static)                                                                                                                                                                                                                                                    | NC_UBYTE    |
|              | <b>valid_max</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 100 (static)                                                                                                                                                                                                                                                  | NC_UBYTE    |
|              | <b>long_name</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 'data quality value' (static)                                                                                                                                                                                                                                 | NC_STRING   |
|              | <b>comment</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 'A continuous quality descriptor, varying between 0 (no data) and 1 (full quality data). Recommend to ignore data with <code>qa_value &lt; 0.5</code> ' (static)                                                                                              | NC_STRING   |

|                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                               |             |           |
|------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------|
|                                                                        | <b>coordinates</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                                                               |             | NC_STRING |
| <b>sulfurdioxide_total_vertical_column</b> in SO2___/PRODUCT           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                               |             |           |
| Description:                                                           | Vertical column density of the sulphur dioxide $SO_2$ product for the polluted scenario (anthropogenic $SO_2$ in the PBL).                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                               |             |           |
| Dimensions:                                                            | time, scanline, ground_pixel.                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                               |             |           |
| Type:                                                                  | NC_FLOAT.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                               |             |           |
| Source:                                                                | Processor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                               |             |           |
| Attributes:                                                            | <i>Name</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <i>Value</i>                                                                                                                                  | <i>Type</i> |           |
|                                                                        | <b>units</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 'mol m-2' (static)                                                                                                                            | NC_STRING   |           |
|                                                                        | <b>standard_name</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 'atmosphere_mole_content_of_sulfur_dioxide' (static)                                                                                          | NC_STRING   |           |
|                                                                        | <b>long_name</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 'total vertical column of sulfur dioxide for the polluted scenario derived from the total slant column' (static)                              | NC_STRING   |           |
|                                                                        | <b>ancillary_variables</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 'sulfurdioxide_total_vertical_column_precision /PRODUCT/SUPPORT_DATA/DETAILED_RESULTS/sulfurdioxide_total_vertical_column_true-ness' (static) | NC_STRING   |           |
|                                                                        | <b>coordinates</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                                                               | NC_STRING   |           |
|                                                                        | The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. This attribute originates from the CF standard.                                                                                                                                                                                                                                                                                                                |                                                                                                                                               |             |           |
|                                                                        | <b>multiplication_factor_to_convert_to_DU</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                    | %(MOLM2_TO_DU)s (static)                                                                                                                      | NC_FLOAT    |           |
|                                                                        | The quantities in Sentinel 5 precursor files are given in SI units. For an integrated column value this means that the unit is $\text{mol m}^{-2}$ . Traditionally the unit for an integrated column is "DU" or Dobson Units. This attribute provides the multiplication factor to calculate the total column in DU from the value in $\text{mol m}^{-2}$ . This is provided as a convenience to users who have tools that work in DU.                                                           |                                                                                                                                               |             |           |
|                                                                        | <b>multiplication_factor_to_convert_to_molecules_percm2</b>                                                                                                                                                                                                                                                                                                                                                                                                                                      | 6.02214e+19 (static)                                                                                                                          | NC_FLOAT    |           |
|                                                                        | The quantities in Sentinel 5 precursor files are given in SI units. For an integrated column value this means that the unit is $\text{mol m}^{-2}$ . Traditionally the unit for an integrated column is "molecules $\text{cm}^{-2}$ ". This attribute provides the multiplication factor to calculate the total column in molecules $\text{cm}^{-2}$ from the value in $\text{mol m}^{-2}$ . This is provided as a convenience to users who have tools that work in molecules $\text{cm}^{-2}$ . |                                                                                                                                               |             |           |
| <b>sulfurdioxide_total_vertical_column_precision</b> in SO2___/PRODUCT |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                               |             |           |
| Description:                                                           | Random error of the vertical column density of the sulphur dioxide $SO_2$ product for the polluted scenario (anthropogenic $SO_2$ in the PBL).                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                               |             |           |
| Dimensions:                                                            | time, scanline, ground_pixel.                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                               |             |           |
| Type:                                                                  | NC_FLOAT.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                               |             |           |
| Source:                                                                | Processor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                               |             |           |
| Attributes:                                                            | <i>Name</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <i>Value</i>                                                                                                                                  | <i>Type</i> |           |
|                                                                        | <b>units</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 'mol m-2' (static)                                                                                                                            | NC_STRING   |           |
|                                                                        | <b>standard_name</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 'atmosphere_mole_content_of_sulfur_dioxide_standard_error' (static)                                                                           | NC_STRING   |           |
|                                                                        | <b>long_name</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 'precision of the total vertical column of sulfur dioxide for the polluted scenario derived from the total slant column' (static)             | NC_STRING   |           |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                 |           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|-----------|
| <b>coordinates</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | '/PRODUCT/longitude /PRODUCT/latitude' (static) | NC_STRING |
| The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. This attribute originates from the CF standard.                                                                                                                                                                                                                                                                                                                |                                                 |           |
| <b>multiplication_factor_to_convert_to_DU</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                    | %(MOLM2_TO_DU)s (static)                        | NC_FLOAT  |
| The quantities in Sentinel 5 precursor files are given in SI units. For an integrated column value this means that the unit is mol m <sup>-2</sup> . Traditionally the unit for an integrated column is "DU" or Dobson Units. This attribute provides the multiplication factor to calculate the total column in DU from the value in mol m <sup>-2</sup> . This is provided as a convenience to users who have tools that work in DU.                                                           |                                                 |           |
| <b>multiplication_factor_to_convert_to_molecules_per_cm2</b>                                                                                                                                                                                                                                                                                                                                                                                                                                     | 6.02214e+19 (static)                            | NC_FLOAT  |
| The quantities in Sentinel 5 precursor files are given in SI units. For an integrated column value this means that the unit is mol m <sup>-2</sup> . Traditionally the unit for an integrated column is "molecules cm <sup>-2</sup> ". This attribute provides the multiplication factor to calculate the total column in molecules cm <sup>-2</sup> from the value in mol m <sup>-2</sup> . This is provided as a convenience to users who have tools that work in molecules cm <sup>-2</sup> . |                                                 |           |

### 12.1.1 Group "SO<sub>2</sub>\_LAYER\_HEIGHT" in "PRODUCT"

#### Variables in SO<sub>2</sub>\_\_\_/PRODUCT/SO<sub>2</sub>\_LAYER\_HEIGHT

##### **sulfurdioxide\_layer\_height** in SO<sub>2</sub>\_\_\_/PRODUCT/SO<sub>2</sub>\_LAYER\_HEIGHT

Description: Retrieved layer height of a sulphur dioxide SO<sub>2</sub> plume.

Dimensions: time, scanline, ground\_pixel.

Type: NC\_FLOAT.

Source: Processor.

|                                                                                                                                                                                   |                            |                                                          |             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|----------------------------------------------------------|-------------|
| Attributes:                                                                                                                                                                       | <i>Name</i>                | <i>Value</i>                                             | <i>Type</i> |
|                                                                                                                                                                                   | <b>units</b>               | 'm' (static)                                             | NC_STRING   |
|                                                                                                                                                                                   | <b>standard_name</b>       | 'height_above_mean_sea_level_of_sulfur_dioxide' (static) | NC_STRING   |
|                                                                                                                                                                                   | <b>long_name</b>           | 'sulfur dioxide layer height' (static)                   | NC_STRING   |
|                                                                                                                                                                                   | <b>ancillary_variables</b> | 'sulfurdioxide_layer_height_precision' (static)          | NC_STRING   |
|                                                                                                                                                                                   | <b>coordinates</b>         | '/PRODUCT/longitude /PRODUCT/latitude' (static)          | NC_STRING   |
| The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. This attribute originates from the CF standard. |                            |                                                          |             |

##### **sulfurdioxide\_layer\_height\_precision** in SO<sub>2</sub>\_\_\_/PRODUCT/SO<sub>2</sub>\_LAYER\_HEIGHT

Description: Random error of the retrieved SO<sub>2</sub> layer height.

Dimensions: time, scanline, ground\_pixel.

Type: NC\_FLOAT.

Source: Processor.

|             |                      |                                                                |             |
|-------------|----------------------|----------------------------------------------------------------|-------------|
| Attributes: | <i>Name</i>          | <i>Value</i>                                                   | <i>Type</i> |
|             | <b>units</b>         | 'm' (static)                                                   | NC_STRING   |
|             | <b>standard_name</b> | 'height_above_mean_sea_level_of_sulfur_dioxide_error' (static) | NC_STRING   |
|             | <b>long_name</b>     | 'random error of sulfur dioxide layer height' (static)         | NC_STRING   |

|                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                 |             |           |
|------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|-------------|-----------|
|                                                                                                      | <b>coordinates</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | ‘/PRODUCT/longitude /PRODUCT/latitude’ (static)                                                                                 |             | NC_STRING |
|                                                                                                      | The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. This attribute originates from the CF standard.                                                                                                                                                                                                                                                                                                                |                                                                                                                                 |             |           |
| <b>sulfurdioxide_total_vertical_column_layer_height</b> in SO2___/PRODUCT/SO2_LAYER_HEIGHT           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                 |             |           |
| Description:                                                                                         | Vertical column density of the sulfur dioxide $SO_2$ product for a sulfur dioxide plume at layer height altitude w.r.t. the sea level.                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                 |             |           |
| Dimensions:                                                                                          | time, scanline, ground_pixel.                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                 |             |           |
| Type:                                                                                                | NC_FLOAT.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                 |             |           |
| Source:                                                                                              | Processor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                 |             |           |
| Attributes:                                                                                          | <i>Name</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <i>Value</i>                                                                                                                    | <i>Type</i> |           |
|                                                                                                      | <b>units</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | ‘mol m-2’ (static)                                                                                                              | NC_STRING   |           |
|                                                                                                      | <b>standard_name</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | ‘atmosphere_mole_content_of_sulfur_dioxide’ (static)                                                                            | NC_STRING   |           |
|                                                                                                      | <b>long_name</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | ‘total vertical column density of sulfur dioxide at the retrieved layer height’ (static)                                        | NC_STRING   |           |
|                                                                                                      | <b>ancillary_variables</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | ‘sulfurdioxide_total_vertical_column_layer_height_precision sulfurdioxide_total_vertical_column_layer_height_trueness’ (static) | NC_STRING   |           |
|                                                                                                      | <b>coordinates</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | ‘/PRODUCT/longitude /PRODUCT/latitude’ (static)                                                                                 |             | NC_STRING |
|                                                                                                      | The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. This attribute originates from the CF standard.                                                                                                                                                                                                                                                                                                                |                                                                                                                                 |             |           |
|                                                                                                      | <b>multiplication_factor_to_convert_to_DU</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2241.15 (static)                                                                                                                | NC_FLOAT    |           |
|                                                                                                      | The quantities in Sentinel 5 precursor files are given in SI units. For an integrated column value this means that the unit is $\text{mol m}^{-2}$ . Traditionally the unit for an integrated column is “DU” or Dobson Units. This attribute provides the multiplication factor to calculate the total column in DU from the value in $\text{mol m}^{-2}$ . This is provided as a convenience to users who have tools that work in DU.                                                           |                                                                                                                                 |             |           |
|                                                                                                      | <b>multiplication_factor_to_convert_to_molecules_per_cm2</b>                                                                                                                                                                                                                                                                                                                                                                                                                                     | 6.02214e+19 (static)                                                                                                            | NC_FLOAT    |           |
|                                                                                                      | The quantities in Sentinel 5 precursor files are given in SI units. For an integrated column value this means that the unit is $\text{mol m}^{-2}$ . Traditionally the unit for an integrated column is “molecules $\text{cm}^{-2}$ ”. This attribute provides the multiplication factor to calculate the total column in molecules $\text{cm}^{-2}$ from the value in $\text{mol m}^{-2}$ . This is provided as a convenience to users who have tools that work in molecules $\text{cm}^{-2}$ . |                                                                                                                                 |             |           |
| <b>sulfurdioxide_total_vertical_column_layer_height_precision</b> in SO2___/PRODUCT/SO2_LAYER_HEIGHT |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                 |             |           |
| Description:                                                                                         | Random error of the vertical column density of the sulfur dioxide $SO_2$ product for a sulfur dioxide plume at layer height altitude w.r.t. the sea level.                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                 |             |           |
| Dimensions:                                                                                          | time, scanline, ground_pixel.                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                 |             |           |
| Type:                                                                                                | NC_FLOAT.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                 |             |           |
| Source:                                                                                              | Processor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                 |             |           |
| Attributes:                                                                                          | <i>Name</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <i>Value</i>                                                                                                                    | <i>Type</i> |           |
|                                                                                                      | <b>units</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | ‘mol m-2’ (static)                                                                                                              | NC_STRING   |           |
|                                                                                                      | <b>standard_name</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | ‘atmosphere_mole_content_of_sulfur_dioxide_standard_error’ (static)                                                             | NC_STRING   |           |
|                                                                                                      | <b>long_name</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | ‘random error of the total vertical column density of sulfur dioxide at the retrieved layer height’ (static)                    | NC_STRING   |           |

|                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                  |             |
|-----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|-------------|
|                                                                                                     | <b>coordinates</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                                  | NC_STRING   |
|                                                                                                     | The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. This attribute originates from the CF standard.                                                                                                                                                                                                                                                                                                                |                                                                                                                  |             |
|                                                                                                     | <b>multiplication_factor_to_convert_to_DU</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2241.15 (static)                                                                                                 | NC_FLOAT    |
|                                                                                                     | The quantities in Sentinel 5 precursor files are given in SI units. For an integrated column value this means that the unit is $\text{mol m}^{-2}$ . Traditionally the unit for an integrated column is "DU" or Dobson Units. This attribute provides the multiplication factor to calculate the total column in DU from the value in $\text{mol m}^{-2}$ . This is provided as a convenience to users who have tools that work in DU.                                                           |                                                                                                                  |             |
|                                                                                                     | <b>multiplication_factor_to_convert_to_molecules_percm2</b>                                                                                                                                                                                                                                                                                                                                                                                                                                      | 6.02214e+19 (static)                                                                                             | NC_FLOAT    |
|                                                                                                     | The quantities in Sentinel 5 precursor files are given in SI units. For an integrated column value this means that the unit is $\text{mol m}^{-2}$ . Traditionally the unit for an integrated column is "molecules $\text{cm}^{-2}$ ". This attribute provides the multiplication factor to calculate the total column in molecules $\text{cm}^{-2}$ from the value in $\text{mol m}^{-2}$ . This is provided as a convenience to users who have tools that work in molecules $\text{cm}^{-2}$ . |                                                                                                                  |             |
| <b>sulfurdioxide_total_vertical_column_layer_height_trueness</b> in SO2___/PRODUCT/SO2_LAYER_HEIGHT |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                  |             |
| Description:                                                                                        | Systematic error of the vertical column density of the sulfur dioxide $\text{SO}_2$ product for a sulfur dioxide plume at layer height altitude w.r.t. the sea level.                                                                                                                                                                                                                                                                                                                            |                                                                                                                  |             |
| Dimensions:                                                                                         | time, scanline, ground_pixel.                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                  |             |
| Type:                                                                                               | NC_FLOAT.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                  |             |
| Source:                                                                                             | Processor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                  |             |
| Attributes:                                                                                         | <i>Name</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <i>Value</i>                                                                                                     | <i>Type</i> |
|                                                                                                     | <b>units</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 'mol m-2' (static)                                                                                               | NC_STRING   |
|                                                                                                     | <b>long_name</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 'systematic error of the total vertical column density of sulfur dioxide at the retrieved layer height' (static) | NC_STRING   |
|                                                                                                     | <b>coordinates</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                                  | NC_STRING   |
|                                                                                                     | The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. This attribute originates from the CF standard.                                                                                                                                                                                                                                                                                                                |                                                                                                                  |             |
|                                                                                                     | <b>multiplication_factor_to_convert_to_DU</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2241.15 (static)                                                                                                 | NC_FLOAT    |
|                                                                                                     | The quantities in Sentinel 5 precursor files are given in SI units. For an integrated column value this means that the unit is $\text{mol m}^{-2}$ . Traditionally the unit for an integrated column is "DU" or Dobson Units. This attribute provides the multiplication factor to calculate the total column in DU from the value in $\text{mol m}^{-2}$ . This is provided as a convenience to users who have tools that work in DU.                                                           |                                                                                                                  |             |
|                                                                                                     | <b>multiplication_factor_to_convert_to_molecules_percm2</b>                                                                                                                                                                                                                                                                                                                                                                                                                                      | 6.02214e+19 (static)                                                                                             | NC_FLOAT    |
|                                                                                                     | The quantities in Sentinel 5 precursor files are given in SI units. For an integrated column value this means that the unit is $\text{mol m}^{-2}$ . Traditionally the unit for an integrated column is "molecules $\text{cm}^{-2}$ ". This attribute provides the multiplication factor to calculate the total column in molecules $\text{cm}^{-2}$ from the value in $\text{mol m}^{-2}$ . This is provided as a convenience to users who have tools that work in molecules $\text{cm}^{-2}$ . |                                                                                                                  |             |
| <b>sulfurdioxide total air mass factor layer height</b> in SO2___/PRODUCT/SO2_LAYER_HEIGHT          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                  |             |

|                                                                                                            |                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                  |             |
|------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description:                                                                                               | Total air mass factor, $M$ for a sulfur dioxide plume at the layer height altitude w.r.t. the sea level.                                                                                                                                                                       |                                                                                                                                                                                                  |             |
| Dimensions:                                                                                                | time, scanline, ground_pixel.                                                                                                                                                                                                                                                  |                                                                                                                                                                                                  |             |
| Type:                                                                                                      | NC_FLOAT.                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                  |             |
| Source:                                                                                                    | Processor.                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                  |             |
| Attributes:                                                                                                | <i>Name</i>                                                                                                                                                                                                                                                                    | <i>Value</i>                                                                                                                                                                                     | <i>Type</i> |
|                                                                                                            | <b>units</b>                                                                                                                                                                                                                                                                   | '1' (static)                                                                                                                                                                                     | NC_STRING   |
|                                                                                                            | <b>long_name</b>                                                                                                                                                                                                                                                               | 'total air mass factor for a sulfur dioxide plume at the retrieved layer height' (static)                                                                                                        | NC_STRING   |
|                                                                                                            | <b>ancillary_variables</b>                                                                                                                                                                                                                                                     | 'sulfurdioxide_total_air_mass_factor_layer_height_precision sulfurdioxide_total_air_mass_factor_layer_height_trueness sulfurdioxide_total_air_mass_factor_layer_height_kernel_trueness' (static) | NC_STRING   |
|                                                                                                            | <b>coordinates</b>                                                                                                                                                                                                                                                             | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                                                                                                                  | NC_STRING   |
|                                                                                                            | The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. Following [ER5, section 5.2] we use this attribute to connect the data with the geolocation. This attribute originates from the CF standard. |                                                                                                                                                                                                  |             |
| <b>sulfurdioxide_total_air_mass_factor_layer_height_precision</b> in SO2___/PRODUCT/SO2_LAYER_HEIGHT       |                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                  |             |
| Description:                                                                                               | Random error of the total air mass factor $M$ for a sulfur dioxide plume at layer height altitude w.r.t. the sea level.                                                                                                                                                        |                                                                                                                                                                                                  |             |
| Dimensions:                                                                                                | time, scanline, ground_pixel.                                                                                                                                                                                                                                                  |                                                                                                                                                                                                  |             |
| Type:                                                                                                      | NC_FLOAT.                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                  |             |
| Source:                                                                                                    | Processor.                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                  |             |
| Attributes:                                                                                                | <i>Name</i>                                                                                                                                                                                                                                                                    | <i>Value</i>                                                                                                                                                                                     | <i>Type</i> |
|                                                                                                            | <b>units</b>                                                                                                                                                                                                                                                                   | '1' (static)                                                                                                                                                                                     | NC_STRING   |
|                                                                                                            | <b>long_name</b>                                                                                                                                                                                                                                                               | 'random error of the total air mass factor for a sulfur dioxide plume at the retrieved layer height' (static)                                                                                    | NC_STRING   |
|                                                                                                            | <b>coordinates</b>                                                                                                                                                                                                                                                             | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                                                                                                                  | NC_STRING   |
|                                                                                                            | The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. Following [ER5, section 5.2] we use this attribute to connect the data with the geolocation. This attribute originates from the CF standard. |                                                                                                                                                                                                  |             |
| <b>sulfurdioxide_total_air_mass_factor_layer_height_trueness</b> in SO2___/PRODUCT/SO2_LAYER_HEIGHT        |                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                  |             |
| Description:                                                                                               | Systematic error of the total air mass factor $M$ for a sulfur dioxide plume at layer height altitude w.r.t. the sea level.                                                                                                                                                    |                                                                                                                                                                                                  |             |
| Dimensions:                                                                                                | time, scanline, ground_pixel.                                                                                                                                                                                                                                                  |                                                                                                                                                                                                  |             |
| Type:                                                                                                      | NC_FLOAT.                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                  |             |
| Source:                                                                                                    | Processor.                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                  |             |
| Attributes:                                                                                                | <i>Name</i>                                                                                                                                                                                                                                                                    | <i>Value</i>                                                                                                                                                                                     | <i>Type</i> |
|                                                                                                            | <b>units</b>                                                                                                                                                                                                                                                                   | '1' (static)                                                                                                                                                                                     | NC_STRING   |
|                                                                                                            | <b>long_name</b>                                                                                                                                                                                                                                                               | 'systematic error of the total air mass factor for a sulfur dioxide plume at the retrieved layer height' (static)                                                                                | NC_STRING   |
|                                                                                                            | <b>coordinates</b>                                                                                                                                                                                                                                                             | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                                                                                                                  | NC_STRING   |
|                                                                                                            | The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. Following [ER5, section 5.2] we use this attribute to connect the data with the geolocation. This attribute originates from the CF standard. |                                                                                                                                                                                                  |             |
| <b>sulfurdioxide_total_air_mass_factor_layer_height_kernel_trueness</b> in SO2___/PRODUCT/SO2_LAYER_HEIGHT |                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                  |             |

|                                                                                                                                                                                                                                                                                |                                                                                                                                                   |                                                                                                                                 |             |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|-------------|
| Description:                                                                                                                                                                                                                                                                   | Systematic error of the total air mass factor $M$ when kernels are used for a sulfur dioxide plume at layer height altitude w.r.t. the sea level. |                                                                                                                                 |             |
| Dimensions:                                                                                                                                                                                                                                                                    | time, scanline, ground_pixel.                                                                                                                     |                                                                                                                                 |             |
| Type:                                                                                                                                                                                                                                                                          | NC_FLOAT.                                                                                                                                         |                                                                                                                                 |             |
| Source:                                                                                                                                                                                                                                                                        | Processor.                                                                                                                                        |                                                                                                                                 |             |
| Attributes:                                                                                                                                                                                                                                                                    | <i>Name</i>                                                                                                                                       | <i>Value</i>                                                                                                                    | <i>Type</i> |
|                                                                                                                                                                                                                                                                                | <b>units</b>                                                                                                                                      | '1' (static)                                                                                                                    | NC_STRING   |
|                                                                                                                                                                                                                                                                                | <b>long_name</b>                                                                                                                                  | 'systematic error of the total air mass factor using kernels for a sulfur dioxide plume at the retrieved layer height' (static) | NC_STRING   |
|                                                                                                                                                                                                                                                                                | <b>coordinates</b>                                                                                                                                | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                                                 | NC_STRING   |
| The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. Following [ER5, section 5.2] we use this attribute to connect the data with the geolocation. This attribute originates from the CF standard. |                                                                                                                                                   |                                                                                                                                 |             |
| <b>sulfurdioxide_clear_air_mass_factor_layer_height</b> in SO2___/PRODUCT/SO2_LAYER_HEIGHT                                                                                                                                                                                     |                                                                                                                                                   |                                                                                                                                 |             |
| Description:                                                                                                                                                                                                                                                                   | Clear sky air mass factor for a sulfur dioxide plume at the layer height altitude w.r.t. the sea level.                                           |                                                                                                                                 |             |
| Dimensions:                                                                                                                                                                                                                                                                    | time, scanline, ground_pixel.                                                                                                                     |                                                                                                                                 |             |
| Type:                                                                                                                                                                                                                                                                          | NC_FLOAT.                                                                                                                                         |                                                                                                                                 |             |
| Source:                                                                                                                                                                                                                                                                        | Processor.                                                                                                                                        |                                                                                                                                 |             |
| Attributes:                                                                                                                                                                                                                                                                    | <i>Name</i>                                                                                                                                       | <i>Value</i>                                                                                                                    | <i>Type</i> |
|                                                                                                                                                                                                                                                                                | <b>units</b>                                                                                                                                      | '1' (static)                                                                                                                    | NC_STRING   |
|                                                                                                                                                                                                                                                                                | <b>long_name</b>                                                                                                                                  | 'clear sky air mass factor for a sulfur dioxide plume at the retrieved layer height' (static)                                   | NC_STRING   |
|                                                                                                                                                                                                                                                                                | <b>coordinates</b>                                                                                                                                | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                                                 | NC_STRING   |
| The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. Following [ER5, section 5.2] we use this attribute to connect the data with the geolocation. This attribute originates from the CF standard. |                                                                                                                                                   |                                                                                                                                 |             |
| <b>sulfurdioxide_cloudy_air_mass_factor_layer_height</b> in SO2___/PRODUCT/SO2_LAYER_HEIGHT                                                                                                                                                                                    |                                                                                                                                                   |                                                                                                                                 |             |
| Description:                                                                                                                                                                                                                                                                   | Cloudy sky air mass factor for a sulfur dioxide plume at layer height altitude w.r.t. the sea level.                                              |                                                                                                                                 |             |
| Dimensions:                                                                                                                                                                                                                                                                    | time, scanline, ground_pixel.                                                                                                                     |                                                                                                                                 |             |
| Type:                                                                                                                                                                                                                                                                          | NC_FLOAT.                                                                                                                                         |                                                                                                                                 |             |
| Source:                                                                                                                                                                                                                                                                        | Processor.                                                                                                                                        |                                                                                                                                 |             |
| Attributes:                                                                                                                                                                                                                                                                    | <i>Name</i>                                                                                                                                       | <i>Value</i>                                                                                                                    | <i>Type</i> |
|                                                                                                                                                                                                                                                                                | <b>units</b>                                                                                                                                      | '1' (static)                                                                                                                    | NC_STRING   |
|                                                                                                                                                                                                                                                                                | <b>long_name</b>                                                                                                                                  | 'cloudy sky air mass factor for a sulfur dioxide plume at the retrieved layer height' (static)                                  | NC_STRING   |
|                                                                                                                                                                                                                                                                                | <b>coordinates</b>                                                                                                                                | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                                                 | NC_STRING   |
| The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. Following [ER5, section 5.2] we use this attribute to connect the data with the geolocation. This attribute originates from the CF standard. |                                                                                                                                                   |                                                                                                                                 |             |
| <b>sulfurdioxide_averaging_kernel_scaling_box_layer_height</b> in SO2___/PRODUCT/SO2_LAYER_HEIGHT                                                                                                                                                                              |                                                                                                                                                   |                                                                                                                                 |             |
| Description:                                                                                                                                                                                                                                                                   | Scaling box of the total air mass factor $M$ for a sulfur dioxide plume at layer height altitude w.r.t. the sea level.                            |                                                                                                                                 |             |
| Dimensions:                                                                                                                                                                                                                                                                    | time, scanline, ground_pixel.                                                                                                                     |                                                                                                                                 |             |
| Type:                                                                                                                                                                                                                                                                          | NC_FLOAT.                                                                                                                                         |                                                                                                                                 |             |
| Source:                                                                                                                                                                                                                                                                        | Processor.                                                                                                                                        |                                                                                                                                 |             |
| Attributes:                                                                                                                                                                                                                                                                    | <i>Name</i>                                                                                                                                       | <i>Value</i>                                                                                                                    | <i>Type</i> |

|                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                              |             |
|---------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
|                                                                           | <b>units</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | '1' (static)                                                                                                                                                 | NC_STRING   |
|                                                                           | <b>long_name</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 'scaling box of the total air mass factor for a sulfur dioxide plume at the retrieved layer height' (static)                                                 | NC_STRING   |
|                                                                           | <b>coordinates</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                                                                              | NC_STRING   |
|                                                                           | The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. Following [ER5, section 5.2] we use this attribute to connect the data with the geolocation. This attribute originates from the CF standard.                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                              |             |
| <b>sulfurdioxide_layer_pressure</b> in SO2___/PRODUCT/SO2_LAYER_HEIGHT    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                              |             |
| Description:                                                              | Pressure at the level of the sulfurdioxide layer height.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                              |             |
| Dimensions:                                                               | time, scanline, ground_pixel.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                              |             |
| Type:                                                                     | NC_FLOAT.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                              |             |
| Source:                                                                   | Processor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                              |             |
| Attributes:                                                               | <i>Name</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <i>Value</i>                                                                                                                                                 | <i>Type</i> |
|                                                                           | <b>units</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 'Pa' (static)                                                                                                                                                | NC_STRING   |
|                                                                           | <b>standard_name</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 'TBD' (static)                                                                                                                                               | NC_STRING   |
|                                                                           | <b>long_name</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 'optical centroid pressure of the sulfur dioxide layer' (static)                                                                                             | NC_STRING   |
|                                                                           | <b>coordinates</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                                                                              | NC_STRING   |
|                                                                           | The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. This attribute originates from the CF standard.                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                              |             |
| <b>qa_value_layer_height</b> in SO2___/PRODUCT/SO2_LAYER_HEIGHT           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                              |             |
| Description:                                                              | A continuous quality descriptor for the layer height retrieval, varying between 0 (no data) and 1 (full quality data). The value will change based on observation conditions and retrieval flags.                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                              |             |
| Dimensions:                                                               | time, scanline, ground_pixel.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                              |             |
| Type:                                                                     | NC_UBYTE.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                              |             |
| Source:                                                                   | Processor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                              |             |
| Attributes:                                                               | <i>Name</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <i>Value</i>                                                                                                                                                 | <i>Type</i> |
|                                                                           | <b>units</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | '1' (static)                                                                                                                                                 | NC_STRING   |
|                                                                           | <b>scale_factor</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 0.01 (static)                                                                                                                                                | NC_FLOAT    |
|                                                                           | <b>add_offset</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 0 (static)                                                                                                                                                   | NC_FLOAT    |
|                                                                           | <b>valid_min</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 0 (static)                                                                                                                                                   | NC_UBYTE    |
|                                                                           | <b>valid_max</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 100 (static)                                                                                                                                                 | NC_UBYTE    |
|                                                                           | <b>long_name</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 'data quality value for the sulfur dioxide layer height retrieval' (static)                                                                                  | NC_STRING   |
|                                                                           | <b>comment</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 'A continuous quality descriptor, varying between 0 (no data) and 1 (full quality data). Recommend to ignore data with qa_value_layer_height < 0.5' (static) | NC_STRING   |
|                                                                           | <b>coordinates</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                                                                              | NC_STRING   |
| <b>sulfurdioxide_layer_height_flag</b> in SO2___/PRODUCT/SO2_LAYER_HEIGHT |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                              |             |
| Description:                                                              | Pixel-wise flag which indicates if a layer height retrieval has been performed and if the quality may be reduced. The value is additive, hence several conditions can apply. Note that only values lower than 32 are successfull retrievals. The flag is: 0 for successfull SO2 layer height retrieval, 1 for exceeded input data range, 2 for layer height exceeding height of 30 km (out of training range), 4 for high aerosol absorption, 8 for negative aerosol absorption (i.e. sulfates), 16 for retrieved VCD below 15 DU (below training range), 32 for retrieved LH below 0 km, 64 for LH retrieval not triggered, 128 for missing input data . |                                                                                                                                                              |             |
| Dimensions:                                                               | time, scanline, ground_pixel.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                              |             |
| Type:                                                                     | NC_USHORT.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                              |             |

|             |                                                                                                                                                                                              |                                                                                                                                                                             |             |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Source:     | Processor.                                                                                                                                                                                   |                                                                                                                                                                             |             |
| Attributes: | <i>Name</i>                                                                                                                                                                                  | <i>Value</i>                                                                                                                                                                | <i>Type</i> |
|             | <b>units</b>                                                                                                                                                                                 | '1' (static)                                                                                                                                                                | NC_STRING   |
|             | <b>long_name</b>                                                                                                                                                                             | 'sulfur dioxide layer height flag' (static)                                                                                                                                 | NC_STRING   |
|             | <b>flag_meanings</b>                                                                                                                                                                         | 'success input_range_exceeded layer_height_ - above_trained_range high_aerosol negative_aerosol low_so2_vcd negative_layer_height no_retrieval input_data_missing' (static) | NC_STRING   |
|             | <b>flag_values</b>                                                                                                                                                                           | 0,1,2,4,8,16,32,64,128 (static)                                                                                                                                             | NC_USHORT   |
|             | <b>coordinates</b>                                                                                                                                                                           | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                                                                                             | NC_STRING   |
|             | The latitude and longitude are in a different group. How to specify the related geospatial coordinates in this case is not specified in the climate and forecast metadata conventions [ER5]. |                                                                                                                                                                             |             |

## 12.1.2 Group "SUPPORT\_DATA" in "PRODUCT"

### 12.1.2.1 Group "GEOLOCATIONS" in "SUPPORT\_DATA"

#### Variables in SO2\_\_\_/PRODUCT/SUPPORT\_DATA/GEOLOCATIONS

##### **satellite\_latitude** in SO2\_\_\_/PRODUCT/SUPPORT\_DATA/GEOLOCATIONS

Description: Latitude of the geodetic sub satellite point on the WGS84 reference ellipsoid.

Dimensions: time, scanline.

Type: NC\_FLOAT.

Source: L1B.

|             |                  |                                                                                          |             |
|-------------|------------------|------------------------------------------------------------------------------------------|-------------|
| Attributes: | <i>Name</i>      | <i>Value</i>                                                                             | <i>Type</i> |
|             | <b>long_name</b> | 'sub satellite latitude' (static)                                                        | NC_STRING   |
|             | <b>units</b>     | 'degrees_north' (static)                                                                 | NC_STRING   |
|             | <b>comment</b>   | 'Latitude of the geodetic sub satellite point on the WGS84 reference ellipsoid' (static) | NC_STRING   |
|             | <b>valid_min</b> | -90.0 (static)                                                                           | NC_FLOAT    |
|             | <b>valid_max</b> | 90.0 (static)                                                                            | NC_FLOAT    |

##### **satellite\_longitude** in SO2\_\_\_/PRODUCT/SUPPORT\_DATA/GEOLOCATIONS

Description: Longitude of the geodetic sub satellite point on the WGS84 reference ellipsoid.

Dimensions: time, scanline.

Type: NC\_FLOAT.

Source: L1B.

|             |                  |                                                                                           |             |
|-------------|------------------|-------------------------------------------------------------------------------------------|-------------|
| Attributes: | <i>Name</i>      | <i>Value</i>                                                                              | <i>Type</i> |
|             | <b>long_name</b> | 'satellite_longitude' (static)                                                            | NC_STRING   |
|             | <b>units</b>     | 'degrees_east' (static)                                                                   | NC_STRING   |
|             | <b>comment</b>   | 'Longitude of the geodetic sub satellite point on the WGS84 reference ellipsoid' (static) | NC_STRING   |
|             | <b>valid_min</b> | -180.0 (static)                                                                           | NC_FLOAT    |
|             | <b>valid_max</b> | 180.0 (static)                                                                            | NC_FLOAT    |

##### **satellite\_altitude** in SO2\_\_\_/PRODUCT/SUPPORT\_DATA/GEOLOCATIONS

Description: The altitude of the satellite with respect to the geodetic sub satellite point on the WGS84 reference ellipsoid.

|                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                 |             |
|--------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|-------------|
| Dimensions:                                                              | time, scanline.                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                 |             |
| Type:                                                                    | NC_FLOAT.                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                 |             |
| Source:                                                                  | L1B.                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                 |             |
| Attributes:                                                              | <i>Name</i>                                                                                                                                                                                                                                                                                                                                                                                                                               | <i>Value</i>                                                                                                                    | <i>Type</i> |
|                                                                          | <b>long_name</b>                                                                                                                                                                                                                                                                                                                                                                                                                          | 'satellite altitude' (static)                                                                                                   | NC_STRING   |
|                                                                          | <b>units</b>                                                                                                                                                                                                                                                                                                                                                                                                                              | 'm' (static)                                                                                                                    | NC_STRING   |
|                                                                          | <b>comment</b>                                                                                                                                                                                                                                                                                                                                                                                                                            | 'The altitude of the satellite with respect to the geodetic sub satellite point on the WGS84 reference ellipsoid' (static)      | NC_STRING   |
|                                                                          | <b>valid_min</b>                                                                                                                                                                                                                                                                                                                                                                                                                          | 700000.0 (static)                                                                                                               | NC_FLOAT    |
|                                                                          | <b>valid_max</b>                                                                                                                                                                                                                                                                                                                                                                                                                          | 900000.0 (static)                                                                                                               | NC_FLOAT    |
| <b>satellite_orbit_phase</b> in SO2___/PRODUCT/SUPPORT_DATA/GEOLOCATIONS |                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                 |             |
| Description:                                                             | Relative offset [0.0, ..., 1.0] of the measurement in the orbit.                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                 |             |
| Dimensions:                                                              | time, scanline.                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                 |             |
| Type:                                                                    | NC_FLOAT.                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                 |             |
| Source:                                                                  | L1B.                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                 |             |
| Attributes:                                                              | <i>Name</i>                                                                                                                                                                                                                                                                                                                                                                                                                               | <i>Value</i>                                                                                                                    | <i>Type</i> |
|                                                                          | <b>long_name</b>                                                                                                                                                                                                                                                                                                                                                                                                                          | 'fractional satellite orbit phase' (static)                                                                                     | NC_STRING   |
|                                                                          | <b>units</b>                                                                                                                                                                                                                                                                                                                                                                                                                              | '1' (static)                                                                                                                    | NC_STRING   |
|                                                                          | <b>comment</b>                                                                                                                                                                                                                                                                                                                                                                                                                            | 'Relative offset [0.0, ..., 1.0] of the measurement in the orbit' (static)                                                      | NC_STRING   |
|                                                                          | <b>valid_min</b>                                                                                                                                                                                                                                                                                                                                                                                                                          | -0.02 (static)                                                                                                                  | NC_FLOAT    |
|                                                                          | <b>valid_max</b>                                                                                                                                                                                                                                                                                                                                                                                                                          | 1.02 (static)                                                                                                                   | NC_FLOAT    |
| <b>solar_zenith_angle</b> in SO2___/PRODUCT/SUPPORT_DATA/GEOLOCATIONS    |                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                 |             |
| Description:                                                             | Solar zenith angle $\vartheta_0$ at the ground pixel location on the reference ellipsoid. Angle is measured away from the vertical. ESA definition of day side: $\vartheta_0 < 92^\circ$ . Pixels are processed when $\vartheta_0 \leq \vartheta_0^{\max}$ with $80^\circ \leq \vartheta_0^{\max} \leq 88^\circ$ , depending on the algorithm. The actual value for $\vartheta_0^{\max}$ can be found in the algorithm metadata settings. |                                                                                                                                 |             |
| Dimensions:                                                              | time, scanline, ground_pixel.                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                 |             |
| Type:                                                                    | NC_FLOAT.                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                 |             |
| Source:                                                                  | L1B.                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                 |             |
| Attributes:                                                              | <i>Name</i>                                                                                                                                                                                                                                                                                                                                                                                                                               | <i>Value</i>                                                                                                                    | <i>Type</i> |
|                                                                          | <b>long_name</b>                                                                                                                                                                                                                                                                                                                                                                                                                          | 'solar zenith angle' (static)                                                                                                   | NC_STRING   |
|                                                                          | <b>standard_name</b>                                                                                                                                                                                                                                                                                                                                                                                                                      | 'solar_zenith_angle' (static)                                                                                                   | NC_STRING   |
|                                                                          | <b>units</b>                                                                                                                                                                                                                                                                                                                                                                                                                              | 'degree' (static)                                                                                                               | NC_STRING   |
|                                                                          | <b>valid_min</b>                                                                                                                                                                                                                                                                                                                                                                                                                          | 0.0 (static)                                                                                                                    | NC_FLOAT    |
|                                                                          | <b>valid_max</b>                                                                                                                                                                                                                                                                                                                                                                                                                          | 180.0 (static)                                                                                                                  | NC_FLOAT    |
|                                                                          | <b>coordinates</b>                                                                                                                                                                                                                                                                                                                                                                                                                        | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                                                 | NC_STRING   |
|                                                                          | The latitude and longitude are in a different group. How to specify the related geospatial coordinates in this case is not specified in the climate and forecast metadata conventions [ER5].                                                                                                                                                                                                                                              |                                                                                                                                 |             |
|                                                                          | <b>comment</b>                                                                                                                                                                                                                                                                                                                                                                                                                            | 'Solar zenith angle at the ground pixel location on the reference ellipsoid. Angle is measured away from the vertical' (static) | NC_STRING   |
| <b>solar_azimuth_angle</b> in SO2___/PRODUCT/SUPPORT_DATA/GEOLOCATIONS   |                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                 |             |
| Description:                                                             | The solar azimuth angle at the ground pixel location on the reference ellipsoid. The angle is measured clockwise from the North (North = 0°, East = 90°, South = ±180°, West = -90°). This is the same definition that is use in both OMI and GOME-2 level 1B files.                                                                                                                                                                      |                                                                                                                                 |             |

|                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                           |             |
|--------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
|                                                                          | See the note on the <code>viewing_azimuth_angle</code> on the calculation of the relative azimuth angle as used in radiative transfer calculations.                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                           |             |
| Dimensions:                                                              | time, scanline, ground_pixel.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                           |             |
| Type:                                                                    | NC_FLOAT.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                           |             |
| Source:                                                                  | L1B.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                           |             |
| Attributes:                                                              | <i>Name</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <i>Value</i>                                                                                                                                                                                                                                              | <i>Type</i> |
|                                                                          | <b>long_name</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 'solar azimuth angle' (static)                                                                                                                                                                                                                            | NC_STRING   |
|                                                                          | <b>standard_name</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 'solar_azimuth_angle' (static)                                                                                                                                                                                                                            | NC_STRING   |
|                                                                          | <b>units</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 'degree' (static)                                                                                                                                                                                                                                         | NC_STRING   |
|                                                                          | <b>valid_min</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | -180.0 (static)                                                                                                                                                                                                                                           | NC_FLOAT    |
|                                                                          | <b>valid_max</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 180.0 (static)                                                                                                                                                                                                                                            | NC_FLOAT    |
|                                                                          | <b>coordinates</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | '/PRODUCT/longitude /PRODUCT/latitude' (static) NC_STRING<br>The latitude and longitude are in a different group. How to specify the related geospatial coordinates in this case is not specified in the climate and forecast metadata conventions [ER5]. |             |
|                                                                          | <b>comment</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 'Solar azimuth angle at the ground pixel location on the reference ellipsoid. Angle is measured clockwise from the North (East = 90, South = +/-180, West = -90)' (static) NC_STRING                                                                      |             |
| <b>viewing_zenith_angle</b> in SO2___/PRODUCT/SUPPORT_DATA/GEOLOCATIONS  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                           |             |
| Description:                                                             | Zenith angle of the satellite $\vartheta$ at the ground pixel location on the reference ellipsoid. Angle is measured away from the vertical.                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                           |             |
| Dimensions:                                                              | time, scanline, ground_pixel.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                           |             |
| Type:                                                                    | NC_FLOAT.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                           |             |
| Source:                                                                  | L1B.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                           |             |
| Attributes:                                                              | <i>Name</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <i>Value</i>                                                                                                                                                                                                                                              | <i>Type</i> |
|                                                                          | <b>long_name</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 'viewing zenith angle' (static)                                                                                                                                                                                                                           | NC_STRING   |
|                                                                          | <b>standard_name</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 'viewing_zenith_angle' (static)                                                                                                                                                                                                                           | NC_STRING   |
|                                                                          | <b>units</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 'degree' (static)                                                                                                                                                                                                                                         | NC_STRING   |
|                                                                          | <b>valid_min</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 0.0 (static)                                                                                                                                                                                                                                              | NC_FLOAT    |
|                                                                          | <b>valid_max</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 180.0 (static)                                                                                                                                                                                                                                            | NC_FLOAT    |
|                                                                          | <b>coordinates</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | '/PRODUCT/longitude /PRODUCT/latitude' (static) NC_STRING<br>The latitude and longitude are in a different group. How to specify the related geospatial coordinates in this case is not specified in the climate and forecast metadata conventions [ER5]. |             |
|                                                                          | <b>comment</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 'Zenith angle of the satellite at the ground pixel location on the reference ellipsoid. Angle is measured away from the vertical' (static) NC_STRING                                                                                                      |             |
| <b>viewing_azimuth_angle</b> in SO2___/PRODUCT/SUPPORT_DATA/GEOLOCATIONS |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                           |             |
| Description:                                                             | The satellite azimuth angle at the ground pixel location on the reference ellipsoid. The angle is measured clockwise from the North (North = 0°, East = 90°, South = ±180°, West = −90°). This is the same definition that is use in both OMI and GOME-2 level 1B files.<br><br>To calculate the azimuth difference $\varphi - \varphi_0$ it is not sufficient to just subtract <code>solar_azimuth_angle</code> from <code>viewing_azimuth_angle</code> . The angle needed for radiative transfer calculations is $(180^\circ - (\varphi - \varphi_0)) \bmod 360^\circ$ . |                                                                                                                                                                                                                                                           |             |
| Dimensions:                                                              | time, scanline, ground_pixel.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                           |             |
| Type:                                                                    | NC_FLOAT.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                           |             |
| Source:                                                                  | L1B.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                           |             |
| Attributes:                                                              | <i>Name</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <i>Value</i>                                                                                                                                                                                                                                              | <i>Type</i> |
|                                                                          | <b>long_name</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 'viewing azimuth angle' (static)                                                                                                                                                                                                                          | NC_STRING   |

|                      |                                                                                                                                                                                                                                                           |           |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>standard_name</b> | 'viewing_azimuth_angle' (static)                                                                                                                                                                                                                          | NC_STRING |
| <b>units</b>         | 'degree' (static)                                                                                                                                                                                                                                         | NC_STRING |
| <b>valid_min</b>     | -180.0 (static)                                                                                                                                                                                                                                           | NC_FLOAT  |
| <b>valid_max</b>     | 180.0 (static)                                                                                                                                                                                                                                            | NC_FLOAT  |
| <b>coordinates</b>   | '/PRODUCT/longitude /PRODUCT/latitude' (static) NC_STRING<br>The latitude and longitude are in a different group. How to specify the related geospatial coordinates in this case is not specified in the climate and forecast metadata conventions [ER5]. |           |
| <b>comment</b>       | 'Satellite azimuth angle at the ground pixel location on the reference ellipsoid. Angle is measured clockwise from the North (East = 90, South = +/- 180, West = -90)' (static) NC_STRING                                                                 |           |

**latitude\_bounds** in SO2\_\_\_/PRODUCT/SUPPORT\_DATA/GEOLOCATIONS

Description: The latitude of the pixel corners of the ground pixels in the data. Latitude, longitude coordinates for the ground pixel center and the ground pixel corners are calculated at the WGS84 ellipsoid.

The order of the pixel corners follows the CF-metadata conventions [ER5, section 7.1], i.e. the ordering is counter-clockwise when viewed from above. A graphical representation is given in figure 7.

Dimensions: time, scanline, ground\_pixel, corner.

Type: NC\_FLOAT.

Source: Processor.

|             |              |                          |             |
|-------------|--------------|--------------------------|-------------|
| Attributes: | <i>Name</i>  | <i>Value</i>             | <i>Type</i> |
|             | <b>units</b> | 'degrees_north' (static) | NC_STRING   |

**longitude\_bounds** in SO2\_\_\_/PRODUCT/SUPPORT\_DATA/GEOLOCATIONS

Description: The longitude of the pixel corners of the ground pixels in the data. Latitude, longitude coordinates for the ground pixel center and the ground pixel corners are calculated at the WGS84 ellipsoid.

The order of the pixel corners follows the CF-metadata conventions [ER5, section 7.1], i.e. the ordering is counter-clockwise when viewed from above. A graphical representation is given in figure 7.

Dimensions: time, scanline, ground\_pixel, corner.

Type: NC\_FLOAT.

Source: Processor.

|             |              |                         |             |
|-------------|--------------|-------------------------|-------------|
| Attributes: | <i>Name</i>  | <i>Value</i>            | <i>Type</i> |
|             | <b>units</b> | 'degrees_east' (static) | NC_STRING   |

**geolocation\_flags** in SO2\_\_\_/PRODUCT/SUPPORT\_DATA/GEOLOCATIONS

Description: Additional flags describing the ground pixel, including the influence of a solar eclipse, the possibility of sun glint, whether we are in the descending part of the orbit, whether we are on the night side of the orbit, whether the pixel crosses the dateline (useful for plotting), or if there was some geolocation error.

Dimensions: time, scanline, ground\_pixel.

Type: NC\_UBYTE.

Source: Processor.

|             |                    |                                                 |             |
|-------------|--------------------|-------------------------------------------------|-------------|
| Attributes: | <i>Name</i>        | <i>Value</i>                                    | <i>Type</i> |
|             | <b>_FillValue</b>  | 255 (static)                                    | NC_UBYTE    |
|             | <b>coordinates</b> | '/PRODUCT/longitude /PRODUCT/latitude' (static) | NC_STRING   |
|             | <b>flag_masks</b>  | 0, 1, 2, 4, 8, 16, 32, 128 (static)             | NC_UBYTE    |

|                      |                                                                                                                                    |           |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>flag_meanings</b> | 'no_error solar_eclipse sun_glint_possible descending night geo_boundary_crossing spacecraft_manoeuvre geolocation_error' (static) | NC_STRING |
| <b>flag_values</b>   | 0, 1, 2, 4, 8, 16, 32, 128 (static)                                                                                                | NC_UBYTE  |
| <b>long_name</b>     | 'geolocation flags' (static)                                                                                                       | NC_STRING |
| <b>max_val</b>       | 254 (static)                                                                                                                       | NC_UBYTE  |
| <b>min_val</b>       | 0 (static)                                                                                                                         | NC_UBYTE  |
| <b>units</b>         | '1' (static)                                                                                                                       | NC_STRING |

#### 12.1.2.2 Group "DETAILED\_RESULTS" in "SUPPORT\_DATA"

##### Dimensions in SO2\_\_\_/PRODUCT/SUPPORT\_DATA/DETAILED\_RESULTS

**number\_of\_slant\_columns\_win1** Fitted slant column densities of all fitted species in fitting window 1.

**size** 1 (dynamic)

**source** Processor.

**number\_of\_slant\_columns\_win2** Fitted slant column densities of all fitted species in fitting window 2.

**size** 1 (dynamic)

**source** Processor.

**number\_of\_slant\_columns\_win3** Fitted slant column densities of all fitted species in fitting window 3.

**size** 1 (dynamic)

**source** Processor.

**number\_of\_doas\_polynomial\_coefficients\_win1** The number of the DOAS polynomial coefficients for fitting window 1

**size** 1 (dynamic)

**source** Processor.

**number\_of\_doas\_polynomial\_coefficients\_win2** The number of the DOAS polynomial coefficients for fitting window 2

**size** 1 (dynamic)

**source** Processor.

**number\_of\_doas\_polynomial\_coefficients\_win3** The number of the DOAS polynomial coefficients for fitting window 3

**size** 1 (dynamic)

**source** Processor.

**debug\_upas2\_levels\_1** Level Dim for debugging porpuse.

**size** -1 (fixed)

##### Variables in SO2\_\_\_/PRODUCT/SUPPORT\_DATA/DETAILED\_RESULTS

**sulfurdioxide\_total\_vertical\_column\_trueness** in SO2\_\_\_/PRODUCT/SUPPORT\_DATA/DETAILED\_RESULTS

**Description:** Systematic error of the vertical column density of the sulphur dioxide SO<sub>2</sub> product for the polluted scenario (anthropogenic SO<sub>2</sub> in the PBL).

**Dimensions:** time, scanline, ground\_pixel.

**Type:** NC\_FLOAT.

**Source:** Processor.

|                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                              |             |
|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|-------------|
| Attributes:                                                                                | <i>Name</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <i>Value</i>                                                                                                 | <i>Type</i> |
|                                                                                            | <b>units</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 'mol m-2' (static)                                                                                           | NC_STRING   |
|                                                                                            | <b>long_name</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 'systematic error of the total vertical column density of sulfur dioxide for the polluted scenario' (static) | NC_STRING   |
|                                                                                            | <b>coordinates</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                              | NC_STRING   |
|                                                                                            | The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. This attribute originates from the CF standard.                                                                                                                                                                                                                                                                                                                |                                                                                                              |             |
|                                                                                            | <b>multiplication_factor_to_convert_to_DU</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                    | %(MOLM2_TO_DU)s (static)                                                                                     | NC_FLOAT    |
|                                                                                            | The quantities in Sentinel 5 precursor files are given in SI units. For an integrated column value this means that the unit is mol m <sup>-2</sup> . Traditionally the unit for an integrated column is "DU" or Dobson Units. This attribute provides the multiplication factor to calculate the total column in DU from the value in mol m <sup>-2</sup> . This is provided as a convenience to users who have tools that work in DU.                                                           |                                                                                                              |             |
|                                                                                            | <b>multiplication_factor_to_convert_to_molecules_percm2</b>                                                                                                                                                                                                                                                                                                                                                                                                                                      | 6.02214e+19 (static)                                                                                         | NC_FLOAT    |
|                                                                                            | The quantities in Sentinel 5 precursor files are given in SI units. For an integrated column value this means that the unit is mol m <sup>-2</sup> . Traditionally the unit for an integrated column is "molecules cm <sup>-2</sup> ". This attribute provides the multiplication factor to calculate the total column in molecules cm <sup>-2</sup> from the value in mol m <sup>-2</sup> . This is provided as a convenience to users who have tools that work in molecules cm <sup>-2</sup> . |                                                                                                              |             |
| <b>fitted_slant_columns_win1</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                              |             |
| Description:                                                                               | Fitted slant column densities $N^s$ of all fitted species in fitting window 1.                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                              |             |
| Dimensions:                                                                                | time, scanline, ground_pixel, number_of_slant_columns_win1.                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                              |             |
| Type:                                                                                      | NC_DOUBLE.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                              |             |
| Source:                                                                                    | Processor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                              |             |
| Attributes:                                                                                | <i>Name</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <i>Value</i>                                                                                                 | <i>Type</i> |
|                                                                                            | <b>units</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 'mol m-2' (static)                                                                                           | NC_STRING   |
|                                                                                            | <b>long_name</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 'slant columns of all absorbers in fitting window 1' (static)                                                | NC_STRING   |
|                                                                                            | <b>ancillary_variables</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 'fitted_slant_columns_win1_precision' (static)                                                               | NC_STRING   |
|                                                                                            | <b>coordinates</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                              | NC_STRING   |
|                                                                                            | The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. This attribute originates from the CF standard.                                                                                                                                                                                                                                                                                                                |                                                                                                              |             |
|                                                                                            | <b>index_meaning</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | '1' (dynamic)                                                                                                | NC_STRING   |
|                                                                                            | This attribute provides the meaning of the indexes for the current variable. Indexes are supposed to be divided by a blank space.                                                                                                                                                                                                                                                                                                                                                                |                                                                                                              |             |
| <b>fitted_slant_columns_win1_precision</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                              |             |
| Description:                                                                               | Random error of the slant column densities $N^s$ of all fitted species in fitting window 1                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                              |             |
| Dimensions:                                                                                | time, scanline, ground_pixel, number_of_slant_columns_win1.                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                              |             |
| Type:                                                                                      | NC_FLOAT.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                              |             |
| Source:                                                                                    | Processor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                              |             |
| Attributes:                                                                                | <i>Name</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <i>Value</i>                                                                                                 | <i>Type</i> |
|                                                                                            | <b>units</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 'mol m-2' (static)                                                                                           | NC_STRING   |
|                                                                                            | <b>long_name</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 'slant column random errors of all absorbers in fitting window 1' (static)                                   | NC_STRING   |
|                                                                                            | <b>coordinates</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                              | NC_STRING   |

|                                                                                                                                                                                   |                                                                                          |                                                                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. This attribute originates from the CF standard. |                                                                                          |                                                                  |
| <b>index_meaning</b>                                                                                                                                                              | '1' (dynamic)                                                                            | NC_STRING                                                        |
| This attribute provides the meaning of the indexes for the current variable. Indexes are supposed to be divided by a blank space.                                                 |                                                                                          |                                                                  |
| <b>fitted_slant_columns_win2</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS                                                                                                  |                                                                                          |                                                                  |
| Description:                                                                                                                                                                      | Slant column densities $N^s$ of all fitted species in fitting window 2                   |                                                                  |
| Dimensions:                                                                                                                                                                       | time, scanline, ground_pixel, number_of_slant_columns_win2.                              |                                                                  |
| Type:                                                                                                                                                                             | NC_DOUBLE.                                                                               |                                                                  |
| Source:                                                                                                                                                                           | Processor.                                                                               |                                                                  |
| Attributes:                                                                                                                                                                       | <i>Name</i>                                                                              | <i>Value</i>                                                     |
|                                                                                                                                                                                   | <b>units</b>                                                                             | 'mol m-2' (static)                                               |
|                                                                                                                                                                                   | <b>long_name</b>                                                                         | 'slant columns of all absorbers in fitting window 2' (static)    |
|                                                                                                                                                                                   | <b>ancillary_variables</b>                                                               | 'fitted_slant_columns_win2_precision' (static)                   |
|                                                                                                                                                                                   | <b>coordinates</b>                                                                       | '/PRODUCT/longitude /PRODUCT/latitude' (static)                  |
| The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. This attribute originates from the CF standard. |                                                                                          |                                                                  |
|                                                                                                                                                                                   | <b>index_meaning</b>                                                                     | '1' (dynamic)                                                    |
| This attribute provides the meaning of the indexes for the current variable. Indexes are supposed to be divided by a blank space.                                                 |                                                                                          |                                                                  |
| <b>fitted_slant_columns_win2_precision</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS                                                                                        |                                                                                          |                                                                  |
| Description:                                                                                                                                                                      | Random error of the slant column density $N^s$ of all fitted species in fitting window 2 |                                                                  |
| Dimensions:                                                                                                                                                                       | time, scanline, ground_pixel, number_of_slant_columns_win2.                              |                                                                  |
| Type:                                                                                                                                                                             | NC_FLOAT.                                                                                |                                                                  |
| Source:                                                                                                                                                                           | Processor.                                                                               |                                                                  |
| Attributes:                                                                                                                                                                       | <i>Name</i>                                                                              | <i>Value</i>                                                     |
|                                                                                                                                                                                   | <b>units</b>                                                                             | 'mol m-2' (static)                                               |
|                                                                                                                                                                                   | <b>long_name</b>                                                                         | 'slant column random all absorbers in fitting window 2' (static) |
|                                                                                                                                                                                   | <b>coordinates</b>                                                                       | '/PRODUCT/longitude /PRODUCT/latitude' (static)                  |
| The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. This attribute originates from the CF standard. |                                                                                          |                                                                  |
|                                                                                                                                                                                   | <b>index_meaning</b>                                                                     | '1' (dynamic)                                                    |
| This attribute provides the meaning of the indexes for the current variable. Indexes are supposed to be divided by a blank space.                                                 |                                                                                          |                                                                  |
| <b>fitted_slant_columns_win3</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS                                                                                                  |                                                                                          |                                                                  |
| Description:                                                                                                                                                                      | Slant column density $N^s$ of all fitted species in fitting window 3                     |                                                                  |
| Dimensions:                                                                                                                                                                       | time, scanline, ground_pixel, number_of_slant_columns_win3.                              |                                                                  |
| Type:                                                                                                                                                                             | NC_DOUBLE.                                                                               |                                                                  |
| Source:                                                                                                                                                                           | Processor.                                                                               |                                                                  |
| Attributes:                                                                                                                                                                       | <i>Name</i>                                                                              | <i>Value</i>                                                     |
|                                                                                                                                                                                   | <b>units</b>                                                                             | 'mol m-2' (static)                                               |
|                                                                                                                                                                                   | <b>long_name</b>                                                                         | 'slant columns of all absorbers in fitting window 3' (static)    |
|                                                                                                                                                                                   | <b>ancillary_variables</b>                                                               | 'fitted_slant_columns_win3_precision' (static)                   |
|                                                                                                                                                                                   | <b>coordinates</b>                                                                       | '/PRODUCT/longitude /PRODUCT/latitude' (static)                  |

|                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. This attribute originates from the CF standard. |                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                       |
| <b>index_meaning</b>                                                                                                                                                              | '1' (dynamic)                                                                                                                                                                                                                                                                                                                                                                  | NC_STRING                                                                                             |
| This attribute provides the meaning of the indexes for the current variable. Indexes are supposed to be divided by a blank space.                                                 |                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                       |
| <b>fitted_slant_columns_win3_precision</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                       |
| Description:                                                                                                                                                                      | Random error of the slant column density $N^s$ of all fitted species in fitting window 3                                                                                                                                                                                                                                                                                       |                                                                                                       |
| Dimensions:                                                                                                                                                                       | time, scanline, ground_pixel, number_of_slant_columns_win3.                                                                                                                                                                                                                                                                                                                    |                                                                                                       |
| Type:                                                                                                                                                                             | NC_FLOAT.                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                       |
| Source:                                                                                                                                                                           | Processor.                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                       |
| Attributes:                                                                                                                                                                       | <i>Name</i>                                                                                                                                                                                                                                                                                                                                                                    | <i>Value</i>                                                                                          |
|                                                                                                                                                                                   | <b>units</b>                                                                                                                                                                                                                                                                                                                                                                   | 'mol m-2' (static)                                                                                    |
|                                                                                                                                                                                   | <b>long_name</b>                                                                                                                                                                                                                                                                                                                                                               | 'slant column random errors of all absorbers in fitting window 3' (static)                            |
|                                                                                                                                                                                   | <b>coordinates</b>                                                                                                                                                                                                                                                                                                                                                             | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                       |
| The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. This attribute originates from the CF standard. |                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                       |
|                                                                                                                                                                                   | <b>index_meaning</b>                                                                                                                                                                                                                                                                                                                                                           | '1' (dynamic)                                                                                         |
| This attribute provides the meaning of the indexes for the current variable. Indexes are supposed to be divided by a blank space.                                                 |                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                       |
| <b>selected_fitting_window_flag</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                       |
| Description:                                                                                                                                                                      | Flag which gives information about the selected window while computing the slant column density from DOAS.                                                                                                                                                                                                                                                                     |                                                                                                       |
| Dimensions:                                                                                                                                                                       | time, scanline, ground_pixel.                                                                                                                                                                                                                                                                                                                                                  |                                                                                                       |
| Type:                                                                                                                                                                             | NC_INT.                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                       |
| Source:                                                                                                                                                                           | Processor.                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                       |
| Attributes:                                                                                                                                                                       | <i>Name</i>                                                                                                                                                                                                                                                                                                                                                                    | <i>Value</i>                                                                                          |
|                                                                                                                                                                                   | <b>units</b>                                                                                                                                                                                                                                                                                                                                                                   | '1' (static)                                                                                          |
|                                                                                                                                                                                   | <b>long_name</b>                                                                                                                                                                                                                                                                                                                                                               | 'flag describing the selected fitting window for sulfur dioxide retrieval' (static)                   |
|                                                                                                                                                                                   | <b>coordinates</b>                                                                                                                                                                                                                                                                                                                                                             | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                       |
| The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. This attribute originates from the CF standard. |                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                       |
|                                                                                                                                                                                   | <b>flag_values</b>                                                                                                                                                                                                                                                                                                                                                             | 1,2,3 (static)                                                                                        |
| <b>sulfurdioxide_slant_column_corrected</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                       |
| Description:                                                                                                                                                                      | SO <sub>2</sub> slant column density $N^s$ for the final selected fitting window, corrected by the background correction algorithm. The global attribute "Status_BG" indicates if the slant column density has been corrected ("Nominal") or untouched ("Fallback"). Note that if the "Status_BG" attribute is set to "Fallback", the data will be set entirely as fill value. |                                                                                                       |
| Dimensions:                                                                                                                                                                       | time, scanline, ground_pixel.                                                                                                                                                                                                                                                                                                                                                  |                                                                                                       |
| Type:                                                                                                                                                                             | NC_FLOAT.                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                       |
| Source:                                                                                                                                                                           | Processor.                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                       |
| Attributes:                                                                                                                                                                       | <i>Name</i>                                                                                                                                                                                                                                                                                                                                                                    | <i>Value</i>                                                                                          |
|                                                                                                                                                                                   | <b>units</b>                                                                                                                                                                                                                                                                                                                                                                   | 'mol m-2' (static)                                                                                    |
|                                                                                                                                                                                   | <b>long_name</b>                                                                                                                                                                                                                                                                                                                                                               | 'background corrected sulfur dioxide slant column density for final selected fitting window' (static) |
|                                                                                                                                                                                   | <b>ancillary_variables</b>                                                                                                                                                                                                                                                                                                                                                     | 'sulfurdioxide_slant_column_corrected_trueness' (static)                                              |
|                                                                                                                                                                                   | <b>coordinates</b>                                                                                                                                                                                                                                                                                                                                                             | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                       |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                          |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. This attribute originates from the CF standard.                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                          |
| <b>multiplication_factor_to_convert_to_DU</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                    | %(MOLM2_TO_DU)s (static)                                                                                                                                                                                                                                                                                                                                                                           | NC_FLOAT                                                                                 |
| The quantities in Sentinel 5 precursor files are given in SI units. For an integrated column value this means that the unit is mol m <sup>-2</sup> . Traditionally the unit for an integrated column is “DU” or Dobson Units. This attribute provides the multiplication factor to calculate the total column in DU from the value in mol m <sup>-2</sup> . This is provided as a convenience to users who have tools that work in DU.                                                           |                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                          |
| <b>multiplication_factor_to_convert_to_molecules_per_cm2</b>                                                                                                                                                                                                                                                                                                                                                                                                                                     | 6.02214e+19 (static)                                                                                                                                                                                                                                                                                                                                                                               | NC_FLOAT                                                                                 |
| The quantities in Sentinel 5 precursor files are given in SI units. For an integrated column value this means that the unit is mol m <sup>-2</sup> . Traditionally the unit for an integrated column is “molecules cm <sup>-2</sup> ”. This attribute provides the multiplication factor to calculate the total column in molecules cm <sup>-2</sup> from the value in mol m <sup>-2</sup> . This is provided as a convenience to users who have tools that work in molecules cm <sup>-2</sup> . |                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                          |
| <b>sulfurdioxide_slant_column_corrected_trueness</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                          |
| Description:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Systematic error of SO <sub>2</sub> slant column density $N^s$ for the final selected fitting window, corrected by the background correction algorithm. The global attribute “Status_BG” indicates if the slant column density has been corrected (“Nominal”) or untouched (“Fallback”). Note that if the “Status_BG” attribute is set to “Fallback”, the data will be set entirely as fill value. |                                                                                          |
| Dimensions:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | time, scanline, ground_pixel.                                                                                                                                                                                                                                                                                                                                                                      |                                                                                          |
| Type:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | NC_FLOAT.                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                          |
| Source:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Processor.                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                          |
| Attributes:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <i>Name</i>                                                                                                                                                                                                                                                                                                                                                                                        | <i>Value</i>                                                                             |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>units</b>                                                                                                                                                                                                                                                                                                                                                                                       | ‘mol m-2’ (static)                                                                       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>long_name</b>                                                                                                                                                                                                                                                                                                                                                                                   | ‘systematic error of the corrected sulfur dioxide slant column’ (static)                 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>coordinates</b>                                                                                                                                                                                                                                                                                                                                                                                 | ‘/PRODUCT/longitude /PRODUCT/latitude’ (static)                                          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. This attribute originates from the CF standard.                                                                                                                                                                                                                  |                                                                                          |
| <b>sulfurdioxide_slant_column_corrected_win1</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                          |
| Description:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | SO <sub>2</sub> slant column density $N^s$ fitted in fitting window 1 and corrected by the background correction algorithm.                                                                                                                                                                                                                                                                        |                                                                                          |
| Dimensions:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | time, scanline, ground_pixel.                                                                                                                                                                                                                                                                                                                                                                      |                                                                                          |
| Type:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | NC_FLOAT.                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                          |
| Source:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Processor.                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                          |
| Attributes:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <i>Name</i>                                                                                                                                                                                                                                                                                                                                                                                        | <i>Value</i>                                                                             |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>units</b>                                                                                                                                                                                                                                                                                                                                                                                       | ‘mol m-2’ (static)                                                                       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>long_name</b>                                                                                                                                                                                                                                                                                                                                                                                   | ‘background corrected sulfur dioxide slant column density for fitting window 1’ (static) |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>coordinates</b>                                                                                                                                                                                                                                                                                                                                                                                 | ‘/PRODUCT/longitude /PRODUCT/latitude’ (static)                                          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. This attribute originates from the CF standard.                                                                                                                                                                                                                  |                                                                                          |
| <b>multiplication_factor_to_convert to DU</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                    | %(MOLM2_TO_DU)s (static)                                                                                                                                                                                                                                                                                                                                                                           | NC_FLOAT                                                                                 |

The quantities in Sentinel 5 precursor files are given in SI units. For an integrated column value this means that the unit is  $\text{mol m}^{-2}$ . Traditionally the unit for an integrated column is “DU” or Dobson Units. This attribute provides the multiplication factor to calculate the total column in DU from the value in  $\text{mol m}^{-2}$ . This is provided as a convenience to users who have tools that work in DU.

|                                                                |                      |          |
|----------------------------------------------------------------|----------------------|----------|
| <b>multiplication_ - factor_to_convert_to_molecules_percm2</b> | 6.02214e+19 (static) | NC_FLOAT |
|----------------------------------------------------------------|----------------------|----------|

The quantities in Sentinel 5 precursor files are given in SI units. For an integrated column value this means that the unit is  $\text{mol m}^{-2}$ . Traditionally the unit for an integrated column is “molecules  $\text{cm}^{-2}$ ”. This attribute provides the multiplication factor to calculate the total column in molecules  $\text{cm}^{-2}$  from the value in  $\text{mol m}^{-2}$ . This is provided as a convenience to users who have tools that work in molecules  $\text{cm}^{-2}$ .

**sulfurdioxide\_slant\_column\_corrected\_win2** in SO<sub>2</sub>\_\_\_/PRODUCT/SUPPORT\_DATA/DETAILED\_RESULTS

Description: SO<sub>2</sub> slant column density  $N^s$  fitted in fitting window 2 and corrected by the background correction algorithm.

Dimensions: time, scanline, ground\_pixel.

Type: NC\_FLOAT.

Source: Processor.

| Attributes: | Name                                                                                                                                                                              | Value                                                                                    | Type      |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|-----------|
|             | <b>units</b>                                                                                                                                                                      | ‘mol m-2’ (static)                                                                       | NC_STRING |
|             | <b>long_name</b>                                                                                                                                                                  | ‘background corrected sulfur dioxide slant column density for fitting window 2’ (static) | NC_STRING |
|             | <b>coordinates</b>                                                                                                                                                                | ‘/PRODUCT/longitude /PRODUCT/latitude’ (static)                                          | NC_STRING |
|             | The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. This attribute originates from the CF standard. |                                                                                          |           |

|                                                  |                          |          |
|--------------------------------------------------|--------------------------|----------|
| <b>multiplication_ - factor_to_convert_to_DU</b> | %(MOLM2_TO_DU)s (static) | NC_FLOAT |
|--------------------------------------------------|--------------------------|----------|

The quantities in Sentinel 5 precursor files are given in SI units. For an integrated column value this means that the unit is  $\text{mol m}^{-2}$ . Traditionally the unit for an integrated column is “DU” or Dobson Units. This attribute provides the multiplication factor to calculate the total column in DU from the value in  $\text{mol m}^{-2}$ . This is provided as a convenience to users who have tools that work in DU.

|                                                                |                      |          |
|----------------------------------------------------------------|----------------------|----------|
| <b>multiplication_ - factor_to_convert_to_molecules_percm2</b> | 6.02214e+19 (static) | NC_FLOAT |
|----------------------------------------------------------------|----------------------|----------|

The quantities in Sentinel 5 precursor files are given in SI units. For an integrated column value this means that the unit is  $\text{mol m}^{-2}$ . Traditionally the unit for an integrated column is “molecules  $\text{cm}^{-2}$ ”. This attribute provides the multiplication factor to calculate the total column in molecules  $\text{cm}^{-2}$  from the value in  $\text{mol m}^{-2}$ . This is provided as a convenience to users who have tools that work in molecules  $\text{cm}^{-2}$ .

**sulfurdioxide\_slant\_column\_corrected\_win3** in SO<sub>2</sub>\_\_\_/PRODUCT/SUPPORT\_DATA/DETAILED\_RESULTS

Description: SO<sub>2</sub> slant column density  $N^s$  fitted in fitting window 3 and corrected by the background correction algorithm.

Dimensions: time, scanline, ground\_pixel.

Type: NC\_FLOAT.

Source: Processor.

|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                          |             |  |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|-------------|--|
| Attributes:  | <i>Name</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <i>Value</i>                                                                             | <i>Type</i> |  |
|              | <b>units</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 'mol m-2' (static)                                                                       | NC_STRING   |  |
|              | <b>long_name</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 'background corrected sulfur dioxide slant column density for fitting window 3' (static) | NC_STRING   |  |
|              | <b>coordinates</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                          | NC_STRING   |  |
|              | The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. This attribute originates from the CF standard.                                                                                                                                                                                                                                                                                                                |                                                                                          |             |  |
|              | <b>multiplication_factor_to_convert_to_DU</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                    | %(MOLM2_TO_DU)s (static)                                                                 | NC_FLOAT    |  |
|              | The quantities in Sentinel 5 precursor files are given in SI units. For an integrated column value this means that the unit is mol m <sup>-2</sup> . Traditionally the unit for an integrated column is "DU" or Dobson Units. This attribute provides the multiplication factor to calculate the total column in DU from the value in mol m <sup>-2</sup> . This is provided as a convenience to users who have tools that work in DU.                                                           |                                                                                          |             |  |
|              | <b>multiplication_factor_to_convert_to_molecules_percm2</b>                                                                                                                                                                                                                                                                                                                                                                                                                                      | 6.02214e+19 (static)                                                                     | NC_FLOAT    |  |
|              | The quantities in Sentinel 5 precursor files are given in SI units. For an integrated column value this means that the unit is mol m <sup>-2</sup> . Traditionally the unit for an integrated column is "molecules cm <sup>-2</sup> ". This attribute provides the multiplication factor to calculate the total column in molecules cm <sup>-2</sup> from the value in mol m <sup>-2</sup> . This is provided as a convenience to users who have tools that work in molecules cm <sup>-2</sup> . |                                                                                          |             |  |
|              | <b>fitted_root_mean_square_win1</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                          |             |  |
| Description: | Root mean square value of doas result for fitting window 1.                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                          |             |  |
| Dimensions:  | time, scanline, ground_pixel.                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                          |             |  |
| Type:        | NC_FLOAT.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                          |             |  |
| Source:      | Processor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                          |             |  |
| Attributes:  | <i>Name</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <i>Value</i>                                                                             | <i>Type</i> |  |
|              | <b>units</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | '1' (static)                                                                             | NC_STRING   |  |
|              | <b>long_name</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 'root mean square residual of the fit in fitting window 1' (static)                      | NC_STRING   |  |
|              | <b>coordinates</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                          | NC_STRING   |  |
|              | The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. This attribute originates from the CF standard.                                                                                                                                                                                                                                                                                                                |                                                                                          |             |  |
|              | <b>fitted_root_mean_square_win2</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                          |             |  |
|              | Description:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Root mean square value of doas result for fitting window 2.                              |             |  |
|              | Dimensions:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | time, scanline, ground_pixel.                                                            |             |  |
|              | Type:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | NC_FLOAT.                                                                                |             |  |
|              | Source:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Processor.                                                                               |             |  |
| Attributes:  | <i>Name</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <i>Value</i>                                                                             | <i>Type</i> |  |
|              | <b>units</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | '1' (static)                                                                             | NC_STRING   |  |
|              | <b>long_name</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 'root mean square residual of the fit in fitting window 2' (static)                      | NC_STRING   |  |
|              | <b>coordinates</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                          | NC_STRING   |  |
|              | The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. This attribute originates from the CF standard.                                                                                                                                                                                                                                                                                                                |                                                                                          |             |  |
|              | <b>fitted_root_mean_square_win3</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                          |             |  |
|              | Description:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Root mean square value of doas result for fitting window 3.                              |             |  |

|                                                                                      |                                         |                                                                                                                                                                                                                                                                                                                                             |             |
|--------------------------------------------------------------------------------------|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Dimensions:                                                                          | time, scanline, ground_pixel.           |                                                                                                                                                                                                                                                                                                                                             |             |
| Type:                                                                                | NC_FLOAT.                               |                                                                                                                                                                                                                                                                                                                                             |             |
| Source:                                                                              | Processor.                              |                                                                                                                                                                                                                                                                                                                                             |             |
| Attributes:                                                                          | <i>Name</i>                             | <i>Value</i>                                                                                                                                                                                                                                                                                                                                | <i>Type</i> |
|                                                                                      | <b>units</b>                            | '1' (static)                                                                                                                                                                                                                                                                                                                                | NC_STRING   |
|                                                                                      | <b>long_name</b>                        | 'root mean square residual of the fit in fitting window 3' (static)                                                                                                                                                                                                                                                                         | NC_STRING   |
|                                                                                      | <b>coordinates</b>                      | '/PRODUCT/longitude /PRODUCT/latitude' (static) NC_STRING<br>The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. This attribute originates from the CF standard.                                                                                              |             |
| <b>averaging_kernel</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS              |                                         |                                                                                                                                                                                                                                                                                                                                             |             |
| Description:                                                                         | Averaging kernel                        |                                                                                                                                                                                                                                                                                                                                             |             |
| Dimensions:                                                                          | time, scanline, ground_pixel, layer.    |                                                                                                                                                                                                                                                                                                                                             |             |
| Type:                                                                                | NC_FLOAT.                               |                                                                                                                                                                                                                                                                                                                                             |             |
| Source:                                                                              | Processor.                              |                                                                                                                                                                                                                                                                                                                                             |             |
| Attributes:                                                                          | <i>Name</i>                             | <i>Value</i>                                                                                                                                                                                                                                                                                                                                | <i>Type</i> |
|                                                                                      | <b>units</b>                            | '1' (static)                                                                                                                                                                                                                                                                                                                                | NC_STRING   |
|                                                                                      | <b>long_name</b>                        | 'averaging kernel' (static)                                                                                                                                                                                                                                                                                                                 | NC_STRING   |
|                                                                                      | <b>coordinates</b>                      | '/PRODUCT/longitude /PRODUCT/latitude' (static) NC_STRING<br>The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. Following [ER5, section 5.2] we use this attribute to connect the data with the geolocation. This attribute originates from the CF standard. |             |
| <b>sulfurdioxide_profile_apriori</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS |                                         |                                                                                                                                                                                                                                                                                                                                             |             |
| Description:                                                                         | Volume mixing ratio profile of $SO_2$ . |                                                                                                                                                                                                                                                                                                                                             |             |
| Dimensions:                                                                          | time, scanline, ground_pixel, layer.    |                                                                                                                                                                                                                                                                                                                                             |             |
| Type:                                                                                | NC_FLOAT.                               |                                                                                                                                                                                                                                                                                                                                             |             |
| Source:                                                                              | Processor.                              |                                                                                                                                                                                                                                                                                                                                             |             |
| Attributes:                                                                          | <i>Name</i>                             | <i>Value</i>                                                                                                                                                                                                                                                                                                                                | <i>Type</i> |
|                                                                                      | <b>units</b>                            | '1' (static)                                                                                                                                                                                                                                                                                                                                | NC_STRING   |
|                                                                                      | <b>long_name</b>                        | 'volume mixing ratio profile of sulfur dioxide' (static)                                                                                                                                                                                                                                                                                    | NC_STRING   |
|                                                                                      | <b>coordinates</b>                      | '/PRODUCT/longitude /PRODUCT/latitude' (static) NC_STRING<br>The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. Following [ER5, section 5.2] we use this attribute to connect the data with the geolocation. This attribute originates from the CF standard. |             |
| <b>fitted_radiance_shift</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS         |                                         |                                                                                                                                                                                                                                                                                                                                             |             |
| Dimensions:                                                                          | time, scanline, ground_pixel.           |                                                                                                                                                                                                                                                                                                                                             |             |
| Type:                                                                                | NC_FLOAT.                               |                                                                                                                                                                                                                                                                                                                                             |             |
| Source:                                                                              | Processor.                              |                                                                                                                                                                                                                                                                                                                                             |             |
| Attributes:                                                                          | <i>Name</i>                             | <i>Value</i>                                                                                                                                                                                                                                                                                                                                | <i>Type</i> |
|                                                                                      | <b>units</b>                            | 'nm' (static)                                                                                                                                                                                                                                                                                                                               | NC_STRING   |
|                                                                                      | <b>long_name</b>                        | 'radiance wavelength shift from the doas fit' (static)                                                                                                                                                                                                                                                                                      | NC_STRING   |
|                                                                                      | <b>coordinates</b>                      | '/PRODUCT/longitude /PRODUCT/latitude' (static) NC_STRING<br>The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. This attribute originates from the CF standard.                                                                                              |             |
| <b>fitted_radiance_squeeze</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS       |                                         |                                                                                                                                                                                                                                                                                                                                             |             |
| Dimensions:                                                                          | time, scanline, ground_pixel.           |                                                                                                                                                                                                                                                                                                                                             |             |
| Type:                                                                                | NC_FLOAT.                               |                                                                                                                                                                                                                                                                                                                                             |             |
| Source:                                                                              | Processor.                              |                                                                                                                                                                                                                                                                                                                                             |             |

|                                                                                                    |                                                                                                 |                                                                              |                                                                                                                                                                                                           |
|----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Attributes:                                                                                        | <i>Name</i>                                                                                     | <i>Value</i>                                                                 | <i>Type</i>                                                                                                                                                                                               |
|                                                                                                    | <b>units</b>                                                                                    | ‘1’ (static)                                                                 | NC_STRING                                                                                                                                                                                                 |
|                                                                                                    | <b>long_name</b>                                                                                | ‘radiance wavelength squeeze from the doas fit’ (static)                     | NC_STRING                                                                                                                                                                                                 |
|                                                                                                    | <b>coordinates</b>                                                                              | ‘/PRODUCT/longitude /PRODUCT/latitude’ (static)                              | NC_STRING<br>The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. This attribute originates from the CF standard.            |
| <b>fitted_radiance_shift_win1</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS                  |                                                                                                 |                                                                              |                                                                                                                                                                                                           |
| Dimensions:                                                                                        | time, scanline, ground_pixel.                                                                   |                                                                              |                                                                                                                                                                                                           |
| Type:                                                                                              | NC_FLOAT.                                                                                       |                                                                              |                                                                                                                                                                                                           |
| Source:                                                                                            | Processor.                                                                                      |                                                                              |                                                                                                                                                                                                           |
| Attributes:                                                                                        | <i>Name</i>                                                                                     | <i>Value</i>                                                                 | <i>Type</i>                                                                                                                                                                                               |
|                                                                                                    | <b>units</b>                                                                                    | ‘nm’ (static)                                                                | NC_STRING                                                                                                                                                                                                 |
|                                                                                                    | <b>long_name</b>                                                                                | ‘radiance wavelength shift from the doas fit in fitting window 1’ (static)   | NC_STRING                                                                                                                                                                                                 |
|                                                                                                    | <b>coordinates</b>                                                                              | ‘/PRODUCT/longitude /PRODUCT/latitude’ (static)                              | NC_STRING<br>The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. This attribute originates from the CF standard.            |
| <b>fitted_radiance_squeeze_win1</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS                |                                                                                                 |                                                                              |                                                                                                                                                                                                           |
| Dimensions:                                                                                        | time, scanline, ground_pixel.                                                                   |                                                                              |                                                                                                                                                                                                           |
| Type:                                                                                              | NC_FLOAT.                                                                                       |                                                                              |                                                                                                                                                                                                           |
| Source:                                                                                            | Processor.                                                                                      |                                                                              |                                                                                                                                                                                                           |
| Attributes:                                                                                        | <i>Name</i>                                                                                     | <i>Value</i>                                                                 | <i>Type</i>                                                                                                                                                                                               |
|                                                                                                    | <b>units</b>                                                                                    | ‘1’ (static)                                                                 | NC_STRING                                                                                                                                                                                                 |
|                                                                                                    | <b>long_name</b>                                                                                | ‘radiance wavelength squeeze from the doas fit in fitting window 1’ (static) | NC_STRING                                                                                                                                                                                                 |
|                                                                                                    | <b>coordinates</b>                                                                              | ‘/PRODUCT/longitude /PRODUCT/latitude’ (static)                              | NC_STRING<br>The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. This attribute originates from the CF standard.            |
| <b>number_of_spectral_points_in_retrieval</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS      |                                                                                                 |                                                                              |                                                                                                                                                                                                           |
| Description:                                                                                       | The number of points in the spectrum that were used in the DOAS retrieval.                      |                                                                              |                                                                                                                                                                                                           |
| Dimensions:                                                                                        | time, scanline, ground_pixel.                                                                   |                                                                              |                                                                                                                                                                                                           |
| Type:                                                                                              | NC_USHORT.                                                                                      |                                                                              |                                                                                                                                                                                                           |
| Source:                                                                                            | Processor.                                                                                      |                                                                              |                                                                                                                                                                                                           |
| Attributes:                                                                                        | <i>Name</i>                                                                                     | <i>Value</i>                                                                 | <i>Type</i>                                                                                                                                                                                               |
|                                                                                                    | <b>long_name</b>                                                                                | ‘Number of spectral points used in the DOAS retrieval’ (static)              | NC_STRING                                                                                                                                                                                                 |
|                                                                                                    | <b>units</b>                                                                                    | ‘1’ (static)                                                                 | NC_STRING                                                                                                                                                                                                 |
|                                                                                                    | <b>coordinates</b>                                                                              | ‘/PRODUCT/longitude /PRODUCT/latitude’ (static)                              | NC_STRING<br>The latitude and longitude are in a different group. How to specify the related geospatial coordinates in this case is not specified in the climate and forecast metadata conventions [ER5]. |
| <b>number_of_spectral_points_in_retrieval_win1</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS |                                                                                                 |                                                                              |                                                                                                                                                                                                           |
| Description:                                                                                       | The number of points in the spectrum that were used in the DOAS retrieval for fitting window 1. |                                                                              |                                                                                                                                                                                                           |
| Dimensions:                                                                                        | time, scanline, ground_pixel.                                                                   |                                                                              |                                                                                                                                                                                                           |
| Type:                                                                                              | NC_USHORT.                                                                                      |                                                                              |                                                                                                                                                                                                           |
| Source:                                                                                            | Processor.                                                                                      |                                                                              |                                                                                                                                                                                                           |

|                                                                                                    |                                                                                         |                                                                                                                                                                                                                                                           |             |
|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Attributes:                                                                                        | <i>Name</i>                                                                             | <i>Value</i>                                                                                                                                                                                                                                              | <i>Type</i> |
|                                                                                                    | <b>units</b>                                                                            | '1' (static)                                                                                                                                                                                                                                              | NC_STRING   |
|                                                                                                    | <b>long_name</b>                                                                        | 'Number of spectral points used in the DOAS retrieval for window 1' (static)                                                                                                                                                                              | NC_STRING   |
|                                                                                                    | <b>coordinates</b>                                                                      | '/PRODUCT/longitude /PRODUCT/latitude' (static) NC_STRING<br>The latitude and longitude are in a different group. How to specify the related geospatial coordinates in this case is not specified in the climate and forecast metadata conventions [ER5]. |             |
| <b>fitted_radiance_shift_win2</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS                  |                                                                                         |                                                                                                                                                                                                                                                           |             |
| Dimensions:                                                                                        | time, scanline, ground_pixel.                                                           |                                                                                                                                                                                                                                                           |             |
| Type:                                                                                              | NC_FLOAT.                                                                               |                                                                                                                                                                                                                                                           |             |
| Source:                                                                                            | Processor.                                                                              |                                                                                                                                                                                                                                                           |             |
| Attributes:                                                                                        | <i>Name</i>                                                                             | <i>Value</i>                                                                                                                                                                                                                                              | <i>Type</i> |
|                                                                                                    | <b>units</b>                                                                            | 'nm' (static)                                                                                                                                                                                                                                             | NC_STRING   |
|                                                                                                    | <b>long_name</b>                                                                        | 'radiance wavelength shift from the doas fit in fitting window 2' (static)                                                                                                                                                                                | NC_STRING   |
|                                                                                                    | <b>coordinates</b>                                                                      | '/PRODUCT/longitude /PRODUCT/latitude' (static) NC_STRING<br>The latitude and longitude are in a different group. How to specify the related geospatial coordinates in this case is not specified in the climate and forecast metadata conventions [ER5]. |             |
| <b>fitted_radiance_squeeze_win2</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS                |                                                                                         |                                                                                                                                                                                                                                                           |             |
| Dimensions:                                                                                        | time, scanline, ground_pixel.                                                           |                                                                                                                                                                                                                                                           |             |
| Type:                                                                                              | NC_FLOAT.                                                                               |                                                                                                                                                                                                                                                           |             |
| Source:                                                                                            | Processor.                                                                              |                                                                                                                                                                                                                                                           |             |
| Attributes:                                                                                        | <i>Name</i>                                                                             | <i>Value</i>                                                                                                                                                                                                                                              | <i>Type</i> |
|                                                                                                    | <b>units</b>                                                                            | '1' (static)                                                                                                                                                                                                                                              | NC_STRING   |
|                                                                                                    | <b>long_name</b>                                                                        | 'radiance wavelength shift from the doas fit in fitting window 2' (static)                                                                                                                                                                                | NC_STRING   |
|                                                                                                    | <b>coordinates</b>                                                                      | '/PRODUCT/longitude /PRODUCT/latitude' (static) NC_STRING<br>The latitude and longitude are in a different group. How to specify the related geospatial coordinates in this case is not specified in the climate and forecast metadata conventions [ER5]. |             |
| <b>number_of_spectral_points_in_retrieval_win2</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS |                                                                                         |                                                                                                                                                                                                                                                           |             |
| Description:                                                                                       | The number of points in the spectrum that were used in the DOAS retrieval for window 2. |                                                                                                                                                                                                                                                           |             |
| Dimensions:                                                                                        | time, scanline, ground_pixel.                                                           |                                                                                                                                                                                                                                                           |             |
| Type:                                                                                              | NC_USHORT.                                                                              |                                                                                                                                                                                                                                                           |             |
| Source:                                                                                            | Processor.                                                                              |                                                                                                                                                                                                                                                           |             |
| Attributes:                                                                                        | <i>Name</i>                                                                             | <i>Value</i>                                                                                                                                                                                                                                              | <i>Type</i> |
|                                                                                                    | <b>units</b>                                                                            | '1' (static)                                                                                                                                                                                                                                              | NC_STRING   |
|                                                                                                    | <b>long_name</b>                                                                        | 'Number of spectral points used in the DOAS retrieval for window 2' (static)                                                                                                                                                                              | NC_STRING   |
|                                                                                                    | <b>coordinates</b>                                                                      | '/PRODUCT/longitude /PRODUCT/latitude' (static) NC_STRING<br>The latitude and longitude are in a different group. How to specify the related geospatial coordinates in this case is not specified in the climate and forecast metadata conventions [ER5]. |             |
| <b>fitted_radiance_shift_win3</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS                  |                                                                                         |                                                                                                                                                                                                                                                           |             |
| Dimensions:                                                                                        | time, scanline, ground_pixel.                                                           |                                                                                                                                                                                                                                                           |             |
| Type:                                                                                              | NC_FLOAT.                                                                               |                                                                                                                                                                                                                                                           |             |

|                                                                                                    |                                                                                                                                                                                                                                                                               |                                                                              |             |
|----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|-------------|
| Source:                                                                                            | Processor.                                                                                                                                                                                                                                                                    |                                                                              |             |
| Attributes:                                                                                        | <i>Name</i>                                                                                                                                                                                                                                                                   | <i>Value</i>                                                                 | <i>Type</i> |
|                                                                                                    | <b>units</b>                                                                                                                                                                                                                                                                  | 'nm' (static)                                                                | NC_STRING   |
|                                                                                                    | <b>long_name</b>                                                                                                                                                                                                                                                              | 'radiance wavelength shift from the doas fit in fitting window 3' (static)   | NC_STRING   |
|                                                                                                    | <b>coordinates</b>                                                                                                                                                                                                                                                            | '/PRODUCT/longitude /PRODUCT/latitude' (static)                              | NC_STRING   |
|                                                                                                    | The latitude and longitude are in a different group. How to specify the related geospatial coordinates in this case is not specified in the climate and forecast metadata conventions [ER5].                                                                                  |                                                                              |             |
| <b>fitted_radiance_squeeze_win3</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS                |                                                                                                                                                                                                                                                                               |                                                                              |             |
| Dimensions:                                                                                        | time, scanline, ground_pixel.                                                                                                                                                                                                                                                 |                                                                              |             |
| Type:                                                                                              | NC_FLOAT.                                                                                                                                                                                                                                                                     |                                                                              |             |
| Source:                                                                                            | Processor.                                                                                                                                                                                                                                                                    |                                                                              |             |
| Attributes:                                                                                        | <i>Name</i>                                                                                                                                                                                                                                                                   | <i>Value</i>                                                                 | <i>Type</i> |
|                                                                                                    | <b>units</b>                                                                                                                                                                                                                                                                  | '1' (static)                                                                 | NC_STRING   |
|                                                                                                    | <b>long_name</b>                                                                                                                                                                                                                                                              | 'radiance wavelength shift from the doas fit in fitting window 3' (static)   | NC_STRING   |
|                                                                                                    | <b>coordinates</b>                                                                                                                                                                                                                                                            | '/PRODUCT/longitude /PRODUCT/latitude' (static)                              | NC_STRING   |
|                                                                                                    | The latitude and longitude are in a different group. How to specify the related geospatial coordinates in this case is not specified in the climate and forecast metadata conventions [ER5].                                                                                  |                                                                              |             |
| <b>number_of_spectral_points_in_retrieval_win3</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS |                                                                                                                                                                                                                                                                               |                                                                              |             |
| Description:                                                                                       | The number of points in the spectrum that were used in the DOAS retrieval for window 3.                                                                                                                                                                                       |                                                                              |             |
| Dimensions:                                                                                        | time, scanline, ground_pixel.                                                                                                                                                                                                                                                 |                                                                              |             |
| Type:                                                                                              | NC_USHORT.                                                                                                                                                                                                                                                                    |                                                                              |             |
| Source:                                                                                            | Processor.                                                                                                                                                                                                                                                                    |                                                                              |             |
| Attributes:                                                                                        | <i>Name</i>                                                                                                                                                                                                                                                                   | <i>Value</i>                                                                 | <i>Type</i> |
|                                                                                                    | <b>units</b>                                                                                                                                                                                                                                                                  | '1' (static)                                                                 | NC_STRING   |
|                                                                                                    | <b>long_name</b>                                                                                                                                                                                                                                                              | 'Number of spectral points used in the DOAS retrieval for window 3' (static) | NC_STRING   |
|                                                                                                    | <b>coordinates</b>                                                                                                                                                                                                                                                            | '/PRODUCT/longitude /PRODUCT/latitude' (static)                              | NC_STRING   |
|                                                                                                    | The latitude and longitude are in a different group. How to specify the related geospatial coordinates in this case is not specified in the climate and forecast metadata conventions [ER5].                                                                                  |                                                                              |             |
| <b>qa_value_box_profile</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS                        |                                                                                                                                                                                                                                                                               |                                                                              |             |
| Description:                                                                                       | A continuous quality descriptor, varying between 0 (no data) and 1 (full quality data). The value will change based on observation conditions and retrieval flags. Detailed quality flags are provided in the <code>processing_quality_flags</code> elsewhere in the product. |                                                                              |             |
| Dimensions:                                                                                        | time, scanline, ground_pixel.                                                                                                                                                                                                                                                 |                                                                              |             |
| Type:                                                                                              | NC_UBYTE.                                                                                                                                                                                                                                                                     |                                                                              |             |
| Source:                                                                                            | Processor.                                                                                                                                                                                                                                                                    |                                                                              |             |
| Attributes:                                                                                        | <i>Name</i>                                                                                                                                                                                                                                                                   | <i>Value</i>                                                                 | <i>Type</i> |
|                                                                                                    | <b>units</b>                                                                                                                                                                                                                                                                  | '1' (static)                                                                 | NC_STRING   |
|                                                                                                    | <b>scale_factor</b>                                                                                                                                                                                                                                                           | 0.01 (static)                                                                | NC_FLOAT    |
|                                                                                                    | <b>add_offset</b>                                                                                                                                                                                                                                                             | 0 (static)                                                                   | NC_FLOAT    |
|                                                                                                    | <b>valid_min</b>                                                                                                                                                                                                                                                              | 0 (static)                                                                   | NC_UBYTE    |
|                                                                                                    | <b>valid_max</b>                                                                                                                                                                                                                                                              | 100 (static)                                                                 | NC_UBYTE    |
|                                                                                                    | <b>long_name</b>                                                                                                                                                                                                                                                              | 'data quality value' (static)                                                | NC_STRING   |

|                                                                                                               |                                                                                                                                                                                                                                                                                |                                                                                                                                                                                      |             |           |
|---------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------|
|                                                                                                               | <b>comment</b>                                                                                                                                                                                                                                                                 | 'A continuous quality descriptor, varying between 0 (no data) and 1 (full quality data). Recommend to ignore data with qa_value < 0.5' (static)                                      |             | NC_STRING |
|                                                                                                               | <b>coordinates</b>                                                                                                                                                                                                                                                             | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                                                                                                      |             | NC_STRING |
| <b>sulfurdioxide_total_air_mass_factor_polluted</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS           |                                                                                                                                                                                                                                                                                |                                                                                                                                                                                      |             |           |
| Description:                                                                                                  | Total air mass factor, $M$ for the polluted case.                                                                                                                                                                                                                              |                                                                                                                                                                                      |             |           |
| Dimensions:                                                                                                   | time, scanline, ground_pixel.                                                                                                                                                                                                                                                  |                                                                                                                                                                                      |             |           |
| Type:                                                                                                         | NC_FLOAT.                                                                                                                                                                                                                                                                      |                                                                                                                                                                                      |             |           |
| Source:                                                                                                       | Processor.                                                                                                                                                                                                                                                                     |                                                                                                                                                                                      |             |           |
| Attributes:                                                                                                   | <i>Name</i>                                                                                                                                                                                                                                                                    | <i>Value</i>                                                                                                                                                                         | <i>Type</i> |           |
|                                                                                                               | <b>units</b>                                                                                                                                                                                                                                                                   | '1' (static)                                                                                                                                                                         | NC_STRING   |           |
|                                                                                                               | <b>long_name</b>                                                                                                                                                                                                                                                               | 'total air mass factor for boundary layer polluted scenario' (static)                                                                                                                | NC_STRING   |           |
|                                                                                                               | <b>ancillary_variables</b>                                                                                                                                                                                                                                                     | 'sulfurdioxide_total_air_mass_factor_polluted_precision sulfurdioxide_total_air_mass_factor_polluted_trueness sulfurdioxide_total_air_mass_factor_polluted_kernel_trueness' (static) | NC_STRING   |           |
|                                                                                                               | <b>coordinates</b>                                                                                                                                                                                                                                                             | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                                                                                                      | NC_STRING   |           |
|                                                                                                               | The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. Following [ER5, section 5.2] we use this attribute to connect the data with the geolocation. This attribute originates from the CF standard. |                                                                                                                                                                                      |             |           |
| <b>sulfurdioxide_total_air_mass_factor_polluted_precision</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS |                                                                                                                                                                                                                                                                                |                                                                                                                                                                                      |             |           |
| Description:                                                                                                  | Random error of the total air mass factor $M$ for the polluted case.                                                                                                                                                                                                           |                                                                                                                                                                                      |             |           |
| Dimensions:                                                                                                   | time, scanline, ground_pixel.                                                                                                                                                                                                                                                  |                                                                                                                                                                                      |             |           |
| Type:                                                                                                         | NC_FLOAT.                                                                                                                                                                                                                                                                      |                                                                                                                                                                                      |             |           |
| Source:                                                                                                       | Processor.                                                                                                                                                                                                                                                                     |                                                                                                                                                                                      |             |           |
| Attributes:                                                                                                   | <i>Name</i>                                                                                                                                                                                                                                                                    | <i>Value</i>                                                                                                                                                                         | <i>Type</i> |           |
|                                                                                                               | <b>units</b>                                                                                                                                                                                                                                                                   | '1' (static)                                                                                                                                                                         | NC_STRING   |           |
|                                                                                                               | <b>long_name</b>                                                                                                                                                                                                                                                               | 'random error of the total air mass factor for the boundary layer polluted scenario' (static)                                                                                        | NC_STRING   |           |
|                                                                                                               | <b>coordinates</b>                                                                                                                                                                                                                                                             | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                                                                                                      | NC_STRING   |           |
|                                                                                                               | The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. Following [ER5, section 5.2] we use this attribute to connect the data with the geolocation. This attribute originates from the CF standard. |                                                                                                                                                                                      |             |           |
| <b>sulfurdioxide_total_air_mass_factor_polluted_trueness</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS  |                                                                                                                                                                                                                                                                                |                                                                                                                                                                                      |             |           |
| Description:                                                                                                  | Systematic error of the total air mass factor $M$ for the polluted case.                                                                                                                                                                                                       |                                                                                                                                                                                      |             |           |
| Dimensions:                                                                                                   | time, scanline, ground_pixel.                                                                                                                                                                                                                                                  |                                                                                                                                                                                      |             |           |
| Type:                                                                                                         | NC_FLOAT.                                                                                                                                                                                                                                                                      |                                                                                                                                                                                      |             |           |
| Source:                                                                                                       | Processor.                                                                                                                                                                                                                                                                     |                                                                                                                                                                                      |             |           |
| Attributes:                                                                                                   | <i>Name</i>                                                                                                                                                                                                                                                                    | <i>Value</i>                                                                                                                                                                         | <i>Type</i> |           |
|                                                                                                               | <b>units</b>                                                                                                                                                                                                                                                                   | '1' (static)                                                                                                                                                                         | NC_STRING   |           |
|                                                                                                               | <b>long_name</b>                                                                                                                                                                                                                                                               | 'systematic error of the total air mass factor for the boundary layer polluted scenario' (static)                                                                                    | NC_STRING   |           |
|                                                                                                               | <b>coordinates</b>                                                                                                                                                                                                                                                             | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                                                                                                      | NC_STRING   |           |
|                                                                                                               | The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. Following [ER5, section 5.2] we use this attribute to connect the data with the geolocation. This attribute originates from the CF standard. |                                                                                                                                                                                      |             |           |

---

**sulfurdioxide\_total\_air\_mass\_factor\_polluted\_kernel\_trueness** in SO2\_\_\_/PRODUCT/SUPPORT\_DATA/DETAILED\_RESULTS

Description: Systematic error of the total air mass factor  $M$  when kernels are used for the polluted case.

Dimensions: time, scanline, ground\_pixel.

Type: NC\_FLOAT.

Source: Processor.

| Attributes: | Name                                                                                                                                                                                                                                                                           | Value                                                                                     | Type      |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|-----------|
|             | <b>units</b>                                                                                                                                                                                                                                                                   | '1' (static)                                                                              | NC_STRING |
|             | <b>long_name</b>                                                                                                                                                                                                                                                               | 'systematic error of the total air mass factor for the kernel polluted scenario' (static) | NC_STRING |
|             | <b>coordinates</b>                                                                                                                                                                                                                                                             | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                           | NC_STRING |
|             | The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. Following [ER5, section 5.2] we use this attribute to connect the data with the geolocation. This attribute originates from the CF standard. |                                                                                           |           |

---

**sulfurdioxide\_clear\_air\_mass\_factor\_polluted** in SO2\_\_\_/PRODUCT/SUPPORT\_DATA/DETAILED\_RESULTS

Description: Clear air mass factor for the polluted case.

Dimensions: time, scanline, ground\_pixel.

Type: NC\_FLOAT.

Source: Processor.

| Attributes: | Name                                                                                                                                                                                                                                                                           | Value                                                                         | Type      |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|-----------|
|             | <b>units</b>                                                                                                                                                                                                                                                                   | '1' (static)                                                                  | NC_STRING |
|             | <b>long_name</b>                                                                                                                                                                                                                                                               | 'clear sky air mass factor for the boundary layer polluted scenario' (static) | NC_STRING |
|             | <b>coordinates</b>                                                                                                                                                                                                                                                             | '/PRODUCT/longitude /PRODUCT/latitude' (static)                               | NC_STRING |
|             | The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. Following [ER5, section 5.2] we use this attribute to connect the data with the geolocation. This attribute originates from the CF standard. |                                                                               |           |

---

**sulfurdioxide\_cloudy\_air\_mass\_factor\_polluted** in SO2\_\_\_/PRODUCT/SUPPORT\_DATA/DETAILED\_RESULTS

Description: Cloudy air mass factor for the polluted case.

Dimensions: time, scanline, ground\_pixel.

Type: NC\_FLOAT.

Source: Processor.

| Attributes: | Name                                                                                                                                                                                                                                                                           | Value                                                                          | Type      |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|-----------|
|             | <b>units</b>                                                                                                                                                                                                                                                                   | '1' (static)                                                                   | NC_STRING |
|             | <b>long_name</b>                                                                                                                                                                                                                                                               | 'cloudy sky air mass factor for the boundary layer polluted scenario' (static) | NC_STRING |
|             | <b>coordinates</b>                                                                                                                                                                                                                                                             | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                | NC_STRING |
|             | The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. Following [ER5, section 5.2] we use this attribute to connect the data with the geolocation. This attribute originates from the CF standard. |                                                                                |           |

---

**sulfurdioxide\_total\_vertical\_column\_1km** in SO2\_\_\_/PRODUCT/SUPPORT\_DATA/DETAILED\_RESULTS

Description: Vertical column density of the Sulphur Dioxide SO<sub>2</sub> product for a sulfur dioxide plume at 1km altitude w.r.t. the topography.

Dimensions: time, scanline, ground\_pixel.

Type: NC\_FLOAT.

Source: Processor.

---

|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                             |             |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-------------|
| Attributes: | <i>Name</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <i>Value</i>                                                                                                                | <i>Type</i> |
|             | <b>units</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 'mol m-2' (static)                                                                                                          | NC_STRING   |
|             | <b>standard_name</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 'atmosphere_mole_content_of_sulfur_dioxide' (static)                                                                        | NC_STRING   |
|             | <b>long_name</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 'total vertical column density of sulfur dioxide for a sulfur dioxide plume at 1km altitude w.r.t. the topography' (static) | NC_STRING   |
|             | <b>ancillary_variables</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 'sulfurdioxide_total_vertical_column_1km_precision_sulfurdioxide_total_vertical_column_1km_trueness' (static)               | NC_STRING   |
|             | <b>coordinates</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                                             | NC_STRING   |
|             | The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. This attribute originates from the CF standard.                                                                                                                                                                                                                                                                                                           |                                                                                                                             |             |
|             | <b>multiplication_factor_to_convert_to_DU</b>                                                                                                                                                                                                                                                                                                                                                                                                                                               | %(MOLM2_TO_DU)s (static)                                                                                                    | NC_FLOAT    |
|             | The quantities in Sentinel 5 precursor files are given in SI units. For an integrated column value this means that the unit is molm <sup>-2</sup> . Traditionally the unit for an integrated column is "DU" or Dobson Units. This attribute provides the multiplication factor to calculate the total column in DU from the value in molm <sup>-2</sup> . This is provided as a convenience to users who have tools that work in DU.                                                        |                                                                                                                             |             |
|             | <b>multiplication_factor_to_convert_to_molecules_percm2</b>                                                                                                                                                                                                                                                                                                                                                                                                                                 | 6.02214e+19 (static)                                                                                                        | NC_FLOAT    |
|             | The quantities in Sentinel 5 precursor files are given in SI units. For an integrated column value this means that the unit is molm <sup>-2</sup> . Traditionally the unit for an integrated column is "moleculescm <sup>-2</sup> ". This attribute provides the multiplication factor to calculate the total column in moleculescm <sup>-2</sup> from the value in molm <sup>-2</sup> . This is provided as a convenience to users who have tools that work in moleculescm <sup>-2</sup> . |                                                                                                                             |             |

|                                                                                                          |                                                                                                                                                              |  |  |
|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|
| <b>sulfurdioxide_total_vertical_column_1km_precision</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS |                                                                                                                                                              |  |  |
| Description:                                                                                             | Random error of the vertical column density of the Sulphur Dioxide SO <sub>2</sub> product for a sulfur dioxide plume at 1km altitude w.r.t. the topography. |  |  |
| Dimensions:                                                                                              | time, scanline, ground_pixel.                                                                                                                                |  |  |
| Type:                                                                                                    | NC_FLOAT.                                                                                                                                                    |  |  |
| Source:                                                                                                  | Processor.                                                                                                                                                   |  |  |

|             |                                                                                                                                                                                   |                                                                                                                                                 |             |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Attributes: | <i>Name</i>                                                                                                                                                                       | <i>Value</i>                                                                                                                                    | <i>Type</i> |
|             | <b>units</b>                                                                                                                                                                      | 'mol m-2' (static)                                                                                                                              | NC_STRING   |
|             | <b>standard_name</b>                                                                                                                                                              | 'atmosphere_mole_content_of_sulfur_dioxide_standard_error' (static)                                                                             | NC_STRING   |
|             | <b>long_name</b>                                                                                                                                                                  | 'random error of the total vertical column density of sulfur dioxide for a sulfur dioxide plume at 1km altitude w.r.t. the topography' (static) | NC_STRING   |
|             | <b>coordinates</b>                                                                                                                                                                | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                                                                 | NC_STRING   |
|             | The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. This attribute originates from the CF standard. |                                                                                                                                                 |             |
|             | <b>multiplication_factor_to_convert to DU</b>                                                                                                                                     | %(MOLM2_TO_DU)s (static)                                                                                                                        | NC_FLOAT    |
|             |                                                                                                                                                                                   |                                                                                                                                                 |             |

The quantities in Sentinel 5 precursor files are given in SI units. For an integrated column value this means that the unit is  $\text{mol m}^{-2}$ . Traditionally the unit for an integrated column is “DU” or Dobson Units. This attribute provides the multiplication factor to calculate the total column in DU from the value in  $\text{mol m}^{-2}$ . This is provided as a convenience to users who have tools that work in DU.

|                                                                |                      |          |
|----------------------------------------------------------------|----------------------|----------|
| <b>multiplication_ - factor_to_convert_to_molecules_percm2</b> | 6.02214e+19 (static) | NC_FLOAT |
|----------------------------------------------------------------|----------------------|----------|

The quantities in Sentinel 5 precursor files are given in SI units. For an integrated column value this means that the unit is  $\text{mol m}^{-2}$ . Traditionally the unit for an integrated column is “molecules  $\text{cm}^{-2}$ ”. This attribute provides the multiplication factor to calculate the total column in molecules  $\text{cm}^{-2}$  from the value in  $\text{mol m}^{-2}$ . This is provided as a convenience to users who have tools that work in molecules  $\text{cm}^{-2}$ .

**sulfurdioxide\_total\_vertical\_column\_1km\_trueness** in SO2\_\_\_/PRODUCT/SUPPORT\_DATA/DETAILED\_RESULTS

Description: Systematic error of the vertical column density of the Sulphur Dioxide SO<sub>2</sub> product for a sulfur dioxide plume at 1km altitude w.r.t. the topography.

Dimensions: time, scanline, ground\_pixel.

Type: NC\_FLOAT.

Source: Processor.

|             |                  |                                                                                                                                                     |             |
|-------------|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Attributes: | <i>Name</i>      | <i>Value</i>                                                                                                                                        | <i>Type</i> |
|             | <b>units</b>     | ‘mol m-2’ (static)                                                                                                                                  | NC_STRING   |
|             | <b>long_name</b> | ‘systematic error of the total vertical column density of sulfur dioxide for a sulfur dioxide plume at 1km altitude w.r.t. the topography’ (static) | NC_STRING   |

|                    |                                                 |           |
|--------------------|-------------------------------------------------|-----------|
| <b>coordinates</b> | ‘/PRODUCT/longitude /PRODUCT/latitude’ (static) | NC_STRING |
|--------------------|-------------------------------------------------|-----------|

The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. This attribute originates from the CF standard.

|                                                  |                          |          |
|--------------------------------------------------|--------------------------|----------|
| <b>multiplication_ - factor_to_convert_to_DU</b> | %(MOLM2_TO_DU)s (static) | NC_FLOAT |
|--------------------------------------------------|--------------------------|----------|

The quantities in Sentinel 5 precursor files are given in SI units. For an integrated column value this means that the unit is  $\text{mol m}^{-2}$ . Traditionally the unit for an integrated column is “DU” or Dobson Units. This attribute provides the multiplication factor to calculate the total column in DU from the value in  $\text{mol m}^{-2}$ . This is provided as a convenience to users who have tools that work in DU.

|                                                                |                      |          |
|----------------------------------------------------------------|----------------------|----------|
| <b>multiplication_ - factor_to_convert_to_molecules_percm2</b> | 6.02214e+19 (static) | NC_FLOAT |
|----------------------------------------------------------------|----------------------|----------|

The quantities in Sentinel 5 precursor files are given in SI units. For an integrated column value this means that the unit is  $\text{mol m}^{-2}$ . Traditionally the unit for an integrated column is “molecules  $\text{cm}^{-2}$ ”. This attribute provides the multiplication factor to calculate the total column in molecules  $\text{cm}^{-2}$  from the value in  $\text{mol m}^{-2}$ . This is provided as a convenience to users who have tools that work in molecules  $\text{cm}^{-2}$ .

**sulfurdioxide\_total\_air\_mass\_factor\_1km** in SO2\_\_\_/PRODUCT/SUPPORT\_DATA/DETAILED\_RESULTS

Description: Total air mass factor, *M* for a sulfur dioxide plume at 1km altitude w.r.t. the topography.

Dimensions: time, scanline, ground\_pixel.

Type: NC\_FLOAT.

Source: Processor.

|                                                                                                                                                                                                                                                                                |                            |                                                                                                                                                                       |             |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Attributes:                                                                                                                                                                                                                                                                    | <i>Name</i>                | <i>Value</i>                                                                                                                                                          | <i>Type</i> |
|                                                                                                                                                                                                                                                                                | <b>units</b>               | '1' (static)                                                                                                                                                          | NC_STRING   |
|                                                                                                                                                                                                                                                                                | <b>long_name</b>           | 'total air mass factor for a sulfur dioxide plume at 1km altitude w.r.t. the topography' (static)                                                                     | NC_STRING   |
|                                                                                                                                                                                                                                                                                | <b>ancillary_variables</b> | 'sulfurdioxide_total_air_mass_factor_1km_precision_sulfurdioxide_total_air_mass_factor_1km_trueness_sulfurdioxide_total_air_mass_factor_1km_kernel_trueness' (static) | NC_STRING   |
|                                                                                                                                                                                                                                                                                | <b>coordinates</b>         | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                                                                                       | NC_STRING   |
| The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. Following [ER5, section 5.2] we use this attribute to connect the data with the geolocation. This attribute originates from the CF standard. |                            |                                                                                                                                                                       |             |

**sulfurdioxide\_total\_air\_mass\_factor\_1km\_precision** in SO<sub>2</sub>\_\_\_/PRODUCT/SUPPORT\_DATA/DETAILED\_RESULTS

Description: Random error of the total air mass factor  $M$  for a sulfur dioxide plume at 1km altitude w.r.t. the topography.

Dimensions: time, scanline, ground\_pixel.

Type: NC\_FLOAT.

Source: Processor.

|                                                                                                                                                                                                                                                                                |                    |                                                                                                                       |             |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-----------------------------------------------------------------------------------------------------------------------|-------------|
| Attributes:                                                                                                                                                                                                                                                                    | <i>Name</i>        | <i>Value</i>                                                                                                          | <i>Type</i> |
|                                                                                                                                                                                                                                                                                | <b>units</b>       | '1' (static)                                                                                                          | NC_STRING   |
|                                                                                                                                                                                                                                                                                | <b>long_name</b>   | 'random error of the total air mass factor for a sulfur dioxide plume at 1km altitude w.r.t. the topography' (static) | NC_STRING   |
|                                                                                                                                                                                                                                                                                | <b>coordinates</b> | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                                       | NC_STRING   |
| The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. Following [ER5, section 5.2] we use this attribute to connect the data with the geolocation. This attribute originates from the CF standard. |                    |                                                                                                                       |             |

**sulfurdioxide\_total\_air\_mass\_factor\_1km\_trueness** in SO<sub>2</sub>\_\_\_/PRODUCT/SUPPORT\_DATA/DETAILED\_RESULTS

Description: Systematic error of the total air mass factor  $M$  for a sulfur dioxide plume at 1km altitude w.r.t. the topography.

Dimensions: time, scanline, ground\_pixel.

Type: NC\_FLOAT.

Source: Processor.

|                                                                                                                                                                                                                                                                                |                    |                                                                                                                           |             |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|---------------------------------------------------------------------------------------------------------------------------|-------------|
| Attributes:                                                                                                                                                                                                                                                                    | <i>Name</i>        | <i>Value</i>                                                                                                              | <i>Type</i> |
|                                                                                                                                                                                                                                                                                | <b>units</b>       | '1' (static)                                                                                                              | NC_STRING   |
|                                                                                                                                                                                                                                                                                | <b>long_name</b>   | 'systematic error of the total air mass factor for a sulfur dioxide plume at 1km altitude w.r.t. the topography' (static) | NC_STRING   |
|                                                                                                                                                                                                                                                                                | <b>coordinates</b> | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                                           | NC_STRING   |
| The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. Following [ER5, section 5.2] we use this attribute to connect the data with the geolocation. This attribute originates from the CF standard. |                    |                                                                                                                           |             |

**sulfurdioxide\_total\_air\_mass\_factor\_1km\_kernel\_trueness** in SO<sub>2</sub>\_\_\_/PRODUCT/SUPPORT\_DATA/DETAILED\_RESULTS

Description: Systematic error of the total air mass factor  $M$  when kernels are used for a sulfur dioxide plume at 1km altitude w.r.t. the topography.

Dimensions: time, scanline, ground\_pixel.

Type: NC\_FLOAT.

|                                                                                                       |                                                                                                                                                                                                                                                                                |                                                                                                                                         |             |
|-------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Source:                                                                                               | Processor.                                                                                                                                                                                                                                                                     |                                                                                                                                         |             |
| Attributes:                                                                                           | <i>Name</i>                                                                                                                                                                                                                                                                    | <i>Value</i>                                                                                                                            | <i>Type</i> |
|                                                                                                       | <b>units</b>                                                                                                                                                                                                                                                                   | '1' (static)                                                                                                                            | NC_STRING   |
|                                                                                                       | <b>long_name</b>                                                                                                                                                                                                                                                               | 'systematic error of the total air mass factor using kernels for a sulfur dioxide plume at 1km altitude w.r.t. the topography' (static) | NC_STRING   |
|                                                                                                       | <b>coordinates</b>                                                                                                                                                                                                                                                             | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                                                         | NC_STRING   |
|                                                                                                       | The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. Following [ER5, section 5.2] we use this attribute to connect the data with the geolocation. This attribute originates from the CF standard. |                                                                                                                                         |             |
| <b>sulfurdioxide_clear_air_mass_factor_1km</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS        |                                                                                                                                                                                                                                                                                |                                                                                                                                         |             |
| Description:                                                                                          | Clear sky air mass factor for a sulfur dioxide plume at 1km altitude w.r.t. the topography.                                                                                                                                                                                    |                                                                                                                                         |             |
| Dimensions:                                                                                           | time, scanline, ground_pixel.                                                                                                                                                                                                                                                  |                                                                                                                                         |             |
| Type:                                                                                                 | NC_FLOAT.                                                                                                                                                                                                                                                                      |                                                                                                                                         |             |
| Source:                                                                                               | Processor.                                                                                                                                                                                                                                                                     |                                                                                                                                         |             |
| Attributes:                                                                                           | <i>Name</i>                                                                                                                                                                                                                                                                    | <i>Value</i>                                                                                                                            | <i>Type</i> |
|                                                                                                       | <b>units</b>                                                                                                                                                                                                                                                                   | '1' (static)                                                                                                                            | NC_STRING   |
|                                                                                                       | <b>long_name</b>                                                                                                                                                                                                                                                               | 'clear sky air mass factor for a sulfur dioxide plume at 1km altitude w.r.t. the topography' (static)                                   | NC_STRING   |
|                                                                                                       | <b>coordinates</b>                                                                                                                                                                                                                                                             | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                                                         | NC_STRING   |
|                                                                                                       | The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. Following [ER5, section 5.2] we use this attribute to connect the data with the geolocation. This attribute originates from the CF standard. |                                                                                                                                         |             |
| <b>sulfurdioxide_cloudy_air_mass_factor_1km</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS       |                                                                                                                                                                                                                                                                                |                                                                                                                                         |             |
| Description:                                                                                          | Cloudy sky air mass factor for a sulfur dioxide plume at 1km altitude w.r.t. the topography.                                                                                                                                                                                   |                                                                                                                                         |             |
| Dimensions:                                                                                           | time, scanline, ground_pixel.                                                                                                                                                                                                                                                  |                                                                                                                                         |             |
| Type:                                                                                                 | NC_FLOAT.                                                                                                                                                                                                                                                                      |                                                                                                                                         |             |
| Source:                                                                                               | Processor.                                                                                                                                                                                                                                                                     |                                                                                                                                         |             |
| Attributes:                                                                                           | <i>Name</i>                                                                                                                                                                                                                                                                    | <i>Value</i>                                                                                                                            | <i>Type</i> |
|                                                                                                       | <b>units</b>                                                                                                                                                                                                                                                                   | '1' (static)                                                                                                                            | NC_STRING   |
|                                                                                                       | <b>long_name</b>                                                                                                                                                                                                                                                               | 'cloudy sky air mass factor for a sulfur dioxide plume at 1km altitude w.r.t. the topography' (static)                                  | NC_STRING   |
|                                                                                                       | <b>coordinates</b>                                                                                                                                                                                                                                                             | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                                                         | NC_STRING   |
|                                                                                                       | The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. Following [ER5, section 5.2] we use this attribute to connect the data with the geolocation. This attribute originates from the CF standard. |                                                                                                                                         |             |
| <b>sulfurdioxide_averaging_kernel_scaling_box_1km</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS |                                                                                                                                                                                                                                                                                |                                                                                                                                         |             |
| Description:                                                                                          | Scaling box of the total air mass factor $M$ for a sulfur dioxide plume at 1km altitude w.r.t. the topography.                                                                                                                                                                 |                                                                                                                                         |             |
| Dimensions:                                                                                           | time, scanline, ground_pixel.                                                                                                                                                                                                                                                  |                                                                                                                                         |             |
| Type:                                                                                                 | NC_FLOAT.                                                                                                                                                                                                                                                                      |                                                                                                                                         |             |
| Source:                                                                                               | Processor.                                                                                                                                                                                                                                                                     |                                                                                                                                         |             |
| Attributes:                                                                                           | <i>Name</i>                                                                                                                                                                                                                                                                    | <i>Value</i>                                                                                                                            | <i>Type</i> |
|                                                                                                       | <b>units</b>                                                                                                                                                                                                                                                                   | '1' (static)                                                                                                                            | NC_STRING   |
|                                                                                                       | <b>long_name</b>                                                                                                                                                                                                                                                               | 'scaling box of the total air mass factor for a sulfur dioxide plume at 1km altitude w.r.t. the topography' (static)                    | NC_STRING   |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                    |                                                                                                                            |             |           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|-------------|-----------|
| <b>coordinates</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                    | ‘/PRODUCT/longitude /PRODUCT/latitude’ (static)                                                                            |             | NC_STRING |
| The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. Following [ER5, section 5.2] we use this attribute to connect the data with the geolocation. This attribute originates from the CF standard.                                                                                                                                                                                                                   |                                                                                                                                                    |                                                                                                                            |             |           |
| <b>sulfurdioxide_total_vertical_column_7km</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                    |                                                                                                                            |             |           |
| Description:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Vertical column density of the sulphur dioxide $SO_2$ product for a sulfur dioxide plume at 7km altitude w.r.t. the sea level.                     |                                                                                                                            |             |           |
| Dimensions:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | time, scanline, ground_pixel.                                                                                                                      |                                                                                                                            |             |           |
| Type:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | NC_FLOAT.                                                                                                                                          |                                                                                                                            |             |           |
| Source:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Processor.                                                                                                                                         |                                                                                                                            |             |           |
| Attributes:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <i>Name</i>                                                                                                                                        | <i>Value</i>                                                                                                               | <i>Type</i> |           |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>units</b>                                                                                                                                       | ‘mol m-2’ (static)                                                                                                         | NC_STRING   |           |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>standard_name</b>                                                                                                                               | ‘atmosphere_mole_content_of_sulfur_dioxide’ (static)                                                                       | NC_STRING   |           |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>long_name</b>                                                                                                                                   | ‘total vertical column density of sulfur dioxide for a sulfur dioxide plume at 7km altitude w.r.t. the sea level’ (static) | NC_STRING   |           |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>ancillary_variables</b>                                                                                                                         | ‘sulfurdioxide_total_vertical_column_7km_precision_sulfurdioxide_total_vertical_column_7km_trueness’ (static)              | NC_STRING   |           |
| <b>coordinates</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                    | ‘/PRODUCT/longitude /PRODUCT/latitude’ (static)                                                                            |             | NC_STRING |
| The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. This attribute originates from the CF standard.                                                                                                                                                                                                                                                                                                                |                                                                                                                                                    |                                                                                                                            |             |           |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>multiplication_factor_to_convert_to_DU</b>                                                                                                      | %(MOLM2_TO_DU)s (static)                                                                                                   | NC_FLOAT    |           |
| The quantities in Sentinel 5 precursor files are given in SI units. For an integrated column value this means that the unit is mol m <sup>-2</sup> . Traditionally the unit for an integrated column is “DU” or Dobson Units. This attribute provides the multiplication factor to calculate the total column in DU from the value in mol m <sup>-2</sup> . This is provided as a convenience to users who have tools that work in DU.                                                           |                                                                                                                                                    |                                                                                                                            |             |           |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>multiplication_factor_to_convert_to_molecules_per_cm2</b>                                                                                       | 6.02214e+19 (static)                                                                                                       | NC_FLOAT    |           |
| The quantities in Sentinel 5 precursor files are given in SI units. For an integrated column value this means that the unit is mol m <sup>-2</sup> . Traditionally the unit for an integrated column is “molecules cm <sup>-2</sup> ”. This attribute provides the multiplication factor to calculate the total column in molecules cm <sup>-2</sup> from the value in mol m <sup>-2</sup> . This is provided as a convenience to users who have tools that work in molecules cm <sup>-2</sup> . |                                                                                                                                                    |                                                                                                                            |             |           |
| <b>sulfurdioxide_total_vertical_column_7km_precision</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                    |                                                                                                                            |             |           |
| Description:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Random error of the vertical column density of the Sulphur Dioxide $SO_2$ product for a sulfur dioxide plume at 7km altitude w.r.t. the sea level. |                                                                                                                            |             |           |
| Dimensions:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | time, scanline, ground_pixel.                                                                                                                      |                                                                                                                            |             |           |
| Type:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | NC_FLOAT.                                                                                                                                          |                                                                                                                            |             |           |
| Source:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Processor.                                                                                                                                         |                                                                                                                            |             |           |
| Attributes:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <i>Name</i>                                                                                                                                        | <i>Value</i>                                                                                                               | <i>Type</i> |           |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>units</b>                                                                                                                                       | ‘mol m-2’ (static)                                                                                                         | NC_STRING   |           |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>standard_name</b>                                                                                                                               | ‘atmosphere_mole_content_of_sulfur_dioxide_standard_error’ (static)                                                        | NC_STRING   |           |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                            |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| <b>long_name</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 'random error of the total vertical column of sulfur dioxide for a sulfur dioxide plume at 7km altitude w.r.t. the sea level' (static)                                                                                                                                                                                                                                                                                                 | NC_STRING                                                                                                                                  |
| <b>coordinates</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                                                                                                                                                                                                                                                                                                                                                        | NC_STRING                                                                                                                                  |
| The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. This attribute originates from the CF standard.                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                            |
| <b>multiplication_factor_to_convert_to_DU</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                    | %(MOLM2_TO_DU)s (static)                                                                                                                                                                                                                                                                                                                                                                                                               | NC_FLOAT                                                                                                                                   |
| The quantities in Sentinel 5 precursor files are given in SI units. For an integrated column value this means that the unit is mol m <sup>-2</sup> . Traditionally the unit for an integrated column is "DU" or Dobson Units. This attribute provides the multiplication factor to calculate the total column in DU from the value in mol m <sup>-2</sup> . This is provided as a convenience to users who have tools that work in DU.                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                            |
| <b>multiplication_factor_to_convert_to_molecules_per_cm2</b>                                                                                                                                                                                                                                                                                                                                                                                                                                     | 6.02214e+19 (static)                                                                                                                                                                                                                                                                                                                                                                                                                   | NC_FLOAT                                                                                                                                   |
| The quantities in Sentinel 5 precursor files are given in SI units. For an integrated column value this means that the unit is mol m <sup>-2</sup> . Traditionally the unit for an integrated column is "molecules cm <sup>-2</sup> ". This attribute provides the multiplication factor to calculate the total column in molecules cm <sup>-2</sup> from the value in mol m <sup>-2</sup> . This is provided as a convenience to users who have tools that work in molecules cm <sup>-2</sup> . |                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                            |
| <b>sulfurdioxide_total_vertical_column_7km_trueness</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                            |
| Description:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Systematic error of the vertical column density of the Sulphur Dioxide SO <sub>2</sub> product for a sulfur dioxide plume at 7km altitude w.r.t. the sea level.                                                                                                                                                                                                                                                                        |                                                                                                                                            |
| Dimensions:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | time, scanline, ground_pixel.                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                            |
| Type:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | NC_FLOAT.                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                            |
| Source:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Processor.                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                            |
| Attributes:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <i>Name</i>                                                                                                                                                                                                                                                                                                                                                                                                                            | <i>Value</i>                                                                                                                               |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>units</b>                                                                                                                                                                                                                                                                                                                                                                                                                           | 'mol m-2' (static)                                                                                                                         |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>long_name</b>                                                                                                                                                                                                                                                                                                                                                                                                                       | 'systematic error of the total vertical column of sulfur dioxide for a sulfur dioxide plume at 7km altitude w.r.t. the sea level' (static) |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>coordinates</b>                                                                                                                                                                                                                                                                                                                                                                                                                     | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                                                            |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. This attribute originates from the CF standard.                                                                                                                                                                                                                                                      |                                                                                                                                            |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>multiplication_factor_to_convert_to_DU</b>                                                                                                                                                                                                                                                                                                                                                                                          | %(MOLM2_TO_DU)s (static)                                                                                                                   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | The quantities in Sentinel 5 precursor files are given in SI units. For an integrated column value this means that the unit is mol m <sup>-2</sup> . Traditionally the unit for an integrated column is "DU" or Dobson Units. This attribute provides the multiplication factor to calculate the total column in DU from the value in mol m <sup>-2</sup> . This is provided as a convenience to users who have tools that work in DU. |                                                                                                                                            |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>multiplication_factor_to_convert_to_molecules_per_cm2</b>                                                                                                                                                                                                                                                                                                                                                                           | 6.02214e+19 (static)                                                                                                                       |

The quantities in Sentinel 5 precursor files are given in SI units. For an integrated column value this means that the unit is  $\text{mol m}^{-2}$ . Traditionally the unit for an integrated column is “ $\text{molecules cm}^{-2}$ ”. This attribute provides the multiplication factor to calculate the total column in  $\text{molecules cm}^{-2}$  from the value in  $\text{mol m}^{-2}$ . This is provided as a convenience to users who have tools that work in  $\text{molecules cm}^{-2}$ .

**sulfurdioxide\_total\_air\_mass\_factor\_7km** in SO2\_\_\_/PRODUCT/SUPPORT\_DATA/DETAILED\_RESULTS

Description: Total air mass factor,  $M$  for a sulfur dioxide plume at 7km altitude w.r.t. the sea level.

Dimensions: time, scanline, ground\_pixel.

Type: NC\_FLOAT.

Source: Processor.

| Attributes: | Name                                                                                                                                                                                                                                                                           | Value                                                                                                                                                                 | Type      |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
|             | <b>units</b>                                                                                                                                                                                                                                                                   | '1' (static)                                                                                                                                                          | NC_STRING |
|             | <b>long_name</b>                                                                                                                                                                                                                                                               | 'total air mass factor for a sulfur dioxide plume at 7km altitude w.r.t. the sea level' (static)                                                                      | NC_STRING |
|             | <b>ancillary_variables</b>                                                                                                                                                                                                                                                     | 'sulfurdioxide_total_air_mass_factor_7km_precision sulfurdioxide_total_air_mass_factor_7km_trueness sulfurdioxide_total_air_mass_factor_7km_kernel_trueness' (static) | NC_STRING |
|             | <b>coordinates</b>                                                                                                                                                                                                                                                             | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                                                                                       | NC_STRING |
|             | The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. Following [ER5, section 5.2] we use this attribute to connect the data with the geolocation. This attribute originates from the CF standard. |                                                                                                                                                                       |           |

**sulfurdioxide\_total\_air\_mass\_factor\_7km\_precision** in SO2\_\_\_/PRODUCT/SUPPORT\_DATA/DETAILED\_RESULTS

Description: Random error of the total air mass factor  $M$  for a sulfur dioxide plume at 7km altitude w.r.t. the sea level.

Dimensions: time, scanline, ground\_pixel.

Type: NC\_FLOAT.

Source: Processor.

| Attributes: | Name                                                                                                                                                                                                                                                                           | Value                                                                                                                | Type      |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|-----------|
|             | <b>units</b>                                                                                                                                                                                                                                                                   | '1' (static)                                                                                                         | NC_STRING |
|             | <b>long_name</b>                                                                                                                                                                                                                                                               | 'random error of the total air mass factor for a sulfur dioxide plume at 7km altitude w.r.t. the sea level' (static) | NC_STRING |
|             | <b>coordinates</b>                                                                                                                                                                                                                                                             | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                                      | NC_STRING |
|             | The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. Following [ER5, section 5.2] we use this attribute to connect the data with the geolocation. This attribute originates from the CF standard. |                                                                                                                      |           |

**sulfurdioxide\_total\_air\_mass\_factor\_7km\_trueness** in SO2\_\_\_/PRODUCT/SUPPORT\_DATA/DETAILED\_RESULTS

Description: Systematic error of the total air mass factor  $M$  for a sulfur dioxide plume at 7km altitude w.r.t. the sea level.

Dimensions: time, scanline, ground\_pixel.

Type: NC\_FLOAT.

Source: Processor.

| Attributes: | Name             | Value                                                                                                                    | Type      |
|-------------|------------------|--------------------------------------------------------------------------------------------------------------------------|-----------|
|             | <b>units</b>     | '1' (static)                                                                                                             | NC_STRING |
|             | <b>long_name</b> | 'systematic error of the total air mass factor for a sulfur dioxide plume at 7km altitude w.r.t. the sea level' (static) | NC_STRING |

|                                                                                                                |                                                                                                                                                                                                                                                                                |                                                                                                                                        |             |           |
|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------|
|                                                                                                                | <b>coordinates</b>                                                                                                                                                                                                                                                             | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                                                        |             | NC_STRING |
|                                                                                                                | The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. Following [ER5, section 5.2] we use this attribute to connect the data with the geolocation. This attribute originates from the CF standard. |                                                                                                                                        |             |           |
| <b>sulfurdioxide_total_air_mass_factor_7km_kernel_trueness</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS |                                                                                                                                                                                                                                                                                |                                                                                                                                        |             |           |
| Description:                                                                                                   | Systematic error of the total air mass factor $M$ when kernels are used for a sulfur dioxide plume at 7km altitude w.r.t. the sea level.                                                                                                                                       |                                                                                                                                        |             |           |
| Dimensions:                                                                                                    | time, scanline, ground_pixel.                                                                                                                                                                                                                                                  |                                                                                                                                        |             |           |
| Type:                                                                                                          | NC_FLOAT.                                                                                                                                                                                                                                                                      |                                                                                                                                        |             |           |
| Source:                                                                                                        | Processor.                                                                                                                                                                                                                                                                     |                                                                                                                                        |             |           |
| Attributes:                                                                                                    | <i>Name</i>                                                                                                                                                                                                                                                                    | <i>Value</i>                                                                                                                           | <i>Type</i> |           |
|                                                                                                                | <b>units</b>                                                                                                                                                                                                                                                                   | '1' (static)                                                                                                                           | NC_STRING   |           |
|                                                                                                                | <b>long_name</b>                                                                                                                                                                                                                                                               | 'systematic error of the total air mass factor using kernels for a sulfur dioxide plume at 7km altitude w.r.t. the sea level' (static) |             | NC_STRING |
|                                                                                                                | <b>coordinates</b>                                                                                                                                                                                                                                                             | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                                                        |             | NC_STRING |
|                                                                                                                | The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. Following [ER5, section 5.2] we use this attribute to connect the data with the geolocation. This attribute originates from the CF standard. |                                                                                                                                        |             |           |
| <b>sulfurdioxide_clear_air_mass_factor_7km</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS                 |                                                                                                                                                                                                                                                                                |                                                                                                                                        |             |           |
| Description:                                                                                                   | Clear sky air mass factor for a sulfur dioxide plume at 7km altitude w.r.t. the sea level.                                                                                                                                                                                     |                                                                                                                                        |             |           |
| Dimensions:                                                                                                    | time, scanline, ground_pixel.                                                                                                                                                                                                                                                  |                                                                                                                                        |             |           |
| Type:                                                                                                          | NC_FLOAT.                                                                                                                                                                                                                                                                      |                                                                                                                                        |             |           |
| Source:                                                                                                        | Processor.                                                                                                                                                                                                                                                                     |                                                                                                                                        |             |           |
| Attributes:                                                                                                    | <i>Name</i>                                                                                                                                                                                                                                                                    | <i>Value</i>                                                                                                                           | <i>Type</i> |           |
|                                                                                                                | <b>units</b>                                                                                                                                                                                                                                                                   | '1' (static)                                                                                                                           | NC_STRING   |           |
|                                                                                                                | <b>long_name</b>                                                                                                                                                                                                                                                               | 'clear sky air mass factor for a sulfur dioxide plume at 7km altitude w.r.t. the sea level' (static)                                   |             | NC_STRING |
|                                                                                                                | <b>coordinates</b>                                                                                                                                                                                                                                                             | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                                                        |             | NC_STRING |
|                                                                                                                | The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. Following [ER5, section 5.2] we use this attribute to connect the data with the geolocation. This attribute originates from the CF standard. |                                                                                                                                        |             |           |
| <b>sulfurdioxide_cloudy_air_mass_factor_7km</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS                |                                                                                                                                                                                                                                                                                |                                                                                                                                        |             |           |
| Description:                                                                                                   | Cloudy sky air mass factor for a sulfur dioxide plume at 7km altitude w.r.t. the sea level.                                                                                                                                                                                    |                                                                                                                                        |             |           |
| Dimensions:                                                                                                    | time, scanline, ground_pixel.                                                                                                                                                                                                                                                  |                                                                                                                                        |             |           |
| Type:                                                                                                          | NC_FLOAT.                                                                                                                                                                                                                                                                      |                                                                                                                                        |             |           |
| Source:                                                                                                        | Processor.                                                                                                                                                                                                                                                                     |                                                                                                                                        |             |           |
| Attributes:                                                                                                    | <i>Name</i>                                                                                                                                                                                                                                                                    | <i>Value</i>                                                                                                                           | <i>Type</i> |           |
|                                                                                                                | <b>units</b>                                                                                                                                                                                                                                                                   | '1' (static)                                                                                                                           | NC_STRING   |           |
|                                                                                                                | <b>long_name</b>                                                                                                                                                                                                                                                               | 'cloudy sky air mass factor for a sulfur dioxide plume at 7km altitude w.r.t. the sea level' (static)                                  |             | NC_STRING |
|                                                                                                                | <b>coordinates</b>                                                                                                                                                                                                                                                             | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                                                        |             | NC_STRING |
|                                                                                                                | The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. Following [ER5, section 5.2] we use this attribute to connect the data with the geolocation. This attribute originates from the CF standard. |                                                                                                                                        |             |           |
| <b>sulfurdioxide_averaging_kernel_scaling_box_7km</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS          |                                                                                                                                                                                                                                                                                |                                                                                                                                        |             |           |

|                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                             |             |
|-------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-------------|
| Description:                                                                                    | Scaling box of the total air mass factor $M$ for a sulfur dioxide plume at 7km altitude w.r.t. the sea level.                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                             |             |
| Dimensions:                                                                                     | time, scanline, ground_pixel.                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                             |             |
| Type:                                                                                           | NC_FLOAT.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                             |             |
| Source:                                                                                         | Processor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                             |             |
| Attributes:                                                                                     | <i>Name</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <i>Value</i>                                                                                                                | <i>Type</i> |
|                                                                                                 | <b>units</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | '1' (static)                                                                                                                | NC_STRING   |
|                                                                                                 | <b>long_name</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 'scaling box of the total air mass factor for a sulfur dioxide plume at 7km altitude w.r.t. the sea level' (static)         | NC_STRING   |
|                                                                                                 | <b>coordinates</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                                             | NC_STRING   |
|                                                                                                 | The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. Following [ER5, section 5.2] we use this attribute to connect the data with the geolocation. This attribute originates from the CF standard.                                                                                                                                                                                                                   |                                                                                                                             |             |
| <b>sulfurdioxide_total_vertical_column_15km</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                             |             |
| Description:                                                                                    | Vertical column density of the sulfur dioxide $SO_2$ product for a sulfur dioxide plume at 15km altitude w.r.t. the sea level.                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                             |             |
| Dimensions:                                                                                     | time, scanline, ground_pixel.                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                             |             |
| Type:                                                                                           | NC_FLOAT.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                             |             |
| Source:                                                                                         | Processor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                             |             |
| Attributes:                                                                                     | <i>Name</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <i>Value</i>                                                                                                                | <i>Type</i> |
|                                                                                                 | <b>units</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 'mol m-2' (static)                                                                                                          | NC_STRING   |
|                                                                                                 | <b>standard_name</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 'atmosphere_mole_content_of_sulfur_dioxide' (static)                                                                        | NC_STRING   |
|                                                                                                 | <b>long_name</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 'total vertical column density of sulfur dioxide for a sulfur dioxide plume at 15km altitude w.r.t. the sea level' (static) | NC_STRING   |
|                                                                                                 | <b>ancillary_variables</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 'sulfurdioxide_total_vertical_column_15km_precision_sulfurdioxide_total_vertical_column_15km_trueness' (static)             | NC_STRING   |
|                                                                                                 | <b>coordinates</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                                             | NC_STRING   |
|                                                                                                 | The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. This attribute originates from the CF standard.                                                                                                                                                                                                                                                                                                                |                                                                                                                             |             |
|                                                                                                 | <b>multiplication_factor_to_convert_to_DU</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                    | %(MOLM2_TO_DU)s (static)                                                                                                    | NC_FLOAT    |
|                                                                                                 | The quantities in Sentinel 5 precursor files are given in SI units. For an integrated column value this means that the unit is $\text{mol m}^{-2}$ . Traditionally the unit for an integrated column is "DU" or Dobson Units. This attribute provides the multiplication factor to calculate the total column in DU from the value in $\text{mol m}^{-2}$ . This is provided as a convenience to users who have tools that work in DU.                                                           |                                                                                                                             |             |
|                                                                                                 | <b>multiplication_factor_to_convert_to_molecules_percm2</b>                                                                                                                                                                                                                                                                                                                                                                                                                                      | 6.02214e+19 (static)                                                                                                        | NC_FLOAT    |
|                                                                                                 | The quantities in Sentinel 5 precursor files are given in SI units. For an integrated column value this means that the unit is $\text{mol m}^{-2}$ . Traditionally the unit for an integrated column is "molecules $\text{cm}^{-2}$ ". This attribute provides the multiplication factor to calculate the total column in molecules $\text{cm}^{-2}$ from the value in $\text{mol m}^{-2}$ . This is provided as a convenience to users who have tools that work in molecules $\text{cm}^{-2}$ . |                                                                                                                             |             |

|                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                     |             |
|-----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| <b>sulfurdioxide_total_vertical_column_15km_precision</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                     |             |
| Description:                                                                                              | Random error of the vertical column density of the sulfur dioxide $SO_2$ product for a sulfur dioxide plume at 15km altitude w.r.t. the sea level.                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                     |             |
| Dimensions:                                                                                               | time, scanline, ground_pixel.                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                     |             |
| Type:                                                                                                     | NC_FLOAT.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                     |             |
| Source:                                                                                                   | Processor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                     |             |
| Attributes:                                                                                               | <i>Name</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <i>Value</i>                                                                                                                                        | <i>Type</i> |
|                                                                                                           | <b>units</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 'mol m-2' (static)                                                                                                                                  | NC_STRING   |
|                                                                                                           | <b>standard_name</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 'atmosphere_mole_content_of_sulfur_dioxide_standard_error' (static)                                                                                 | NC_STRING   |
|                                                                                                           | <b>long_name</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 'random error of the total vertical column density of sulfur dioxide for a sulfur dioxide plume at 15km altitude w.r.t. the sea level' (static)     | NC_STRING   |
|                                                                                                           | <b>coordinates</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                                                                     | NC_STRING   |
|                                                                                                           | The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. This attribute originates from the CF standard.                                                                                                                                                                                                                                                                                                                |                                                                                                                                                     |             |
|                                                                                                           | <b>multiplication_factor_to_convert_to_DU</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                    | %(MOLM2_TO_DU)s (static)                                                                                                                            | NC_FLOAT    |
|                                                                                                           | The quantities in Sentinel 5 precursor files are given in SI units. For an integrated column value this means that the unit is $\text{mol m}^{-2}$ . Traditionally the unit for an integrated column is "DU" or Dobson Units. This attribute provides the multiplication factor to calculate the total column in DU from the value in $\text{mol m}^{-2}$ . This is provided as a convenience to users who have tools that work in DU.                                                           |                                                                                                                                                     |             |
|                                                                                                           | <b>multiplication_factor_to_convert_to_molecules_per_cm2</b>                                                                                                                                                                                                                                                                                                                                                                                                                                     | 6.02214e+19 (static)                                                                                                                                | NC_FLOAT    |
|                                                                                                           | The quantities in Sentinel 5 precursor files are given in SI units. For an integrated column value this means that the unit is $\text{mol m}^{-2}$ . Traditionally the unit for an integrated column is "molecules $\text{cm}^{-2}$ ". This attribute provides the multiplication factor to calculate the total column in molecules $\text{cm}^{-2}$ from the value in $\text{mol m}^{-2}$ . This is provided as a convenience to users who have tools that work in molecules $\text{cm}^{-2}$ . |                                                                                                                                                     |             |
| <b>sulfurdioxide_total_vertical_column_15km_trueness</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                     |             |
| Description:                                                                                              | Systematic error of the vertical column density of the sulfur dioxide $SO_2$ product for a sulfur dioxide plume at 15km altitude w.r.t. the sea level.                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                     |             |
| Dimensions:                                                                                               | time, scanline, ground_pixel.                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                     |             |
| Type:                                                                                                     | NC_FLOAT.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                     |             |
| Source:                                                                                                   | Processor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                     |             |
| Attributes:                                                                                               | <i>Name</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <i>Value</i>                                                                                                                                        | <i>Type</i> |
|                                                                                                           | <b>units</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 'mol m-2' (static)                                                                                                                                  | NC_STRING   |
|                                                                                                           | <b>long_name</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 'systematic error of the total vertical column density of sulfur dioxide for a sulfur dioxide plume at 15km altitude w.r.t. the sea level' (static) | NC_STRING   |
|                                                                                                           | <b>coordinates</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                                                                     | NC_STRING   |
|                                                                                                           | The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. This attribute originates from the CF standard.                                                                                                                                                                                                                                                                                                                |                                                                                                                                                     |             |
|                                                                                                           | <b>multiplication_factor_to_convert to DU</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2241.15 (static)                                                                                                                                    | NC_FLOAT    |

The quantities in Sentinel 5 precursor files are given in SI units. For an integrated column value this means that the unit is  $\text{mol m}^{-2}$ . Traditionally the unit for an integrated column is “DU” or Dobson Units. This attribute provides the multiplication factor to calculate the total column in DU from the value in  $\text{mol m}^{-2}$ . This is provided as a convenience to users who have tools that work in DU.

**multiplication\_factor\_to\_convert\_to\_molecules\_per\_cm2** 6.02214e+19 (static) NC\_FLOAT

The quantities in Sentinel 5 precursor files are given in SI units. For an integrated column value this means that the unit is  $\text{mol m}^{-2}$ . Traditionally the unit for an integrated column is “molecules  $\text{cm}^{-2}$ ”. This attribute provides the multiplication factor to calculate the total column in molecules  $\text{cm}^{-2}$  from the value in  $\text{mol m}^{-2}$ . This is provided as a convenience to users who have tools that work in molecules  $\text{cm}^{-2}$ .

**sulfurdioxide\_total\_air\_mass\_factor\_15km** in SO2\_\_\_/PRODUCT/SUPPORT\_DATA/DETAILED\_RESULTS

Description: Total air mass factor,  $M$  for a sulfur dioxide plume at 15km altitude w.r.t. the sea level.

Dimensions: time, scanline, ground\_pixel.

Type: NC\_FLOAT.

Source: Processor.

| Attributes: | Name                                                                                                                                                                                                                                                                           | Value                                                                                                                                                                    | Type      |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
|             | <b>units</b>                                                                                                                                                                                                                                                                   | ‘1’ (static)                                                                                                                                                             | NC_STRING |
|             | <b>long_name</b>                                                                                                                                                                                                                                                               | ‘total air mass factor for a sulfur dioxide plume at 15km altitude w.r.t. the sea level’ (static)                                                                        | NC_STRING |
|             | <b>ancillary_variables</b>                                                                                                                                                                                                                                                     | ‘sulfurdioxide_total_air_mass_factor_15km_precision sulfurdioxide_total_air_mass_factor_15km_trueness sulfurdioxide_total_air_mass_factor_15km_kernel_trueness’ (static) | NC_STRING |
|             | <b>coordinates</b>                                                                                                                                                                                                                                                             | ‘/PRODUCT/longitude /PRODUCT/latitude’ (static)                                                                                                                          | NC_STRING |
|             | The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. Following [ER5, section 5.2] we use this attribute to connect the data with the geolocation. This attribute originates from the CF standard. |                                                                                                                                                                          |           |

**sulfurdioxide\_total\_air\_mass\_factor\_15km\_precision** in SO2\_\_\_/PRODUCT/SUPPORT\_DATA/DETAILED\_RESULTS

Description: Random error of the total air mass factor  $M$  for a sulfur dioxide plume at 15km altitude w.r.t. the sea level.

Dimensions: time, scanline, ground\_pixel.

Type: NC\_FLOAT.

Source: Processor.

| Attributes: | Name                                                                                                                                                                                                                                                                           | Value                                                                                                                 | Type      |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-----------|
|             | <b>units</b>                                                                                                                                                                                                                                                                   | ‘1’ (static)                                                                                                          | NC_STRING |
|             | <b>long_name</b>                                                                                                                                                                                                                                                               | ‘random error of the total air mass factor for a sulfur dioxide plume at 15km altitude w.r.t. the sea level’ (static) | NC_STRING |
|             | <b>coordinates</b>                                                                                                                                                                                                                                                             | ‘/PRODUCT/longitude /PRODUCT/latitude’ (static)                                                                       | NC_STRING |
|             | The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. Following [ER5, section 5.2] we use this attribute to connect the data with the geolocation. This attribute originates from the CF standard. |                                                                                                                       |           |

**sulfurdioxide\_total\_air\_mass\_factor\_15km\_trueness** in SO2\_\_\_/PRODUCT/SUPPORT\_DATA/DETAILED\_RESULTS

Description: Systematic error of the total air mass factor  $M$  for a sulfur dioxide plume at 15km altitude w.r.t. the sea level.

|                                                                                                                 |                                                                                                                                                                                                                                                                                |                                                                                                                                         |             |
|-----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Dimensions:                                                                                                     | time, scanline, ground_pixel.                                                                                                                                                                                                                                                  |                                                                                                                                         |             |
| Type:                                                                                                           | NC_FLOAT.                                                                                                                                                                                                                                                                      |                                                                                                                                         |             |
| Source:                                                                                                         | Processor.                                                                                                                                                                                                                                                                     |                                                                                                                                         |             |
| Attributes:                                                                                                     | <i>Name</i>                                                                                                                                                                                                                                                                    | <i>Value</i>                                                                                                                            | <i>Type</i> |
|                                                                                                                 | <b>units</b>                                                                                                                                                                                                                                                                   | '1' (static)                                                                                                                            | NC_STRING   |
|                                                                                                                 | <b>long_name</b>                                                                                                                                                                                                                                                               | 'systematic error of the total air mass factor for a sulfur dioxide plume at 15km altitude w.r.t. the sea level' (static)               | NC_STRING   |
|                                                                                                                 | <b>coordinates</b>                                                                                                                                                                                                                                                             | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                                                         | NC_STRING   |
|                                                                                                                 | The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. Following [ER5, section 5.2] we use this attribute to connect the data with the geolocation. This attribute originates from the CF standard. |                                                                                                                                         |             |
| <b>sulfurdioxide_total_air_mass_factor_15km_kernel_trueness</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS |                                                                                                                                                                                                                                                                                |                                                                                                                                         |             |
| Description:                                                                                                    | Systematic error of the total air mass factor <i>M</i> when kernels are used for a sulfur dioxide plume at 15km altitude w.r.t. the sea level.                                                                                                                                 |                                                                                                                                         |             |
| Dimensions:                                                                                                     | time, scanline, ground_pixel.                                                                                                                                                                                                                                                  |                                                                                                                                         |             |
| Type:                                                                                                           | NC_FLOAT.                                                                                                                                                                                                                                                                      |                                                                                                                                         |             |
| Source:                                                                                                         | Processor.                                                                                                                                                                                                                                                                     |                                                                                                                                         |             |
| Attributes:                                                                                                     | <i>Name</i>                                                                                                                                                                                                                                                                    | <i>Value</i>                                                                                                                            | <i>Type</i> |
|                                                                                                                 | <b>units</b>                                                                                                                                                                                                                                                                   | '1' (static)                                                                                                                            | NC_STRING   |
|                                                                                                                 | <b>long_name</b>                                                                                                                                                                                                                                                               | 'systematic error of the total air mass factor using kernels for a sulfur dioxide plume at 15km altitude w.r.t. the sea level' (static) | NC_STRING   |
|                                                                                                                 | <b>coordinates</b>                                                                                                                                                                                                                                                             | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                                                         | NC_STRING   |
|                                                                                                                 | The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. Following [ER5, section 5.2] we use this attribute to connect the data with the geolocation. This attribute originates from the CF standard. |                                                                                                                                         |             |
| <b>sulfurdioxide_clear_air_mass_factor_15km</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS                 |                                                                                                                                                                                                                                                                                |                                                                                                                                         |             |
| Description:                                                                                                    | Clear sky air mass factor for a sulfur dioxide plume at 15km altitude w.r.t. the sea level.                                                                                                                                                                                    |                                                                                                                                         |             |
| Dimensions:                                                                                                     | time, scanline, ground_pixel.                                                                                                                                                                                                                                                  |                                                                                                                                         |             |
| Type:                                                                                                           | NC_FLOAT.                                                                                                                                                                                                                                                                      |                                                                                                                                         |             |
| Source:                                                                                                         | Processor.                                                                                                                                                                                                                                                                     |                                                                                                                                         |             |
| Attributes:                                                                                                     | <i>Name</i>                                                                                                                                                                                                                                                                    | <i>Value</i>                                                                                                                            | <i>Type</i> |
|                                                                                                                 | <b>units</b>                                                                                                                                                                                                                                                                   | '1' (static)                                                                                                                            | NC_STRING   |
|                                                                                                                 | <b>long_name</b>                                                                                                                                                                                                                                                               | 'clear sky air mass factor for a sulfur dioxide plume at 15km altitude w.r.t. the sea level' (static)                                   | NC_STRING   |
|                                                                                                                 | <b>coordinates</b>                                                                                                                                                                                                                                                             | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                                                         | NC_STRING   |
|                                                                                                                 | The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. Following [ER5, section 5.2] we use this attribute to connect the data with the geolocation. This attribute originates from the CF standard. |                                                                                                                                         |             |
| <b>sulfurdioxide_cloudy_air_mass_factor_15km</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS                |                                                                                                                                                                                                                                                                                |                                                                                                                                         |             |
| Description:                                                                                                    | Cloudy sky air mass factor for a sulfur dioxide plume at 15km altitude w.r.t. the sea level.                                                                                                                                                                                   |                                                                                                                                         |             |
| Dimensions:                                                                                                     | time, scanline, ground_pixel.                                                                                                                                                                                                                                                  |                                                                                                                                         |             |
| Type:                                                                                                           | NC_FLOAT.                                                                                                                                                                                                                                                                      |                                                                                                                                         |             |
| Source:                                                                                                         | Processor.                                                                                                                                                                                                                                                                     |                                                                                                                                         |             |
| Attributes:                                                                                                     | <i>Name</i>                                                                                                                                                                                                                                                                    | <i>Value</i>                                                                                                                            | <i>Type</i> |
|                                                                                                                 | <b>units</b>                                                                                                                                                                                                                                                                   | '1' (static)                                                                                                                            | NC_STRING   |

|                                                                                                        |                                                                                                                                                                                                                                                                                                     |                                                                                                                      |             |           |
|--------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|-------------|-----------|
|                                                                                                        | <b>long_name</b>                                                                                                                                                                                                                                                                                    | 'cloudy sky air mass factor for a sulfur dioxide plume at 15km altitude w.r.t. the sea level' (static)               |             | NC_STRING |
|                                                                                                        | <b>coordinates</b>                                                                                                                                                                                                                                                                                  | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                                      |             | NC_STRING |
|                                                                                                        | The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. Following [ER5, section 5.2] we use this attribute to connect the data with the geolocation. This attribute originates from the CF standard.                      |                                                                                                                      |             |           |
| <b>sulfurdioxide_averaging_kernel_scaling_box_15km</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS |                                                                                                                                                                                                                                                                                                     |                                                                                                                      |             |           |
| Description:                                                                                           | Scaling box of the total air mass factor $M$ for a sulfur dioxide plume at 15km altitude w.r.t. the sea level.                                                                                                                                                                                      |                                                                                                                      |             |           |
| Dimensions:                                                                                            | time, scanline, ground_pixel.                                                                                                                                                                                                                                                                       |                                                                                                                      |             |           |
| Type:                                                                                                  | NC_FLOAT.                                                                                                                                                                                                                                                                                           |                                                                                                                      |             |           |
| Source:                                                                                                | Processor.                                                                                                                                                                                                                                                                                          |                                                                                                                      |             |           |
| Attributes:                                                                                            | <i>Name</i>                                                                                                                                                                                                                                                                                         | <i>Value</i>                                                                                                         | <i>Type</i> |           |
|                                                                                                        | <b>units</b>                                                                                                                                                                                                                                                                                        | '1' (static)                                                                                                         | NC_STRING   |           |
|                                                                                                        | <b>long_name</b>                                                                                                                                                                                                                                                                                    | 'scaling box of the total air mass factor for a sulfur dioxide plume at 15km altitude w.r.t. the sea level' (static) |             | NC_STRING |
|                                                                                                        | <b>coordinates</b>                                                                                                                                                                                                                                                                                  | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                                      |             | NC_STRING |
|                                                                                                        | The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. Following [ER5, section 5.2] we use this attribute to connect the data with the geolocation. This attribute originates from the CF standard.                      |                                                                                                                      |             |           |
| <b>sulfurdioxide_slant_column_correction_flag</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS      |                                                                                                                                                                                                                                                                                                     |                                                                                                                      |             |           |
| Description:                                                                                           | Pixel-wise flag which indicates if the background correction has been applied to the slant column density. Note that in case that the Status_BG global attribute is set to "Fallback", the flag will be always 0.                                                                                   |                                                                                                                      |             |           |
| Dimensions:                                                                                            | time, scanline, ground_pixel.                                                                                                                                                                                                                                                                       |                                                                                                                      |             |           |
| Type:                                                                                                  | NC_UBYTE.                                                                                                                                                                                                                                                                                           |                                                                                                                      |             |           |
| Source:                                                                                                | Processor.                                                                                                                                                                                                                                                                                          |                                                                                                                      |             |           |
| Attributes:                                                                                            | <i>Name</i>                                                                                                                                                                                                                                                                                         | <i>Value</i>                                                                                                         | <i>Type</i> |           |
|                                                                                                        | <b>units</b>                                                                                                                                                                                                                                                                                        | '1' (static)                                                                                                         | NC_STRING   |           |
|                                                                                                        | <b>long_name</b>                                                                                                                                                                                                                                                                                    | 'sulfur dioxide slant column density background correction flag' (static)                                            |             | NC_STRING |
|                                                                                                        | <b>flag_meanings</b>                                                                                                                                                                                                                                                                                | 'not-corrected,corrected' (static)                                                                                   |             | NC_STRING |
|                                                                                                        | <b>flag_values</b>                                                                                                                                                                                                                                                                                  | 0, 1 (static)                                                                                                        |             | NC_UBYTE  |
|                                                                                                        | <b>coordinates</b>                                                                                                                                                                                                                                                                                  | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                                      |             | NC_STRING |
|                                                                                                        | The latitude and longitude are in a different group. How to specify the related geospatial coordinates in this case is not specified in the climate and forecast metadata conventions [ER5].                                                                                                        |                                                                                                                      |             |           |
| <b>sulfurdioxide_detection_flag</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS                    |                                                                                                                                                                                                                                                                                                     |                                                                                                                      |             |           |
| Description:                                                                                           | Pixel-wise flag which indicates if enhanced SO2 has been detected from a potential volcanic eruption. The flag is 0 for no detection, 1 SO2 detection, 2 for clear volcanic detection, 3 for detection close to known anthropogenic source, 4 for detection at high SZA (potential false-positive). |                                                                                                                      |             |           |
| Dimensions:                                                                                            | time, scanline, ground_pixel.                                                                                                                                                                                                                                                                       |                                                                                                                      |             |           |
| Type:                                                                                                  | NC_INT.                                                                                                                                                                                                                                                                                             |                                                                                                                      |             |           |
| Source:                                                                                                | Processor.                                                                                                                                                                                                                                                                                          |                                                                                                                      |             |           |
| Attributes:                                                                                            | <i>Name</i>                                                                                                                                                                                                                                                                                         | <i>Value</i>                                                                                                         | <i>Type</i> |           |
|                                                                                                        | <b>units</b>                                                                                                                                                                                                                                                                                        | '1' (static)                                                                                                         | NC_STRING   |           |

|                                                                                               |                                                                                                                                                                                              |                                                                                                                                                    |             |
|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
|                                                                                               | <b>long_name</b>                                                                                                                                                                             | ‘sulfur dioxide volcano activity flag’ (static)                                                                                                    | NC_STRING   |
|                                                                                               | <b>flag_meanings</b>                                                                                                                                                                         | ‘no detection,detection,clear detection close to known volcano,clear detection close to known anthropogenic source,detection at high SZA’ (static) | NC_STRING   |
|                                                                                               | <b>flag_values</b>                                                                                                                                                                           | 0,1,2,3,4 (static)                                                                                                                                 | NC_INT      |
|                                                                                               | <b>coordinates</b>                                                                                                                                                                           | ‘/PRODUCT/longitude /PRODUCT/latitude’ (static)                                                                                                    | NC_STRING   |
|                                                                                               | The latitude and longitude are in a different group. How to specify the related geospatial coordinates in this case is not specified in the climate and forecast metadata conventions [ER5]. |                                                                                                                                                    |             |
| <b>number_of_iterations_in_retrieval</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS      |                                                                                                                                                                                              |                                                                                                                                                    |             |
| Dimensions:                                                                                   | time, scanline, ground_pixel.                                                                                                                                                                |                                                                                                                                                    |             |
| Type:                                                                                         | NC_USHORT.                                                                                                                                                                                   |                                                                                                                                                    |             |
| Source:                                                                                       | Processor.                                                                                                                                                                                   |                                                                                                                                                    |             |
| Attributes:                                                                                   | <i>Name</i>                                                                                                                                                                                  | <i>Value</i>                                                                                                                                       | <i>Type</i> |
|                                                                                               | <b>units</b>                                                                                                                                                                                 | ‘1’ (static)                                                                                                                                       | NC_STRING   |
|                                                                                               | <b>long_name</b>                                                                                                                                                                             | ‘number of iterations used in the retrieval’ (static)                                                                                              | NC_STRING   |
|                                                                                               | <b>coordinates</b>                                                                                                                                                                           | ‘/PRODUCT/longitude /PRODUCT/latitude’ (static)                                                                                                    | NC_STRING   |
| <b>number_of_iterations_in_retrieval_win1</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS |                                                                                                                                                                                              |                                                                                                                                                    |             |
| Dimensions:                                                                                   | time, scanline, ground_pixel.                                                                                                                                                                |                                                                                                                                                    |             |
| Type:                                                                                         | NC_USHORT.                                                                                                                                                                                   |                                                                                                                                                    |             |
| Source:                                                                                       | Processor.                                                                                                                                                                                   |                                                                                                                                                    |             |
| Attributes:                                                                                   | <i>Name</i>                                                                                                                                                                                  | <i>Value</i>                                                                                                                                       | <i>Type</i> |
|                                                                                               | <b>units</b>                                                                                                                                                                                 | ‘1’ (static)                                                                                                                                       | NC_STRING   |
|                                                                                               | <b>long_name</b>                                                                                                                                                                             | ‘number of iterations used in the retrieval for window 1’ (static)                                                                                 | NC_STRING   |
|                                                                                               | <b>coordinates</b>                                                                                                                                                                           | ‘/PRODUCT/longitude /PRODUCT/latitude’ (static)                                                                                                    | NC_STRING   |
| <b>number_of_iterations_in_retrieval_win2</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS |                                                                                                                                                                                              |                                                                                                                                                    |             |
| Dimensions:                                                                                   | time, scanline, ground_pixel.                                                                                                                                                                |                                                                                                                                                    |             |
| Type:                                                                                         | NC_USHORT.                                                                                                                                                                                   |                                                                                                                                                    |             |
| Source:                                                                                       | Processor.                                                                                                                                                                                   |                                                                                                                                                    |             |
| Attributes:                                                                                   | <i>Name</i>                                                                                                                                                                                  | <i>Value</i>                                                                                                                                       | <i>Type</i> |
|                                                                                               | <b>units</b>                                                                                                                                                                                 | ‘1’ (static)                                                                                                                                       | NC_STRING   |
|                                                                                               | <b>long_name</b>                                                                                                                                                                             | ‘number of iterations used in the retrieval for window 2’ (static)                                                                                 | NC_STRING   |
|                                                                                               | <b>coordinates</b>                                                                                                                                                                           | ‘/PRODUCT/longitude /PRODUCT/latitude’ (static)                                                                                                    | NC_STRING   |
| <b>number_of_iterations_in_retrieval_win3</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS |                                                                                                                                                                                              |                                                                                                                                                    |             |
| Dimensions:                                                                                   | time, scanline, ground_pixel.                                                                                                                                                                |                                                                                                                                                    |             |
| Type:                                                                                         | NC_USHORT.                                                                                                                                                                                   |                                                                                                                                                    |             |
| Source:                                                                                       | Processor.                                                                                                                                                                                   |                                                                                                                                                    |             |
| Attributes:                                                                                   | <i>Name</i>                                                                                                                                                                                  | <i>Value</i>                                                                                                                                       | <i>Type</i> |
|                                                                                               | <b>units</b>                                                                                                                                                                                 | ‘1’ (static)                                                                                                                                       | NC_STRING   |
|                                                                                               | <b>long_name</b>                                                                                                                                                                             | ‘number of iterations used in the retrieval for window 3’ (static)                                                                                 | NC_STRING   |
|                                                                                               | <b>coordinates</b>                                                                                                                                                                           | ‘/PRODUCT/longitude /PRODUCT/latitude’ (static)                                                                                                    | NC_STRING   |
| <b>fitted_root_mean_square</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS                |                                                                                                                                                                                              |                                                                                                                                                    |             |
| Dimensions:                                                                                   | time, scanline, ground_pixel.                                                                                                                                                                |                                                                                                                                                    |             |
| Type:                                                                                         | NC_FLOAT.                                                                                                                                                                                    |                                                                                                                                                    |             |
| Source:                                                                                       | Processor.                                                                                                                                                                                   |                                                                                                                                                    |             |

|                                                                                          |                                                                                |                                                                                   |             |
|------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------|
| Attributes:                                                                              | <i>Name</i>                                                                    | <i>Value</i>                                                                      | <i>Type</i> |
|                                                                                          | <b>units</b>                                                                   | '1' (static)                                                                      | NC_STRING   |
|                                                                                          | <b>long_name</b>                                                               | 'root mean square of the sulfur dioxide slant column' (static)                    | NC_STRING   |
|                                                                                          | <b>coordinates</b>                                                             | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                   | NC_STRING   |
| <b>doas_polynomial_coefficients_win1</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS |                                                                                |                                                                                   |             |
| Dimensions:                                                                              | time, scanline, ground_pixel, number_of_doas_polynomial_coefficients_win1.     |                                                                                   |             |
| Type:                                                                                    | NC_FLOAT.                                                                      |                                                                                   |             |
| Source:                                                                                  | Processor.                                                                     |                                                                                   |             |
| Attributes:                                                                              | <i>Name</i>                                                                    | <i>Value</i>                                                                      | <i>Type</i> |
|                                                                                          | <b>units</b>                                                                   | '1' (static)                                                                      | NC_STRING   |
|                                                                                          | <b>long_name</b>                                                               | 'DOAS polynomial coefficients' (static)                                           | NC_STRING   |
|                                                                                          | <b>coordinates</b>                                                             | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                   | NC_STRING   |
|                                                                                          | <b>comment</b>                                                                 | 'Values of the DOAS polynomial coefficients' (static)                             | NC_STRING   |
| <b>doas_polynomial_coefficients_win2</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS |                                                                                |                                                                                   |             |
| Dimensions:                                                                              | time, scanline, ground_pixel, number_of_doas_polynomial_coefficients_win2.     |                                                                                   |             |
| Type:                                                                                    | NC_FLOAT.                                                                      |                                                                                   |             |
| Source:                                                                                  | Processor.                                                                     |                                                                                   |             |
| Attributes:                                                                              | <i>Name</i>                                                                    | <i>Value</i>                                                                      | <i>Type</i> |
|                                                                                          | <b>units</b>                                                                   | '1' (static)                                                                      | NC_STRING   |
|                                                                                          | <b>long_name</b>                                                               | 'DOAS polynomial coefficients' (static)                                           | NC_STRING   |
|                                                                                          | <b>coordinates</b>                                                             | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                   | NC_STRING   |
|                                                                                          | <b>comment</b>                                                                 | 'Values of the DOAS polynomial coefficients' (static)                             | NC_STRING   |
| <b>doas_polynomial_coefficients_win3</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS |                                                                                |                                                                                   |             |
| Dimensions:                                                                              | time, scanline, ground_pixel, number_of_doas_polynomial_coefficients_win3.     |                                                                                   |             |
| Type:                                                                                    | NC_FLOAT.                                                                      |                                                                                   |             |
| Source:                                                                                  | Processor.                                                                     |                                                                                   |             |
| Attributes:                                                                              | <i>Name</i>                                                                    | <i>Value</i>                                                                      | <i>Type</i> |
|                                                                                          | <b>units</b>                                                                   | '1' (static)                                                                      | NC_STRING   |
|                                                                                          | <b>long_name</b>                                                               | 'DOAS polynomial coefficients' (static)                                           | NC_STRING   |
|                                                                                          | <b>coordinates</b>                                                             | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                   | NC_STRING   |
|                                                                                          | <b>comment</b>                                                                 | 'Values of the DOAS polynomial coefficients' (static)                             | NC_STRING   |
| <b>sulfurdioxide_slant_column_cobra</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS  |                                                                                |                                                                                   |             |
| Description:                                                                             | SO <sub>2</sub> slant column density $N^s$ retrieved with the COBRA algorithm. |                                                                                   |             |
| Dimensions:                                                                              | time, scanline, ground_pixel.                                                  |                                                                                   |             |
| Type:                                                                                    | NC_FLOAT.                                                                      |                                                                                   |             |
| Source:                                                                                  | Processor.                                                                     |                                                                                   |             |
| Attributes:                                                                              | <i>Name</i>                                                                    | <i>Value</i>                                                                      | <i>Type</i> |
|                                                                                          | <b>units</b>                                                                   | 'mol m <sup>-2</sup> ' (static)                                                   | NC_STRING   |
|                                                                                          | <b>long_name</b>                                                               | 'sulfur dioxide slant column density retrieved with the COBRA algorithm' (static) | NC_STRING   |
|                                                                                          | <b>ancillary_variables</b>                                                     | 'sulfurdioxide_slant_column_cobra_precision' (static)                             | NC_STRING   |
|                                                                                          | <b>coordinates</b>                                                             | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                   | NC_STRING   |

| The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. This attribute originates from the CF standard.                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |           |      |       |      |              |                    |           |                  |                                                                                               |           |                      |                                                                               |           |                    |                  |          |                    |                                                 |           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|------|-------|------|--------------|--------------------|-----------|------------------|-----------------------------------------------------------------------------------------------|-----------|----------------------|-------------------------------------------------------------------------------|-----------|--------------------|------------------|----------|--------------------|-------------------------------------------------|-----------|
| <b>multiplication_factor_to_convert_to_DU</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                    | %(MOLM2_TO_DU)s (static)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | NC_FLOAT  |      |       |      |              |                    |           |                  |                                                                                               |           |                      |                                                                               |           |                    |                  |          |                    |                                                 |           |
| The quantities in Sentinel 5 precursor files are given in SI units. For an integrated column value this means that the unit is mol m <sup>-2</sup> . Traditionally the unit for an integrated column is “DU” or Dobson Units. This attribute provides the multiplication factor to calculate the total column in DU from the value in mol m <sup>-2</sup> . This is provided as a convenience to users who have tools that work in DU.                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |           |      |       |      |              |                    |           |                  |                                                                                               |           |                      |                                                                               |           |                    |                  |          |                    |                                                 |           |
| <b>multiplication_factor_to_convert_to_molecules_per_cm2</b>                                                                                                                                                                                                                                                                                                                                                                                                                                     | 6.02214e+19 (static)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | NC_FLOAT  |      |       |      |              |                    |           |                  |                                                                                               |           |                      |                                                                               |           |                    |                  |          |                    |                                                 |           |
| The quantities in Sentinel 5 precursor files are given in SI units. For an integrated column value this means that the unit is mol m <sup>-2</sup> . Traditionally the unit for an integrated column is “molecules cm <sup>-2</sup> ”. This attribute provides the multiplication factor to calculate the total column in molecules cm <sup>-2</sup> from the value in mol m <sup>-2</sup> . This is provided as a convenience to users who have tools that work in molecules cm <sup>-2</sup> . |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |           |      |       |      |              |                    |           |                  |                                                                                               |           |                      |                                                                               |           |                    |                  |          |                    |                                                 |           |
| <b>sulfurdioxide_slant_column_cobra_precision</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |           |      |       |      |              |                    |           |                  |                                                                                               |           |                      |                                                                               |           |                    |                  |          |                    |                                                 |           |
| Description:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Random error of SO <sub>2</sub> slant column density <i>N<sup>s</sup></i> retrieved with the COBRA algorithm.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |           |      |       |      |              |                    |           |                  |                                                                                               |           |                      |                                                                               |           |                    |                  |          |                    |                                                 |           |
| Dimensions:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | time, scanline, ground_pixel.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |           |      |       |      |              |                    |           |                  |                                                                                               |           |                      |                                                                               |           |                    |                  |          |                    |                                                 |           |
| Type:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | NC_FLOAT.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |           |      |       |      |              |                    |           |                  |                                                                                               |           |                      |                                                                               |           |                    |                  |          |                    |                                                 |           |
| Source:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Processor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |           |      |       |      |              |                    |           |                  |                                                                                               |           |                      |                                                                               |           |                    |                  |          |                    |                                                 |           |
| Attributes:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <table> <tr> <th>Name</th><th>Value</th><th>Type</th></tr> <tr> <td><b>units</b></td><td>‘mol m-2’ (static)</td><td>NC_STRING</td></tr> <tr> <td><b>long_name</b></td><td>‘random error of the sulfur dioxide slant column retrieved with the COBRA algorithm’ (static)</td><td>NC_STRING</td></tr> <tr> <td><b>coordinates</b></td><td>‘/PRODUCT/longitude /PRODUCT/latitude’ (static)</td><td>NC_STRING</td></tr> </table> <p>The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. This attribute originates from the CF standard.</p>                                                                                                                                                                                                    |           | Name | Value | Type | <b>units</b> | ‘mol m-2’ (static) | NC_STRING | <b>long_name</b> | ‘random error of the sulfur dioxide slant column retrieved with the COBRA algorithm’ (static) | NC_STRING | <b>coordinates</b>   | ‘/PRODUCT/longitude /PRODUCT/latitude’ (static)                               | NC_STRING |                    |                  |          |                    |                                                 |           |
| Name                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Type      |      |       |      |              |                    |           |                  |                                                                                               |           |                      |                                                                               |           |                    |                  |          |                    |                                                 |           |
| <b>units</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | ‘mol m-2’ (static)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | NC_STRING |      |       |      |              |                    |           |                  |                                                                                               |           |                      |                                                                               |           |                    |                  |          |                    |                                                 |           |
| <b>long_name</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | ‘random error of the sulfur dioxide slant column retrieved with the COBRA algorithm’ (static)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | NC_STRING |      |       |      |              |                    |           |                  |                                                                                               |           |                      |                                                                               |           |                    |                  |          |                    |                                                 |           |
| <b>coordinates</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | ‘/PRODUCT/longitude /PRODUCT/latitude’ (static)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | NC_STRING |      |       |      |              |                    |           |                  |                                                                                               |           |                      |                                                                               |           |                    |                  |          |                    |                                                 |           |
| <b>sulfurdioxide_slant_column_cobra_flag</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |           |      |       |      |              |                    |           |                  |                                                                                               |           |                      |                                                                               |           |                    |                  |          |                    |                                                 |           |
| Description:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Pixel-wise flag which indicates if the COBRA algorithm was applied successfully.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |           |      |       |      |              |                    |           |                  |                                                                                               |           |                      |                                                                               |           |                    |                  |          |                    |                                                 |           |
| Dimensions:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | time, scanline, ground_pixel.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |           |      |       |      |              |                    |           |                  |                                                                                               |           |                      |                                                                               |           |                    |                  |          |                    |                                                 |           |
| Type:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | NC_UBYTE.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |           |      |       |      |              |                    |           |                  |                                                                                               |           |                      |                                                                               |           |                    |                  |          |                    |                                                 |           |
| Source:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Processor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |           |      |       |      |              |                    |           |                  |                                                                                               |           |                      |                                                                               |           |                    |                  |          |                    |                                                 |           |
| Attributes:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <table> <tr> <th>Name</th><th>Value</th><th>Type</th></tr> <tr> <td><b>units</b></td><td>‘1’ (static)</td><td>NC_STRING</td></tr> <tr> <td><b>long_name</b></td><td>‘sulfur dioxide slant column COBRA algorithm flag’ (static)</td><td>NC_STRING</td></tr> <tr> <td><b>flag_meanings</b></td><td>‘0: not applied, 1: not all iterations done, 2: all iterations done’ (static)</td><td>NC_STRING</td></tr> <tr> <td><b>flag_values</b></td><td>0, 1, 2 (static)</td><td>NC_UBYTE</td></tr> <tr> <td><b>coordinates</b></td><td>‘/PRODUCT/longitude /PRODUCT/latitude’ (static)</td><td>NC_STRING</td></tr> </table> <p>The latitude and longitude are in a different group. How to specify the related geospatial coordinates in this case is not specified in the climate and forecast metadata conventions [ER5].</p> |           | Name | Value | Type | <b>units</b> | ‘1’ (static)       | NC_STRING | <b>long_name</b> | ‘sulfur dioxide slant column COBRA algorithm flag’ (static)                                   | NC_STRING | <b>flag_meanings</b> | ‘0: not applied, 1: not all iterations done, 2: all iterations done’ (static) | NC_STRING | <b>flag_values</b> | 0, 1, 2 (static) | NC_UBYTE | <b>coordinates</b> | ‘/PRODUCT/longitude /PRODUCT/latitude’ (static) | NC_STRING |
| Name                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Type      |      |       |      |              |                    |           |                  |                                                                                               |           |                      |                                                                               |           |                    |                  |          |                    |                                                 |           |
| <b>units</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | ‘1’ (static)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | NC_STRING |      |       |      |              |                    |           |                  |                                                                                               |           |                      |                                                                               |           |                    |                  |          |                    |                                                 |           |
| <b>long_name</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | ‘sulfur dioxide slant column COBRA algorithm flag’ (static)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | NC_STRING |      |       |      |              |                    |           |                  |                                                                                               |           |                      |                                                                               |           |                    |                  |          |                    |                                                 |           |
| <b>flag_meanings</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | ‘0: not applied, 1: not all iterations done, 2: all iterations done’ (static)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | NC_STRING |      |       |      |              |                    |           |                  |                                                                                               |           |                      |                                                                               |           |                    |                  |          |                    |                                                 |           |
| <b>flag_values</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 0, 1, 2 (static)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | NC_UBYTE  |      |       |      |              |                    |           |                  |                                                                                               |           |                      |                                                                               |           |                    |                  |          |                    |                                                 |           |
| <b>coordinates</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | ‘/PRODUCT/longitude /PRODUCT/latitude’ (static)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | NC_STRING |      |       |      |              |                    |           |                  |                                                                                               |           |                      |                                                                               |           |                    |                  |          |                    |                                                 |           |
| <b>fitted_root_mean_square_cobra</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |           |      |       |      |              |                    |           |                  |                                                                                               |           |                      |                                                                               |           |                    |                  |          |                    |                                                 |           |
| Description:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Root mean square value of COBRA retrieval result.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |           |      |       |      |              |                    |           |                  |                                                                                               |           |                      |                                                                               |           |                    |                  |          |                    |                                                 |           |
| Dimensions:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | time, scanline, ground_pixel.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |           |      |       |      |              |                    |           |                  |                                                                                               |           |                      |                                                                               |           |                    |                  |          |                    |                                                 |           |

|                                                                                                    |                                                                                                                                                                                                                                                                                                                                                          |                                                                  |             |
|----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|-------------|
| Type:                                                                                              | NC_FLOAT.                                                                                                                                                                                                                                                                                                                                                |                                                                  |             |
| Source:                                                                                            | Processor.                                                                                                                                                                                                                                                                                                                                               |                                                                  |             |
| Attributes:                                                                                        | <i>Name</i>                                                                                                                                                                                                                                                                                                                                              | <i>Value</i>                                                     | <i>Type</i> |
|                                                                                                    | <b>units</b>                                                                                                                                                                                                                                                                                                                                             | '1' (static)                                                     | NC_STRING   |
|                                                                                                    | <b>long_name</b>                                                                                                                                                                                                                                                                                                                                         | 'root mean square residual of the cobra retrieval' (static)      | NC_STRING   |
|                                                                                                    | <b>coordinates</b>                                                                                                                                                                                                                                                                                                                                       | '/PRODUCT/longitude /PRODUCT/latitude' (static)                  | NC_STRING   |
|                                                                                                    | The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. This attribute originates from the CF standard.                                                                                                                                                                        |                                                                  |             |
| <b>cloud_fraction_intensity_weighted</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS           |                                                                                                                                                                                                                                                                                                                                                          |                                                                  |             |
| Description:                                                                                       | VCD clear sky vs. cloudy weighting factor.                                                                                                                                                                                                                                                                                                               |                                                                  |             |
| Dimensions:                                                                                        | time, scanline, ground_pixel.                                                                                                                                                                                                                                                                                                                            |                                                                  |             |
| Type:                                                                                              | NC_FLOAT.                                                                                                                                                                                                                                                                                                                                                |                                                                  |             |
| Source:                                                                                            | Processor.                                                                                                                                                                                                                                                                                                                                               |                                                                  |             |
| Attributes:                                                                                        | <i>Name</i>                                                                                                                                                                                                                                                                                                                                              | <i>Value</i>                                                     | <i>Type</i> |
|                                                                                                    | <b>units</b>                                                                                                                                                                                                                                                                                                                                             | '1' (static)                                                     | NC_STRING   |
|                                                                                                    | Dimensionless unit. This attribute originates from the NUG, CF standards.                                                                                                                                                                                                                                                                                |                                                                  |             |
|                                                                                                    | <b>long_name</b>                                                                                                                                                                                                                                                                                                                                         | 'cloud fraction intensity weighted' (static)                     | NC_STRING   |
|                                                                                                    | <b>valid_min</b>                                                                                                                                                                                                                                                                                                                                         | 0.0 (static)                                                     | NC_FLOAT    |
|                                                                                                    | <b>valid_max</b>                                                                                                                                                                                                                                                                                                                                         | 1.0 (static)                                                     | NC_FLOAT    |
|                                                                                                    | <b>comment</b>                                                                                                                                                                                                                                                                                                                                           | 'VCD clear sky vs. cloudy weighting factor.' (static)            | NC_STRING   |
|                                                                                                    | <b>ancillary_variables</b>                                                                                                                                                                                                                                                                                                                               | 'cloud_fraction_intensity_weighted_precision' (static)           | NC_STRING   |
|                                                                                                    | <b>coordinates</b>                                                                                                                                                                                                                                                                                                                                       | '/PRODUCT/longitude /PRODUCT/latitude' (static)                  | NC_STRING   |
|                                                                                                    | The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. This attribute originates from the CF standard.                                                                                                                                                                        |                                                                  |             |
| <b>cloud_fraction_intensity_weighted_precision</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS |                                                                                                                                                                                                                                                                                                                                                          |                                                                  |             |
| Description:                                                                                       | Error of the VCD clear sky vs. cloudy weighting factor.                                                                                                                                                                                                                                                                                                  |                                                                  |             |
| Dimensions:                                                                                        | time, scanline, ground_pixel.                                                                                                                                                                                                                                                                                                                            |                                                                  |             |
| Type:                                                                                              | NC_FLOAT.                                                                                                                                                                                                                                                                                                                                                |                                                                  |             |
| Source:                                                                                            | Processor.                                                                                                                                                                                                                                                                                                                                               |                                                                  |             |
| Attributes:                                                                                        | <i>Name</i>                                                                                                                                                                                                                                                                                                                                              | <i>Value</i>                                                     | <i>Type</i> |
|                                                                                                    | <b>units</b>                                                                                                                                                                                                                                                                                                                                             | '1' (static)                                                     | NC_STRING   |
|                                                                                                    | Dimensionless unit. This attribute originates from the NUG, CF standards.                                                                                                                                                                                                                                                                                |                                                                  |             |
|                                                                                                    | <b>long_name</b>                                                                                                                                                                                                                                                                                                                                         | 'random error of the cloud fraction intensity weighted' (static) | NC_STRING   |
|                                                                                                    | <b>valid_min</b>                                                                                                                                                                                                                                                                                                                                         | 0.0 (static)                                                     | NC_FLOAT    |
|                                                                                                    | <b>valid_max</b>                                                                                                                                                                                                                                                                                                                                         | 1.0 (static)                                                     | NC_FLOAT    |
|                                                                                                    | <b>coordinates</b>                                                                                                                                                                                                                                                                                                                                       | '/PRODUCT/longitude /PRODUCT/latitude' (static)                  | NC_STRING   |
|                                                                                                    | The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. This attribute originates from the CF standard.                                                                                                                                                                        |                                                                  |             |
| <b>processing_quality_flags</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS                    |                                                                                                                                                                                                                                                                                                                                                          |                                                                  |             |
| Description:                                                                                       | Processing quality flag. This flag indicates processing errors or reasons for not processing a particular pixel (collectively 'errors', leading to a fill value in the output) and warnings that occurred while processing this pixel (warnings which may affect the quality of the retrieval result). A detailed description is provided in appendix A. |                                                                  |             |
| Dimensions:                                                                                        | time, scanline, ground_pixel.                                                                                                                                                                                                                                                                                                                            |                                                                  |             |
| Type:                                                                                              | NC_UINT.                                                                                                                                                                                                                                                                                                                                                 |                                                                  |             |

|             |                  |                                                                              |             |
|-------------|------------------|------------------------------------------------------------------------------|-------------|
| Source:     | Processor.       |                                                                              |             |
| Attributes: | <i>Name</i>      | <i>Value</i>                                                                 | <i>Type</i> |
|             | <b>long_name</b> | 'Processing quality flags' (static)                                          | NC_STRING   |
|             | <b>units</b>     | '1' (static)                                                                 | NC_STRING   |
|             | <b>comment</b>   | 'Flags indicating conditions that affect quality of the retrieval.' (static) | NC_STRING   |

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |           |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>flag_meanings</b> | 'success radiance_missing irradiance_missing<br>input_spectrum_missing reflectance_range_error<br>ler_range_error snr_range_error sza_range_error<br>vza_range_error lut_range_error ozone_range_<br>error wavelength_offset_error initialization_error<br>memory_error assertion_error io_error numer-<br>ical_error lut_error ISRF_error convergence_error<br>cloud_filter_convergence_error max_iteration_<br>convergence_error aot_lower_boundary_conver-<br>gence_error other_boundary_convergence_error<br>geolocation_error ch4_noscat_zero_error h2o_<br>noscat_zero_error max_optical_thickness_error<br>aerosol_boundary_error boundary_hit_error<br>chi2_error svd_error dfs_error radiative_trans-<br>fer_error optimal_estimation_error profile_error<br>cloud_error model_error number_of_input_<br>data_points_too_low_error cloud_pressure_<br>spread_too_low_error cloud_too_low_level_error<br>generic_range_error generic_exception input_<br>spectrum_alignment_error abort_error wrong_<br>input_type_error wavelength_calibration_error<br>coregistration_error slant_column_density_error<br>airmass_factor_error vertical_column_density_er-<br>ror signal_to_noise_ratio_error configuration_error<br>key_error saturation_error max_num_outlier_<br>exceeded_error solar_eclipse_filter cloud_filter<br>altitude_consistency_filter altitude_roughness_<br>filter sun_glint_filter mixed_surface_type_filter<br>snow_ice_filter aai_filter cloud_fraction_fresco_<br>filter aai_scene_albedo_filter small_pixel_radi-<br>ance_std_filter cloud_fraction_viirs_filter cirrus_<br>reflectance_viirs_filter cf_viirs_swir_fovc_filter<br>cf_viirs_swir_ofova_filter cf_viirs_swir_ofovb_filter<br>cf_viirs_swir_ofovc_filter cf_viirs_nir_fovc_filter<br>cf_viirs_nir_ofova_filter cf_viirs_nir_ofovb_filter<br>cf_viirs_nir_ofovc_filter refl_cirrus_viirs_swir_filter<br>refl_cirrus_viirs_nir_filter diff_refl_cirrus_viirs_filter<br>ch4_noscat_ratio_filter ch4_noscat_ratio_std_filter<br>h2o_noscat_ratio_filter h2o_noscat_ratio_std_filter<br>diff_psurf_fresco_ecmwf_filter psurf_fresco_<br>stdv_filter ocean_filter time_range_filter pixel_<br>or_scanline_index_filter geographic_region_filter<br>input_spectrum_warning wavelength_calibration_<br>warning extrapolation_warning sun_glint_warning<br>south_atlantic_anomaly_warning sun_glint_<br>correction snow_ice_warning cloud_warning<br>AAI_warning pixel_level_input_data_missing<br>data_range_warning low_cloud_fraction_warn-<br>ing altitude_consistency_warning signal_to_<br>noise_ratio_warning deconvolution_warning<br>so2_volcanic_origin_likely_warning so2_volcanic_<br>origin_certain_warning interpolation_warning<br>saturation_warning high_sza_warning cloud_re-<br>trieval_warning cloud_inhomogeneity_warning<br>thermal_instability_warning' (static) | NC_STRING |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|

|                                                                             |                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |             |
|-----------------------------------------------------------------------------|--------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
|                                                                             | <b>flag_masks</b>                                      | 255, 255, 255, 255, 255, 255, 255, 255, 255,<br>255, 255, 255, 255, 255, 255, 255, 255, 255,<br>255, 255, 255, 255, 255, 255, 255, 255, 255,<br>255, 255, 255, 255, 255, 255, 255, 255, 255,<br>255, 255, 255, 255, 255, 255, 255, 255, 255,<br>255, 255, 255, 255, 255, 255, 255, 255, 255,<br>255, 255, 255, 255, 255, 255, 255, 255, 255,<br>255, 255, 255, 255, 256, 512, 1024, 2048, 4096, 8192,<br>16384, 32768, 65536, 131072, 262144, 524288,<br>1048576, 2097152, 4194304, 8388608, 16777216,<br>33554432, 67108864, 134217728, 268435456,<br>536870912, 1073741824 (static)             | NC_UINT     |
|                                                                             | <b>flag_values</b>                                     | 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16,<br>17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30,<br>31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43,<br>44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 64, 65,<br>66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79,<br>80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93,<br>94, 95, 96, 97, 256, 512, 1024, 2048, 4096, 8192,<br>16384, 32768, 65536, 131072, 262144, 524288,<br>1048576, 2097152, 4194304, 8388608, 16777216,<br>33554432, 67108864, 134217728, 268435456,<br>536870912, 1073741824 (static) | NC_UINT     |
|                                                                             | <b>coordinates</b>                                     | '/PRODUCT/longitude /PRODUCT/latitude' (static)<br>The latitude and longitude are in a different group. How to specify the related geospatial coordinates in this case is not specified in the climate and forecast metadata conventions [ER5].                                                                                                                                                                                                                                                                                                                                                   | NC_STRING   |
| <b>debug_upas2_float1D</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS  |                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |             |
| Description:                                                                | Debug field, not available in operational environment. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |             |
| Dimensions:                                                                 | time, scanline.                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |             |
| Type:                                                                       | NC_FLOAT.                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |             |
| Source:                                                                     | Processor.                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |             |
| Attributes:                                                                 | <i>Name</i>                                            | <i>Value</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <i>Type</i> |
|                                                                             | <b>units</b>                                           | '1' (static)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | NC_STRING   |
|                                                                             | <b>coordinates</b>                                     | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | NC_STRING   |
| <b>debug_upas2_double1D</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS |                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |             |
| Description:                                                                | Debug field, not available in operational environment. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |             |
| Dimensions:                                                                 | time, scanline.                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |             |
| Type:                                                                       | NC_DOUBLE.                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |             |
| Source:                                                                     | Processor.                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |             |
| Attributes:                                                                 | <i>Name</i>                                            | <i>Value</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <i>Type</i> |
|                                                                             | <b>units</b>                                           | '1' (static)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | NC_STRING   |
|                                                                             | <b>coordinates</b>                                     | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | NC_STRING   |
| <b>debug_upas2_int1D</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS    |                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |             |
| Description:                                                                | Debug field, not available in operational environment. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |             |
| Dimensions:                                                                 | time, scanline.                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |             |
| Type:                                                                       | NC_INT.                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |             |
| Source:                                                                     | Processor.                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |             |
| Attributes:                                                                 | <i>Name</i>                                            | <i>Value</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <i>Type</i> |

|                                                                              |                                                        |                                                 |             |
|------------------------------------------------------------------------------|--------------------------------------------------------|-------------------------------------------------|-------------|
|                                                                              | <b>units</b>                                           | '1' (static)                                    | NC_STRING   |
|                                                                              | <b>coordinates</b>                                     | '/PRODUCT/longitude /PRODUCT/latitude' (static) | NC_STRING   |
| <b>debug_upas2_ubyte1D</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS   |                                                        |                                                 |             |
| Description:                                                                 | Debug field, not available in operational environment. |                                                 |             |
| Dimensions:                                                                  | time, scanline.                                        |                                                 |             |
| Type:                                                                        | NC_UBYTE.                                              |                                                 |             |
| Source:                                                                      | Processor.                                             |                                                 |             |
| Attributes:                                                                  | <i>Name</i>                                            | <i>Value</i>                                    | <i>Type</i> |
|                                                                              | <b>units</b>                                           | '1' (static)                                    | NC_STRING   |
|                                                                              | <b>coordinates</b>                                     | '/PRODUCT/longitude /PRODUCT/latitude' (static) | NC_STRING   |
| <b>debug_upas2_byte1D</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS    |                                                        |                                                 |             |
| Description:                                                                 | Debug field, not available in operational environment. |                                                 |             |
| Dimensions:                                                                  | time, scanline.                                        |                                                 |             |
| Type:                                                                        | NC_BYTE.                                               |                                                 |             |
| Source:                                                                      | Processor.                                             |                                                 |             |
| Attributes:                                                                  | <i>Name</i>                                            | <i>Value</i>                                    | <i>Type</i> |
|                                                                              | <b>units</b>                                           | '1' (static)                                    | NC_STRING   |
|                                                                              | <b>coordinates</b>                                     | '/PRODUCT/longitude /PRODUCT/latitude' (static) | NC_STRING   |
| <b>debug_upas2_ushort1D</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS  |                                                        |                                                 |             |
| Description:                                                                 | Debug field, not available in operational environment. |                                                 |             |
| Dimensions:                                                                  | time, scanline.                                        |                                                 |             |
| Type:                                                                        | NC_USHORT.                                             |                                                 |             |
| Source:                                                                      | Processor.                                             |                                                 |             |
| Attributes:                                                                  | <i>Name</i>                                            | <i>Value</i>                                    | <i>Type</i> |
|                                                                              | <b>units</b>                                           | '1' (static)                                    | NC_STRING   |
|                                                                              | <b>coordinates</b>                                     | '/PRODUCT/longitude /PRODUCT/latitude' (static) | NC_STRING   |
| <b>debug_upas2_float2D_1</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS |                                                        |                                                 |             |
| Description:                                                                 | Debug field, not available in operational environment. |                                                 |             |
| Dimensions:                                                                  | time, scanline, ground_pixel.                          |                                                 |             |
| Type:                                                                        | NC_FLOAT.                                              |                                                 |             |
| Source:                                                                      | Processor.                                             |                                                 |             |
| Attributes:                                                                  | <i>Name</i>                                            | <i>Value</i>                                    | <i>Type</i> |
|                                                                              | <b>units</b>                                           | '1' (static)                                    | NC_STRING   |
|                                                                              | <b>coordinates</b>                                     | '/PRODUCT/longitude /PRODUCT/latitude' (static) | NC_STRING   |
| <b>debug_upas2_float2D_2</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS |                                                        |                                                 |             |
| Description:                                                                 | Debug field, not available in operational environment. |                                                 |             |
| Dimensions:                                                                  | time, scanline, ground_pixel.                          |                                                 |             |
| Type:                                                                        | NC_FLOAT.                                              |                                                 |             |
| Source:                                                                      | Processor.                                             |                                                 |             |
| Attributes:                                                                  | <i>Name</i>                                            | <i>Value</i>                                    | <i>Type</i> |
|                                                                              | <b>units</b>                                           | '1' (static)                                    | NC_STRING   |
|                                                                              | <b>coordinates</b>                                     | '/PRODUCT/longitude /PRODUCT/latitude' (static) | NC_STRING   |
| <b>debug_upas2_float2D_3</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS |                                                        |                                                 |             |
| Description:                                                                 | Debug field, not available in operational environment. |                                                 |             |
| Dimensions:                                                                  | time, scanline, ground_pixel.                          |                                                 |             |
| Type:                                                                        | NC_FLOAT.                                              |                                                 |             |
| Source:                                                                      | Processor.                                             |                                                 |             |

|                                                                               |                                                        |                                                 |             |
|-------------------------------------------------------------------------------|--------------------------------------------------------|-------------------------------------------------|-------------|
| Attributes:                                                                   | <i>Name</i>                                            | <i>Value</i>                                    | <i>Type</i> |
|                                                                               | <b>units</b>                                           | '1' (static)                                    | NC_STRING   |
|                                                                               | <b>coordinates</b>                                     | '/PRODUCT/longitude /PRODUCT/latitude' (static) | NC_STRING   |
| <b>debug_upas2_double2D_1</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS |                                                        |                                                 |             |
| Description:                                                                  | Debug field, not available in operational environment. |                                                 |             |
| Dimensions:                                                                   | time, scanline, ground_pixel.                          |                                                 |             |
| Type:                                                                         | NC_DOUBLE.                                             |                                                 |             |
| Source:                                                                       | Processor.                                             |                                                 |             |
| Attributes:                                                                   | <i>Name</i>                                            | <i>Value</i>                                    | <i>Type</i> |
|                                                                               | <b>units</b>                                           | '1' (static)                                    | NC_STRING   |
|                                                                               | <b>coordinates</b>                                     | '/PRODUCT/longitude /PRODUCT/latitude' (static) | NC_STRING   |
| <b>debug_upas2_double2D_2</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS |                                                        |                                                 |             |
| Description:                                                                  | Debug field, not available in operational environment. |                                                 |             |
| Dimensions:                                                                   | time, scanline, ground_pixel.                          |                                                 |             |
| Type:                                                                         | NC_DOUBLE.                                             |                                                 |             |
| Source:                                                                       | Processor.                                             |                                                 |             |
| Attributes:                                                                   | <i>Name</i>                                            | <i>Value</i>                                    | <i>Type</i> |
|                                                                               | <b>units</b>                                           | '1' (static)                                    | NC_STRING   |
|                                                                               | <b>coordinates</b>                                     | '/PRODUCT/longitude /PRODUCT/latitude' (static) | NC_STRING   |
| <b>debug_upas2_double2D_3</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS |                                                        |                                                 |             |
| Description:                                                                  | Debug field, not available in operational environment. |                                                 |             |
| Dimensions:                                                                   | time, scanline, ground_pixel.                          |                                                 |             |
| Type:                                                                         | NC_DOUBLE.                                             |                                                 |             |
| Source:                                                                       | Processor.                                             |                                                 |             |
| Attributes:                                                                   | <i>Name</i>                                            | <i>Value</i>                                    | <i>Type</i> |
|                                                                               | <b>units</b>                                           | '1' (static)                                    | NC_STRING   |
|                                                                               | <b>coordinates</b>                                     | '/PRODUCT/longitude /PRODUCT/latitude' (static) | NC_STRING   |
| <b>debug_upas2_int2D_1</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS    |                                                        |                                                 |             |
| Description:                                                                  | Debug field, not available in operational environment. |                                                 |             |
| Dimensions:                                                                   | time, scanline, ground_pixel.                          |                                                 |             |
| Type:                                                                         | NC_INT.                                                |                                                 |             |
| Source:                                                                       | Processor.                                             |                                                 |             |
| Attributes:                                                                   | <i>Name</i>                                            | <i>Value</i>                                    | <i>Type</i> |
|                                                                               | <b>units</b>                                           | '1' (static)                                    | NC_STRING   |
|                                                                               | <b>coordinates</b>                                     | '/PRODUCT/longitude /PRODUCT/latitude' (static) | NC_STRING   |
| <b>debug_upas2_int2D_2</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS    |                                                        |                                                 |             |
| Description:                                                                  | Debug field, not available in operational environment. |                                                 |             |
| Dimensions:                                                                   | time, scanline, ground_pixel.                          |                                                 |             |
| Type:                                                                         | NC_INT.                                                |                                                 |             |
| Source:                                                                       | Processor.                                             |                                                 |             |
| Attributes:                                                                   | <i>Name</i>                                            | <i>Value</i>                                    | <i>Type</i> |
|                                                                               | <b>units</b>                                           | '1' (static)                                    | NC_STRING   |
|                                                                               | <b>coordinates</b>                                     | '/PRODUCT/longitude /PRODUCT/latitude' (static) | NC_STRING   |
| <b>debug_upas2_int2D_3</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS    |                                                        |                                                 |             |
| Description:                                                                  | Debug field, not available in operational environment. |                                                 |             |
| Dimensions:                                                                   | time, scanline, ground_pixel.                          |                                                 |             |
| Type:                                                                         | NC_INT.                                                |                                                 |             |

|                                                                              |                                                        |                                                 |             |
|------------------------------------------------------------------------------|--------------------------------------------------------|-------------------------------------------------|-------------|
| Source:                                                                      | Processor.                                             |                                                 |             |
| Attributes:                                                                  | <i>Name</i>                                            | <i>Value</i>                                    | <i>Type</i> |
|                                                                              | <b>units</b>                                           | '1' (static)                                    | NC_STRING   |
|                                                                              | <b>coordinates</b>                                     | '/PRODUCT/longitude /PRODUCT/latitude' (static) | NC_STRING   |
| <b>debug_upas2_ubyte2D_1</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS |                                                        |                                                 |             |
| Description:                                                                 | Debug field, not available in operational environment. |                                                 |             |
| Dimensions:                                                                  | time, scanline, ground_pixel.                          |                                                 |             |
| Type:                                                                        | NC_UBYTE.                                              |                                                 |             |
| Source:                                                                      | Processor.                                             |                                                 |             |
| Attributes:                                                                  | <i>Name</i>                                            | <i>Value</i>                                    | <i>Type</i> |
|                                                                              | <b>units</b>                                           | '1' (static)                                    | NC_STRING   |
|                                                                              | <b>coordinates</b>                                     | '/PRODUCT/longitude /PRODUCT/latitude' (static) | NC_STRING   |
| <b>debug_upas2_ubyte2D_2</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS |                                                        |                                                 |             |
| Description:                                                                 | Debug field, not available in operational environment. |                                                 |             |
| Dimensions:                                                                  | time, scanline, ground_pixel.                          |                                                 |             |
| Type:                                                                        | NC_UBYTE.                                              |                                                 |             |
| Source:                                                                      | Processor.                                             |                                                 |             |
| Attributes:                                                                  | <i>Name</i>                                            | <i>Value</i>                                    | <i>Type</i> |
|                                                                              | <b>units</b>                                           | '1' (static)                                    | NC_STRING   |
|                                                                              | <b>coordinates</b>                                     | '/PRODUCT/longitude /PRODUCT/latitude' (static) | NC_STRING   |
| <b>debug_upas2_ubyte2D_3</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS |                                                        |                                                 |             |
| Description:                                                                 | Debug field, not available in operational environment. |                                                 |             |
| Dimensions:                                                                  | time, scanline, ground_pixel.                          |                                                 |             |
| Type:                                                                        | NC_UBYTE.                                              |                                                 |             |
| Source:                                                                      | Processor.                                             |                                                 |             |
| Attributes:                                                                  | <i>Name</i>                                            | <i>Value</i>                                    | <i>Type</i> |
|                                                                              | <b>units</b>                                           | '1' (static)                                    | NC_STRING   |
|                                                                              | <b>coordinates</b>                                     | '/PRODUCT/longitude /PRODUCT/latitude' (static) | NC_STRING   |
| <b>debug_upas2_byte2D_1</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS  |                                                        |                                                 |             |
| Description:                                                                 | Debug field, not available in operational environment. |                                                 |             |
| Dimensions:                                                                  | time, scanline, ground_pixel.                          |                                                 |             |
| Type:                                                                        | NC_BYTE.                                               |                                                 |             |
| Source:                                                                      | Processor.                                             |                                                 |             |
| Attributes:                                                                  | <i>Name</i>                                            | <i>Value</i>                                    | <i>Type</i> |
|                                                                              | <b>units</b>                                           | '1' (static)                                    | NC_STRING   |
|                                                                              | <b>coordinates</b>                                     | '/PRODUCT/longitude /PRODUCT/latitude' (static) | NC_STRING   |
| <b>debug_upas2_byte2D_2</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS  |                                                        |                                                 |             |
| Description:                                                                 | Debug field, not available in operational environment. |                                                 |             |
| Dimensions:                                                                  | time, scanline, ground_pixel.                          |                                                 |             |
| Type:                                                                        | NC_BYTE.                                               |                                                 |             |
| Source:                                                                      | Processor.                                             |                                                 |             |
| Attributes:                                                                  | <i>Name</i>                                            | <i>Value</i>                                    | <i>Type</i> |
|                                                                              | <b>units</b>                                           | '1' (static)                                    | NC_STRING   |
|                                                                              | <b>coordinates</b>                                     | '/PRODUCT/longitude /PRODUCT/latitude' (static) | NC_STRING   |
| <b>debug_upas2_byte2D_3</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS  |                                                        |                                                 |             |
| Description:                                                                 | Debug field, not available in operational environment. |                                                 |             |
| Dimensions:                                                                  | time, scanline, ground_pixel.                          |                                                 |             |

|                                                                               |                                                        |                                                 |             |
|-------------------------------------------------------------------------------|--------------------------------------------------------|-------------------------------------------------|-------------|
| Type:                                                                         | NC_BYTE.                                               |                                                 |             |
| Source:                                                                       | Processor.                                             |                                                 |             |
| Attributes:                                                                   | <i>Name</i>                                            | <i>Value</i>                                    | <i>Type</i> |
|                                                                               | <b>units</b>                                           | '1' (static)                                    | NC_STRING   |
|                                                                               | <b>coordinates</b>                                     | '/PRODUCT/longitude /PRODUCT/latitude' (static) | NC_STRING   |
| <b>debug_upas2_ushort2D_1</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS |                                                        |                                                 |             |
| Description:                                                                  | Debug field, not available in operational environment. |                                                 |             |
| Dimensions:                                                                   | time, scanline, ground_pixel.                          |                                                 |             |
| Type:                                                                         | NC_USHORT.                                             |                                                 |             |
| Source:                                                                       | Processor.                                             |                                                 |             |
| Attributes:                                                                   | <i>Name</i>                                            | <i>Value</i>                                    | <i>Type</i> |
|                                                                               | <b>units</b>                                           | '1' (static)                                    | NC_STRING   |
|                                                                               | <b>coordinates</b>                                     | '/PRODUCT/longitude /PRODUCT/latitude' (static) | NC_STRING   |
| <b>debug_upas2_ushort2D_2</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS |                                                        |                                                 |             |
| Description:                                                                  | Debug field, not available in operational environment. |                                                 |             |
| Dimensions:                                                                   | time, scanline, ground_pixel.                          |                                                 |             |
| Type:                                                                         | NC_USHORT.                                             |                                                 |             |
| Source:                                                                       | Processor.                                             |                                                 |             |
| Attributes:                                                                   | <i>Name</i>                                            | <i>Value</i>                                    | <i>Type</i> |
|                                                                               | <b>units</b>                                           | '1' (static)                                    | NC_STRING   |
|                                                                               | <b>coordinates</b>                                     | '/PRODUCT/longitude /PRODUCT/latitude' (static) | NC_STRING   |
| <b>debug_upas2_ushort2D_3</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS |                                                        |                                                 |             |
| Description:                                                                  | Debug field, not available in operational environment. |                                                 |             |
| Dimensions:                                                                   | time, scanline, ground_pixel.                          |                                                 |             |
| Type:                                                                         | NC_USHORT.                                             |                                                 |             |
| Source:                                                                       | Processor.                                             |                                                 |             |
| Attributes:                                                                   | <i>Name</i>                                            | <i>Value</i>                                    | <i>Type</i> |
|                                                                               | <b>units</b>                                           | '1' (static)                                    | NC_STRING   |
|                                                                               | <b>coordinates</b>                                     | '/PRODUCT/longitude /PRODUCT/latitude' (static) | NC_STRING   |
| <b>debug_upas2_float3D_1</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS  |                                                        |                                                 |             |
| Description:                                                                  | Debug field, not available in operational environment. |                                                 |             |
| Dimensions:                                                                   | time, scanline, ground_pixel, debug_upas2_levels_1.    |                                                 |             |
| Type:                                                                         | NC_FLOAT.                                              |                                                 |             |
| Source:                                                                       | Processor.                                             |                                                 |             |
| Attributes:                                                                   | <i>Name</i>                                            | <i>Value</i>                                    | <i>Type</i> |
|                                                                               | <b>units</b>                                           | '1' (static)                                    | NC_STRING   |
|                                                                               | <b>coordinates</b>                                     | '/PRODUCT/longitude /PRODUCT/latitude' (static) | NC_STRING   |
| <b>debug_upas2_float3D_2</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS  |                                                        |                                                 |             |
| Description:                                                                  | Debug field, not available in operational environment. |                                                 |             |
| Dimensions:                                                                   | time, scanline, ground_pixel, debug_upas2_levels_1.    |                                                 |             |
| Type:                                                                         | NC_FLOAT.                                              |                                                 |             |
| Source:                                                                       | Processor.                                             |                                                 |             |
| Attributes:                                                                   | <i>Name</i>                                            | <i>Value</i>                                    | <i>Type</i> |
|                                                                               | <b>units</b>                                           | '1' (static)                                    | NC_STRING   |
|                                                                               | <b>coordinates</b>                                     | '/PRODUCT/longitude /PRODUCT/latitude' (static) | NC_STRING   |
| <b>debug_upas2_float3D_3</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS  |                                                        |                                                 |             |
| Description:                                                                  | Debug field, not available in operational environment. |                                                 |             |

|                                                                              |                                                        |                                                 |             |
|------------------------------------------------------------------------------|--------------------------------------------------------|-------------------------------------------------|-------------|
| Dimensions:                                                                  | time, scanline, ground_pixel, debug_upas2_levels_1.    |                                                 |             |
| Type:                                                                        | NC_FLOAT.                                              |                                                 |             |
| Source:                                                                      | Processor.                                             |                                                 |             |
| Attributes:                                                                  | <i>Name</i>                                            | <i>Value</i>                                    | <i>Type</i> |
|                                                                              | <b>units</b>                                           | '1' (static)                                    | NC_STRING   |
|                                                                              | <b>coordinates</b>                                     | '/PRODUCT/longitude /PRODUCT/latitude' (static) | NC_STRING   |
| <b>debug_upas2_float3D_4</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS |                                                        |                                                 |             |
| Description:                                                                 | Debug field, not available in operational environment. |                                                 |             |
| Dimensions:                                                                  | time, scanline, ground_pixel, debug_upas2_levels_1.    |                                                 |             |
| Type:                                                                        | NC_FLOAT.                                              |                                                 |             |
| Source:                                                                      | Processor.                                             |                                                 |             |
| Attributes:                                                                  | <i>Name</i>                                            | <i>Value</i>                                    | <i>Type</i> |
|                                                                              | <b>units</b>                                           | '1' (static)                                    | NC_STRING   |
|                                                                              | <b>coordinates</b>                                     | '/PRODUCT/longitude /PRODUCT/latitude' (static) | NC_STRING   |
| <b>debug_upas2_float3D_5</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS |                                                        |                                                 |             |
| Description:                                                                 | Debug field, not available in operational environment. |                                                 |             |
| Dimensions:                                                                  | time, scanline, ground_pixel, debug_upas2_levels_1.    |                                                 |             |
| Type:                                                                        | NC_FLOAT.                                              |                                                 |             |
| Source:                                                                      | Processor.                                             |                                                 |             |
| Attributes:                                                                  | <i>Name</i>                                            | <i>Value</i>                                    | <i>Type</i> |
|                                                                              | <b>units</b>                                           | '1' (static)                                    | NC_STRING   |
|                                                                              | <b>coordinates</b>                                     | '/PRODUCT/longitude /PRODUCT/latitude' (static) | NC_STRING   |
| <b>debug_upas2_float3D_6</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS |                                                        |                                                 |             |
| Description:                                                                 | Debug field, not available in operational environment. |                                                 |             |
| Dimensions:                                                                  | time, scanline, ground_pixel, debug_upas2_levels_1.    |                                                 |             |
| Type:                                                                        | NC_FLOAT.                                              |                                                 |             |
| Source:                                                                      | Processor.                                             |                                                 |             |
| Attributes:                                                                  | <i>Name</i>                                            | <i>Value</i>                                    | <i>Type</i> |
|                                                                              | <b>units</b>                                           | '1' (static)                                    | NC_STRING   |
|                                                                              | <b>coordinates</b>                                     | '/PRODUCT/longitude /PRODUCT/latitude' (static) | NC_STRING   |

### 12.1.2.3 Group “WAVELENGTH\_CALIBRATIONS” in “DETAILED\_RESULTS”

#### Dimensions in SO2\_\_\_/PRODUCT/SUPPORT\_DATA/DETAILED\_RESULTS/WAVELENGTH\_CALIBRATIONS

**number\_of\_calibrations** The number of the calibrations depending on the solar spectrum.

**size** 1 (dynamic)

**source** Processor.

**degrees\_of\_polynomial\_shift\_win1** Dimension relative to the degrees of the polynomial shift. It may have multiple windows.

**size** 1 (dynamic)

**source** Processor.

**number\_of\_subwindows\_win1** The number of subwindows used in order to calculate the shift. It may have multiple windows.

**size** 1 (dynamic)

**source** Processor.

**degrees\_of\_polynomial\_shift\_win2** Dimension relative to the degrees of the polynomial shift. It may have multiple windows.

**size** 1 (dynamic)

**source** Processor.

**number\_of\_subwindows\_win2** The number of subwindows used in order to calculate the shift. It may have multiple windows.

**size** 1 (dynamic)

**source** Processor.

**degrees\_of\_polynomial\_shift\_win3** Dimension relative to the degrees of the polynomial shift. It may have multiple windows.

**size** 1 (dynamic)

**source** Processor.

**number\_of\_subwindows\_win3** The number of subwindows used in order to calculate the shift. It may have multiple windows.

**size** 1 (dynamic)

**source** Processor.

#### Variables in SO2\_\_\_/PRODUCT/SUPPORT\_DATA/DETAILED\_RESULTS/WAVELENGTH\_CALIBRATIONS

**calibration\_polynomial\_coefficients\_win1** in SO2\_\_\_/PRODUCT/SUPPORT\_DATA/DETAILED\_RESULTS/WAVELENGTH\_CALIBRATIONS

Description: Computed coefficients of the polynomial function. It may have multiple windows.

Dimensions: number\_of\_calibrations, degrees\_of\_polynomial\_shift\_win1.

Type: NC\_FLOAT.

Source: Processor.

| Attributes: | Name             | Value                                                                           | Type      |
|-------------|------------------|---------------------------------------------------------------------------------|-----------|
|             | <b>units</b>     | '1' (static)                                                                    | NC_STRING |
|             | <b>long_name</b> | 'computed coefficients of the polynomial function in fitting window 1' (static) | NC_STRING |

**calibration\_subwindows\_shift\_win1** in SO2\_\_\_/PRODUCT/SUPPORT\_DATA/DETAILED\_RESULTS/WAVELENGTH\_CALIBRATIONS

Description: Computed wavelengths shift values per subwindow. It may have multiple windows.

Dimensions: number\_of\_calibrations, number\_of\_subwindows\_win1.

Type: NC\_FLOAT.

Source: Processor.

| Attributes: | Name             | Value                                                                            | Type      |
|-------------|------------------|----------------------------------------------------------------------------------|-----------|
|             | <b>units</b>     | 'nm' (static)                                                                    | NC_STRING |
|             | <b>long_name</b> | 'irradiance wavelengths shift values per subwindow in fitting window 1' (static) | NC_STRING |

**calibration\_subwindows\_squeeze\_win1** in SO2\_\_\_/PRODUCT/SUPPORT\_DATA/DETAILED\_RESULTS/WAVELENGTH\_CALIBRATIONS

Description: Computed wavelengths squeeze values per subwindow. It may have multiple windows.

Dimensions: number\_of\_calibrations, number\_of\_subwindows\_win1.

Type: NC\_FLOAT.

Source: Processor.

| Attributes: | Name             | Value                                                                                     | Type      |
|-------------|------------------|-------------------------------------------------------------------------------------------|-----------|
|             | <b>units</b>     | '1' (static)                                                                              | NC_STRING |
|             | <b>long_name</b> | 'irradiance wavelengths squeeze fitted values per subwindow in fitting window 1' (static) | NC_STRING |

**calibration\_subwindows\_root\_mean\_square\_win1** in SO2\_\_\_/PRODUCT/SUPPORT\_DATA/DETAILED\_RESULTS/WAVELENGTH\_CALIBRATIONS

Description: Computed RMS values per subwindow. It may have multiple windows.

Dimensions: number\_of\_calibrations, number\_of\_subwindows\_win1.

Type: NC\_FLOAT.

Source: Processor.

| Attributes: | Name             | Value                                                        | Type      |
|-------------|------------------|--------------------------------------------------------------|-----------|
|             | <b>units</b>     | '1' (static)                                                 | NC_STRING |
|             | <b>long_name</b> | 'calibration rms per subwindow in fitting window 1' (static) | NC_STRING |

**calibration\_subwindows\_wavelength\_win1** in SO2\_\_\_/PRODUCT/SUPPORT\_DATA/DETAILED\_RESULTS/WAVELENGTH\_CALIBRATIONS

Description: Calibration wavelength center in each subwindow. It may have multiple windows.

Dimensions: number\_of\_subwindows\_win1.

Type: NC\_FLOAT.

Source: Processor.

| Attributes: | Name             | Value                                                                          | Type      |
|-------------|------------------|--------------------------------------------------------------------------------|-----------|
|             | <b>units</b>     | 'nm' (static)                                                                  | NC_STRING |
|             | <b>long_name</b> | 'calibration wavelength center in each subwindow in fitting window 1' (static) | NC_STRING |

**calibration\_polynomial\_coefficients\_win2** in SO2\_\_\_/PRODUCT/SUPPORT\_DATA/DETAILED\_RESULTS/WAVELENGTH\_CALIBRATIONS

Description: Computed coefficients of the polynomial function. It may have multiple windows.

Dimensions: number\_of\_calibrations, degrees\_of\_polynomial\_shift\_win2.

Type: NC\_FLOAT.

Source: Processor.

| Attributes: | Name             | Value                                                                           | Type      |
|-------------|------------------|---------------------------------------------------------------------------------|-----------|
|             | <b>units</b>     | '1' (static)                                                                    | NC_STRING |
|             | <b>long_name</b> | 'computed coefficients of the polynomial function in fitting window 2' (static) | NC_STRING |

**calibration\_subwindows\_shift\_win2** in SO2\_\_\_/PRODUCT/SUPPORT\_DATA/DETAILED\_RESULTS/WAVELENGTH\_CALIBRATIONS

Description: Computed wavelengths shift values per subwindow. It may have multiple windows.

Dimensions: number\_of\_calibrations, number\_of\_subwindows\_win2.

Type: NC\_FLOAT.

Source: Processor.

| Attributes: | Name             | Value                                                                                   | Type      |
|-------------|------------------|-----------------------------------------------------------------------------------------|-----------|
|             | <b>units</b>     | 'nm' (static)                                                                           | NC_STRING |
|             | <b>long_name</b> | 'irradiance wavelengths shift fitted values per subwindow in fitting window 2' (static) | NC_STRING |

**calibration\_subwindows\_squeeze\_win2** in SO2\_\_\_/PRODUCT/SUPPORT\_DATA/DETAILED\_RESULTS/WAVELENGTH\_CALIBRATIONS

Description: Computed wavelengths squeeze values per subwindow. It may have multiple windows.

Dimensions: number\_of\_calibrations, number\_of\_subwindows\_win2.

Type: NC\_FLOAT.

Source: Processor.

| Attributes: | Name         | Value        | Type      |
|-------------|--------------|--------------|-----------|
|             | <b>units</b> | '1' (static) | NC_STRING |

|                                                                                                                             |                                                                                  |                                                                                           |             |           |
|-----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|-------------|-----------|
|                                                                                                                             | <b>long_name</b>                                                                 | 'irradiance wavelengths squeeze fitted values per subwindow in fitting window 1' (static) |             | NC_STRING |
| <b>calibration_subwindows_root_mean_square_win2</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS/WAVELENGTH_CALIBRATIONS |                                                                                  |                                                                                           |             |           |
| Description:                                                                                                                | Computed RMS values per subwindow. It may have multiple windows.                 |                                                                                           |             |           |
| Dimensions:                                                                                                                 | number_of_calibrations, number_of_subwindows_win2.                               |                                                                                           |             |           |
| Type:                                                                                                                       | NC_FLOAT.                                                                        |                                                                                           |             |           |
| Source:                                                                                                                     | Processor.                                                                       |                                                                                           |             |           |
| Attributes:                                                                                                                 | <i>Name</i>                                                                      | <i>Value</i>                                                                              | <i>Type</i> |           |
|                                                                                                                             | <b>units</b>                                                                     | '1' (static)                                                                              | NC_STRING   |           |
|                                                                                                                             | <b>long_name</b>                                                                 | 'calibration rms per subwindow in fitting window 2' (static)                              | NC_STRING   |           |
| <b>calibration_subwindows_wavelength_win2</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS/WAVELENGTH_CALIBRATIONS       |                                                                                  |                                                                                           |             |           |
| Description:                                                                                                                | Calibration wavelength center in each subwindow. It may have multiple windows.   |                                                                                           |             |           |
| Dimensions:                                                                                                                 | number_of_subwindows_win2.                                                       |                                                                                           |             |           |
| Type:                                                                                                                       | NC_FLOAT.                                                                        |                                                                                           |             |           |
| Source:                                                                                                                     | Processor.                                                                       |                                                                                           |             |           |
| Attributes:                                                                                                                 | <i>Name</i>                                                                      | <i>Value</i>                                                                              | <i>Type</i> |           |
|                                                                                                                             | <b>units</b>                                                                     | 'nm' (static)                                                                             | NC_STRING   |           |
|                                                                                                                             | <b>long_name</b>                                                                 | 'calibration wavelength center in each subwindow in fitting window 2' (static)            | NC_STRING   |           |
| <b>calibration_polynomial_coefficients_win3</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS/WAVELENGTH_CALIBRATIONS     |                                                                                  |                                                                                           |             |           |
| Description:                                                                                                                | Computed coefficients of the polynomial function. It may have multiple windows.  |                                                                                           |             |           |
| Dimensions:                                                                                                                 | number_of_calibrations, degrees_of_polynomial_shift_win3.                        |                                                                                           |             |           |
| Type:                                                                                                                       | NC_FLOAT.                                                                        |                                                                                           |             |           |
| Source:                                                                                                                     | Processor.                                                                       |                                                                                           |             |           |
| Attributes:                                                                                                                 | <i>Name</i>                                                                      | <i>Value</i>                                                                              | <i>Type</i> |           |
|                                                                                                                             | <b>units</b>                                                                     | '1' (static)                                                                              | NC_STRING   |           |
|                                                                                                                             | <b>long_name</b>                                                                 | 'computed coefficients of the polynomial function in fitting window 3' (static)           | NC_STRING   |           |
| <b>calibration_subwindows_shift_win3</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS/WAVELENGTH_CALIBRATIONS            |                                                                                  |                                                                                           |             |           |
| Description:                                                                                                                | Computed wavelengths shift values per subwindow. It may have multiple windows.   |                                                                                           |             |           |
| Dimensions:                                                                                                                 | number_of_calibrations, number_of_subwindows_win3.                               |                                                                                           |             |           |
| Type:                                                                                                                       | NC_FLOAT.                                                                        |                                                                                           |             |           |
| Source:                                                                                                                     | Processor.                                                                       |                                                                                           |             |           |
| Attributes:                                                                                                                 | <i>Name</i>                                                                      | <i>Value</i>                                                                              | <i>Type</i> |           |
|                                                                                                                             | <b>units</b>                                                                     | 'nm' (static)                                                                             | NC_STRING   |           |
|                                                                                                                             | <b>long_name</b>                                                                 | 'irradiance wavelengths shift values per subwindow in fitting window 3' (static)          | NC_STRING   |           |
| <b>calibration_subwindows_squeeze_win3</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS/WAVELENGTH_CALIBRATIONS          |                                                                                  |                                                                                           |             |           |
| Description:                                                                                                                | Computed wavelengths squeeze values per subwindow. It may have multiple windows. |                                                                                           |             |           |
| Dimensions:                                                                                                                 | number_of_calibrations, number_of_subwindows_win3.                               |                                                                                           |             |           |
| Type:                                                                                                                       | NC_FLOAT.                                                                        |                                                                                           |             |           |
| Source:                                                                                                                     | Processor.                                                                       |                                                                                           |             |           |
| Attributes:                                                                                                                 | <i>Name</i>                                                                      | <i>Value</i>                                                                              | <i>Type</i> |           |

|                                                                                                                             |                                                                                |                                                                                                      |             |
|-----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|-------------|
|                                                                                                                             | <b>units</b>                                                                   | '1' (static)                                                                                         | NC_STRING   |
|                                                                                                                             | <b>long_name</b>                                                               | 'irradiance wavelengths squeeze fitted values per subwindow in fitting window 3' (static)            | NC_STRING   |
| <b>calibration_subwindows_root_mean_square_win3</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS/WAVELENGTH_CALIBRATIONS |                                                                                |                                                                                                      |             |
| Description:                                                                                                                | Computed RMS values per subwindow. It may have multiple windows.               |                                                                                                      |             |
| Dimensions:                                                                                                                 | number_of_calibrations, number_of_subwindows_win3.                             |                                                                                                      |             |
| Type:                                                                                                                       | NC_FLOAT.                                                                      |                                                                                                      |             |
| Source:                                                                                                                     | Processor.                                                                     |                                                                                                      |             |
| Attributes:                                                                                                                 | <i>Name</i>                                                                    | <i>Value</i>                                                                                         | <i>Type</i> |
|                                                                                                                             | <b>units</b>                                                                   | '1' (static)                                                                                         | NC_STRING   |
|                                                                                                                             | <b>long_name</b>                                                               | 'calibration rms per subwindow in fitting window 3' (static)                                         | NC_STRING   |
| <b>calibration_subwindows_wavelength_win3</b> in SO2___/PRODUCT/SUPPORT_DATA/DETAILED_RESULTS/WAVELENGTH_CALIBRATIONS       |                                                                                |                                                                                                      |             |
| Description:                                                                                                                | Calibration wavelength center in each subwindow. It may have multiple windows. |                                                                                                      |             |
| Dimensions:                                                                                                                 | number_of_subwindows_win3.                                                     |                                                                                                      |             |
| Type:                                                                                                                       | NC_FLOAT.                                                                      |                                                                                                      |             |
| Source:                                                                                                                     | Processor.                                                                     |                                                                                                      |             |
| Attributes:                                                                                                                 | <i>Name</i>                                                                    | <i>Value</i>                                                                                         | <i>Type</i> |
|                                                                                                                             | <b>units</b>                                                                   | 'nm' (static)                                                                                        | NC_STRING   |
|                                                                                                                             | <b>long_name</b>                                                               | 'calibration wavelength center in each subwindow of the wavelength calibration in window 3' (static) | NC_STRING   |

#### 12.1.2.4 Group "INPUT\_DATA" in "SUPPORT\_DATA"

#### Variables in SO2\_\_\_/PRODUCT/SUPPORT\_DATA/INPUT\_DATA

|                                                                     |                                                                                                                                                                                  |                                                               |             |
|---------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|-------------|
| <b>snow_ice_flag_nise</b> in SO2___/PRODUCT/SUPPORT_DATA/INPUT_DATA |                                                                                                                                                                                  |                                                               |             |
| Description:                                                        | This is the original snow/ice classification data field from NSIDC/NISE. In case this auxiliary data was not available while processing, only FillValue are present in the data. |                                                               |             |
| Dimensions:                                                         | time, scanline, ground_pixel.                                                                                                                                                    |                                                               |             |
| Type:                                                               | NC_UBYTE.                                                                                                                                                                        |                                                               |             |
| Source:                                                             | Processor.                                                                                                                                                                       |                                                               |             |
| Attributes:                                                         | <i>Name</i>                                                                                                                                                                      | <i>Value</i>                                                  | <i>Type</i> |
|                                                                     | <b>units</b>                                                                                                                                                                     | '1' (static)                                                  | NC_STRING   |
|                                                                     | <b>long_name</b>                                                                                                                                                                 | 'snow-ice mask' (static)                                      | NC_STRING   |
|                                                                     | <b>_FillValue</b>                                                                                                                                                                | '254UB' (static)                                              | NC_STRING   |
|                                                                     | <b>comment</b>                                                                                                                                                                   | 'flag indicating snow/ice at center of ground pixel' (static) | NC_STRING   |
|                                                                     | <b>source</b>                                                                                                                                                                    | 'NSIDC/NISE' (static)                                         | NC_STRING   |

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |           |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>flag_meanings</b> | 'snow-free_land sea_ice_1_percent sea_ice_2_percent sea_ice_3_percent sea_ice_4_percent sea_ice_5_percent sea_ice_6_percent sea_ice_7_percent sea_ice_8_percent sea_ice_9_percent sea_ice_10_percent sea_ice_11_percent sea_ice_12_percent sea_ice_13_percent sea_ice_14_percent sea_ice_15_percent sea_ice_16_percent sea_ice_17_percent sea_ice_18_percent sea_ice_19_percent sea_ice_20_percent sea_ice_21_percent sea_ice_22_percent sea_ice_23_percent sea_ice_24_percent sea_ice_25_percent sea_ice_26_percent sea_ice_27_percent sea_ice_28_percent sea_ice_29_percent sea_ice_30_percent sea_ice_31_percent sea_ice_32_percent sea_ice_33_percent sea_ice_34_percent sea_ice_35_percent sea_ice_36_percent sea_ice_37_percent sea_ice_38_percent sea_ice_39_percent sea_ice_40_percent sea_ice_41_percent sea_ice_42_percent sea_ice_43_percent sea_ice_44_percent sea_ice_45_percent sea_ice_46_percent sea_ice_47_percent sea_ice_48_percent sea_ice_49_percent sea_ice_50_percent sea_ice_51_percent sea_ice_52_percent sea_ice_53_percent sea_ice_54_percent sea_ice_55_percent sea_ice_56_percent sea_ice_57_percent sea_ice_58_percent sea_ice_59_percent sea_ice_60_percent sea_ice_61_percent sea_ice_62_percent sea_ice_63_percent sea_ice_64_percent sea_ice_65_percent sea_ice_66_percent sea_ice_67_percent sea_ice_68_percent sea_ice_69_percent sea_ice_70_percent sea_ice_71_percent sea_ice_72_percent sea_ice_73_percent sea_ice_74_percent sea_ice_75_percent sea_ice_76_percent sea_ice_77_percent sea_ice_78_percent sea_ice_79_percent sea_ice_80_percent sea_ice_81_percent sea_ice_82_percent sea_ice_83_percent sea_ice_84_percent sea_ice_85_percent sea_ice_86_percent sea_ice_87_percent sea_ice_88_percent sea_ice_89_percent sea_ice_90_percent sea_ice_91_percent sea_ice_92_percent sea_ice_93_percent sea_ice_94_percent sea_ice_95_percent sea_ice_96_percent sea_ice_97_percent sea_ice_98_percent sea_ice_99_percent sea_ice_100_percent permanent_ice snow mixed_pixels_at_coastlines suspect_ice_value corners ocean' (static) | NC_STRING |
| <b>flag_values</b>   | 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 103, 252, 253, 254, 255 (static)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | NC_UBYTE  |
| <b>coordinates</b>   | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | NC_STRING |

The latitude and longitude are in a different group. How to specify the related geospatial coordinates in this case is not specified in the climate and forecast metadata conventions [ER5].

**snow\_ice\_flag** in SO2\_\_\_/PRODUCT/SUPPORT\_DATA/INPUT\_DATA

Description: This is binary snow/ice classification flag. It is computed internally in the processor based on external dynamic data (e.g. NSIDC/NISE or climatology). In case the original value of the pixel is greater than 30 percent, the flag is set to 1 (snow/ice presence), otherwise 0 (snow/ice free).

Dimensions: time, scanline, ground\_pixel.

Type: NC\_UBYTE.

Source: Processor.

| Attributes: | Name                                                                                                                                                                                         | Value                                                         | Type      |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|-----------|
|             | <b>units</b>                                                                                                                                                                                 | '1' (static)                                                  | NC_STRING |
|             | <b>threshold</b>                                                                                                                                                                             | '0.3' (static)                                                | NC_STRING |
|             | The threshold in percentage to identify the pixel as snow/ice or snow free.                                                                                                                  |                                                               |           |
|             | <b>long_name</b>                                                                                                                                                                             | 'snow-ice mask' (static)                                      | NC_STRING |
|             | <b>_FillValue</b>                                                                                                                                                                            | '254UB' (static)                                              | NC_STRING |
|             | <b>comment</b>                                                                                                                                                                               | 'flag indicating snow/ice at center of ground pixel' (static) | NC_STRING |
|             | <b>source</b>                                                                                                                                                                                |                                                               | NC_STRING |
|             | Possible values: NSIDC/NISE, Fallback_climatology                                                                                                                                            |                                                               |           |
|             | <b>flag_meanings</b>                                                                                                                                                                         | 'snow_free snow_ice' (static)                                 | NC_STRING |
|             | <b>flag_values</b>                                                                                                                                                                           | 0, 1 (static)                                                 | NC_UBYTE  |
|             | <b>coordinates</b>                                                                                                                                                                           | '/PRODUCT/longitude /PRODUCT/latitude' (static)               | NC_STRING |
|             | The latitude and longitude are in a different group. How to specify the related geospatial coordinates in this case is not specified in the climate and forecast metadata conventions [ER5]. |                                                               |           |

**snow\_cover** in SO2\_\_\_/PRODUCT/SUPPORT\_DATA/INPUT\_DATA

Description: The snow cover in the region of the pixel

Dimensions: time, scanline, ground\_pixel.

Type: NC\_FLOAT.

Source: Processor.

| Attributes: | Name                                                                                                                                                                                         | Value                                           | Type      |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|-----------|
|             | <b>units</b>                                                                                                                                                                                 | '1' (static)                                    | NC_STRING |
|             | <b>long_name</b>                                                                                                                                                                             | 'snow-cover' (static)                           | NC_STRING |
|             | <b>source</b>                                                                                                                                                                                | 'ECMWF' (static)                                | NC_STRING |
|             | <b>coordinates</b>                                                                                                                                                                           | '/PRODUCT/longitude /PRODUCT/latitude' (static) | NC_STRING |
|             | The latitude and longitude are in a different group. How to specify the related geospatial coordinates in this case is not specified in the climate and forecast metadata conventions [ER5]. |                                                 |           |

**sea\_ice\_cover** in SO2\_\_\_/PRODUCT/SUPPORT\_DATA/INPUT\_DATA

Description: The sea-ice cover in the region of the pixel

Dimensions: time, scanline, ground\_pixel.

Type: NC\_FLOAT.

Source: Processor.

| Attributes: | Name             | Value                    | Type      |
|-------------|------------------|--------------------------|-----------|
|             | <b>units</b>     | '1' (static)             | NC_STRING |
|             | <b>long_name</b> | 'sea-ice-cover' (static) | NC_STRING |
|             | <b>source</b>    | 'ECMWF' (static)         | NC_STRING |

|                                                                                                                                                                                              |                                                                                              |                                                                                                   |             |           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|-------------|-----------|
|                                                                                                                                                                                              | <b>coordinates</b>                                                                           | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                   |             | NC_STRING |
| The latitude and longitude are in a different group. How to specify the related geospatial coordinates in this case is not specified in the climate and forecast metadata conventions [ER5]. |                                                                                              |                                                                                                   |             |           |
| <b>cloud_fraction_crb</b> in SO2___/PRODUCT/SUPPORT_DATA/INPUT_DATA                                                                                                                          |                                                                                              |                                                                                                   |             |           |
| Description:                                                                                                                                                                                 | Retrieved effective radiometric cloud fraction using the OCRA/ROCINN CRB model.              |                                                                                                   |             |           |
| Dimensions:                                                                                                                                                                                  | time, scanline, ground_pixel.                                                                |                                                                                                   |             |           |
| Type:                                                                                                                                                                                        | NC_FLOAT.                                                                                    |                                                                                                   |             |           |
| Source:                                                                                                                                                                                      | Processor.                                                                                   |                                                                                                   |             |           |
| Attributes:                                                                                                                                                                                  | <i>Name</i>                                                                                  | <i>Value</i>                                                                                      | <i>Type</i> |           |
|                                                                                                                                                                                              | <b>units</b>                                                                                 | '1' (static)                                                                                      | NC_STRING   |           |
| Dimensionless unit. This attribute originates from the NUG, CF standards.                                                                                                                    |                                                                                              |                                                                                                   |             |           |
|                                                                                                                                                                                              | <b>standard_name</b>                                                                         | 'TBD' (static)                                                                                    | NC_STRING   |           |
|                                                                                                                                                                                              | <b>long_name</b>                                                                             | 'effective radiometric cloud fraction from the CRB model' (static)                                | NC_STRING   |           |
|                                                                                                                                                                                              | <b>source</b>                                                                                | 'crb' (static)                                                                                    | NC_STRING   |           |
|                                                                                                                                                                                              | <b>comment</b>                                                                               | 'Coregistered effective radiometric cloud fraction using the OCRA/ROCINN CRB model.'              | NC_STRING   |           |
|                                                                                                                                                                                              | <b>coordinates</b>                                                                           | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                   |             | NC_STRING |
| The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. This attribute originates from the CF standard.            |                                                                                              |                                                                                                   |             |           |
| <b>cloud_fraction_crb_precision</b> in SO2___/PRODUCT/SUPPORT_DATA/INPUT_DATA                                                                                                                |                                                                                              |                                                                                                   |             |           |
| Description:                                                                                                                                                                                 | Error of the retrieved effective radiometric cloud fraction using the OCRA/ROCINN CRB model. |                                                                                                   |             |           |
| Dimensions:                                                                                                                                                                                  | time, scanline, ground_pixel.                                                                |                                                                                                   |             |           |
| Type:                                                                                                                                                                                        | NC_FLOAT.                                                                                    |                                                                                                   |             |           |
| Source:                                                                                                                                                                                      | Processor.                                                                                   |                                                                                                   |             |           |
| Attributes:                                                                                                                                                                                  | <i>Name</i>                                                                                  | <i>Value</i>                                                                                      | <i>Type</i> |           |
|                                                                                                                                                                                              | <b>units</b>                                                                                 | '1' (static)                                                                                      | NC_STRING   |           |
| Dimensionless unit. This attribute originates from the NUG, CF standards.                                                                                                                    |                                                                                              |                                                                                                   |             |           |
|                                                                                                                                                                                              | <b>standard_name</b>                                                                         | 'TBD' (static)                                                                                    | NC_STRING   |           |
|                                                                                                                                                                                              | <b>long_name</b>                                                                             | 'effective radiometric cloud fraction precision from the CRB model' (static)                      | NC_STRING   |           |
|                                                                                                                                                                                              | <b>source</b>                                                                                | 'crb' (static)                                                                                    | NC_STRING   |           |
|                                                                                                                                                                                              | <b>comment</b>                                                                               | 'Error of the coregistered effective radiometric cloud fraction using the OCRA/ROCINN CRB model.' | NC_STRING   |           |
|                                                                                                                                                                                              | <b>coordinates</b>                                                                           | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                   |             | NC_STRING |
| The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. This attribute originates from the CF standard.            |                                                                                              |                                                                                                   |             |           |
| <b>cloud_pressure_crb</b> in SO2___/PRODUCT/SUPPORT_DATA/INPUT_DATA                                                                                                                          |                                                                                              |                                                                                                   |             |           |
| Description:                                                                                                                                                                                 | Retrieved atmospheric pressure at the level of cloud using the OCRA/ROCINN CRB model.        |                                                                                                   |             |           |
| Dimensions:                                                                                                                                                                                  | time, scanline, ground_pixel.                                                                |                                                                                                   |             |           |
| Type:                                                                                                                                                                                        | NC_FLOAT.                                                                                    |                                                                                                   |             |           |
| Source:                                                                                                                                                                                      | Processor.                                                                                   |                                                                                                   |             |           |
| Attributes:                                                                                                                                                                                  | <i>Name</i>                                                                                  | <i>Value</i>                                                                                      | <i>Type</i> |           |
|                                                                                                                                                                                              | <b>units</b>                                                                                 | 'Pa' (static)                                                                                     | NC_STRING   |           |
|                                                                                                                                                                                              | <b>standard_name</b>                                                                         | 'TBD' (static)                                                                                    | NC_STRING   |           |

|                                                                               |                                                                                                           |                                                                                                                                                                                                                                      |             |
|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
|                                                                               | <b>long_name</b>                                                                                          | 'cloud radiometric optical centroid pressure from the CRB model' (static)                                                                                                                                                            | NC_STRING   |
|                                                                               | <b>source</b>                                                                                             | 'crb' (static)                                                                                                                                                                                                                       | NC_STRING   |
|                                                                               | <b>comment</b>                                                                                            | 'Coregistered and converted atmospheric pressure at the level of cloud using the OCRA/ROCINN CRB model.' (static)                                                                                                                    | NC_STRING   |
|                                                                               | <b>coordinates</b>                                                                                        | '/PRODUCT/longitude /PRODUCT/latitude' (static)<br>The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. This attribute originates from the CF standard. | NC_STRING   |
| <b>cloud_pressure_crb_precision</b> in SO2___/PRODUCT/SUPPORT_DATA/INPUT_DATA |                                                                                                           |                                                                                                                                                                                                                                      |             |
| Description:                                                                  | Error of the retrieved atmospheric pressure at the level of cloud using the OCRA/ROCINN CRB model.        |                                                                                                                                                                                                                                      |             |
| Dimensions:                                                                   | time, scanline, ground_pixel.                                                                             |                                                                                                                                                                                                                                      |             |
| Type:                                                                         | NC_FLOAT.                                                                                                 |                                                                                                                                                                                                                                      |             |
| Source:                                                                       | Processor.                                                                                                |                                                                                                                                                                                                                                      |             |
| Attributes:                                                                   | <i>Name</i>                                                                                               | <i>Value</i>                                                                                                                                                                                                                         | <i>Type</i> |
|                                                                               | <b>units</b>                                                                                              | 'Pa' (static)                                                                                                                                                                                                                        | NC_STRING   |
|                                                                               | <b>standard_name</b>                                                                                      | 'TBD' (static)                                                                                                                                                                                                                       | NC_STRING   |
|                                                                               | <b>long_name</b>                                                                                          | 'cloud radiometric optical centroid pressure precision from the CRB model' (static)                                                                                                                                                  | NC_STRING   |
|                                                                               | <b>source</b>                                                                                             | 'crb' (static)                                                                                                                                                                                                                       | NC_STRING   |
|                                                                               | <b>comment</b>                                                                                            | 'Error of the coregistered and converted atmospheric pressure at the level of cloud using the OCRA/ROCINN CRB model.' (static)                                                                                                       | NC_STRING   |
|                                                                               | <b>coordinates</b>                                                                                        | '/PRODUCT/longitude /PRODUCT/latitude' (static)<br>The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. This attribute originates from the CF standard. | NC_STRING   |
| <b>cloud_height_crb</b> in SO2___/PRODUCT/SUPPORT_DATA/INPUT_DATA             |                                                                                                           |                                                                                                                                                                                                                                      |             |
| Description:                                                                  | Retrieved height at the level of cloud w.r.t. the geoid/MSL using the OCRA/ROCINN CRB model.              |                                                                                                                                                                                                                                      |             |
| Dimensions:                                                                   | time, scanline, ground_pixel.                                                                             |                                                                                                                                                                                                                                      |             |
| Type:                                                                         | NC_FLOAT.                                                                                                 |                                                                                                                                                                                                                                      |             |
| Source:                                                                       | Processor.                                                                                                |                                                                                                                                                                                                                                      |             |
| Attributes:                                                                   | <i>Name</i>                                                                                               | <i>Value</i>                                                                                                                                                                                                                         | <i>Type</i> |
|                                                                               | <b>units</b>                                                                                              | 'm' (static)                                                                                                                                                                                                                         | NC_STRING   |
|                                                                               | <b>standard_name</b>                                                                                      | 'TBD' (static)                                                                                                                                                                                                                       | NC_STRING   |
|                                                                               | <b>long_name</b>                                                                                          | 'cloud radiometric optical centroid height from the CRB model' (static)                                                                                                                                                              | NC_STRING   |
|                                                                               | <b>source</b>                                                                                             | 'crb' (static)                                                                                                                                                                                                                       | NC_STRING   |
|                                                                               | <b>comment</b>                                                                                            | 'Coregistered height at the level of cloud w.r.t. the geoid/MSL using the OCRA/ROCINN CRB model.' (static)                                                                                                                           | NC_STRING   |
|                                                                               | <b>coordinates</b>                                                                                        | '/PRODUCT/longitude /PRODUCT/latitude' (static)<br>The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. This attribute originates from the CF standard. | NC_STRING   |
| <b>cloud_height_crb_precision</b> in SO2___/PRODUCT/SUPPORT_DATA/INPUT_DATA   |                                                                                                           |                                                                                                                                                                                                                                      |             |
| Description:                                                                  | Error of the retrieved height at the level of cloud w.r.t. the geoid/MSL using the OCRA/ROCINN CRB model. |                                                                                                                                                                                                                                      |             |
| Dimensions:                                                                   | time, scanline, ground_pixel.                                                                             |                                                                                                                                                                                                                                      |             |
| Type:                                                                         | NC_FLOAT.                                                                                                 |                                                                                                                                                                                                                                      |             |

|                                                                              |                                                                                                                                                                                   |                                                                                                                         |             |
|------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|-------------|
| Source:                                                                      | Processor.                                                                                                                                                                        |                                                                                                                         |             |
| Attributes:                                                                  | <i>Name</i>                                                                                                                                                                       | <i>Value</i>                                                                                                            | <i>Type</i> |
|                                                                              | <b>units</b>                                                                                                                                                                      | 'm' (static)                                                                                                            | NC_STRING   |
|                                                                              | <b>standard_name</b>                                                                                                                                                              | 'TBD' (static)                                                                                                          | NC_STRING   |
|                                                                              | <b>long_name</b>                                                                                                                                                                  | 'cloud radiometric optical centroid height precision from the CRB model' (static)                                       | NC_STRING   |
|                                                                              | <b>source</b>                                                                                                                                                                     | 'crb' (static)                                                                                                          | NC_STRING   |
|                                                                              | <b>comment</b>                                                                                                                                                                    | 'Error of the coregistered height at the level of cloud w.r.t. the geoid/MSL using the OCRA/ROCINN CRB model.' (static) | NC_STRING   |
|                                                                              | <b>coordinates</b>                                                                                                                                                                | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                                         | NC_STRING   |
|                                                                              | The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. This attribute originates from the CF standard. |                                                                                                                         |             |
| <b>cloud_albedo_crb</b> in SO2___/PRODUCT/SUPPORT_DATA/INPUT_DATA            |                                                                                                                                                                                   |                                                                                                                         |             |
| Description:                                                                 | Albedo of cloud using the OCRA/ROCINN CRB model.                                                                                                                                  |                                                                                                                         |             |
| Dimensions:                                                                  | time, scanline, ground_pixel.                                                                                                                                                     |                                                                                                                         |             |
| Type:                                                                        | NC_FLOAT.                                                                                                                                                                         |                                                                                                                         |             |
| Source:                                                                      | Processor.                                                                                                                                                                        |                                                                                                                         |             |
| Attributes:                                                                  | <i>Name</i>                                                                                                                                                                       | <i>Value</i>                                                                                                            | <i>Type</i> |
|                                                                              | <b>units</b>                                                                                                                                                                      | '1' (static)                                                                                                            | NC_STRING   |
|                                                                              | Dimensionless unit. This attribute originates from the NUG, CF standards.                                                                                                         |                                                                                                                         |             |
|                                                                              | <b>standard_name</b>                                                                                                                                                              | 'cloud_albedo' (static)                                                                                                 | NC_STRING   |
|                                                                              | <b>long_name</b>                                                                                                                                                                  | 'cloud albedo from the CRB model' (static)                                                                              | NC_STRING   |
|                                                                              | <b>source</b>                                                                                                                                                                     | 'crb' (static)                                                                                                          | NC_STRING   |
|                                                                              | <b>comment</b>                                                                                                                                                                    | 'Coregistered cloud albedo based on the OCRA/ROCINN CRB model.' (static)                                                | NC_STRING   |
|                                                                              | <b>coordinates</b>                                                                                                                                                                | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                                         | NC_STRING   |
|                                                                              | The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. This attribute originates from the CF standard. |                                                                                                                         |             |
| <b>cloud_albedo_crb_precision</b> in SO2___/PRODUCT/SUPPORT_DATA/INPUT_DATA  |                                                                                                                                                                                   |                                                                                                                         |             |
| Description:                                                                 | Error of the albedo of cloud using the OCRA/ROCINN CRB model.                                                                                                                     |                                                                                                                         |             |
| Dimensions:                                                                  | time, scanline, ground_pixel.                                                                                                                                                     |                                                                                                                         |             |
| Type:                                                                        | NC_FLOAT.                                                                                                                                                                         |                                                                                                                         |             |
| Source:                                                                      | Processor.                                                                                                                                                                        |                                                                                                                         |             |
| Attributes:                                                                  | <i>Name</i>                                                                                                                                                                       | <i>Value</i>                                                                                                            | <i>Type</i> |
|                                                                              | <b>units</b>                                                                                                                                                                      | '1' (static)                                                                                                            | NC_STRING   |
|                                                                              | Dimensionless unit. This attribute originates from the NUG, CF standards.                                                                                                         |                                                                                                                         |             |
|                                                                              | <b>standard_name</b>                                                                                                                                                              | 'cloud_albedo_standard_error' (static)                                                                                  | NC_STRING   |
|                                                                              | <b>long_name</b>                                                                                                                                                                  | 'cloud albedo precision from the CRB model' (static)                                                                    | NC_STRING   |
|                                                                              | <b>source</b>                                                                                                                                                                     | 'crb' (static)                                                                                                          | NC_STRING   |
|                                                                              | <b>comment</b>                                                                                                                                                                    | 'Error of the coregistered cloud albedo based on the OCRA/ROCINN CRB model.' (static)                                   | NC_STRING   |
|                                                                              | <b>coordinates</b>                                                                                                                                                                | '/PRODUCT/longitude /PRODUCT/latitude' (static)                                                                         | NC_STRING   |
|                                                                              | The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. This attribute originates from the CF standard. |                                                                                                                         |             |
| <b>ozone_total_vertical_column</b> in SO2___/PRODUCT/SUPPORT_DATA/INPUT_DATA |                                                                                                                                                                                   |                                                                                                                         |             |

|                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                   |             |
|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|-------------|
| Description:                                                                           | Main output data of O <sub>3</sub> Total column product calculated with DOAS algorithm for near real time processing, while for offline and reprocessing the O <sub>3</sub> is calculated with GODfit algorithm.                                                                                                                                                                                                                                                                                 |                                                   |             |
| Dimensions:                                                                            | time, scanline, ground_pixel.                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                   |             |
| Type:                                                                                  | NC_FLOAT.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                   |             |
| Source:                                                                                | Processor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                   |             |
| Attributes:                                                                            | <i>Name</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <i>Value</i>                                      | <i>Type</i> |
|                                                                                        | <b>units</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 'mol m-2' (static)                                | NC_STRING   |
|                                                                                        | <b>standard_name</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 'atmosphere_mole_content_of_ozone' (static)       | NC_STRING   |
|                                                                                        | <b>long_name</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 'total ozone column' (static)                     | NC_STRING   |
|                                                                                        | <b>coordinates</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | '/PRODUCT/longitude /PRODUCT/latitude' (static)   | NC_STRING   |
|                                                                                        | The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. This attribute originates from the CF standard.                                                                                                                                                                                                                                                                                                                |                                                   |             |
|                                                                                        | <b>multiplication_factor_to_convert_to_DU</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                    | %(MOLM2_TO_DU)s (static)                          | NC_FLOAT    |
|                                                                                        | The quantities in Sentinel 5 precursor files are given in SI units. For an integrated column value this means that the unit is mol m <sup>-2</sup> . Traditionally the unit for an integrated column is "DU" or Dobson Units. This attribute provides the multiplication factor to calculate the total column in DU from the value in mol m <sup>-2</sup> . This is provided as a convenience to users who have tools that work in DU.                                                           |                                                   |             |
|                                                                                        | <b>multiplication_factor_to_convert_to_molecules_per_cm2</b>                                                                                                                                                                                                                                                                                                                                                                                                                                     | 6.02214e+19 (static)                              | NC_FLOAT    |
|                                                                                        | The quantities in Sentinel 5 precursor files are given in SI units. For an integrated column value this means that the unit is mol m <sup>-2</sup> . Traditionally the unit for an integrated column is "molecules cm <sup>-2</sup> ". This attribute provides the multiplication factor to calculate the total column in molecules cm <sup>-2</sup> from the value in mol m <sup>-2</sup> . This is provided as a convenience to users who have tools that work in molecules cm <sup>-2</sup> . |                                                   |             |
| <b>ozone_total_vertical_column_precision</b> in SO2___/PRODUCT/SUPPORT_DATA/INPUT_DATA |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                   |             |
| Description:                                                                           | Precision data of O <sub>3</sub> Total column product calculated with DOAS algorithm for near real time processing, while for offline and reprocessing the O <sub>3</sub> is calculated with GODfit algorithm.                                                                                                                                                                                                                                                                                   |                                                   |             |
| Dimensions:                                                                            | time, scanline, ground_pixel.                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                   |             |
| Type:                                                                                  | NC_FLOAT.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                   |             |
| Source:                                                                                | Processor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                   |             |
| Attributes:                                                                            | <i>Name</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <i>Value</i>                                      | <i>Type</i> |
|                                                                                        | <b>units</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 'mol m-2' (static)                                | NC_STRING   |
|                                                                                        | <b>standard_name</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 'atmosphere_mole_content_of_ozone error' (static) | NC_STRING   |
|                                                                                        | <b>long_name</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 'total ozone column random error' (static)        | NC_STRING   |
|                                                                                        | <b>coordinates</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | '/PRODUCT/longitude /PRODUCT/latitude' (static)   | NC_STRING   |
|                                                                                        | The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. This attribute originates from the CF standard.                                                                                                                                                                                                                                                                                                                |                                                   |             |
|                                                                                        | <b>multiplication_factor_to_convert_to_DU</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                    | %(MOLM2_TO_DU)s (static)                          | NC_FLOAT    |
|                                                                                        | The quantities in Sentinel 5 precursor files are given in SI units. For an integrated column value this means that the unit is mol m <sup>-2</sup> . Traditionally the unit for an integrated column is "DU" or Dobson Units. This attribute provides the multiplication factor to calculate the total column in DU from the value in mol m <sup>-2</sup> . This is provided as a convenience to users who have tools that work in DU.                                                           |                                                   |             |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                               |                                                                                                                                                                                                                                   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>multiplication_factor_to_convert_to_molecules_per_cm2</b>                                                                                                                                                                                                                                                                                                                                                                                                                                   | 6.02214e+19 (static)                                                                                                                                                                                          | NC_FLOAT                                                                                                                                                                                                                          |
| <p>The quantities in Sentinel 5 precursor files are given in SI units. For an integrated column value this means that the unit is mol m<sup>-2</sup>. Traditionally the unit for an integrated column is “molecules cm<sup>-2</sup>”. This attribute provides the multiplication factor to calculate the total column in molecules cm<sup>-2</sup> from the value in mol m<sup>-2</sup>. This is provided as a convenience to users who have tools that work in molecules cm<sup>-2</sup>.</p> |                                                                                                                                                                                                               |                                                                                                                                                                                                                                   |
| <b>surface_altitude</b> in SO2___/PRODUCT/SUPPORT_DATA/INPUT_DATA                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                               |                                                                                                                                                                                                                                   |
| Description:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | The mean of the sub-pixels of the surface altitude above the reference geoid (WGS84) within the approximate field of view, based on the GMTED2010 surface elevation database.                                 |                                                                                                                                                                                                                                   |
| Dimensions:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | time, scanline, ground_pixel.                                                                                                                                                                                 |                                                                                                                                                                                                                                   |
| Type:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | NC_FLOAT.                                                                                                                                                                                                     |                                                                                                                                                                                                                                   |
| Source:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | surface elevation database.                                                                                                                                                                                   |                                                                                                                                                                                                                                   |
| Attributes:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <i>Name</i>                                                                                                                                                                                                   | <i>Value</i>                                                                                                                                                                                                                      |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>long_name</b>                                                                                                                                                                                              | ‘surface altitude’ (static) NC_STRING                                                                                                                                                                                             |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>standard_name</b>                                                                                                                                                                                          | ‘surface_altitude’ (static) NC_STRING                                                                                                                                                                                             |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>units</b>                                                                                                                                                                                                  | ‘m’ (static) NC_STRING                                                                                                                                                                                                            |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>coordinates</b>                                                                                                                                                                                            | ‘/PRODUCT/longitude /PRODUCT/latitude’ (static) NC_STRING                                                                                                                                                                         |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | The latitude and longitude are in a different group. How to specify the related geospatial coordinates in this case is not specified in the climate and forecast metadata conventions [ER5].                  |                                                                                                                                                                                                                                   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>source</b>                                                                                                                                                                                                 | ‘http://topotools.cr.usgs.gov/gmted_viewer/’ (static) NC_STRING                                                                                                                                                                   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>comment</b>                                                                                                                                                                                                | ‘The mean of the sub-pixels of the surface altitude above the reference geoid (WGS84) within the approximate field of view, based on the GMTED2010 surface elevation database’ (static) NC_STRING                                 |
| <b>surface_altitude_precision</b> in SO2___/PRODUCT/SUPPORT_DATA/INPUT_DATA                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                               |                                                                                                                                                                                                                                   |
| Description:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | The standard deviation of sub-pixels used in calculating the mean surface altitude above the reference geoid (WGS84) within the approximate field of view, based on the GMTED2010 surface elevation database. |                                                                                                                                                                                                                                   |
| Dimensions:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | time, scanline, ground_pixel.                                                                                                                                                                                 |                                                                                                                                                                                                                                   |
| Type:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | NC_FLOAT.                                                                                                                                                                                                     |                                                                                                                                                                                                                                   |
| Source:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | surface elevation database.                                                                                                                                                                                   |                                                                                                                                                                                                                                   |
| Attributes:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <i>Name</i>                                                                                                                                                                                                   | <i>Value</i>                                                                                                                                                                                                                      |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>long_name</b>                                                                                                                                                                                              | ‘surface altitude precision’ (static) NC_STRING                                                                                                                                                                                   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>standard_name</b>                                                                                                                                                                                          | ‘surface_altitude standard_error’ (static) NC_STRING                                                                                                                                                                              |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>units</b>                                                                                                                                                                                                  | ‘m’ (static) NC_STRING                                                                                                                                                                                                            |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>standard_error_multiplier</b>                                                                                                                                                                              | 1.0 (static) NC_FLOAT                                                                                                                                                                                                             |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>coordinates</b>                                                                                                                                                                                            | ‘/PRODUCT/longitude /PRODUCT/latitude’ (static) NC_STRING                                                                                                                                                                         |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | The latitude and longitude are in a different group. How to specify the related geospatial coordinates in this case is not specified in the climate and forecast metadata conventions [ER5].                  |                                                                                                                                                                                                                                   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>source</b>                                                                                                                                                                                                 | ‘http://topotools.cr.usgs.gov/gmted_viewer/’ (static) NC_STRING                                                                                                                                                                   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>comment</b>                                                                                                                                                                                                | ‘The standard deviation of sub-pixels used in calculating the mean surface altitude above the reference geoid (WGS84) within the approximate field of view, based on the GMTED2010 surface elevation database’ (static) NC_STRING |

|                                                                                      |                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>surface_classification</b> in SO2___/PRODUCT/SUPPORT_DATA/INPUT_DATA              |                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Description:                                                                         | This is a combined land/water mask and surface classification data field.                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Dimensions:                                                                          | time, scanline, ground_pixel.                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Type:                                                                                | NC_UBYTE.                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Source:                                                                              | surface elevation database (including flag attributes).                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Attributes:                                                                          | <i>Name</i>                                                                                                                                                                                  | <i>Value</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                                                                                      | <b>units</b>                                                                                                                                                                                 | '1' (static) NC_STRING                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                                                                      | <b>long_name</b>                                                                                                                                                                             | 'land-water mask' (static) NC_STRING                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                                                                      | <b>comment</b>                                                                                                                                                                               | 'flag indicating land/water and further surface classifications for the ground pixel' (static) NC_STRING                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                                                                      | <b>source</b>                                                                                                                                                                                | 'USGS ( <a href="http://edc2.usgs.gov/glcc/globdoc2_0.php">http://edc2.usgs.gov/glcc/globdoc2_0.php</a> ) and NASA SDP toolkit ( <a href="http://newsroom.gsfc.nasa.gov/sdptoolkit/toolkit.html">http://newsroom.gsfc.nasa.gov/sdptoolkit/toolkit.html</a> )' (static) NC_STRING                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                                                                      | <b>flag_meanings</b>                                                                                                                                                                         | 'land, water, some_water, coast, value_covers_majority_of_pixel, water+shallow_ocean, water+shallow_inland_water, water+ocean_coastline-lake_shoreline, water+intermittent_water, water+deep_inland_water, water+continental_shelf_ocean, water+deep_ocean, land+urban_and_built-up_land, land+dryland_cropland_and_pasture, land+irrigated_cropland_and_pasture, land+mixed_dryland-irrigated_cropland_and_pasture, land+cropland-grassland_mosaic, land+cropland-woodland_mosaic, land+grassland, land+shrubland, land+mixed_shrubland-grassland, land+savanna, land+deciduous_broadleaf_forest, land+deciduous_needleleaf_forest, land+evergreen_broadleaf_forest, land+evergreen_needleleaf_forest, land+mixed_forest, land+herbaceous_wetland, land+wooded_wetland, land+barren_or_sparsely_vegetated, land+herbaceous_tundra, land+wooded_tundra, land+mixed_tundra, land+bare_ground_tundra, land+snow_or_ice' (static) NC_STRING |
|                                                                                      | <b>flag_values</b>                                                                                                                                                                           | 0, 1, 2, 3, 4, 9, 17, 25, 33, 41, 49, 57, 8, 16, 24, 32, 40, 48, 56, 64, 72, 80, 88, 96, 104, 112, 120, 128, 136, 144, 152, 160, 168, 176, 184 (static) NC_UBYTE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                                                                      | <b>flag_masks</b>                                                                                                                                                                            | 3, 3, 3, 3, 4, 249, 249, 249, 249, 249, 249, 249, 249, 249, 249, 249, 249, 249, 249, 249, 249, 249, 249, 249 (static) NC_UBYTE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                                                                      | <b>coordinates</b>                                                                                                                                                                           | '/PRODUCT/longitude /PRODUCT/latitude' (static) NC_STRING                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                                                                                      | The latitude and longitude are in a different group. How to specify the related geospatial coordinates in this case is not specified in the climate and forecast metadata conventions [ER5]. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>instrument_configuration_identifier</b> in SO2___/PRODUCT/SUPPORT_DATA/INPUT_DATA |                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

**Description:** The lcID from the instrument configuration in the Level 1B data product. The TROPOMI instrument has many configurable parameters. For example, the exposure time, co-addition period, gains and (for UVN-DEMs) the binning factors can be varied. As a result, the instrument can be operated in many different modes or configurations. Each combination of instrument settings is referred to as an instrument configuration and is identified by an instrument configuration ID, a number in the range [1,65535]. This instrument configuration ID, or lcID, is primarily used by the instrument, where it identifies an entry in the instrument configuration tables. On ground, the lcID is used to determine the intended purpose of a measurement and is used in the L0 to 1b data processing to determine the processing path.

**Dimensions:** time, scanline.

**Type:** NC\_INT.

**Source:** L1B.

| Attributes: | Name             | Value                                                                                                                                                                                                                   | Type      |
|-------------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
|             | <b>long_name</b> | 'lcID' (static)                                                                                                                                                                                                         | NC_STRING |
|             | <b>comment</b>   | 'The Instrument Configuration ID defines the type of measurement and its purpose. The number of instrument configuration IDs will increase over the mission as new types of measurements are created and used' (static) | NC_STRING |

#### **instrument\_configuration\_version** in SO2\_\_\_/PRODUCT/SUPPORT\_DATA/INPUT\_DATA

**Description:** For an lcID (see the `instrument_configuration_identifier` above), it is possible to have multiple versions, identified by the instrument configuration version or lcVersion. The combination of lcID and lcVersion uniquely identifies the set of configuration settings of the instrument. At a given time, only one lcVersion of an lcID can be active within the instrument. The lcVersion allows to have multiple versions of a measurement with the same purpose, but with different settings. As a result of, for example, instrument degradation, it may be required to change the settings for a measurement. In that case, it is not necessary to create a new lcID, instead the same lcID can be using with a new lcVersion.

**Dimensions:** time, scanline.

**Type:** NC\_SHORT.

**Source:** L1B.

| Attributes: | Name             | Value                                                         | Type      |
|-------------|------------------|---------------------------------------------------------------|-----------|
|             | <b>long_name</b> | 'lcVersion' (static)                                          | NC_STRING |
|             | <b>comment</b>   | 'Version of the instrument_configuration_identifier' (static) | NC_STRING |

#### **scaled\_small\_pixel\_variance** in SO2\_\_\_/PRODUCT/SUPPORT\_DATA/INPUT\_DATA

**Description:** The scaled variance of the small pixel values for each ground pixel.

$$\langle R(t, r, c) \rangle = \frac{1}{N_{\text{small pixels}}} \sum_{i=0}^{N_{\text{small pixels}}-1} R(t, r, c, i) \quad (3)$$

$$V(t, r, c) = \frac{1}{N_{\text{small pixels}}} \sum_{i=0}^{N_{\text{small pixels}}-1} (R(t, r, c, i) - \langle R(t, r, c) \rangle)^2 \quad (4)$$

$$V_{\text{scaled}}(t, r, c) = \frac{V(t, r, c)}{\langle R(t, r, c) \rangle^2} \quad (5)$$

with  $\langle R(t, r, c) \rangle$  the mean reflectance for small pixels of ground pixel  $(t, r, c)$ ,  $V(t, r, c)$  the variance of the small pixels,  $V_{\text{scaled}}(t, r, c)$  the scaled small pixel variance, and  $R(t, r, c, i)$  with  $i = [0, \dots, N_{\text{small pixels}} - 1]$  the small pixel reflectance of ground pixel  $(t, r, c)$ . The reflectance  $R$  is calculated as  $R = (\pi I) / (\mu_0 E_0)$ , with  $I$  the radiance,  $E_0$  the irradiance and  $\mu_0 = \cos(\vartheta_0)$ , where  $\vartheta_0$  is the solar zenith angle.

**Dimensions:** time, scanline, ground\_pixel.

**Type:** NC\_FLOAT.

|                                                                                                                                                        |                                                                                                                                                                                              |                                                                        |             |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|-------------|
| Source:                                                                                                                                                | Processor.                                                                                                                                                                                   |                                                                        |             |
| Attributes:                                                                                                                                            | <i>Name</i>                                                                                                                                                                                  | <i>Value</i>                                                           | <i>Type</i> |
|                                                                                                                                                        | <b>long_name</b>                                                                                                                                                                             | 'scaled small pixel variance' (static)                                 | NC_STRING   |
|                                                                                                                                                        | <b>units</b>                                                                                                                                                                                 | '1' (static)                                                           | NC_STRING   |
|                                                                                                                                                        | <b>coordinates</b>                                                                                                                                                                           | '/PRODUCT/longitude /PRODUCT/latitude' (static)                        | NC_STRING   |
|                                                                                                                                                        | The latitude and longitude are in a different group. How to specify the related geospatial coordinates in this case is not specified in the climate and forecast metadata conventions [ER5]. |                                                                        |             |
|                                                                                                                                                        | <b>comment</b>                                                                                                                                                                               | 'The scaled variance of the reflectances of the small pixels' (static) | NC_STRING   |
|                                                                                                                                                        | <b>radiation_wavelength</b>                                                                                                                                                                  |                                                                        | NC_FLOAT    |
| The approximate wavelength of the small pixel column in nm. Note that due to the spectral smile this wavelength will depend on the ground_pixel index. |                                                                                                                                                                                              |                                                                        |             |
| <b>surface_pressure</b> in SO2___/PRODUCT/SUPPORT_DATA/INPUT_DATA                                                                                      |                                                                                                                                                                                              |                                                                        |             |
| Description:                                                                                                                                           | Surface pressure from ECMWF model data.                                                                                                                                                      |                                                                        |             |
| Dimensions:                                                                                                                                            | time, scanline, ground_pixel.                                                                                                                                                                |                                                                        |             |
| Type:                                                                                                                                                  | NC_FLOAT.                                                                                                                                                                                    |                                                                        |             |
| Source:                                                                                                                                                | Processor.                                                                                                                                                                                   |                                                                        |             |
| Attributes:                                                                                                                                            | <i>Name</i>                                                                                                                                                                                  | <i>Value</i>                                                           | <i>Type</i> |
|                                                                                                                                                        | <b>units</b>                                                                                                                                                                                 | 'Pa' (static)                                                          | NC_STRING   |
|                                                                                                                                                        | <b>standard_name</b>                                                                                                                                                                         | 'surface_air_pressure' (static)                                        | NC_STRING   |
|                                                                                                                                                        | <b>long_name</b>                                                                                                                                                                             | 'surface_air_pressure' (static)                                        | NC_STRING   |
|                                                                                                                                                        | <b>source</b>                                                                                                                                                                                |                                                                        | NC_STRING   |
|                                                                                                                                                        | <b>coordinates</b>                                                                                                                                                                           | '/PRODUCT/longitude /PRODUCT/latitude' (static)                        | NC_STRING   |
|                                                                                                                                                        | The latitude and longitude are in a different group. How to specify the related geospatial coordinates in this case is not specified in the climate and forecast metadata conventions [ER5]. |                                                                        |             |
| <b>surface_temperature</b> in SO2___/PRODUCT/SUPPORT_DATA/INPUT_DATA                                                                                   |                                                                                                                                                                                              |                                                                        |             |
| Description:                                                                                                                                           | Surface pressure from ECMWF model data.                                                                                                                                                      |                                                                        |             |
| Dimensions:                                                                                                                                            | time, scanline, ground_pixel.                                                                                                                                                                |                                                                        |             |
| Type:                                                                                                                                                  | NC_FLOAT.                                                                                                                                                                                    |                                                                        |             |
| Source:                                                                                                                                                | Processor.                                                                                                                                                                                   |                                                                        |             |
| Attributes:                                                                                                                                            | <i>Name</i>                                                                                                                                                                                  | <i>Value</i>                                                           | <i>Type</i> |
|                                                                                                                                                        | <b>units</b>                                                                                                                                                                                 | 'K' (static)                                                           | NC_STRING   |
|                                                                                                                                                        | <b>standard_name</b>                                                                                                                                                                         | 'surface_air_temperature' (static)                                     | NC_STRING   |
|                                                                                                                                                        | <b>long_name</b>                                                                                                                                                                             | 'surface_air_temperature' (static)                                     | NC_STRING   |
|                                                                                                                                                        | <b>source</b>                                                                                                                                                                                |                                                                        | NC_STRING   |
|                                                                                                                                                        | <b>coordinates</b>                                                                                                                                                                           | '/PRODUCT/longitude /PRODUCT/latitude' (static)                        | NC_STRING   |
|                                                                                                                                                        | The latitude and longitude are in a different group. How to specify the related geospatial coordinates in this case is not specified in the climate and forecast metadata conventions [ER5]. |                                                                        |             |
| <b>northward_wind</b> in SO2___/PRODUCT/SUPPORT_DATA/INPUT_DATA                                                                                        |                                                                                                                                                                                              |                                                                        |             |
| Description:                                                                                                                                           | 10 metre V wind component                                                                                                                                                                    |                                                                        |             |
| Dimensions:                                                                                                                                            | time, scanline, ground_pixel.                                                                                                                                                                |                                                                        |             |
| Type:                                                                                                                                                  | NC_FLOAT.                                                                                                                                                                                    |                                                                        |             |
| Source:                                                                                                                                                | Processor.                                                                                                                                                                                   |                                                                        |             |
| Attributes:                                                                                                                                            | <i>Name</i>                                                                                                                                                                                  | <i>Value</i>                                                           | <i>Type</i> |
|                                                                                                                                                        | <b>units</b>                                                                                                                                                                                 | 'm s-1' (static)                                                       | NC_STRING   |

|                                                                                                                                                                                                                                                                                |                                                                                                    |                            |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|----------------------------|
| Velocity in meters per second This attribute originates from the NUG, CF standards.                                                                                                                                                                                            |                                                                                                    |                            |
| <b>standard_name</b>                                                                                                                                                                                                                                                           | 'northward_wind' (static)                                                                          | NC_STRING                  |
| <b>long_name</b>                                                                                                                                                                                                                                                               | 'Northward wind from ECMWF at 10 meter height level' (static)                                      | NC_STRING                  |
| <b>coordinates</b>                                                                                                                                                                                                                                                             | '/PRODUCT/longitude /PRODUCT/latitude' (static) NC_STRING                                          |                            |
| The latitude and longitude are in a different group. How to specify the related geospatial coordinates in this case is not specified in the climate and forecast metadata conventions [ER5].                                                                                   |                                                                                                    |                            |
| <b>eastward_wind</b> in SO2___/PRODUCT/SUPPORT_DATA/INPUT_DATA                                                                                                                                                                                                                 |                                                                                                    |                            |
| Description:                                                                                                                                                                                                                                                                   | 10 metre U wind component                                                                          |                            |
| Dimensions:                                                                                                                                                                                                                                                                    | time, scanline, ground_pixel.                                                                      |                            |
| Type:                                                                                                                                                                                                                                                                          | NC_FLOAT.                                                                                          |                            |
| Source:                                                                                                                                                                                                                                                                        | Processor.                                                                                         |                            |
| Attributes:                                                                                                                                                                                                                                                                    | <i>Name</i>                                                                                        | <i>Value</i>               |
|                                                                                                                                                                                                                                                                                | <b>units</b>                                                                                       | 'm s-1' (static) NC_STRING |
| Velocity in meters per second This attribute originates from the NUG, CF standards.                                                                                                                                                                                            |                                                                                                    |                            |
| <b>standard_name</b>                                                                                                                                                                                                                                                           | 'eastward_wind' (static)                                                                           | NC_STRING                  |
| <b>long_name</b>                                                                                                                                                                                                                                                               | 'Eastward wind from ECMWF at 10 meter height level' (static)                                       | NC_STRING                  |
| <b>coordinates</b>                                                                                                                                                                                                                                                             | '/PRODUCT/longitude /PRODUCT/latitude' (static) NC_STRING                                          |                            |
| The latitude and longitude are in a different group. How to specify the related geospatial coordinates in this case is not specified in the climate and forecast metadata conventions [ER5].                                                                                   |                                                                                                    |                            |
| <b>tm5_constant_a</b> in SO2___/PRODUCT/SUPPORT_DATA/INPUT_DATA                                                                                                                                                                                                                |                                                                                                    |                            |
| Dimensions:                                                                                                                                                                                                                                                                    | time, layer.                                                                                       |                            |
| Type:                                                                                                                                                                                                                                                                          | NC_FLOAT.                                                                                          |                            |
| Source:                                                                                                                                                                                                                                                                        | Processor.                                                                                         |                            |
| Attributes:                                                                                                                                                                                                                                                                    | <i>Name</i>                                                                                        | <i>Value</i>               |
|                                                                                                                                                                                                                                                                                | <b>units</b>                                                                                       | 'Pa' (static) NC_STRING    |
| <b>tm5_constant_b</b> in SO2___/PRODUCT/SUPPORT_DATA/INPUT_DATA                                                                                                                                                                                                                |                                                                                                    |                            |
| Dimensions:                                                                                                                                                                                                                                                                    | time, layer.                                                                                       |                            |
| Type:                                                                                                                                                                                                                                                                          | NC_FLOAT.                                                                                          |                            |
| Source:                                                                                                                                                                                                                                                                        | Processor.                                                                                         |                            |
| Attributes:                                                                                                                                                                                                                                                                    | <i>Name</i>                                                                                        | <i>Value</i>               |
|                                                                                                                                                                                                                                                                                | <b>units</b>                                                                                       | '1' (static) NC_STRING     |
| <b>tm5_tropopause_layer_index</b> in SO2___/PRODUCT/SUPPORT_DATA/INPUT_DATA                                                                                                                                                                                                    |                                                                                                    |                            |
| Dimensions:                                                                                                                                                                                                                                                                    | time, scanline, ground_pixel.                                                                      |                            |
| Type:                                                                                                                                                                                                                                                                          | NC_INT.                                                                                            |                            |
| Source:                                                                                                                                                                                                                                                                        | Processor.                                                                                         |                            |
| Attributes:                                                                                                                                                                                                                                                                    | <i>Name</i>                                                                                        | <i>Value</i>               |
|                                                                                                                                                                                                                                                                                | <b>units</b>                                                                                       | '1' (static) NC_STRING     |
| <b>coordinates</b>                                                                                                                                                                                                                                                             | '/PRODUCT/longitude /PRODUCT/latitude' (static) NC_STRING                                          |                            |
| The latitude and longitude coordinates of the TROPOMI swath is not defined as a Cartesian product of latitude and longitude axes. Following [ER5, section 5.2] we use this attribute to connect the data with the geolocation. This attribute originates from the CF standard. |                                                                                                    |                            |
| <b>aerosol_index_340_380</b> in SO2___/PRODUCT/SUPPORT_DATA/INPUT_DATA                                                                                                                                                                                                         |                                                                                                    |                            |
| Description:                                                                                                                                                                                                                                                                   | Aerosol index from L2 __ AER _ AI (at wavelengths 340/380, i.e. the TOMS pair). Only in OFFL mode. |                            |
| Dimensions:                                                                                                                                                                                                                                                                    | time, scanline, ground_pixel.                                                                      |                            |

|                                                                       |                                                                  |                                                 |             |
|-----------------------------------------------------------------------|------------------------------------------------------------------|-------------------------------------------------|-------------|
| Type:                                                                 | NC_FLOAT.                                                        |                                                 |             |
| Source:                                                               | Processor.                                                       |                                                 |             |
| Attributes:                                                           | <i>Name</i>                                                      | <i>Value</i>                                    | <i>Type</i> |
|                                                                       | <b>units</b>                                                     | '1' (static)                                    | NC_STRING   |
|                                                                       | <b>standard_name</b>                                             | 'ultraviolet_aerosol_index' (static)            | NC_STRING   |
|                                                                       | <b>comment</b>                                                   | 'Aerosol index from 380 and 340 nm' (static)    | NC_STRING   |
|                                                                       | <b>long_name</b>                                                 | 'aerosol index from 380 and 340 nm' (static)    | NC_STRING   |
|                                                                       | <b>radiation - wavelength</b>                                    | 340.0, 380.0 (static)                           | NC_FLOAT    |
|                                                                       | The wavelengths used for the determination of the aerosol index. |                                                 |             |
|                                                                       | <b>coordinates</b>                                               | '/PRODUCT/longitude /PRODUCT/latitude' (static) | NC_STRING   |
| <b>surface_albedo_328nm</b> in SO2___/PRODUCT/SUPPORT_DATA/INPUT_DATA |                                                                  |                                                 |             |
| Description:                                                          | Surface Albedo from OMI database for fit window 1+2 at 328nm     |                                                 |             |
| Dimensions:                                                           | time, scanline, ground_pixel.                                    |                                                 |             |
| Type:                                                                 | NC_FLOAT.                                                        |                                                 |             |
| Source:                                                               | Processor.                                                       |                                                 |             |
| Attributes:                                                           | <i>Name</i>                                                      | <i>Value</i>                                    | <i>Type</i> |
|                                                                       | <b>units</b>                                                     | '1' (static)                                    | NC_STRING   |
|                                                                       | <b>standard_name</b>                                             | 'surface_albedo' (static)                       | NC_STRING   |
|                                                                       | <b>long_name</b>                                                 | 'surface albdeo at 328nm' (static)              | NC_STRING   |
|                                                                       | <b>coordinates</b>                                               | '/PRODUCT/longitude /PRODUCT/latitude' (static) | NC_STRING   |
| <b>surface_albedo_376nm</b> in SO2___/PRODUCT/SUPPORT_DATA/INPUT_DATA |                                                                  |                                                 |             |
| Description:                                                          | Surface Albedo from OMI database for fit window 3 at 376nm       |                                                 |             |
| Dimensions:                                                           | time, scanline, ground_pixel.                                    |                                                 |             |
| Type:                                                                 | NC_FLOAT.                                                        |                                                 |             |
| Source:                                                               | Processor.                                                       |                                                 |             |
| Attributes:                                                           | <i>Name</i>                                                      | <i>Value</i>                                    | <i>Type</i> |
|                                                                       | <b>units</b>                                                     | '1' (static)                                    | NC_STRING   |
|                                                                       | <b>standard_name</b>                                             | 'surface_albedo' (static)                       | NC_STRING   |
|                                                                       | <b>long_name</b>                                                 | 'surface albdeo at 376nm' (static)              | NC_STRING   |
|                                                                       | <b>coordinates</b>                                               | '/PRODUCT/longitude /PRODUCT/latitude' (static) | NC_STRING   |

#### 12.1.2.5 Group “BACKGROUND\_CORRECTION” in “INPUT\_DATA”

This is the main group containing the background correction SO<sub>2</sub>.

#### Attributes in SO2\_\_\_/PRODUCT/SUPPORT\_DATA/INPUT\_DATA/BACKGROUND\_CORRECTION

|                                                    |                                                                                                                                                   |                                         |             |
|----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|-------------|
| Group attributes attached to BACKGROUND_CORRECTION |                                                                                                                                                   |                                         |             |
|                                                    | <i>Name</i>                                                                                                                                       | <i>Value</i>                            | <i>Type</i> |
|                                                    | <b>time_coverage_start</b>                                                                                                                        | 'YYYY-MM-DDTHH:MM:SS.mmmmmmZ' (dynamic) | NC_STRING   |
|                                                    | Start of the data granule in UTC as an ISO 8601 [RD35] string. See the discussion of the <code>delta_time</code> variable on page 41 for details. |                                         |             |
|                                                    | <b>time_coverage_end</b>                                                                                                                          | 'YYYY-MM-DDTHH:MM:SS.mmmmmmZ' (dynamic) | NC_STRING   |
|                                                    | End of the data granule in UTC as an ISO 8601 [RD35] string. See the discussion of the <code>delta_time</code> variable on page 41 for details.   |                                         |             |
|                                                    | <b>orbit</b>                                                                                                                                      | 0 (dynamic)                             | NC_INT      |
|                                                    | The absolute orbit number which indicates the first minor orbit of the given set of L2 products used for computing the background correction.     |                                         |             |
|                                                    | <b>earthshine_reference_valid</b>                                                                                                                 | 'false' (dynamic)                       | NC_STRING   |

Flag indicating if the computation of the L1b earthshine was successful (true) or not (false).

**background\_correction\_-** 'false' (dynamic) NC\_STRING  
**valid**

Flag indicating if the computation of the Background Correction was successful (true) or not (false).

#### Dimensions in SO2\_\_\_/PRODUCT/SUPPORT\_DATA/INPUT\_DATA/BACKGROUND\_CORRECTION

**lat\_grid** Number of latitude bins used.

**size** 36 (fixed)

**o3\_grid** Number of ozone bins used.

**size** 52 (fixed)

**detector\_rows** Cross-pixel dimension.

**size** 450 (fixed)

**source** L1B.

**wavelengths** Wavelength dimension for the calculation of the earthshine spectra.

**size** 9000 (dynamic)

**source** Processor.

#### Variables in SO2\_\_\_/PRODUCT/SUPPORT\_DATA/INPUT\_DATA/BACKGROUND\_CORRECTION

**o3\_grid** in SO2\_\_\_/PRODUCT/SUPPORT\_DATA/INPUT\_DATA/BACKGROUND\_CORRECTION

Description: Values of ozone bins.

Dimensions: o3\_grid (coordinate variable).

Type: NC\_FLOAT.

Source: Processor.

| Attributes: | Name         | Value         | Type      |
|-------------|--------------|---------------|-----------|
|             | <b>units</b> | 'DU' (static) | NC_STRING |

**lat\_grid** in SO2\_\_\_/PRODUCT/SUPPORT\_DATA/INPUT\_DATA/BACKGROUND\_CORRECTION

Description: Values of the latitude bins.

Dimensions: lat\_grid (coordinate variable).

Type: NC\_FLOAT.

Source: Processor.

| Attributes: | Name         | Value                    | Type      |
|-------------|--------------|--------------------------|-----------|
|             | <b>units</b> | 'degrees_north' (static) | NC_STRING |

**window1\_north** in SO2\_\_\_/PRODUCT/SUPPORT\_DATA/INPUT\_DATA/BACKGROUND\_CORRECTION

Description: Values of the SO2 Background Correction of the first fitting window of SO2 for the northern hemisphere.

Dimensions: o3\_grid, detector\_rows.

Type: NC\_FLOAT.

Source: Processor.

| Attributes: | Name         | Value              | Type      |
|-------------|--------------|--------------------|-----------|
|             | <b>units</b> | 'mol m-2' (static) | NC_STRING |

**window1\_south** in SO2\_\_\_/PRODUCT/SUPPORT\_DATA/INPUT\_DATA/BACKGROUND\_CORRECTION

Description: Values of the SO2 Background Correction of the first fitting window of SO2 for the southern hemisphere.

Dimensions: o3\_grid, detector\_rows.

Type: NC\_FLOAT.

Source: Processor.

|                                                                                                        |                                                                              |                                  |             |
|--------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|----------------------------------|-------------|
| Attributes:                                                                                            | <i>Name</i>                                                                  | <i>Value</i>                     | <i>Type</i> |
|                                                                                                        | <b>units</b>                                                                 | 'mol m-2' (static)               | NC_STRING   |
| <b>window2</b> in SO2___/PRODUCT/SUPPORT_DATA/INPUT_DATA/BACKGROUND_CORRECTION                         |                                                                              |                                  |             |
| Description:                                                                                           | Values of the SO2 Background Correction of the second fitting window of SO2. |                                  |             |
| Dimensions:                                                                                            | lat_grid, detector_rows.                                                     |                                  |             |
| Type:                                                                                                  | NC_FLOAT.                                                                    |                                  |             |
| Source:                                                                                                | Processor.                                                                   |                                  |             |
| Attributes:                                                                                            | <i>Name</i>                                                                  | <i>Value</i>                     | <i>Type</i> |
|                                                                                                        | <b>units</b>                                                                 | 'mol m-2' (static)               | NC_STRING   |
| <b>window3</b> in SO2___/PRODUCT/SUPPORT_DATA/INPUT_DATA/BACKGROUND_CORRECTION                         |                                                                              |                                  |             |
| Description:                                                                                           | Values of the SO2 Background Correction of the third fitting window of SO2.  |                                  |             |
| Dimensions:                                                                                            | lat_grid, detector_rows.                                                     |                                  |             |
| Type:                                                                                                  | NC_FLOAT.                                                                    |                                  |             |
| Source:                                                                                                | Processor.                                                                   |                                  |             |
| Attributes:                                                                                            | <i>Name</i>                                                                  | <i>Value</i>                     | <i>Type</i> |
|                                                                                                        | <b>units</b>                                                                 | 'mol m-2' (static)               | NC_STRING   |
| <b>earthshine_reference_wavelength</b> in SO2___/PRODUCT/SUPPORT_DATA/INPUT_DATA/BACKGROUND_CORRECTION |                                                                              |                                  |             |
| Description:                                                                                           | Wavelength grid of the calculated earthshine spectra.                        |                                  |             |
| Dimensions:                                                                                            | wavelengths.                                                                 |                                  |             |
| Type:                                                                                                  | NC_FLOAT.                                                                    |                                  |             |
| Source:                                                                                                | Processor.                                                                   |                                  |             |
| Attributes:                                                                                            | <i>Name</i>                                                                  | <i>Value</i>                     | <i>Type</i> |
|                                                                                                        | <b>units</b>                                                                 | 'nm' (static)                    | NC_STRING   |
| <b>earthshine_reference_radiance</b> in SO2___/PRODUCT/SUPPORT_DATA/INPUT_DATA/BACKGROUND_CORRECTION   |                                                                              |                                  |             |
| Description:                                                                                           | Calculated earthshine spectra.                                               |                                  |             |
| Dimensions:                                                                                            | detector_rows, wavelengths.                                                  |                                  |             |
| Type:                                                                                                  | NC_FLOAT.                                                                    |                                  |             |
| Source:                                                                                                | Processor.                                                                   |                                  |             |
| Attributes:                                                                                            | <i>Name</i>                                                                  | <i>Value</i>                     | <i>Type</i> |
|                                                                                                        | <b>units</b>                                                                 | 'mol.m-2.nm-1.sr-1.s-1' (static) | NC_STRING   |

#### 12.1.2.6 Group "METADATA" in "\_bgso2"

Metadata Description.

Attributes in SO2\_\_\_/PRODUCT/SUPPORT\_DATA/INPUT\_DATA/BACKGROUND\_CORRECTION/METADATA

|                                                                                                               |                         |                       |             |
|---------------------------------------------------------------------------------------------------------------|-------------------------|-----------------------|-------------|
| Group attributes attached to METADATA                                                                         |                         |                       |             |
|                                                                                                               | <i>Name</i>             | <i>Value</i>          | <i>Type</i> |
|                                                                                                               | <b>ProductShortName</b> | 'AUX_BGSO2_' (static) | NC_STRING   |
| The short product name. For the auxiliary background correction SO2. output this is fixed to "AUX _ BGSO2 _". |                         |                       |             |
|                                                                                                               | <b>input_orbits_l2</b>  |                       | NC_STRING   |
| List of L2 orbits ID that were used to produce Background correction file.                                    |                         |                       |             |
|                                                                                                               | <b>input_files_l2</b>   |                       | NC_STRING   |
| Absolute path of L2 products which were used to produce Background correction file.                           |                         |                       |             |
|                                                                                                               | <b>input_orbits_l1b</b> |                       | NC_STRING   |

List of L1b orbits ID that were used to calculate the L1b Earthshine spectra.

|                                                                                                                   |           |
|-------------------------------------------------------------------------------------------------------------------|-----------|
| <b>input_files_l1b</b>                                                                                            | NC_STRING |
| Absolute path of band 3 L1B products that were used to calculate the L1b Earthshine spectra.                      |           |
| <b>processingMode</b>                                                                                             | NC_STRING |
| Processor MODE (NRTI, OFFL, RPRO or TEST).                                                                        |           |
| <b>fallback_reference_date</b>                                                                                    | NC_STRING |
| Reference date that was selected in case the fallback background correction data was used.                        |           |
| <b>fallback_reference_date_difference</b>                                                                         | NC_STRING |
| Time difference to the reference date that was selected in case the fallback background correction data was used. |           |
| <b>fallback_aux_bg_source_filename</b>                                                                            | NC_STRING |
| Original filename of the fallback background generation file that has been used.                                  |           |
| <b>fallback_aux_bg_source_processingMode</b>                                                                      | NC_STRING |
| Processing mode of the original background correction file that has been used as fallback.                        |           |
| <b>fallback_l2_source_filename</b>                                                                                | NC_STRING |
| Original filename of the level-2 file from which the fallback background correction was retrieved.                |           |
| <b>fallback_l2_source_processingMode</b>                                                                          | NC_STRING |
| Processing mode of the level-2 file from which the fallback background correction was retrieved.                  |           |
| <b>fallback_l2_source_orbit</b>                                                                                   | NC_STRING |
| Orbit of the level-2 file from which the fallback background correction was retrieved.                            |           |
| <b>fallback_l2_source_sensing_time_start</b>                                                                      | NC_STRING |
| Sensing start time of the level-2 file from which the fallback background correction was retrieved.               |           |
| <b>fallback_l2_source_sensing_time_end</b>                                                                        | NC_STRING |
| Sensing end time of the level-2 file from which the fallback background correction was retrieved.                 |           |

#### 12.1.2.7 Group “PROCESSOR” in “input\_data”

The `processing_configuration` attribute of the PROCESSOR group aims at tracking the original configuration used for processing the current L2 product. It is also used in the latest version of the S5P L1b product.

#### Attributes in SO<sub>2</sub>\_\_\_/PRODUCT/SUPPORT\_DATA/INPUT\_DATA/PROCESSOR

| Group attributes attached to PROCESSOR |                                                                          |           |
|----------------------------------------|--------------------------------------------------------------------------|-----------|
| Name                                   | Value                                                                    | Type      |
| <b>processing_configuration</b>        | ‘Processing configuration used to generate the current product’ (static) | NC_STRING |

## 12.2 Group “METADATA” in “SO<sub>2</sub>\_\_\_”

This is a group to collect metadata items, such as the items that also appear in the header file and items required by Inspire [ER4]. Most metadata will be stored as attributes. Grouping attributes that belong to a specific standard is done by using sub-groups in the Metadata group.

Included in this group are the granule description and quality assurance parameters.

Note that some metadata attributes are required to be attached to the global level by convention, such as the CF-Metadata convention [ER5] and the NetCDF user guide [ER7].

### 12.2.1 Group “QA\_STATISTICS” in “METADATA”

Quality assurance statistics are gathered in variables located in this group. These can include histograms of the main parameters and event occurrence statistics. The contents of this group is under discussion. Note that the QA statistics may be stored as scalar variables rather than attributes. The former allow attributes to be attached to them, providing a more meaningful description than just the name.

703 **Attributes in SO<sub>2</sub>\_\_\_/METADATA/QA\_STATISTICS**

| Group attributes attached to QA_STATISTICS                                                                                                                                                                                                                                                                                          |              |             |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-------------|
| <i>Name</i>                                                                                                                                                                                                                                                                                                                         | <i>Value</i> | <i>Type</i> |
| <b>number_of_groundpixels</b>                                                                                                                                                                                                                                                                                                       | 0 (static)   | NC_INT      |
| Number of ground pixels in the file.                                                                                                                                                                                                                                                                                                |              |             |
| <b>number_of_processed_pixels</b>                                                                                                                                                                                                                                                                                                   | 0 (static)   | NC_INT      |
| Number of ground pixels where a retrieval was attempted. This is the <code>number_of_groundpixels</code> minus the pixels that were rejected based on time or configuration (range and step-size in scanline or ground_pixel index).                                                                                                |              |             |
| <b>number_of_successfully_processed_pixels</b>                                                                                                                                                                                                                                                                                      | 0 (static)   | NC_INT      |
| Number of ground pixels where a retrieval was successful.                                                                                                                                                                                                                                                                           |              |             |
| <b>number_of_rejected_pixels_not_enough_spectrum</b>                                                                                                                                                                                                                                                                                | 0 (static)   | NC_INT      |
| Number of pixels where processing was not attempted because after filtering for bad and missing pixels there were not enough spectral pixels left in either the radiance, irradiance or after calculating the reflectance.                                                                                                          |              |             |
| <b>number_of_failed_retrievals</b>                                                                                                                                                                                                                                                                                                  | 0 (static)   | NC_INT      |
| Number of pixels where processing failed for whatever reason.                                                                                                                                                                                                                                                                       |              |             |
| <b>number_of_ground_pixels_with_warnings</b>                                                                                                                                                                                                                                                                                        | 0 (static)   | NC_INT      |
| Number of pixels with one or more warnings.                                                                                                                                                                                                                                                                                         |              |             |
| <b>number_of_radiance_missing_occurrences</b>                                                                                                                                                                                                                                                                                       | 0 (static)   | NC_INT      |
| Number of ground pixels where processing error “the number of spectral pixels in the radiance due to flagging is too small to perform the fitting” occurred, i.e. where the lower 8 bits of the <code>processing_quality_flags</code> have the value “1”.                                                                           |              |             |
| <b>number_of_irradiance_missing_occurrences</b>                                                                                                                                                                                                                                                                                     | 0 (static)   | NC_INT      |
| Number of ground pixels where processing error “the number of spectral pixels in the irradiance due to flagging is too small to perform the fitting” occurred, i.e. where the lower 8 bits of the <code>processing_quality_flags</code> have the value “2”.                                                                         |              |             |
| <b>number_of_input_spectrum_missing_occurrences</b>                                                                                                                                                                                                                                                                                 | 0 (static)   | NC_INT      |
| Number of ground pixels where processing error “the reflectance spectrum does not contain enough points to perform the retrieval. This is different from (ir)radiance_missing in that the missing points may not be aligned” occurred, i.e. where the lower 8 bits of the <code>processing_quality_flags</code> have the value “3”. |              |             |
| <b>number_of_reflectance_range_error_occurrences</b>                                                                                                                                                                                                                                                                                | 0 (static)   | NC_INT      |
| <b>number_of_ler_range_error_occurrences</b>                                                                                                                                                                                                                                                                                        | 0 (static)   | NC_INT      |
| Number of ground pixels where processing error “lambert-equivalent reflectivity out of range error” occurred, i.e. where the lower 8 bits of the <code>processing_quality_flags</code> have the value “5”.                                                                                                                          |              |             |
| <b>number_of_snr_range_error_occurrences</b>                                                                                                                                                                                                                                                                                        | 0 (static)   | NC_INT      |
| Number of ground pixels where processing error “too low signal to noise to perform retrieval” occurred, i.e. where the lower 8 bits of the <code>processing_quality_flags</code> have the value “6”.                                                                                                                                |              |             |
| <b>number_of_sza_range_error_occurrences</b>                                                                                                                                                                                                                                                                                        | 0 (static)   | NC_INT      |

Number of ground pixels where processing error “solar zenith angle out of range, maximum value from configuration” occurred, i.e. where the lower 8 bits of the `processing_quality_flags` have the value “7”.

|                                              |            |        |
|----------------------------------------------|------------|--------|
| <b>number_of_vza_range_error_occurrences</b> | 0 (static) | NC_INT |
|----------------------------------------------|------------|--------|

Number of ground pixels where processing error “viewing zenith angle out of range, maximum value from configuration” occurred, i.e. where the lower 8 bits of the `processing_quality_flags` have the value “8”.

|                                              |            |        |
|----------------------------------------------|------------|--------|
| <b>number_of_lut_range_error_occurrences</b> | 0 (static) | NC_INT |
|----------------------------------------------|------------|--------|

Number of ground pixels where processing error “extrapolation in lookup table (airmass factor, cloud radiances)” occurred, i.e. where the lower 8 bits of the `processing_quality_flags` have the value “9”.

|                                                |            |        |
|------------------------------------------------|------------|--------|
| <b>number_of_ozone_range_error_occurrences</b> | 0 (static) | NC_INT |
|------------------------------------------------|------------|--------|

Number of ground pixels where processing error “ozone column significantly out of range of profile climatology” occurred, i.e. where the lower 8 bits of the `processing_quality_flags` have the value “10”.

|                                                      |            |        |
|------------------------------------------------------|------------|--------|
| <b>number_of_wavelength_offset_error_occurrences</b> | 0 (static) | NC_INT |
|------------------------------------------------------|------------|--------|

Number of ground pixels where processing error “wavelength offset exceeds maximum from configuration” occurred, i.e. where the lower 8 bits of the `processing_quality_flags` have the value “11”.

|                                                   |            |        |
|---------------------------------------------------|------------|--------|
| <b>number_of_initialization_error_occurrences</b> | 0 (static) | NC_INT |
|---------------------------------------------------|------------|--------|

Number of ground pixels where processing error “an error occurred during the processing of the pixel, no output was generated. The following errors raise this flag: Mismatch between irradiance and radiance wavelengths; The on-ground distance between band 1 and band 2 ground pixels exceeds a threshold set in the configuration. Derived a-priori information does not validate, no processing is possible” occurred, i.e. where the lower 8 bits of the `processing_quality_flags` have the value “12”.

|                                           |            |        |
|-------------------------------------------|------------|--------|
| <b>number_of_memory_error_occurrences</b> | 0 (static) | NC_INT |
|-------------------------------------------|------------|--------|

Number of ground pixels where processing error “memory allocation or deallocation error” occurred, i.e. where the lower 8 bits of the `processing_quality_flags` have the value “13”.

|                                              |            |        |
|----------------------------------------------|------------|--------|
| <b>number_of_assertion_error_occurrences</b> | 0 (static) | NC_INT |
|----------------------------------------------|------------|--------|

Number of ground pixels where processing error “error in algorithm detected during assertion” occurred, i.e. where the lower 8 bits of the `processing_quality_flags` have the value “14”.

|                                       |            |        |
|---------------------------------------|------------|--------|
| <b>number_of_io_error_occurrences</b> | 0 (static) | NC_INT |
|---------------------------------------|------------|--------|

Number of ground pixels where processing error “error detected during transfer of data between algorithm and framework” occurred, i.e. where the lower 8 bits of the `processing_quality_flags` have the value “15”.

|                                              |            |        |
|----------------------------------------------|------------|--------|
| <b>number_of_numerical_error_occurrences</b> | 0 (static) | NC_INT |
|----------------------------------------------|------------|--------|

Number of ground pixels where processing error “general fatal numerical error occurred during inversion” occurred, i.e. where the lower 8 bits of the `processing_quality_flags` have the value “16”.

|                                        |            |        |
|----------------------------------------|------------|--------|
| <b>number_of_lut_error_occurrences</b> | 0 (static) | NC_INT |
|----------------------------------------|------------|--------|

Number of ground pixels where processing error “error in accessing the lookup table” occurred, i.e. where the lower 8 bits of the `processing_quality_flags` have the value “17”.

|                                         |            |        |
|-----------------------------------------|------------|--------|
| <b>number_of_ISRF_error_occurrences</b> | 0 (static) | NC_INT |
|-----------------------------------------|------------|--------|

Number of ground pixels where processing error “error detected in the input instrument spectral response function input data” occurred, i.e. where the lower 8 bits of the `processing_quality_flags` have the value “18”.

|                                                |            |        |
|------------------------------------------------|------------|--------|
| <b>number_of_convergence_error_occurrences</b> | 0 (static) | NC_INT |
|------------------------------------------------|------------|--------|

Number of ground pixels where processing error “the main algorithm did not converge” occurred, i.e. where the lower 8 bits of the `processing_quality_flags` have the value “19”.

|                                                             |            |        |
|-------------------------------------------------------------|------------|--------|
| <b>number_of_cloud_filter_convergence_error_occurrences</b> | 0 (static) | NC_INT |
|-------------------------------------------------------------|------------|--------|

Number of ground pixels where processing error “the cloud filter did not converge” occurred, i.e. where the lower 8 bits of the `processing_quality_flags` have the value “20”.

|                                                              |            |        |
|--------------------------------------------------------------|------------|--------|
| <b>number_of_max_iteration_convergence_error_occurrences</b> | 0 (static) | NC_INT |
|--------------------------------------------------------------|------------|--------|

Number of ground pixels where processing error “no convergence because retrieval exceeds maximum number of iterations. Maximum value from configuration” occurred, i.e. where the lower 8 bits of the `processing_quality_flags` have the value “21”.

|                                                                   |            |        |
|-------------------------------------------------------------------|------------|--------|
| <b>number_of_aot_lower_boundary_convergence_error_occurrences</b> | 0 (static) | NC_INT |
|-------------------------------------------------------------------|------------|--------|

Number of ground pixels where processing error “no convergence because the aerosol optical thickness crosses lower boundary twice in succession” occurred, i.e. where the lower 8 bits of the `processing_quality_flags` have the value “22”.

|                                                               |            |        |
|---------------------------------------------------------------|------------|--------|
| <b>number_of_other_boundary_convergence_error_occurrences</b> | 0 (static) | NC_INT |
|---------------------------------------------------------------|------------|--------|

Number of ground pixels where processing error “no convergence because a state vector element crosses boundary twice in succession. Note that a separate failure flag is defined for non-convergence due to crossing of lower AOT boundary” occurred, i.e. where the lower 8 bits of the `processing_quality_flags` have the value “23”.

|                                                |            |        |
|------------------------------------------------|------------|--------|
| <b>number_of_geolocation_error_occurrences</b> | 0 (static) | NC_INT |
|------------------------------------------------|------------|--------|

Number of ground pixels where processing error “geolocation out of range” occurred, i.e. where the lower 8 bits of the `processing_quality_flags` have the value “24”.

|                                                    |            |        |
|----------------------------------------------------|------------|--------|
| <b>number_of_ch4_noscat_zero_error_occurrences</b> | 0 (static) | NC_INT |
|----------------------------------------------------|------------|--------|

Number of ground pixels where processing error “the CH<sub>4</sub> column retrieved by the non-scattering CO algorithm from the weak band or strong band is 0” occurred, i.e. where the lower 8 bits of the `processing_quality_flags` have the value “25”.

|                                                    |            |        |
|----------------------------------------------------|------------|--------|
| <b>number_of_h2o_noscat_zero_error_occurrences</b> | 0 (static) | NC_INT |
|----------------------------------------------------|------------|--------|

Number of ground pixels where processing error “the H<sub>2</sub>O column retrieved by the non-scattering CO algorithm from the weak band or strong band is 0” occurred, i.e. where the lower 8 bits of the `processing_quality_flags` have the value “26”.

|                                                          |            |        |
|----------------------------------------------------------|------------|--------|
| <b>number_of_max_optical_thickness_error_occurrences</b> | 0 (static) | NC_INT |
|----------------------------------------------------------|------------|--------|

Number of ground pixels where processing error “maximum optical thickness exceeded during iterations” occurred, i.e. where the lower 8 bits of the `processing_quality_flags` have the value “27”.

|                                                     |            |        |
|-----------------------------------------------------|------------|--------|
| <b>number_of_aerosol_boundary_error_occurrences</b> | 0 (static) | NC_INT |
|-----------------------------------------------------|------------|--------|

|                                                                                                                                                                                                                                                   |            |        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--------|
| Number of ground pixels where processing error “boundary hit of aerosol parameters at last iteration” occurred, i.e. where the lower 8 bits of the <code>processing_quality_flags</code> have the value “28”.                                     |            |        |
| <b>number_of_boundary_hit_error_occurrences</b>                                                                                                                                                                                                   | 0 (static) | NC_INT |
| Number of ground pixels where processing error “fatal boundary hit during iterations” occurred, i.e. where the lower 8 bits of the <code>processing_quality_flags</code> have the value “29”.                                                     |            |        |
| <b>number_of_chi2_error_occurrences</b>                                                                                                                                                                                                           | 0 (static) | NC_INT |
| Number of ground pixels where processing error “ $\chi^2$ is not-a-number or larger than $10^{10}$ ” occurred, i.e. where the lower 8 bits of the <code>processing_quality_flags</code> have the value “30”.                                      |            |        |
| <b>number_of_svd_error_occurrences</b>                                                                                                                                                                                                            | 0 (static) | NC_INT |
| Number of ground pixels where processing error “singular value decomposition failure” occurred, i.e. where the lower 8 bits of the <code>processing_quality_flags</code> have the value “31”.                                                     |            |        |
| <b>number_of_dfs_error_occurrences</b>                                                                                                                                                                                                            | 0 (static) | NC_INT |
| Number of ground pixels where processing error “degree of freedom is not-a-number” occurred, i.e. where the lower 8 bits of the <code>processing_quality_flags</code> have the value “32”.                                                        |            |        |
| <b>number_of_radiative_transfer_error_occurrences</b>                                                                                                                                                                                             | 0 (static) | NC_INT |
| Number of ground pixels where processing error “errors occurred during the radiative transfer computations, no processing possible” occurred, i.e. where the lower 8 bits of the <code>processing_quality_flags</code> have the value “33”.       |            |        |
| <b>number_of_optimal_estimation_error_occurrences</b>                                                                                                                                                                                             | 0 (static) | NC_INT |
| Number of ground pixels where processing error “errors occurred during the optimal estimation, processing has been terminated” occurred, i.e. where the lower 8 bits of the <code>processing_quality_flags</code> have the value “34”.            |            |        |
| <b>number_of_profile_error_occurrences</b>                                                                                                                                                                                                        | 0 (static) | NC_INT |
| Number of ground pixels where processing error “flag that indicates if there were any errors during the computation of the ozone profile” occurred, i.e. where the lower 8 bits of the <code>processing_quality_flags</code> have the value “35”. |            |        |
| <b>number_of_cloud_error_occurrences</b>                                                                                                                                                                                                          | 0 (static) | NC_INT |
| Number of ground pixels where processing error “no cloud data” occurred, i.e. where the lower 8 bits of the <code>processing_quality_flags</code> have the value “36”.                                                                            |            |        |
| <b>number_of_model_error_occurrences</b>                                                                                                                                                                                                          | 0 (static) | NC_INT |
| Number of ground pixels where processing error “forward model failure” occurred, i.e. where the lower 8 bits of the <code>processing_quality_flags</code> have the value “37”.                                                                    |            |        |
| <b>number_of_number_of_input_data_points_too_low_error_occurrences</b>                                                                                                                                                                            | 0 (static) | NC_INT |
| Number of ground pixels where processing error “not enough input ozone columns to calculate a tropospheric column” occurred, i.e. where the lower 8 bits of the <code>processing_quality_flags</code> have the value “38”.                        |            |        |
| <b>number_of_cloud_pressure_spread_too_low_error_occurrences</b>                                                                                                                                                                                  | 0 (static) | NC_INT |
| Number of ground pixels where processing error “cloud pressure variability too low to estimate a tropospheric column” occurred, i.e. where the lower 8 bits of the <code>processing_quality_flags</code> have the value “39”.                     |            |        |

|                                                                                                                                                                                                                                       |            |        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--------|
| <b>number_of_cloud_too_low_level_error_occurrences</b>                                                                                                                                                                                | 0 (static) | NC_INT |
| Number of ground pixels where processing error “clouds are too low in the atmosphere to assume sufficient shielding” occurred, i.e. where the lower 8 bits of the <code>processing_quality_flags</code> have the value “40”.          |            |        |
| <b>number_of_generic_range_error_occurrences</b>                                                                                                                                                                                      | 0 (static) | NC_INT |
| Number of ground pixels where processing error “generic range error” occurred, i.e. where the lower 8 bits of the <code>processing_quality_flags</code> have the value “41”.                                                          |            |        |
| <b>number_of_generic_exception_occurrences</b>                                                                                                                                                                                        | 0 (static) | NC_INT |
| Number of ground pixels where processing error “catch all generic error” occurred, i.e. where the lower 8 bits of the <code>processing_quality_flags</code> have the value “42”.                                                      |            |        |
| <b>number_of_input_spectrum_alignment_error_occurrences</b>                                                                                                                                                                           | 0 (static) | NC_INT |
| Number of ground pixels where processing error “input radiance and irradiance spectra are not aligned correctly” occurred, i.e. where the lower 8 bits of the <code>processing_quality_flags</code> have the value “43”.              |            |        |
| <b>number_of_abort_error_occurrences</b>                                                                                                                                                                                              | 0 (static) | NC_INT |
| Number of ground pixels where processing error “not processed because processor aborted prematurely (time out or user abort)” occurred, i.e. where the lower 8 bits of the <code>processing_quality_flags</code> have the value “44”. |            |        |
| <b>number_of_wrong_input_type_error_occurrences</b>                                                                                                                                                                                   | 0 (static) | NC_INT |
| Number of ground pixels where processing error “wrong input type error, mismatch between expectation and received data” occurred, i.e. where the lower 8 bits of the <code>processing_quality_flags</code> have the value “45”.       |            |        |
| <b>number_of_wavelength_calibration_error_occurrences</b>                                                                                                                                                                             | 0 (static) | NC_INT |
| Number of ground pixels where processing error “an error occurred in the wavelength calibration of this pixel” occurred, i.e. where the lower 8 bits of the <code>processing_quality_flags</code> have the value “46”.                |            |        |
| <b>number_of_coregistration_error_occurrences</b>                                                                                                                                                                                     | 0 (static) | NC_INT |
| Number of ground pixels where processing error “no colocated pixels found in a supporting band” occurred, i.e. where the lower 8 bits of the <code>processing_quality_flags</code> have the value “47”.                               |            |        |
| <b>number_of_slant_column_density_error_occurrences</b>                                                                                                                                                                               | 0 (static) | NC_INT |
| Number of ground pixels where processing error “slant column fit returned error, no values can be compute” occurred, i.e. where the lower 8 bits of the <code>processing_quality_flags</code> have the value “48”.                    |            |        |
| <b>number_of_airmass_factor_error_occurrences</b>                                                                                                                                                                                     | 0 (static) | NC_INT |
| Number of ground pixels where processing error “airmass factor could not be compute” occurred, i.e. where the lower 8 bits of the <code>processing_quality_flags</code> have the value “49”.                                          |            |        |
| <b>number_of_vertical_column_density_error_occurrences</b>                                                                                                                                                                            | 0 (static) | NC_INT |
| Number of ground pixels where processing error “vertical column density could not be compute” occurred, i.e. where the lower 8 bits of the <code>processing_quality_flags</code> have the value “50”.                                 |            |        |
| <b>number_of_signal_to_noise_ratio_error_occurrences</b>                                                                                                                                                                              | 0 (static) | NC_INT |

|                                                                                                                                                                                                                                                                                              |            |        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--------|
| Number of ground pixels where processing error “the signal to noise ratio for this spectrum is too low for processin” occurred, i.e. where the lower 8 bits of the <code>processing_quality_flags</code> have the value “51”.                                                                |            |        |
| <b>number_of_configuration_error_occurrences</b>                                                                                                                                                                                                                                             | 0 (static) | NC_INT |
| Number of ground pixels where processing error “error while parsing the configuratio” occurred, i.e. where the lower 8 bits of the <code>processing_quality_flags</code> have the value “52”.                                                                                                |            |        |
| <b>number_of_key_error_occurrences</b>                                                                                                                                                                                                                                                       | 0 (static) | NC_INT |
| Number of ground pixels where processing error “key does not exis” occurred, i.e. where the lower 8 bits of the <code>processing_quality_flags</code> have the value “53”.                                                                                                                   |            |        |
| <b>number_of_saturation_error_occurrences</b>                                                                                                                                                                                                                                                | 0 (static) | NC_INT |
| Number of ground pixels where processing error “saturation in input spectru” occurred, i.e. where the lower 8 bits of the <code>processing_quality_flags</code> have the value “54”.                                                                                                         |            |        |
| <b>number_of_solar_eclipse_filter_occurrences</b>                                                                                                                                                                                                                                            | 0 (static) | NC_INT |
| Number of ground pixels where input filter “solar eclipse” occurred, i.e. where the lower 8 bits of the <code>processing_quality_flags</code> have the value “64”.                                                                                                                           |            |        |
| <b>number_of_cloud_filter_occurrences</b>                                                                                                                                                                                                                                                    | 0 (static) | NC_INT |
| Number of ground pixels where input filter “the cloud filter triggered causing the pixel to be skipped” occurred, i.e. where the lower 8 bits of the <code>processing_quality_flags</code> have the value “65”.                                                                              |            |        |
| <b>number_of_altitude_consistency_filter_occurrences</b>                                                                                                                                                                                                                                     | 0 (static) | NC_INT |
| Number of ground pixels where input filter “too large difference between ECMWF altitude and DEM altitude value” occurred, i.e. where the lower 8 bits of the <code>processing_quality_flags</code> have the value “66”.                                                                      |            |        |
| <b>number_of_altitude_roughness_filter_occurrences</b>                                                                                                                                                                                                                                       | 0 (static) | NC_INT |
| Number of ground pixels where input filter “too large standard deviation of altitude in DEM” occurred, i.e. where the lower 8 bits of the <code>processing_quality_flags</code> have the value “67”.                                                                                         |            |        |
| <b>number_of_sun_glint_filter_occurrences</b>                                                                                                                                                                                                                                                | 0 (static) | NC_INT |
| Number of ground pixels where input filter “for pixels over water, viewing direction inside sun glint region. Definition of sun glint angle and threshold value from ATBD” occurred, i.e. where the lower 8 bits of the <code>processing_quality_flags</code> have the value “68”.           |            |        |
| <b>number_of_mixed_surface_type_filter_occurrences</b>                                                                                                                                                                                                                                       | 0 (static) | NC_INT |
| Number of ground pixels where input filter “pixel contains land and water areas (e.g. coastal pixel)” occurred, i.e. where the lower 8 bits of the <code>processing_quality_flags</code> have the value “69”.                                                                                |            |        |
| <b>number_of_snow_ice_filter_occurrences</b>                                                                                                                                                                                                                                                 | 0 (static) | NC_INT |
| Number of ground pixels where input filter “pixel contains snow/ice: Snow/ice flag according to dynamic input OR climatological surface albedo at VIS wavelength is larger than 0.5” occurred, i.e. where the lower 8 bits of the <code>processing_quality_flags</code> have the value “70”. |            |        |
| <b>number_of_aai_filter_occurrences</b>                                                                                                                                                                                                                                                      | 0 (static) | NC_INT |
| Number of ground pixels where input filter “aAI smaller than 2.0” occurred, i.e. where the lower 8 bits of the <code>processing_quality_flags</code> have the value “71”.                                                                                                                    |            |        |
| <b>number_of_cloud_fraction_fresco_filter_occurrences</b>                                                                                                                                                                                                                                    | 0 (static) | NC_INT |

Number of ground pixels where input filter “pixel contains clouds: The FRESCO effective cloud fraction is larger than threshold. Threshold value from ATBD” occurred, i.e. where the lower 8 bits of the `processing_quality_flags` have the value “72”.

|                                                      |            |        |
|------------------------------------------------------|------------|--------|
| <b>number_of_aai_scene_albedo_filter_occurrences</b> | 0 (static) | NC_INT |
|------------------------------------------------------|------------|--------|

Number of ground pixels where input filter “pixel contains clouds: The difference between scene albedo at 380 nm from AAI calculation and the climatological surface albedo exceeds threshold. Threshold value from ATBD. This test filters out clouds” occurred, i.e. where the lower 8 bits of the `processing_quality_flags` have the value “73”.

|                                                              |            |        |
|--------------------------------------------------------------|------------|--------|
| <b>number_of_small_pixel_radiance_std_filter_occurrences</b> | 0 (static) | NC_INT |
|--------------------------------------------------------------|------------|--------|

Number of ground pixels where input filter “pixel contains clouds: Standard deviation of radiances in small-pixel column exceeds threshold. Threshold value from ATBD” occurred, i.e. where the lower 8 bits of the `processing_quality_flags` have the value “74”.

|                                                          |            |        |
|----------------------------------------------------------|------------|--------|
| <b>number_of_cloud_fraction_viirs_filter_occurrences</b> | 0 (static) | NC_INT |
|----------------------------------------------------------|------------|--------|

Number of ground pixels where input filter “pixel contains clouds: The cloud fraction from VIIRS / NPP exceeds threshold. Threshold value from ATBD” occurred, i.e. where the lower 8 bits of the `processing_quality_flags` have the value “75”.

|                                                              |            |        |
|--------------------------------------------------------------|------------|--------|
| <b>number_of_cirrus_reflectance_viirs_filter_occurrences</b> | 0 (static) | NC_INT |
|--------------------------------------------------------------|------------|--------|

Number of ground pixels where input filter “pixel contains clouds: Cirrus reflectance from VIIRS / NPP exceeds threshold. Threshold value from ATBD” occurred, i.e. where the lower 8 bits of the `processing_quality_flags` have the value “76”.

|                                                        |            |        |
|--------------------------------------------------------|------------|--------|
| <b>number_of_cf_viirs_swir_ifov_filter_occurrences</b> | 0 (static) | NC_INT |
|--------------------------------------------------------|------------|--------|

Number of ground pixels where input filter “fraction of cloudy VIIRS pixels within S5P SWIR ground pixel exceeds a priori threshold from configuration” occurred, i.e. where the lower 8 bits of the `processing_quality_flags` have the value “77”.

|                                                         |            |        |
|---------------------------------------------------------|------------|--------|
| <b>number_of_cf_viirs_swir_ofova_filter_occurrences</b> | 0 (static) | NC_INT |
|---------------------------------------------------------|------------|--------|

Number of ground pixels where input filter “fraction of cloudy VIIRS pixels within S5P SWIR OFOVa exceeds a priori threshold from configuration” occurred, i.e. where the lower 8 bits of the `processing_quality_flags` have the value “78”.

|                                                         |            |        |
|---------------------------------------------------------|------------|--------|
| <b>number_of_cf_viirs_swir_ofovb_filter_occurrences</b> | 0 (static) | NC_INT |
|---------------------------------------------------------|------------|--------|

Number of ground pixels where input filter “fraction of cloudy VIIRS pixels within S5P SWIR OFOVb exceeds a priori threshold from configuration” occurred, i.e. where the lower 8 bits of the `processing_quality_flags` have the value “79”.

|                                                         |            |        |
|---------------------------------------------------------|------------|--------|
| <b>number_of_cf_viirs_swir_ofovc_filter_occurrences</b> | 0 (static) | NC_INT |
|---------------------------------------------------------|------------|--------|

Number of ground pixels where input filter “fraction of cloudy VIIRS pixels within S5P SWIR OFOVc exceeds a priori threshold from configuration” occurred, i.e. where the lower 8 bits of the `processing_quality_flags` have the value “80”.

|                                                       |            |        |
|-------------------------------------------------------|------------|--------|
| <b>number_of_cf_viirs_nir_ifov_filter_occurrences</b> | 0 (static) | NC_INT |
|-------------------------------------------------------|------------|--------|

Number of ground pixels where input filter “fraction of cloudy VIIRS pixels within S5P NIR ground pixel exceeds a priori threshold from configuration” occurred, i.e. where the lower 8 bits of the `processing_quality_flags` have the value “81”.

|                                                        |            |        |
|--------------------------------------------------------|------------|--------|
| <b>number_of_cf_viirs_nir_ofova_filter_occurrences</b> | 0 (static) | NC_INT |
|--------------------------------------------------------|------------|--------|

Number of ground pixels where input filter “fraction of cloudy VIIRS pixels within S5P NIR OFOVa exceeds a priori threshold from configuration” occurred, i.e. where the lower 8 bits of the `processing_quality_flags` have the value “82”.

|                                                        |            |        |
|--------------------------------------------------------|------------|--------|
| <b>number_of_cf_viirs_nir_ofovb_filter_occurrences</b> | 0 (static) | NC_INT |
|--------------------------------------------------------|------------|--------|

Number of ground pixels where input filter “fraction of cloudy VIIRS pixels within S5P NIR OFOVb exceeds a priori threshold from configuration” occurred, i.e. where the lower 8 bits of the `processing_quality_flags` have the value “83”.

|                                                        |            |        |
|--------------------------------------------------------|------------|--------|
| <b>number_of_cf_viirs_nir_ofovc_filter_occurrences</b> | 0 (static) | NC_INT |
|--------------------------------------------------------|------------|--------|

Number of ground pixels where input filter “fraction of cloudy VIIRS pixels within S5P NIR OFOVc exceeds a priori threshold from configuration” occurred, i.e. where the lower 8 bits of the `processing_quality_flags` have the value “84”.

|                                                            |            |        |
|------------------------------------------------------------|------------|--------|
| <b>number_of_refl_cirrus_viirs_swir_filter_occurrences</b> | 0 (static) | NC_INT |
|------------------------------------------------------------|------------|--------|

Number of ground pixels where input filter “average VIIRS cirrus reflectance within SWIR ground pixel exceeds a priori threshold from configuration” occurred, i.e. where the lower 8 bits of the `processing_quality_flags` have the value “85”.

|                                                           |            |        |
|-----------------------------------------------------------|------------|--------|
| <b>number_of_refl_cirrus_viirs_nir_filter_occurrences</b> | 0 (static) | NC_INT |
|-----------------------------------------------------------|------------|--------|

Number of ground pixels where input filter “average VIIRS cirrus reflectance within NIR ground pixel exceeds a priori threshold from configuration” occurred, i.e. where the lower 8 bits of the `processing_quality_flags` have the value “86”.

|                                                            |            |        |
|------------------------------------------------------------|------------|--------|
| <b>number_of_diff_refl_cirrus_viirs_filter_occurrences</b> | 0 (static) | NC_INT |
|------------------------------------------------------------|------------|--------|

Number of ground pixels where input filter “difference in VIIRS average cirrus reflectance between SWIR and NIR ground pixel exceeds a priori threshold from configuration” occurred, i.e. where the lower 8 bits of the `processing_quality_flags` have the value “87”.

|                                                      |            |        |
|------------------------------------------------------|------------|--------|
| <b>number_of_ch4_noscat_ratio_filter_occurrences</b> | 0 (static) | NC_INT |
|------------------------------------------------------|------------|--------|

Number of ground pixels where input filter “the ratio between  $[CH_4]_{weak}$  and  $[CH_4]_{strong}$  is below or exceeds a priori thresholds from configuration” occurred, i.e. where the lower 8 bits of the `processing_quality_flags` have the value “88”.

|                                                          |            |        |
|----------------------------------------------------------|------------|--------|
| <b>number_of_ch4_noscat_ratio_std_filter_occurrences</b> | 0 (static) | NC_INT |
|----------------------------------------------------------|------------|--------|

Number of ground pixels where input filter “the standard deviation of  $[CH_4]_{weak}/[CH_4]_{strong}$  within the SWIR pixel and the 8 neighbouring pixels exceeds a priori threshold from configuration” occurred, i.e. where the lower 8 bits of the `processing_quality_flags` have the value “89”.

|                                                      |            |        |
|------------------------------------------------------|------------|--------|
| <b>number_of_h2o_noscat_ratio_filter_occurrences</b> | 0 (static) | NC_INT |
|------------------------------------------------------|------------|--------|

Number of ground pixels where input filter “the ratio between  $[H_2O]_{weak}$  and  $[H_2O]_{strong}$  is below or exceeds a priori thresholds from configuration” occurred, i.e. where the lower 8 bits of the `processing_quality_flags` have the value “90”.

|                                                          |            |        |
|----------------------------------------------------------|------------|--------|
| <b>number_of_h2o_noscat_ratio_std_filter_occurrences</b> | 0 (static) | NC_INT |
|----------------------------------------------------------|------------|--------|

Number of ground pixels where input filter “the standard deviation of  $[H_2O]_{weak}/[H_2O]_{strong}$  within the SWIR pixel and the 8 neighbouring pixels exceeds a priori threshold from configuration” occurred, i.e. where the lower 8 bits of the `processing_quality_flags` have the value “91”.

|                                                             |            |        |
|-------------------------------------------------------------|------------|--------|
| <b>number_of_diff_psurf_fresco_ecmwf_filter_occurrences</b> | 0 (static) | NC_INT |
|-------------------------------------------------------------|------------|--------|

Number of ground pixels where input filter “difference between the FRESCO apparent surface pressure and the ECMWF surface pressure exceeds a priori threshold from configuration” occurred, i.e. where the lower 8 bits of the `processing_quality_flags` have the value “92”.

|                                                       |            |        |
|-------------------------------------------------------|------------|--------|
| <b>number_of_psurf_fresco_stdv_filter_occurrences</b> | 0 (static) | NC_INT |
|-------------------------------------------------------|------------|--------|

Number of ground pixels where input filter “the standard deviation of the FRESCO apparent surface pressure in the NIR pixel and the 8 surrounding pixels exceeds a priori threshold from configuration” occurred, i.e. where the lower 8 bits of the `processing_quality_flags` have the value “93”.

|                                           |            |        |
|-------------------------------------------|------------|--------|
| <b>number_of_ocean_filter_occurrences</b> | 0 (static) | NC_INT |
|-------------------------------------------|------------|--------|

Number of ground pixels where input filter “the ground pixel is over ocean (and ocean glint retrievals are not switched on)” occurred, i.e. where the lower 8 bits of the `processing_quality_flags` have the value “94”.

|                                                |            |        |
|------------------------------------------------|------------|--------|
| <b>number_of_time_range_filter_occurrences</b> | 0 (static) | NC_INT |
|------------------------------------------------|------------|--------|

Number of ground pixels where input filter “time is out of the range that is to be processed” occurred, i.e. where the lower 8 bits of the `processing_quality_flags` have the value “95”.

|                                                              |            |        |
|--------------------------------------------------------------|------------|--------|
| <b>number_of_pixel_or_scan_line_index_filter_occurrences</b> | 0 (static) | NC_INT |
|--------------------------------------------------------------|------------|--------|

Number of ground pixels where input filter “not processed because pixel index does not match general selection criteria” occurred, i.e. where the lower 8 bits of the `processing_quality_flags` have the value “96”.

|                                                       |            |        |
|-------------------------------------------------------|------------|--------|
| <b>number_of_geographic_region_filter_occurrences</b> | 0 (static) | NC_INT |
|-------------------------------------------------------|------------|--------|

Number of ground pixels where input filter “pixel falls outside the specified regions of interest” occurred, i.e. where the lower 8 bits of the `processing_quality_flags` have the value “97”.

|                                                     |            |        |
|-----------------------------------------------------|------------|--------|
| <b>number_of_input_spectrum_warning_occurrences</b> | 0 (static) | NC_INT |
|-----------------------------------------------------|------------|--------|

Number of ground pixels where processing warning “number of good pixels in radiance, irradiance or calculated reflectance below threshold from configuration” occurred, i.e. where bit 8 in the `processing_quality_flags` is set to “1”.

|                                                             |            |        |
|-------------------------------------------------------------|------------|--------|
| <b>number_of_wavelength_calibration_warning_occurrences</b> | 0 (static) | NC_INT |
|-------------------------------------------------------------|------------|--------|

Number of ground pixels where processing warning “offset from wavelength fit is larger than limit set in configuration” occurred, i.e. where bit 9 in the `processing_quality_flags` is set to “1”.

|                                                    |            |        |
|----------------------------------------------------|------------|--------|
| <b>number_of_extrapolation_warning_occurrences</b> | 0 (static) | NC_INT |
|----------------------------------------------------|------------|--------|

Number of ground pixels where processing warning “pressure or temperature outside cross section LUT range, other lookup table extrapolation” occurred, i.e. where bit 10 in the `processing_quality_flags` is set to “1”.

|                                                |            |        |
|------------------------------------------------|------------|--------|
| <b>number_of_sun_glint_warning_occurrences</b> | 0 (static) | NC_INT |
|------------------------------------------------|------------|--------|

Number of ground pixels where processing warning “sun glint possibility warning” occurred, i.e. where bit 11 in the `processing_quality_flags` is set to “1”.

|                                                             |            |        |
|-------------------------------------------------------------|------------|--------|
| <b>number_of_south_atlantic_anomaly_warning_occurrences</b> | 0 (static) | NC_INT |
|-------------------------------------------------------------|------------|--------|

Number of ground pixels where processing warning “tROPOMI is inside the south Atlantic anomaly while taking these measurements” occurred, i.e. where bit 12 in the `processing_quality_flags` is set to “1”.

|                                                   |            |        |
|---------------------------------------------------|------------|--------|
| <b>number_of_sun_glint_correction_occurrences</b> | 0 (static) | NC_INT |
|---------------------------------------------------|------------|--------|

Number of ground pixels where processing warning “a sun glint correction has been applied” occurred, i.e. where bit 13 in the `processing_quality_flags` is set to “1”.

|                                               |            |        |
|-----------------------------------------------|------------|--------|
| <b>number_of_snow_ice_warning_occurrences</b> | 0 (static) | NC_INT |
|-----------------------------------------------|------------|--------|

Number of ground pixels where processing warning “snow/ice flag is set, i.e. using scene data from the cloud support product” occurred, i.e. where bit 14 in the `processing_quality_flags` is set to “1”.

|                                            |            |        |
|--------------------------------------------|------------|--------|
| <b>number_of_cloud_warning_occurrences</b> | 0 (static) | NC_INT |
|--------------------------------------------|------------|--------|

Number of ground pixels where processing warning “cloud filter based on FRESCO apparent surface pressure (VIIRS not available), cloud fraction above threshold or cloud pressure adjusted to force cloud above surface” occurred, i.e. where bit 15 in the `processing_quality_flags` is set to “1”.

|                                          |            |        |
|------------------------------------------|------------|--------|
| <b>number_of_AAI_warning_occurrences</b> | 0 (static) | NC_INT |
|------------------------------------------|------------|--------|

Number of ground pixels where processing warning “possible aerosol contamination as indicated by the AAI” occurred, i.e. where bit 16 in the `processing_quality_flags` is set to “1”.

|                                                             |            |        |
|-------------------------------------------------------------|------------|--------|
| <b>number_of_pixel_level_input_data_missing_occurrences</b> | 0 (static) | NC_INT |
|-------------------------------------------------------------|------------|--------|

Number of ground pixels where processing warning “dynamic auxiliary input data (e.g.. cloud) is missing for this ground pixel. A fallback option is used” occurred, i.e. where bit 17 in the `processing_quality_flags` is set to “1”.

|                                                 |            |        |
|-------------------------------------------------|------------|--------|
| <b>number_of_data_range_warning_occurrences</b> | 0 (static) | NC_INT |
|-------------------------------------------------|------------|--------|

Number of ground pixels where processing warning “carbon monoxide column tends to negative values; Water column tends to negative values; Heavy water (HDO) column tends to negative values; others” occurred, i.e. where bit 18 in the `processing_quality_flags` is set to “1”.

|                                                         |            |        |
|---------------------------------------------------------|------------|--------|
| <b>number_of_low_cloud_fraction_warning_occurrences</b> | 0 (static) | NC_INT |
|---------------------------------------------------------|------------|--------|

Number of ground pixels where processing warning “low cloud fraction, therefore no cloud pressure retrieved” occurred, i.e. where bit 19 in the `processing_quality_flags` is set to “1”.

|                                                           |            |        |
|-----------------------------------------------------------|------------|--------|
| <b>number_of_altitude_consistency_warning_occurrences</b> | 0 (static) | NC_INT |
|-----------------------------------------------------------|------------|--------|

Number of ground pixels where processing warning “difference between ECMWF surface elevation and high-resolution surface elevation exceeds threshold from configuration” occurred, i.e. where bit 20 in the `processing_quality_flags` is set to “1”.

|                                                            |            |        |
|------------------------------------------------------------|------------|--------|
| <b>number_of_signal_to_noise_ratio_warning_occurrences</b> | 0 (static) | NC_INT |
|------------------------------------------------------------|------------|--------|

Number of ground pixels where processing warning “signal to noise ratio in SWIR and/or NIR band below threshold from configuration” occurred, i.e. where bit 21 in the `processing_quality_flags` is set to “1”.

|                                                    |            |        |
|----------------------------------------------------|------------|--------|
| <b>number_of_deconvolution_warning_occurrences</b> | 0 (static) | NC_INT |
|----------------------------------------------------|------------|--------|

Number of ground pixels where processing warning “failed deconvolution irradiance spectrum (not pixel-specific, but row-specific)” occurred, i.e. where bit 22 in the `processing_quality_flags` is set to “1”.

|                                                                 |            |        |
|-----------------------------------------------------------------|------------|--------|
| <b>number_of_so2_volcanic_origin_likely_warning_occurrences</b> | 0 (static) | NC_INT |
|-----------------------------------------------------------------|------------|--------|

Number of ground pixels where processing warning “warning for SO<sub>2</sub> BL product, UTLS products: volcanic origin except for heavily polluted sites” occurred, i.e. where bit 23 in the `processing_quality_flags` is set to “1”.

|                                                                                                                                                                                                                                                                              |                 |           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----------|
| <b>number_of_so2_volcanic_origin_certain_warning_occurrences</b>                                                                                                                                                                                                             | 0 (static)      | NC_INT    |
| Number of ground pixels where processing warning “warning for SO <sub>2</sub> BL product, UTLS products: volcanic origin certain” occurred, i.e. where bit 24 in the <code>processing_quality_flags</code> is set to “1”.                                                    |                 |           |
| <b>number_of_interpolation_warning_occurrences</b>                                                                                                                                                                                                                           | 0 (static)      | NC_INT    |
| Number of ground pixels where processing warning “warning for interpolation on partially missing data. In this case the valid available data is used, potentially leading to a bias” occurred, i.e. where bit 25 in the <code>processing_quality_flags</code> is set to “1”. |                 |           |
| <b>number_of_saturation_warning_occurrences</b>                                                                                                                                                                                                                              | 0 (static)      | NC_INT    |
| Number of ground pixels where processing warning “saturation occurred spectrum, possibly causing biases in the retrieval” occurred, i.e. where bit 26 in the <code>processing_quality_flags</code> is set to “1”.                                                            |                 |           |
| <b>number_of_high_sza_warning_occurrences</b>                                                                                                                                                                                                                                | 0 (static)      | NC_INT    |
| Number of ground pixels where processing warning “warning for high solar zenith angle. In this case, the processing can be performed with less final quality” occurred, i.e. where bit 27 in the <code>processing_quality_flags</code> is set to “1”.                        |                 |           |
| <b>number_of_cloud_retrieval_warning_occurrences</b>                                                                                                                                                                                                                         | 0 (static)      | NC_INT    |
| Number of ground pixels where processing warning “warning occurring when the retrieval diagnostic indicates a degraded quality of the cloud retrieval” occurred, i.e. where bit 28 in the <code>processing_quality_flags</code> is set to “1”.                               |                 |           |
| <b>number_of_cloud_inhomogeneity_warning_occurrences</b>                                                                                                                                                                                                                     | 0 (static)      | NC_INT    |
| Number of ground pixels where processing warning “the cloud coregistration inhomogeneity parameter is above a given threshold” occurred, i.e. where bit 29 in the <code>processing_quality_flags</code> is set to “1”.                                                       |                 |           |
| <b>global_processing_warnings</b>                                                                                                                                                                                                                                            | ‘None’ (static) | NC_STRING |
| All warning messages, separated by newlines, with duplicates removed.                                                                                                                                                                                                        |                 |           |
| <b>time_for_algorithm_initialization</b>                                                                                                                                                                                                                                     | -1.0 (static)   | NC_DOUBLE |
| Time in seconds needed for initialization.                                                                                                                                                                                                                                   |                 |           |
| <b>time_for_processing</b>                                                                                                                                                                                                                                                   | -1.0 (static)   | NC_DOUBLE |
| Time in seconds needed for processing.                                                                                                                                                                                                                                       |                 |           |
| <b>time_per_pixel</b>                                                                                                                                                                                                                                                        | -1.0 (static)   | NC_DOUBLE |
| Time per pixel in seconds needed for processing.                                                                                                                                                                                                                             |                 |           |
| <b>time_standard_deviation_per_pixel</b>                                                                                                                                                                                                                                     | -1.0 (static)   | NC_DOUBLE |
| Standard deviation of the time per pixel in seconds needed for processing.                                                                                                                                                                                                   |                 |           |
| <b>number_of_processed_pixels_sulfurdioxide_layer_height</b>                                                                                                                                                                                                                 | 0 (static)      | NC_INT    |
| Number of ground pixels where a sulfurdioxide layer height retrieval was attempted.                                                                                                                                                                                          |                 |           |

#### Dimensions in SO<sub>2</sub>\_\_\_/METADATA/QA\_STATISTICS

**vertices** For the histogram boundaries.

**size** 2 (fixed)

**histogram\_axis** Histogram axis.

**size** 100 (fixed)

**pdf\_axis** Probability density function axis.

**size** 400 (fixed)

## Variables in SO2\_\_\_/METADATA/QA\_STATISTICS

### **histogram\_axis** in SO2\_\_\_/METADATA/QA\_STATISTICS

Description: Horizontal axis for the histograms of the main parameter.

Dimensions: histogram\_axis (coordinate variable).

Type: NC\_FLOAT.

Source: Processor.

| Attributes: | Name                                                                                                                                                                             | Value                       | Type      |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|-----------|
|             | <b>units</b>                                                                                                                                                                     | '1' (dynamic)               | NC_STRING |
|             | Same unit as the main parameter. Other attributes – standard_name, long_name – are to be copied from the main parameter as well. This attribute originates from the CF standard. |                             |           |
|             | <b>bounds</b>                                                                                                                                                                    | 'histogram_bounds' (static) | NC_STRING |

### **pdf\_axis** in SO2\_\_\_/METADATA/QA\_STATISTICS

Description: Horizontal axis for the probability distribution functions of the main parameter.

Dimensions: pdf\_axis (coordinate variable).

Type: NC\_FLOAT.

Source: Processor.

| Attributes: | Name                                                                                                                                                                             | Value                 | Type      |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-----------|
|             | <b>units</b>                                                                                                                                                                     | '1' (dynamic)         | NC_STRING |
|             | Same unit as the main parameter. Other attributes – standard_name, long_name – are to be copied from the main parameter as well. This attribute originates from the CF standard. |                       |           |
|             | <b>bounds</b>                                                                                                                                                                    | 'pdf_bounds' (static) | NC_STRING |

### **sulfurdioxide\_total\_column\_histogram** in SO2\_\_\_/METADATA/QA\_STATISTICS

Description: Histogram of the SO<sub>2</sub> values in the current granule.

Dimensions: histogram\_axis.

Type: NC\_INT.

Source: Processor.

| Attributes: | Name           | Value                                                               | Type      |
|-------------|----------------|---------------------------------------------------------------------|-----------|
|             | <b>units</b>   | '1' (static)                                                        | NC_STRING |
|             | <b>comment</b> | 'Histogram of the total column SO2 in the current granule' (static) | NC_STRING |

### **sulfurdioxide\_total\_column\_pdf** in SO2\_\_\_/METADATA/QA\_STATISTICS

Description: Probability density function of the SO<sub>2</sub> values in the current granule. The values are weighted with  $\cos(\delta_{\text{geo}})$  and spread out using the error estimate.

Dimensions: pdf\_axis.

Type: NC\_FLOAT.

Source: Processor.

| Attributes: | Name           | Value                                                                                  | Type      |
|-------------|----------------|----------------------------------------------------------------------------------------|-----------|
|             | <b>units</b>   | '1' (static)                                                                           | NC_STRING |
|             | <b>comment</b> | 'Probability density function of the total column SO2 in the current granule' (static) | NC_STRING |

## 12.2.2 Group “ALGORITHM\_SETTINGS” in “METADATA”

The algorithm settings are attached as attributes to this group. The current settings are listed here, each item in the list is a string attribute.

## 12.2.3 Group “GRANULE\_DESCRIPTION” in “METADATA”

Common granule level metadata.

### Attributes in SO<sub>2</sub>\_\_\_/METADATA/GRANULE\_DESCRIPTION

| Group attributes attached to GRANULE_DESCRIPTION                                                                                                     |                                       |           |
|------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|-----------|
| Name                                                                                                                                                 | Value                                 | Type      |
| <b>GranuleStart</b>                                                                                                                                  |                                       | NC_STRING |
| Start of the granule as ISO date/time string in UTC: YYYY-MM-DDTHH:MM:SS.mmmmmmZ. The formal definition of ISO date/time strings is given in [RD35]. |                                       |           |
| <b>GranuleEnd</b>                                                                                                                                    |                                       | NC_STRING |
| End of the granule as ISO date/time string in UTC: YYYY-MM-DDTHH:MM:SS.mmmmmmZ. The formal definition of ISO date/time strings is given in [RD35].   |                                       |           |
| <b>InstrumentName</b>                                                                                                                                | 'TROPOMI' (static)                    | NC_STRING |
| The name of the instrument, fixed to “TROPOMI”.                                                                                                      |                                       |           |
| <b>MissionName</b>                                                                                                                                   | 'Sentinel-5 precursor' (static)       | NC_STRING |
| The name of the mission, fixed to “Sentinel-5 precursor”.                                                                                            |                                       |           |
| <b>MissionShortName</b>                                                                                                                              | 'S5P' (static)                        | NC_STRING |
| The short name of the mission, fixed to “S5P”.                                                                                                       |                                       |           |
| <b>ProcessLevel</b>                                                                                                                                  | '2' (static)                          | NC_STRING |
| This is a level 2 product.                                                                                                                           |                                       |           |
| <b>ProcessingCenter</b>                                                                                                                              | '%(processingcenter)s' (dynamic)      | NC_STRING |
| Where was the processor run? The source is the probably the joborder, the most likely value for operational use is “DLR/Oberpfaffenhofen”.           |                                       |           |
| <b>ProcessingNode</b>                                                                                                                                |                                       | NC_STRING |
| The name of the machine that processed the data. This may aid in diagnosing failures in the processing.                                              |                                       |           |
| <b>ProcessorVersion</b>                                                                                                                              | '%(version)s' (dynamic)               | NC_STRING |
| The version number of the processor used to produce the file. This is a string formatted as “major.minor.bugfix”.                                    |                                       |           |
| <b>ProductFormatVersion</b>                                                                                                                          | 1 (static)                            | NC_INT    |
| The version of the format of the product file. This should be incremented whenever a datafield is added to the files.                                |                                       |           |
| <b>ProcessingMode</b>                                                                                                                                |                                       | NC_STRING |
| This attribute indicates the mode of the processor.<br>Possible values: Near-realtime, Offline, Reprocessing, Test, SyntheticTest                    |                                       |           |
| <b>CollectionIdentifier</b>                                                                                                                          | '%(collection_identifier)s' (dynamic) | NC_STRING |
| Identification of the processing collection, i.e. the group of products that can be used together as a consistent data set.                          |                                       |           |
| <b>ProductShortName</b>                                                                                                                              | 'L2_SO2___' (static)                  | NC_STRING |
| The short product name. For the SO <sub>2</sub> product this is fixed to “L2 __ SO2 ___”.                                                            |                                       |           |

### 12.2.3.1 Group “ESA\_METADATA” in “ESA\_metadata”

Metadata defined in the ESA file format standard [RD26].

### 12.2.3.2 Group “earth\_explorer\_header” in “ESA\_METADATA”

## Attributes in SO2\_\_\_/METADATA/ESA\_METADATA/earth\_explorer\_header

Group attributes attached to earth\_explorer\_header

| Name              | Value                            | Type      |
|-------------------|----------------------------------|-----------|
| <b>objectType</b> | 'Earth_Explorer_Header' (static) | NC_STRING |

### 12.2.3.3 Group “fixed\_header” in “earth\_explorer\_header”

The fixed header. We do not use a variable header, so only the fixed header is present.

## Attributes in SO2\_\_\_/METADATA/ESA\_METADATA/earth\_explorer\_header/fixed\_header

Group attributes attached to fixed\_header

| Name                    | Value                                                                                                                                                                                                                                                                                                | Type      |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>objectType</b>       | 'Fixed_Header' (static)                                                                                                                                                                                                                                                                              | NC_STRING |
| <b>File_Name</b>        | '%(logical_filename)s' (dynamic)<br>The <i>logical</i> file name, i.e. the file name without extension.                                                                                                                                                                                              | NC_STRING |
| <b>File_Description</b> |                                                                                                                                                                                                                                                                                                      | NC_STRING |
|                         | This is a copy of the global “title” attribute.                                                                                                                                                                                                                                                      |           |
| <b>Notes</b>            |                                                                                                                                                                                                                                                                                                      | NC_STRING |
|                         | This is a copy of the global “comment” attribute.                                                                                                                                                                                                                                                    |           |
| <b>Mission</b>          | 'S5P' (static)<br>The mission identifier for the Sentinel 5-precursor mission is “S5P”.                                                                                                                                                                                                              | NC_STRING |
| <b>File_Class</b>       |                                                                                                                                                                                                                                                                                                      | NC_STRING |
|                         | The file class of the output. Values are taken from the tailoring of the EO file format tailoring for S5P [RD25, section 4.1.2].                                                                                                                                                                     |           |
| <b>File_Type</b>        | '%(shortname)s' (dynamic)<br>Following the EO file format tailoring for S5P [RD25, sections 4.1.3.1 and 4.1.3.2].                                                                                                                                                                                    | NC_STRING |
| <b>File_Version</b>     | 0 (dynamic)<br>The file version information is not part of the file name conventions for S5P. If a file version number is to be recorded in this attribute, then it has to be provided by the PDGS via the job order. If provided, then the value is $\geq 1$ . If not provided the fill value is 0. | NC_INT    |

### 12.2.3.4 Group “validity\_period” in “fixed\_header”

## Attributes in SO2\_\_\_/METADATA/ESA\_METADATA/earth\_explorer\_header/fixed\_header/validity\_period

Group attributes attached to validity\_period

| Name                  | Value                                                                                                                                                                                                                         | Type      |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>objectType</b>     | 'Validity_Period' (static)                                                                                                                                                                                                    | NC_STRING |
| <b>Validity_Start</b> |                                                                                                                                                                                                                               | NC_STRING |
|                       | The value is the string “UTC=” concatenated with the <code>time_coverage_start</code> global attribute. This attribute corresponds to the “Validity_Start” element in the “Validity_Period” XML structure in the header file. |           |
| <b>Validity_Stop</b>  |                                                                                                                                                                                                                               | NC_STRING |
|                       | The value is the string “UTC=” concatenated with the <code>time_coverage_end</code> global attribute. This attribute corresponds to the “Validity_Stop” element in the “Validity_Period” XML structure in the header file.    |           |

### 12.2.3.5 Group “source” in “fixed\_header”

## Attributes in SO2\_\_\_/METADATA/ESA\_METADATA/earth\_explorer\_header/fixed\_header/source

| Group attributes attached to source                                                                                                                                                                                                            |                                  |             |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-------------|
| <i>Name</i>                                                                                                                                                                                                                                    | <i>Value</i>                     | <i>Type</i> |
| <b>objectType</b>                                                                                                                                                                                                                              | 'Source' (static)                | NC_STRING   |
| <b>System</b>                                                                                                                                                                                                                                  | '%(processingcenter)s' (dynamic) | NC_STRING   |
| Name of the Ground Segment element creating the file. For Level 2 files, this is the PDGS, but for testing a different value may be used. This attribute corresponds to the "System" element in the "Source" XML structure in the header file. |                                  |             |
| <b>Creator</b>                                                                                                                                                                                                                                 | '%(processor_name)s' (dynamic)   | NC_STRING   |
| Name of the facility or tool, within the Ground Segment element, creating the file. This attribute corresponds to the "Creator" element in the "Source" XML structure in the header file.                                                      |                                  |             |
| <b>Creator_Version</b>                                                                                                                                                                                                                         | '%(version)s' (dynamic)          | NC_STRING   |
| Version number of the tool that created the file. This attribute corresponds to the "Creator_Version" element in the "Source" XML structure in the header file.                                                                                |                                  |             |
| <b>Creation_Date</b>                                                                                                                                                                                                                           |                                  | NC_STRING   |
| The start date and time of processing, as a string: "UTC=YYYY-MM-DDThh:mm:ss". This attribute corresponds to the "Creation_Date" element in the "Source" XML structure in the header file.                                                     |                                  |             |

### 12.2.3.6 Group "variable\_header" in "earth\_explorer\_header"

#### Attributes in SO2\_\_\_/METADATA/ESA\_METADATA/earth\_explorer\_header/variable\_header

| Group attributes attached to variable_header |                            |             |
|----------------------------------------------|----------------------------|-------------|
| <i>Name</i>                                  | <i>Value</i>               | <i>Type</i> |
| <b>objectType</b>                            | 'Variable_Header' (static) | NC_STRING   |

### 12.2.3.7 Group "gmd:lineage" in "variable\_header"

Non-quantitative quality information about the lineage of the data specified by the scope.

#### Attributes in SO2\_\_\_/METADATA/ESA\_METADATA/earth\_explorer\_header/variable\_header/gmd:lineage

| Group attributes attached to gmd:lineage                                                                                                                                      |                                                                                                      |             |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|-------------|
| <i>Name</i>                                                                                                                                                                   | <i>Value</i>                                                                                         | <i>Type</i> |
| <b>objectType</b>                                                                                                                                                             | 'gmd:LI_Lineage' (static)                                                                            | NC_STRING   |
| <b>gmd:statement</b>                                                                                                                                                          | 'L2 %(product)s dataset produced by %(processingcenter)s from the S5P/TROPOMI L1B product' (dynamic) | NC_STRING   |
| General explanation of the data producer's knowledge about the lineage of a dataset. Insert short description of the actual Level 2 product in this string (at the %(... )s). |                                                                                                      |             |

### 12.2.3.8 Group "gmd:processStep" in "gmd:lineage"

Information about an event or transformation in the life of the dataset including details of the algorithm and software used for processing.

#### Attributes in SO2\_\_\_/METADATA/ESA\_METADATA/earth\_explorer\_header/variable\_header/gmd:lineage/gmd:processStep

| Group attributes attached to gmd:processStep |                                                                                                                                |             |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|-------------|
| <i>Name</i>                                  | <i>Value</i>                                                                                                                   | <i>Type</i> |
| <b>objectType</b>                            | 'gmi:LE_ProcessStep' (static)                                                                                                  | NC_STRING   |
| <b>gmd:description</b>                       | 'Processing of L1b to L2 %(product)s data for orbit %(orbit)d using the %(institute)s processor version %(version)s' (dynamic) | NC_STRING   |

Description of the event, including related parameters or tolerances. Insert short description of the actual Level 2 product, the orbit number, the name of the institute responsible for the CFI and the software version in this string (at the respective %(... )s and %(... )d).

#### 12.2.3.9 Group “gmi:output” in “gmd:processStep”

Description of the output.

**Attributes in SO2\_\_\_/METADATA/ESA\_METADATA/earth\_explorer\_header/variable\_header/gmd:lineage/gmd:processStep/gmi:output**

Group attributes attached to gmi:output

| Name                                                                     | Value                    | Type      |
|--------------------------------------------------------------------------|--------------------------|-----------|
| <b>gmd:description</b>                                                   |                          | NC_STRING |
| Short description of the output, a copy of the global ‘title’ attribute. |                          |           |
| <b>objectType</b>                                                        | ‘gmi:LE_Source’ (static) | NC_STRING |

#### 12.2.3.10 Group “gmd:sourceCitation” in “gmi:output”

Reference to the actual filename of the output data and production date and time.

**Attributes in SO2\_\_\_/METADATA/ESA\_METADATA/earth\_explorer\_header/variable\_header/gmd:lineage/gmd:processStep/gmi:output/gmd:sourceCitation**

Group attributes attached to gmd:sourceCitation

| Name                                | Value                            | Type      |
|-------------------------------------|----------------------------------|-----------|
| <b>gmd:title</b>                    | ‘%(logical_filename)s’ (dynamic) | NC_STRING |
| Output file name without extension. |                                  |           |
| <b>objectType</b>                   | ‘gmd:CI_Citation’ (static)       | NC_STRING |

#### 12.2.3.11 Group “gmd:date” in “gmd:sourceCitation”

Production date and time of the output file.

**Attributes in SO2\_\_\_/METADATA/ESA\_METADATA/earth\_explorer\_header/variable\_header/gmd:lineage/gmd:processStep/gmi:output/gmd:sourceCitation/gmd:date**

Group attributes attached to gmd:date

| Name                                                                                                                                                        | Value                      | Type      |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|-----------|
| <b>gmd:date</b>                                                                                                                                             |                            | NC_STRING |
| Production date and time of the output file. Note that the definition in the XML schema appears to allow the use of a “CI_DateTime” instead of a “CI_Date”. |                            |           |
| <b>objectType</b>                                                                                                                                           | ‘gmd:CI_DateTime’ (static) | NC_STRING |

#### 12.2.3.12 Group “gmd:dateType” in “gmd:date”

Meaning of the reference date for the cited resource.

**Attributes in SO2\_\_\_/METADATA/ESA\_METADATA/earth\_explorer\_header/variable\_header/gmd:lineage/gmd:processStep/gmi:output/gmd:sourceCitation/gmd:date/gmd:dateType**

Group attributes attached to gmd:dateType

| Name                 | Value                                                                                       | Type      |
|----------------------|---------------------------------------------------------------------------------------------|-----------|
| <b>codeList</b>      | ‘http://www.isotc211.org/2005/resources/Codelist/gmxCodeLists.xml#CI_DateTypeCode’ (static) | NC_STRING |
| <b>codeListValue</b> | ‘creation’ (static)                                                                         | NC_STRING |
| <b>objectType</b>    | ‘gmd:CI_DateTypeCode’ (static)                                                              | NC_STRING |

### 12.2.3.13 Group “gmd:identifier” in “gmd:sourceCitation”

Identification of the output product.

**Attributes in SO2\_\_\_/METADATA/ESA\_METADATA/earth\_explorer\_header/variable\_header/gmd:lineage/  
gmd:processStep/gmi:output/gmd:sourceCitation/gmd:identifier**

| Group attributes attached to gmd:identifier                                                            |                              |           |
|--------------------------------------------------------------------------------------------------------|------------------------------|-----------|
| Name                                                                                                   | Value                        | Type      |
| <b>gmd:code</b>                                                                                        | '%(shortname)s' (dynamic)    | NC_STRING |
| The product short name, a copy of the 'ProductShortName' attribute in '/METADATA/GRANULE_DESCRIPTION'. |                              |           |
| <b>objectType</b>                                                                                      | 'gmd:MD_Identifier' (static) | NC_STRING |

### 12.2.3.14 Group “gmi:processedLevel” in “gmi:output”

Process level of the output file.

**Attributes in SO2\_\_\_/METADATA/ESA\_METADATA/earth\_explorer\_header/variable\_header/gmd:lineage/  
gmd:processStep/gmi:output/gmi:processedLevel**

| Group attributes attached to gmi:processedLevel |                              |           |
|-------------------------------------------------|------------------------------|-----------|
| Name                                            | Value                        | Type      |
| <b>gmd:code</b>                                 | 'L2' (static)                | NC_STRING |
| <b>objectType</b>                               | 'gmd:MD_Identifier' (static) | NC_STRING |

### 12.2.3.15 Group “gmi:processingInformation” in “gmd:processStep”

Description of the processor in more detail.

**Attributes in SO2\_\_\_/METADATA/ESA\_METADATA/earth\_explorer\_header/variable\_header/gmd:lineage/  
gmd:processStep/gmi:processingInformation**

| Group attributes attached to gmi:processingInformation |                              |           |
|--------------------------------------------------------|------------------------------|-----------|
| Name                                                   | Value                        | Type      |
| <b>objectType</b>                                      | 'gmi:LE_Processing' (static) | NC_STRING |

### 12.2.3.16 Group “gmi:identifier” in “gmi:processingInformation”

Identification of the processor.

**Attributes in SO2\_\_\_/METADATA/ESA\_METADATA/earth\_explorer\_header/variable\_header/gmd:lineage/  
gmd:processStep/gmi:processingInformation/gmi:identifier**

| Group attributes attached to gmi:identifier                                                                                                                    |                                                                         |           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|-----------|
| Name                                                                                                                                                           | Value                                                                   | Type      |
| <b>gmd:code</b>                                                                                                                                                | '%(institute)s L2 %(product)s processor, version %(version)s' (dynamic) | NC_STRING |
| Descriptive name of the processor, with the %(. . . )s placeholders replaced with the responsible institute's name, product name and software release version. |                                                                         |           |
| <b>objectType</b>                                                                                                                                              | 'gmd:MD_Identifier' (static)                                            | NC_STRING |

### 12.2.3.17 Group “gmi:softwareReference” in “gmi:processingInformation”

Reference to document describing processing software.

**Attributes in SO2\_\_\_/METADATA/ESA\_METADATA/earth\_explorer\_header/variable\_header/gmd:lineage/  
gmd:processStep/gmi:processingInformation/gmi:softwareReference**

Group attributes attached to gmi:softwareReference

| Name                            | Value                                     | Type      |
|---------------------------------|-------------------------------------------|-----------|
| <b>gmd:title</b>                | 'UPAS L2 %(product)s processor' (dynamic) | NC_STRING |
| Title of processor description. |                                           |           |
| <b>objectType</b>               | 'gmd:CI_Citation' (static)                | NC_STRING |

#### 12.2.3.18 Group “gmd:date” in “gmi:softwareReference”

Release date (compile date) of the processor.

**Attributes in SO2\_\_\_/METADATA/ESA\_METADATA/earth\_explorer\_header/variable\_header/gmd:lineage/  
gmd:processStep/gmi:processingInformation/gmi:softwareReference/gmd:date**

Group attributes attached to gmd:date

| Name                                                                       | Value                      | Type      |
|----------------------------------------------------------------------------|----------------------------|-----------|
| <b>gmd:date</b>                                                            |                            | NC_STRING |
| Release date of the processor expressed as an ISO 8601 date string [RD35]. |                            |           |
| <b>objectType</b>                                                          | 'gmd:CI_DateTime' (static) | NC_STRING |

#### 12.2.3.19 Group “gmd:dateType” in “gmd:date”

Confirm that this is the release date of the processor.

**Attributes in SO2\_\_\_/METADATA/ESA\_METADATA/earth\_explorer\_header/variable\_header/gmd:lineage/  
gmd:processStep/gmi:processingInformation/gmi:softwareReference/gmd:date/gmd:dateType**

Group attributes attached to gmd:dateType

| Name                 | Value                                                                                           | Type      |
|----------------------|-------------------------------------------------------------------------------------------------|-----------|
| <b>codeList</b>      | 'http://www.isotc211.org/2005/resources/Codelist/<br>gmxCodelists.xml#CI_DateTypeCode' (static) | NC_STRING |
| <b>codeListValue</b> | 'creation' (static)                                                                             | NC_STRING |
| <b>objectType</b>    | 'gmd:CI_DateTypeCode' (static)                                                                  | NC_STRING |

#### 12.2.3.20 Group “gmi:documentation#1” in “gmi:processingInformation”

Reference to the ATBD of the product.

**Attributes in SO2\_\_\_/METADATA/ESA\_METADATA/earth\_explorer\_header/variable\_header/gmd:lineage/  
gmd:processStep/gmi:processingInformation/gmi:documentation#1**

Group attributes attached to gmi:documentation#1

| Name                                                                    | Value                      | Type      |
|-------------------------------------------------------------------------|----------------------------|-----------|
| <b>objectType</b>                                                       | 'gmd:CI_Citation' (static) | NC_STRING |
| <b>gmd:title</b>                                                        | '%(title_atbd)s' (dynamic) | NC_STRING |
| The filename of the current release of the ATBD of the current product. |                            |           |

#### 12.2.3.21 Group “gmd:date” in “gmi:documentation#1”

Release date of the ATBD.

**Attributes in SO2\_\_\_/METADATA/ESA\_METADATA/earth\_explorer\_header/variable\_header/gmd:lineage/  
gmd:processStep/gmi:processingInformation/gmi:documentation#1/gmd:date**

Group attributes attached to gmd:date

| Name | Value | Type |
|------|-------|------|
|------|-------|------|

|                                                                       |                           |           |
|-----------------------------------------------------------------------|---------------------------|-----------|
| <b>gmd:date</b>                                                       | '%(date_atbd)s' (dynamic) | NC_STRING |
| Release date of the ATBD expressed as an ISO 8601 date string [RD35]. |                           |           |
| <b>objectType</b>                                                     | 'gmd:CI_Date' (static)    | NC_STRING |

#### 12.2.3.22 Group “gmd:dateType” in “gmd:date”

Confirm that this is the date of publication.

**Attributes in SO2\_\_\_/METADATA/ESA\_METADATA/earth\_explorer\_header/variable\_header/gmd:lineage/gmd:processStep/gmi:processingInformation/gmi:documentation#1/gmd:date/gmd:dateType**

| Group attributes attached to gmd:dateType |                                                                                             |           |
|-------------------------------------------|---------------------------------------------------------------------------------------------|-----------|
| Name                                      | Value                                                                                       | Type      |
| <b>codeList</b>                           | 'http://www.isotc211.org/2005/resources/Codelist/gmxCodelists.xml#CI_DateTypeCode' (static) | NC_STRING |
| <b>codeListValue</b>                      | 'publication' (static)                                                                      | NC_STRING |
| <b>objectType</b>                         | 'gmd:CI_DateTypeCode' (static)                                                              | NC_STRING |

#### 12.2.3.23 Group “gmi:documentation#2” in “gmi:processingInformation”

Reference to the PUM of the product.

**Attributes in SO2\_\_\_/METADATA/ESA\_METADATA/earth\_explorer\_header/variable\_header/gmd:lineage/gmd:processStep/gmi:processingInformation/gmi:documentation#2**

| Group attributes attached to gmi:documentation#2                       |                            |           |
|------------------------------------------------------------------------|----------------------------|-----------|
| Name                                                                   | Value                      | Type      |
| <b>objectType</b>                                                      | 'gmd:CI_Citation' (static) | NC_STRING |
| <b>gmd:title</b>                                                       | '%(title_pum)s' (dynamic)  | NC_STRING |
| The filename of the current release of the PUM of the current product. |                            |           |

#### 12.2.3.24 Group “gmd:date” in “gmi:documentation#2”

Release date of the PUM.

**Attributes in SO2\_\_\_/METADATA/ESA\_METADATA/earth\_explorer\_header/variable\_header/gmd:lineage/gmd:processStep/gmi:processingInformation/gmi:documentation#2/gmd:date**

| Group attributes attached to gmd:date                                |                          |           |
|----------------------------------------------------------------------|--------------------------|-----------|
| Name                                                                 | Value                    | Type      |
| <b>gmd:date</b>                                                      | '%(date_pum)s' (dynamic) | NC_STRING |
| Release date of the PUM expressed as an ISO 8601 date string [RD35]. |                          |           |
| <b>objectType</b>                                                    | 'gmd:CI_Date' (static)   | NC_STRING |

#### 12.2.3.25 Group “gmd:dateType” in “gmd:date”

Confirm that this is the date of publication.

**Attributes in SO2\_\_\_/METADATA/ESA\_METADATA/earth\_explorer\_header/variable\_header/gmd:lineage/gmd:processStep/gmi:processingInformation/gmi:documentation#2/gmd:date/gmd:dateType**

| Group attributes attached to gmd:dateType |                                                                                             |           |
|-------------------------------------------|---------------------------------------------------------------------------------------------|-----------|
| Name                                      | Value                                                                                       | Type      |
| <b>codeList</b>                           | 'http://www.isotc211.org/2005/resources/Codelist/gmxCodelists.xml#CI_DateTypeCode' (static) | NC_STRING |
| <b>codeListValue</b>                      | 'publication' (static)                                                                      | NC_STRING |

|                   |                                |           |
|-------------------|--------------------------------|-----------|
| <b>objectType</b> | 'gmd:CI_DateTypeCode' (static) | NC_STRING |
|-------------------|--------------------------------|-----------|

### 12.2.3.26 Group “gmi:report” in “gmd:processStep”

Short report of what occurred during the process step.

**Attributes in SO<sub>2</sub>\_\_\_/METADATA/ESA\_METADATA/earth\_explorer\_header/variable\_header/gmd:lineage/gmd:processStep/gmi:report**

| Group attributes attached to gmi:report                                                                                   |                                                                                                                    |           |
|---------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|-----------|
| Name                                                                                                                      | Value                                                                                                              | Type      |
| <b>gmi:description</b>                                                                                                    | 'Sentinel 5-precursor TROPOMI L1b processed to L2 data using the %(institute)s L2 %(product)s processor' (dynamic) | NC_STRING |
| Textual description of what occurred during the process step. Replace %(... )s as indicated.                              |                                                                                                                    |           |
| <b>gmi:fileType</b>                                                                                                       | 'netCDF-4' (static)                                                                                                | NC_STRING |
| Type of file that contains the processing report, in our case the processing report is contained in the main output file. |                                                                                                                    |           |
| <b>gmi:name</b>                                                                                                           | '%(logical_filename)s.nc' (dynamic)                                                                                | NC_STRING |
| <b>objectType</b>                                                                                                         | 'gmi:LE_ProcessStepReport' (dynamic)                                                                               | NC_STRING |

### 12.2.3.27 Group “gmd:source#1” in “gmd:processStep”

Information about the source data used in creating the data specified by the scope. Repeat group as needed, incrementing the number of the source (after the # mark).

**Attributes in SO<sub>2</sub>\_\_\_/METADATA/ESA\_METADATA/earth\_explorer\_header/variable\_header/gmd:lineage/gmd:processStep/gmd:source#1**

| Group attributes attached to gmd:source#1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                          |           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|-----------|
| Name                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Value                    | Type      |
| objectType                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 'gmi:LE_Source' (static) | NC_STRING |
| gmd:description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | NC_STRING                |           |
| Description of the input data, including L1B, L2, dynamic auxiliary input data and semi-static auxiliary input data. Base strings are “TROPOMI L1B %s radiance product”, “TROPOMI L1B %s irradiance product”, “TROPOMI L2 %s product”, “Auxiliary ECMWF %s Meteorological forecast data”, “Processor %s configuration file”, “Auxiliary %s reference data”, “Auxiliary %s algorithm lookup table”, “Auxiliary CTM %s model input data”, “Auxiliary snow and ice input data” and “Auxiliary NPP/VIIRS cloud screening input data”. The %s to be replaced with specific descriptors. |                          |           |

### 12.2.3.28 Group “gmi:processedLevel” in “gmd:source#1”

**Attributes in SO<sub>2</sub>\_\_\_/METADATA/ESA\_METADATA/earth\_explorer\_header/variable\_header/gmd:lineage/gmd:processStep/gmd:source#1/gmi:processedLevel**

| Group attributes attached to gmi:processedLevel |                              |           |
|-------------------------------------------------|------------------------------|-----------|
| Name                                            | Value                        | Type      |
| <b>gmd:code</b>                                 | Empty!                       | NC_STRING |
| <b>objectType</b>                               | 'gmd:MD_Identifier' (static) | NC_STRING |

### 12.2.3.29 Group “gmd:sourceCitation” in “gmd:source#1”

Reference to the actual filename of the input data.

**Attributes in SO<sub>2</sub>\_\_\_/METADATA/ESA\_METADATA/earth\_explorer\_header/variable\_header/gmd:lineage/gmd:processStep/gmd:source#1/gmd:sourceCitation**

Group attributes attached to gmd:sourceCitation

| Name              | Value                      | Type      |
|-------------------|----------------------------|-----------|
| <b>objectType</b> | 'gmd:CI_Citation' (static) | NC_STRING |

### 12.2.3.30 Group “gmd:date” in “gmd:sourceCitation”

Attributes in SO<sub>2</sub>\_\_\_/METADATA/ESA\_METADATA/earth\_explorer\_header/variable\_header/gmd:lineage/  
gmd:processStep/gmd:source#1/gmd:sourceCitation/gmd:date

Group attributes attached to gmd:date

| Name                                                                                                                                                                                                                         | Value                  | Type      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|-----------|
| <b>gmd:date</b>                                                                                                                                                                                                              |                        | NC_STRING |
| Production date and time of the input file(s) in this group expressed as an ISO 8601 date-time string [RD35]. Note that the definition in the XML schema appears to allow the use of a “CI_DateTime” instead of a “CI_Date”. |                        |           |
| <b>objectType</b>                                                                                                                                                                                                            | 'gmd:CI_Date' (static) | NC_STRING |

### 12.2.3.31 Group “gmd:dateType” in “gmd:date”

Meaning of the reference date for the cited resource.

Attributes in SO<sub>2</sub>\_\_\_/METADATA/ESA\_METADATA/earth\_explorer\_header/variable\_header/gmd:lineage/  
gmd:processStep/gmd:source#1/gmd:sourceCitation/gmd:date/gmd:dateType

Group attributes attached to gmd:dateType

| Name                 | Value                                                                                       | Type      |
|----------------------|---------------------------------------------------------------------------------------------|-----------|
| <b>codeList</b>      | 'http://www.isotc211.org/2005/resources/Codelist/gmxCodeLists.xml#CI_DateTypeCode' (static) | NC_STRING |
| <b>codeListValue</b> | 'creation' (static)                                                                         | NC_STRING |
| <b>objectType</b>    | 'gmd:CI_DateTypeCode' (static)                                                              | NC_STRING |

### 12.2.3.32 Group “gmd:title” in “gmd:sourceCitation”

Attributes in SO<sub>2</sub>\_\_\_/METADATA/ESA\_METADATA/earth\_explorer\_header/variable\_header/gmd:lineage/  
gmd:processStep/gmd:source#1/gmd:sourceCitation/gmd:title

Group attributes attached to gmd:title

| Name                                                                                                                  | Value | Type      |
|-----------------------------------------------------------------------------------------------------------------------|-------|-----------|
| <b>gco:characterString</b>                                                                                            |       | NC_STRING |
| Textual description of the input file group (same as the “gmd:description” attribute in the “gmi:LE_Sources” object). |       |           |

### 12.2.3.33 Group “gmd:alternateTitle#1” in “gmd:sourceCitation”

All filenames in this group, in case more files of a particular file type are delivered, for instance for meteorological or model input. Repeat group as needed, incrementing the number of the input file (after the # mark).

Attributes in SO<sub>2</sub>\_\_\_/METADATA/ESA\_METADATA/earth\_explorer\_header/variable\_header/gmd:lineage/  
gmd:processStep/gmd:source#1/gmd:sourceCitation/gmd:alternateTitle#1

Group attributes attached to gmd:alternateTitle#1

| Name | Value | Type |
|------|-------|------|
|------|-------|------|

|                                 |               |           |
|---------------------------------|---------------|-----------|
| <b>gmx:FileName</b>             | <i>Empty!</i> | NC_STRING |
| The basename of the input file. |               |           |

#### 12.2.3.34 Group “EOP\_METADATA” in “EOP\_metadata”

Based on the OGC 10-025 standard for Observations & Measurements [RD42], an Earth Observation Product (EOP) schema was developed which refines an observation into the feature type earth observation. This schema was then extended with sensor-specific thematic schemas.

##### Attributes in SO<sub>2</sub>\_\_\_/METADATA/EOP\_METADATA

| Group attributes attached to EOP_METADATA                                                                                                 |                                     |           |
|-------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|-----------|
| Name                                                                                                                                      | Value                               | Type      |
| <b>gml:id</b>                                                                                                                             | ‘%(logical_filename)s.ID’ (dynamic) | NC_STRING |
| Unique ID for this “atm:EarthObservation” object. Constructed from the logical output filename and the extension “ID” separated by a dot. |                                     |           |
| <b>objectType</b>                                                                                                                         | ‘atm:EarthObservation’ (static)     | NC_STRING |

#### 12.2.3.35 Group “om:phenomenonTime” in “EOP\_METADATA”

Time coverage of the granule.

##### Attributes in SO<sub>2</sub>\_\_\_/METADATA/EOP\_METADATA/om:phenomenonTime

| Group attributes attached to om:phenomenonTime                                                      |                           |           |
|-----------------------------------------------------------------------------------------------------|---------------------------|-----------|
| Name                                                                                                | Value                     | Type      |
| <b>gml:beginPosition</b>                                                                            |                           | NC_STRING |
| Start of time coverage of the data in the granule expressed as an ISO 8601 date-time string [RD35]. |                           |           |
| <b>gml:endPosition</b>                                                                              |                           | NC_STRING |
| End of time coverage of the data in the granule expressed as an ISO 8601 date-time string [RD35].   |                           |           |
| <b>objectType</b>                                                                                   | ‘gml:TimePeriod’ (static) | NC_STRING |

#### 12.2.3.36 Group “om:procedure” in “EOP\_METADATA”

Platform, instrument and sensor used for the acquisition and the acquisition parameters.

##### Attributes in SO<sub>2</sub>\_\_\_/METADATA/EOP\_METADATA/om:procedure

| Group attributes attached to om:procedure                                                                                                           |                                          |           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|-----------|
| Name                                                                                                                                                | Value                                    | Type      |
| <b>gml:id</b>                                                                                                                                       | ‘%(logical_filename)s.EOE’ (dynamic)     | NC_STRING |
| Unique ID for this “eop:EarthObservationEquipment” object. Constructed from the logical output filename and the extension “EOE” separated by a dot. |                                          |           |
| <b>objectType</b>                                                                                                                                   | ‘eop:EarthObservationEquipment’ (static) | NC_STRING |

#### 12.2.3.37 Group “eop:platform” in “om:procedure”

Platform name and orbit type.

##### Attributes in SO<sub>2</sub>\_\_\_/METADATA/EOP\_METADATA/om:procedure/eop:platform

| Group attributes attached to eop:platform |                         |           |
|-------------------------------------------|-------------------------|-----------|
| Name                                      | Value                   | Type      |
| <b>eop:shortName</b>                      | ‘Sentinel-5p’ (static)  | NC_STRING |
| <b>objectType</b>                         | ‘eop:Platform’ (static) | NC_STRING |

### 12.2.3.38 Group “eop:instrument” in “om:procedure”

Instrument descriptor.

#### Attributes in SO2\_\_\_/METADATA/EOP\_METADATA/om:procedure/eop:instrument

| Group attributes attached to eop:instrument |                           |           |
|---------------------------------------------|---------------------------|-----------|
| Name                                        | Value                     | Type      |
| <b>eop:shortName</b>                        | ‘TROPOMI’ (static)        | NC_STRING |
| <b>objectType</b>                           | ‘eop:Instrument’ (static) | NC_STRING |

### 12.2.3.39 Group “eop:sensor” in “om:procedure”

Sensor description.

#### Attributes in SO2\_\_\_/METADATA/EOP\_METADATA/om:procedure/eop:sensor

| Group attributes attached to eop:sensor |                        |           |
|-----------------------------------------|------------------------|-----------|
| Name                                    | Value                  | Type      |
| <b>eop:sensorType</b>                   | ‘ATMOSPHERIC’ (static) | NC_STRING |
| <b>objectType</b>                       | ‘eop:Sensor’ (static)  | NC_STRING |

### 12.2.3.40 Group “eop:acquisitionParameters” in “om:procedure”

Additional parameters describing the data acquisition. Only an orbit number is used here.

#### Attributes in SO2\_\_\_/METADATA/EOP\_METADATA/om:procedure/eop:acquisitionParameters

| Group attributes attached to eop:acquisitionParameters |                            |           |
|--------------------------------------------------------|----------------------------|-----------|
| Name                                                   | Value                      | Type      |
| <b>eop:orbitNumber</b>                                 | % (orbit)d (dynamic)       | NC_INT    |
| <b>objectType</b>                                      | ‘eop:Acquisition’ (static) | NC_STRING |

### 12.2.3.41 Group “om:observedProperty” in “EOP\_METADATA”

An xlink to the observed property definition.

#### Attributes in SO2\_\_\_/METADATA/EOP\_METADATA/om:observedProperty

| Group attributes attached to om:observedProperty                  |                          |           |
|-------------------------------------------------------------------|--------------------------|-----------|
| Name                                                              | Value                    | Type      |
| <b>nilReason</b>                                                  | ‘inapplicable’ (dynamic) | NC_STRING |
| This element should use the attribute ‘nilReason=“inapplicable”’. |                          |           |

### 12.2.3.42 Group “om:featureOfInterest” in “EOP\_METADATA”

#### Attributes in SO2\_\_\_/METADATA/EOP\_METADATA/om:featureOfInterest

| Group attributes attached to om:featureOfInterest                                                                                  |                                     |           |
|------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|-----------|
| Name                                                                                                                               | Value                               | Type      |
| <b>objectType</b>                                                                                                                  | ‘eop:FootPrint’ (static)            | NC_STRING |
| <b>gml:id</b>                                                                                                                      | % (logical_filename)s.FP’ (dynamic) | NC_STRING |
| Unique ID for this “eop:FootPrint” object. Constructed from the logical output filename and the extension “FP” separated by a dot. |                                     |           |

#### 12.2.3.43 Group “eop:multiExtentOf” in “om:featureOfInterest”

Acquisition footprint coordinates, described by a closed polygon – the last point is equal to the first point, using latitude, longitude pairs. The expected structure is “gml:Polygon/gml:exterior/gml:LinearRing/gml:posList”.

##### Attributes in SO2\_\_\_/METADATA/EOP\_METADATA/om:featureOfInterest/eop:multiExtentOf

| Group attributes attached to eop:multiExtentOf |                             |           |
|------------------------------------------------|-----------------------------|-----------|
| Name                                           | Value                       | Type      |
| <b>objectType</b>                              | ‘gml:MultiSurface’ (static) | NC_STRING |

#### 12.2.3.44 Group “gml:surfaceMembers” in “eop:multiExtentOf”

##### Attributes in SO2\_\_\_/METADATA/EOP\_METADATA/om:featureOfInterest/eop:multiExtentOf/gml:surfaceMembers

| Group attributes attached to gml:surfaceMembers |                        |           |
|-------------------------------------------------|------------------------|-----------|
| Name                                            | Value                  | Type      |
| <b>objectType</b>                               | ‘gml:Polygon’ (static) | NC_STRING |

#### 12.2.3.45 Group “gml:exterior” in “gml:surfaceMembers”

##### Attributes in SO2\_\_\_/METADATA/EOP\_METADATA/om:featureOfInterest/eop:multiExtentOf/gml:surfaceMembers/gml:exterior

| Group attributes attached to gml:exterior                                                                                                                                                                                                         |                           |           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|-----------|
| Name                                                                                                                                                                                                                                              | Value                     | Type      |
| <b>gml:posList</b>                                                                                                                                                                                                                                |                           | NC_STRING |
| The Polygon geometry shall be encoded in the EPSG:4326 geographic coordinate reference system (WGS-84) and the coordinate pairs shall be ordered as latitude/longitude. Polygons enclose areas with points listed in counter-clockwise direction. |                           |           |
| <b>objectType</b>                                                                                                                                                                                                                                 | ‘gml:LinearRing’ (static) | NC_STRING |

#### 12.2.3.46 Group “eop:metaDataProperty” in “EOP\_METADATA”

This group contains all the metadata relative to the Earth observation product that do not fit inside one of the other groups, i.e. metadata that do not describe the time, the mechanism, the location or the result of the observation.

These metadata are mainly the EarthObservation identifier, the acquisition type and information relative to the downlink and archiving centers.

##### Attributes in SO2\_\_\_/METADATA/EOP\_METADATA/eop:metaDataProperty

| Group attributes attached to eop:metaDataProperty                                                                                                                                                                                                    |                                         |           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|-----------|
| Name                                                                                                                                                                                                                                                 | Value                                   | Type      |
| <b>objectType</b>                                                                                                                                                                                                                                    | ‘eop:EarthObservationMetaData’ (static) | NC_STRING |
| <b>eop:acquisitionType</b>                                                                                                                                                                                                                           | ‘NOMINAL’ (dynamic)                     | NC_STRING |
| Used to distinguish at a high level the appropriateness of the acquisition for “general” use, whether the product is a nominal acquisition, special calibration product or other. Copy from L1b. For Level 2 this should <i>always</i> be ‘NOMINAL’. |                                         |           |
| <b>eop:identifier</b>                                                                                                                                                                                                                                | ‘%(logical_filename)s’ (dynamic)        | NC_STRING |
| Logical file name.                                                                                                                                                                                                                                   |                                         |           |
| <b>eop:doi</b>                                                                                                                                                                                                                                       | ‘%(product_doi)s’ (dynamic)             | NC_STRING |
| Digital Object Identifier identifying the product (see <a href="http://www.datacite.org">http://www.datacite.org</a> for DOIs for datasets).                                                                                                         |                                         |           |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                       |           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|-----------|
| <b>eop:parentIdentifier</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 'urn:ogc:def:EOP:ESA:SENTINEL.S5P_TROP_ -<br>%(shortname)s' (dynamic) | NC_STRING |
| Unique collection identifier for metadata file, see the Level 1B metadata specification [RD32, table 5] for a discussion of the value.<br>This is a copy of the "gmd:fileIdentifier" attribute in the "/METADATA/ISO_METADATA" group.                                                                                                                                                                                                                                                                                                                                                         |                                                                       |           |
| <b>eop:productType</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 'S5P_%(mode)s_%(product)s' (dynamic)                                  | NC_STRING |
| Product type identifier. Replace %(mode)s with the operational mode the processor is running in ('NRTI', 'OFFL' or 'RPRO', as per [RD25]) and %(product)s with the 10 character output file name semantic descriptors as given in [RD43, RD44, RD45].                                                                                                                                                                                                                                                                                                                                         |                                                                       |           |
| <b>eop:status</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 'ACQUIRED' (dynamic)                                                  | NC_STRING |
| Refers to product status. Values listed in the standard: 'ARCHIVED', 'ACQUIRED', 'CANCELLED', 'FAILED', 'PLANNED', 'POTENTIAL', 'REJECTED', 'QUALITY-DEGRADED'. Copied from L1B.                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                       |           |
| <b>eop:productQualityStatus</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 'NOMINAL' (dynamic)                                                   | NC_STRING |
| Indicator that specifies whether the product quality is degraded or not. Allowed values: 'DEGRADED', 'NOMINAL'.                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                       |           |
| <b>eop:productQualityDegradationText</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 'NOT APPLICABLE' (dynamic)                                            | NC_STRING |
| Contains further textual information concerning the quality degradation. According to the metadata standards it shall be provided <i>only</i> if "eop:productQualityStatus" value is set to 'DEGRADED'. Because the way we generate out output files, this attribute will always be present, even when "eop:productQualityStatus" value is 'NOMINAL'. In those cases the value shall be set to "NOT APPLICABLE".<br>Possible values are "MISSING AUXILIARY INPUT" and "NOT APPLICABLE". Note that Level 1B does not set this value, so only problems detectable in the processor are covered. |                                                                       |           |

#### 12.2.3.47 Group "eop:processing" in "eop:metaDataProperty"

Processing information.

#### Attributes in SO2\_\_\_/METADATA/EOP\_METADATA/eop:metaDataProperty/eop:processing

| Group attributes attached to eop:processing                                                                                                                                            |                                      |           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|-----------|
| Name                                                                                                                                                                                   | Value                                | Type      |
| <b>objectType</b>                                                                                                                                                                      | 'eop:ProcessingInformation' (static) | NC_STRING |
| <b>eop:processingCenter</b>                                                                                                                                                            | '%(processingcenter)s' (dynamic)     | NC_STRING |
| The processing center, taken from the "Processing_Station" key in the joborder.                                                                                                        |                                      |           |
| <b>eop:processingDate</b>                                                                                                                                                              | 'YYYY-mm-ddTHH:MM:SSZ' (dynamic)     | NC_STRING |
| The processing date, as an ISO 8601 date-time string [RD35].                                                                                                                           |                                      |           |
| <b>eop:processingLevel</b>                                                                                                                                                             | 'L2' (static)                        | NC_STRING |
| These are all Level 2 products.                                                                                                                                                        |                                      |           |
| <b>eop:processorName</b>                                                                                                                                                               | '%(processor_name)s' (static)        | NC_STRING |
| The name of the processor, "tropnll2dp.exe" for KNMI and "upas-12" for DLR.                                                                                                            |                                      |           |
| <b>eop:processorVersion</b>                                                                                                                                                            | '%(version)s' (dynamic)              | NC_STRING |
| Version of the processor, as "major.minor.bugfix".                                                                                                                                     |                                      |           |
| <b>eop:nativeProductFormat</b>                                                                                                                                                         | 'netCDF-4' (static)                  | NC_STRING |
| Native product format.                                                                                                                                                                 |                                      |           |
| <b>eop:processingMode</b>                                                                                                                                                              | '%(mode)s' (dynamic)                 | NC_STRING |
| Processing mode taken from mission specific code list. For S5P we use the <i>File Class</i> identifiers [RD25, section 4.1.2]: 'TEST', 'OGCA', 'GSOV', 'OPER', 'NRTI', 'OFFL', 'RPRO'. |                                      |           |

#### 12.2.3.48 Group "ISO\_METADATA" in "iso\_metadata"

Metadata that is structured following the ISO metadata standards [RD27, RD40], especially part 2. The metadata in this group is structured using the methods from Level 1B, which is described in the Level 1B metadata specification [RD32].

All “objectType” attributes indicate the XML object when generating an ISO 19139 [RD40] compliant XML metadata file.

Note that this group is meant to be treated as a ‘black box’. The information is collected here so that it can be extracted into XML side-files for ingestion into data search tools and metadata collections.

#### Attributes in SO2\_\_\_/METADATA/ISO\_METADATA

| Group attributes attached to ISO_METADATA                                                                                   |                                                                                                            |           |
|-----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|-----------|
| Name                                                                                                                        | Value                                                                                                      | Type      |
| <b>gmd:dateStamp</b>                                                                                                        | ‘2015-10-16’ (static)                                                                                      | NC_STRING |
| Date of creation of the metadata, as ISO 8601 [RD35] string specifying year, month and day.                                 |                                                                                                            |           |
| <b>gmd:fileIdentifier</b>                                                                                                   | ‘urn:ogc:def:EOP:ESA:SENTINEL.S5P_TROP_-<br>%(shortname)s’ (dynamic)                                       | NC_STRING |
| Unique identifier for metadata file, see the Level 1B metadata specification [RD32, table 5] for a discussion of the value. |                                                                                                            |           |
| Replace %(...s with the “ProductShortName” value from the Level 2 “/METADATA/GRANULE_-DESCRIPTION” metadata group.          |                                                                                                            |           |
| <b>gmd:hierarchyLevelName</b>                                                                                               | ‘EO Product Collection’ (static)                                                                           | NC_STRING |
| Name of the hierarchy levels for which the metadata is provided.                                                            |                                                                                                            |           |
| <b>gmd:metadataStandardName</b>                                                                                             | ‘ISO 19115-2 Geographic Information - Metadata Part<br>2 Extensions for imagery and gridded data’ (static) | NC_STRING |
| Name of the metadata standard.                                                                                              |                                                                                                            |           |
| <b>gmd:metadataStandardVersion</b>                                                                                          | ‘ISO 19115-2:2009(E), S5P profile’ (static)                                                                | NC_STRING |
| Version (profile) of the metadata standard used.                                                                            |                                                                                                            |           |
| <b>objectType</b>                                                                                                           | ‘gmi:MI_Metadata’ (static)                                                                                 | NC_STRING |
| Name of the metadata class [RD32, table 5].                                                                                 |                                                                                                            |           |

#### 12.2.3.49 Group “gmd:language” in “ISO\_METADATA”

Language used for the metadata, fixed to English.

#### Attributes in SO2\_\_\_/METADATA/ISO\_METADATA/gmd:language

| Group attributes attached to gmd:language |                                                   |           |
|-------------------------------------------|---------------------------------------------------|-----------|
| Name                                      | Value                                             | Type      |
| <b>codeList</b>                           | ‘http://www.loc.gov/standards/iso639-2/’ (static) | NC_STRING |
| <b>codeListValue</b>                      | ‘eng’ (static)                                    | NC_STRING |
| <b>objectType</b>                         | ‘gmd:LanguageCode’ (static)                       | NC_STRING |

#### 12.2.3.50 Group “gmd:characterSet” in “ISO\_METADATA”

The character encoding used for the metadata. This is fixed to UTF-8, but the climate and forecasting conventions, version 1.6 limits this further to 7-bit ASCII (which is a subset of UTF-8).

#### Attributes in SO2\_\_\_/METADATA/ISO\_METADATA/gmd:characterSet

| Group attributes attached to gmd:characterSet |                                                                                                     |           |
|-----------------------------------------------|-----------------------------------------------------------------------------------------------------|-----------|
| Name                                          | Value                                                                                               | Type      |
| <b>codeList</b>                               | ‘http://www.isotc211.org/2005/resources/Codelist/<br>gmxCodelists.xml#MD_CharacterSetCode’ (static) | NC_STRING |
| <b>codeListValue</b>                          | ‘utf8’ (static)                                                                                     | NC_STRING |
| <b>objectType</b>                             | ‘gmd:MD_CharacterSetCode’ (static)                                                                  | NC_STRING |

### 12.2.3.51 Group “gmd:hierarchyLevel” in “ISO\_METADATA”

Scope to which metadata applies.

#### Attributes in SO<sub>2</sub>\_\_\_/METADATA/ISO\_METADATA/gmd:hierarchyLevel

| Group attributes attached to gmd:hierarchyLevel |                                                                                          |           |
|-------------------------------------------------|------------------------------------------------------------------------------------------|-----------|
| Name                                            | Value                                                                                    | Type      |
| <b>codeList</b>                                 | ‘http://www.isotc211.org/2005/resources/Codelist/gmxCodelists.xml#MD_ScopeCode’ (static) | NC_STRING |
| <b>codeListValue</b>                            | ‘series’ (static)                                                                        | NC_STRING |
| <b>objectType</b>                               | ‘gmd:MD_ScopeCode’ (static)                                                              | NC_STRING |

### 12.2.3.52 Group “gmd:contact” in “ISO\_METADATA”

Contact information for the product.

#### Attributes in SO<sub>2</sub>\_\_\_/METADATA/ISO\_METADATA/gmd:contact

| Group attributes attached to gmd:contact |                                                                                               |           |
|------------------------------------------|-----------------------------------------------------------------------------------------------|-----------|
| Name                                     | Value                                                                                         | Type      |
| <b>gmd:organisationName</b>              | ‘Copernicus Space Component Data Access System, ESA, Services Coordinated Interface’ (static) | NC_STRING |
| <b>objectType</b>                        | ‘gmd:CI_ResponsibleParty’ (static)                                                            | NC_STRING |

### 12.2.3.53 Group “gmd:contactInfo” in “gmd:contact”

The detailed contact information.

#### Attributes in SO<sub>2</sub>\_\_\_/METADATA/ISO\_METADATA/gmd:contact/gmd:contactInfo

| Group attributes attached to gmd:contactInfo |                           |           |
|----------------------------------------------|---------------------------|-----------|
| Name                                         | Value                     | Type      |
| <b>objectType</b>                            | ‘gmd:CI_Contact’ (static) | NC_STRING |

### 12.2.3.54 Group “gmd:address” in “gmd:contactInfo”

The actual email address.

#### Attributes in SO<sub>2</sub>\_\_\_/METADATA/ISO\_METADATA/gmd:contact/gmd:contactInfo/gmd:address

| Group attributes attached to gmd:address |                                         |           |
|------------------------------------------|-----------------------------------------|-----------|
| Name                                     | Value                                   | Type      |
| <b>gmd:electronicMailAddress</b>         | ‘EOSupport@copernicus.esa.int’ (static) | NC_STRING |
| <b>objectType</b>                        | ‘gmd:CI_Address’ (static)               | NC_STRING |

### 12.2.3.55 Group “gmd:role” in “gmd:contact”

The role of the address provided in this group.

#### Attributes in SO<sub>2</sub>\_\_\_/METADATA/ISO\_METADATA/gmd:contact/gmd:role

| Group attributes attached to gmd:role |                                                                                         |           |
|---------------------------------------|-----------------------------------------------------------------------------------------|-----------|
| Name                                  | Value                                                                                   | Type      |
| <b>codeList</b>                       | ‘http://www.isotc211.org/2005/resources/Codelist/gmxCodelists.xml#CI_RoleCode’ (static) | NC_STRING |
| <b>codeListValue</b>                  | ‘pointOfContact’ (static)                                                               | NC_STRING |
| <b>objectType</b>                     | ‘gmd:CI_RoleCode’ (static)                                                              | NC_STRING |

#### 12.2.3.56 Group “gmd:identificationInfo” in “ISO\_METADATA”

Identification information contains information to uniquely identify the data. Identification information includes information about the citation for the resource, an abstract, the purpose, credit, the status and points of contact. The MD\_Identification entity is mandatory. The MD\_Identification entity is specified (subclassed) as MD\_DataIdentification because in this case it is used to identify data.

#### Attributes in SO<sub>2</sub>\_\_\_/METADATA/ISO\_METADATA/gmd:identificationInfo

Group attributes attached to gmd:identificationInfo

| <i>Name</i>         | <i>Value</i> | <i>Type</i> |
|---------------------|--------------|-------------|
| <b>gmd:abstract</b> |              | NC_STRING   |

Brief narrative summary of the content of the resource. This is product specific.

**L2\_\_AER\_AI (KNMI)** Aerosol index with a spatial resolution of either  $7.2 \times 3.6 \text{ km}^2$  or  $5.6 \times 3.6 \text{ km}^2$  (for the small pixels since 6th of august 2019) observed at about 13:30 local solar time from spectra measured by TROPOMI

**L2\_\_AER\_LH (KNMI)** Altitude of elevated aerosol layer for cloud-free observations with a spatial resolution of either  $7.2 \times 3.6 \text{ km}^2$  or  $5.6 \times 3.6 \text{ km}^2$  (for the small pixels since 6th of august 2019) observed at about 13:30 local solar time from spectra measured by TROPOMI

**L2\_\_NO2\_\_ (KNMI)** Nitrogen dioxide tropospheric column with a spatial resolution of either  $7.2 \times 3.6 \text{ km}^2$  or  $5.6 \times 3.6 \text{ km}^2$  (for the small pixels since 6th of august 2019) observed at about 13:30 local solar time from spectra measured by TROPOMI

**L2\_\_O3\_\_PR (KNMI)** Ozone profile with a vertical resolution of 6 km and a horizontal resolution of  $28 \times 21 \text{ km}^2$  observed at about 13:30 local solar time from spectra measured by TROPOMI

**L2\_\_O3\_TPR (KNMI)** Tropospheric ozone profile with a vertical resolution of 6 km and a horizontal resolution of either  $7.2 \times 3.6 \text{ km}^2$  or  $5.6 \times 3.6 \text{ km}^2$  (for the small pixels since 6th of august 2019) observed at about 13:30 local solar time from spectra measured by TROPOMI

**L2\_\_CH4\_\_ (SRON)** Dry-air mixing ratio of methane for cloud-free observations over land with a spatial resolution of either  $7.2 \times 3.6 \text{ km}^2$  or  $5.6 \times 3.6 \text{ km}^2$  (for the small pixels since 6th of august 2019) observed at about 13:30 local solar time from spectra measured by TROPOMI

**L2\_\_CO\_\_ (SRON)** Carbon monoxide column over land with a spatial resolution of either  $7.2 \times 3.6 \text{ km}^2$  or  $5.6 \times 3.6 \text{ km}^2$  (for the small pixels since 6th of august 2019) observed at about 13:30 local solar time from spectra measured by TROPOMI

**L2\_\_FRESCO (KNMI)** Cloud fraction and cloud pressure with a spatial resolution of either  $7.2 \times 3.6 \text{ km}^2$  or  $5.6 \times 3.6 \text{ km}^2$  (for the small pixels since 6th of august 2019) observed at about 13:30 local solar time from spectra measured by TROPOMI (KNMI cloud support product)

**L2\_\_CLOUD\_ (DLR)** Cloud fraction, cloud pressure and cloud albedo with a spatial resolution of either  $7.2 \times 3.6 \text{ km}^2$  or  $5.6 \times 3.6 \text{ km}^2$  (for the small pixels since 6th of august 2019) observed at about 13:30 local solar time from spectra measured by TROPOMI

**L2\_\_HCHO\_ (BIRA)** Formaldehyde tropospheric column with a spatial resolution of either  $7.2 \times 3.6 \text{ km}^2$  or  $5.6 \times 3.6 \text{ km}^2$  (for the small pixels since 6th of august 2019) observed at about 13:30 local solar time from spectra measured by TROPOMI

**L2\_\_O3\_\_ (DLR/BIRA)** Ozone total column with a spatial resolution of either  $7.2 \times 3.6 \text{ km}^2$  or  $5.6 \times 3.6 \text{ km}^2$  (for the small pixels since 6th of august 2019) observed at about 13:30 local solar time from spectra measured by TROPOMI

**L2\_\_O3\_TCL (DLR/IUP)** Tropospheric ozone with a spatial resolution of either  $7.2 \times 3.6 \text{ km}^2$  or  $5.6 \times 3.6 \text{ km}^2$  (for the small pixels since 6th of august 2019) observed at about 13:30 local solar time from spectra measured by TROPOMI

**L2\_\_SO2\_\_ (BIRA)** Sulfur dioxide column with a spatial resolution of either  $7.2 \times 3.6 \text{ km}^2$  or  $5.6 \times 3.6 \text{ km}^2$  (for the small pixels since 6th of august 2019) observed at about 13:30 local solar time from spectra measured by TROPOMI

**L2\_\_NP\_BD(3,6,7)** Regrided NPP-VIIRS data with a spatial resolution of either  $7.2 \times 3.6 \text{ km}^2$  or  $5.6 \times 3.6 \text{ km}^2$  (for the small pixels since 6th of august 2019)

|                                                          |                                             |           |
|----------------------------------------------------------|---------------------------------------------|-----------|
| <b>gmd:credit</b>                                        | '%(credit)s' (static)                       | NC_STRING |
| Recognition of those who contributed to the resource(s). |                                             |           |
| <b>gmd:language</b>                                      | 'eng' (static)                              | NC_STRING |
| <b>gmd:topicCategory</b>                                 | 'climatologyMeteorologyAtmosphere' (static) | NC_STRING |
| Main theme(s) of the dataset.                            |                                             |           |

|                                              |                                      |           |
|----------------------------------------------|--------------------------------------|-----------|
| <b>objectType</b>                            | 'gmd:MD_DataIdentification' (static) | NC_STRING |
| Name of the metadata class [RD32, table 10]. |                                      |           |

### 12.2.3.57 Group “gmd:citation” in “gmd:identificationInfo”

Citation data for the resource.

#### Attributes in SO2\_\_\_/METADATA/ISO\_METADATA/gmd:identificationInfo/gmd:citation

| Group attributes attached to gmd:citation                                                    |                            |           |
|----------------------------------------------------------------------------------------------|----------------------------|-----------|
| Name                                                                                         | Value                      | Type      |
| <b>gmd:title</b>                                                                             |                            | NC_STRING |
| Name by which the cited resource is known. This is the same as the global “title” attribute. |                            |           |
| <b>objectType</b>                                                                            | 'gmd:CI_Citation' (static) | NC_STRING |
| Name of the metadata class [RD32, table 11].                                                 |                            |           |

### 12.2.3.58 Group “gmd:date” in “gmd:citation”

#### Attributes in SO2\_\_\_/METADATA/ISO\_METADATA/gmd:identificationInfo/gmd:citation/gmd:date

| Group attributes attached to gmd:date |                                       |           |
|---------------------------------------|---------------------------------------|-----------|
| Name                                  | Value                                 | Type      |
| <b>gmd:date</b>                       | '%(processor_release_date)s' (static) | NC_STRING |
| <b>objectType</b>                     | 'gmd:CI_Date' (static)                | NC_STRING |

### 12.2.3.59 Group “gmd:dateType” in “gmd:date”

Event used for reference date.

#### Attributes in SO2\_\_\_/METADATA/ISO\_METADATA/gmd:identificationInfo/gmd:citation/gmd:date/gmd:dateType

| Group attributes attached to gmd:dateType |                                                                                             |           |
|-------------------------------------------|---------------------------------------------------------------------------------------------|-----------|
| Name                                      | Value                                                                                       | Type      |
| <b>codeList</b>                           | 'http://www.isotc211.org/2005/resources/Codelist/gmxCodeLists.xml#CI_DateTypeCode' (static) | NC_STRING |
| <b>codeListValue</b>                      | 'creation' (static)                                                                         | NC_STRING |
| <b>objectType</b>                         | 'gmd:CI_DateTypeCode' (static)                                                              | NC_STRING |

### 12.2.3.60 Group “gmd:identifier” in “gmd:citation”

Unique identifier for metadata file, see the Level 1B metadata specification [RD32, table 5] for a discussion of the value.

#### Attributes in SO2\_\_\_/METADATA/ISO\_METADATA/gmd:identificationInfo/gmd:citation/gmd:identifier

| Group attributes attached to gmd:identifier                                                                                     |                                                                      |           |
|---------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|-----------|
| Name                                                                                                                            | Value                                                                | Type      |
| <b>gmd:code</b>                                                                                                                 | 'urn:ogc:def:EOP:ESA:SENTINEL.S5P_TROP_-<br>%(shortname)s' (dynamic) | NC_STRING |
| Replace “%(shortname)s” with the “ProductShortName” value from the Level 2 “/METADATA/GRANULE_-<br>DESCRIPTION” metadata group. |                                                                      |           |
| <b>objectType</b>                                                                                                               | 'gmd:MD_Identifier' (static)                                         | NC_STRING |

### 12.2.3.61 Group “gmd:pointOfContact” in “gmd:identificationInfo”

See description of the “gmd:contact” attribute above.

#### Attributes in SO2\_\_\_/METADATA/ISO\_METADATA/gmd:identificationInfo/gmd:pointOfContact

| Group attributes attached to gmd:pointOfContact |                                                                                               |           |
|-------------------------------------------------|-----------------------------------------------------------------------------------------------|-----------|
| Name                                            | Value                                                                                         | Type      |
| <b>gmd:organisationName</b>                     | ‘Copernicus Space Component Data Access System, ESA, Services Coordinated Interface’ (static) | NC_STRING |
| <b>objectType</b>                               | ‘gmd:CI_ResponsibleParty’ (static)                                                            | NC_STRING |

### 12.2.3.62 Group “gmd:contactInfo” in “gmd:pointOfContact”

#### Attributes in SO2\_\_\_/METADATA/ISO\_METADATA/gmd:identificationInfo/gmd:pointOfContact/gmd:contactInfo

| Group attributes attached to gmd:contactInfo |                           |           |
|----------------------------------------------|---------------------------|-----------|
| Name                                         | Value                     | Type      |
| <b>objectType</b>                            | ‘gmd:CI_Contact’ (static) | NC_STRING |

### 12.2.3.63 Group “gmd:address” in “gmd:contactInfo”

#### Attributes in SO2\_\_\_/METADATA/ISO\_METADATA/gmd:identificationInfo/gmd:pointOfContact/gmd:contactInfo/gmd:address

| Group attributes attached to gmd:address |                                         |           |
|------------------------------------------|-----------------------------------------|-----------|
| Name                                     | Value                                   | Type      |
| <b>gmd:electronicMailAddress</b>         | ‘EOSupport@copernicus.esa.int’ (static) | NC_STRING |
| <b>objectType</b>                        | ‘gmd:CI_Address’ (static)               | NC_STRING |

### 12.2.3.64 Group “gmd:role” in “gmd:pointOfContact”

#### Attributes in SO2\_\_\_/METADATA/ISO\_METADATA/gmd:identificationInfo/gmd:pointOfContact/gmd:role

| Group attributes attached to gmd:role |                                                                                         |           |
|---------------------------------------|-----------------------------------------------------------------------------------------|-----------|
| Name                                  | Value                                                                                   | Type      |
| <b>codeList</b>                       | ‘http://www.isotc211.org/2005/resources/Codelist/gmxCodelists.xml#CI_RoleCode’ (static) | NC_STRING |
| <b>codeListValue</b>                  | ‘distributor’ (static)                                                                  | NC_STRING |
| <b>objectType</b>                     | ‘gmd:CI_RoleCode’ (static)                                                              | NC_STRING |

### 12.2.3.65 Group “gmd:descriptiveKeywords#1” in “gmd:identificationInfo”

Provides category keywords, their type, and reference source. Within the framework of GEMET the choice of keywords is very limited. More meaningful keywords can be derived from the Climate and Forecast metadata conventions’ standard name list, see “gmd:descriptiveKeywords#2” below.

#### Attributes in SO2\_\_\_/METADATA/ISO\_METADATA/gmd:identificationInfo/gmd:descriptiveKeywords#1

| Group attributes attached to gmd:descriptiveKeywords#1 |                                   |           |
|--------------------------------------------------------|-----------------------------------|-----------|
| Name                                                   | Value                             | Type      |
| <b>gmd:keyword#1</b>                                   | ‘Atmospheric conditions’ (static) | NC_STRING |
| <b>objectType</b>                                      | ‘gmd:MD_Keywords’ (static)        | NC_STRING |

### 12.2.3.66 Group “gmd:type” in “gmd:descriptiveKeywords#1”

Subject matter used to group similar keywords.

**Attributes in SO<sub>2</sub>\_\_\_/METADATA/ISO\_METADATA/gmd:identificationInfo/gmd:descriptiveKeywords#1/gmd:type**

| Group attributes attached to gmd:type |                                                                                                |           |
|---------------------------------------|------------------------------------------------------------------------------------------------|-----------|
| Name                                  | Value                                                                                          | Type      |
| <b>codeList</b>                       | 'http://www.isotc211.org/2005/resources/Codelist/gmxCodelists.xml#MD_KeywordTypeCode' (static) | NC_STRING |
| <b>codeListValue</b>                  | 'theme' (static)                                                                               | NC_STRING |
| <b>objectType</b>                     | 'gmd:MD_KeywordTypeCode' (static)                                                              | NC_STRING |

### 12.2.3.67 Group “gmd:thesaurusName” in “gmd:descriptiveKeywords#1”

Name by which the cited resource is known.

**Attributes in SO<sub>2</sub>\_\_\_/METADATA/ISO\_METADATA/gmd:identificationInfo/gmd:descriptiveKeywords#1/gmd:thesaurusName**

| Group attributes attached to gmd:thesaurusName |                                                |           |
|------------------------------------------------|------------------------------------------------|-----------|
| Name                                           | Value                                          | Type      |
| <b>gmd:title</b>                               | 'GEMET - INSPIRE themes, version 1.0' (static) | NC_STRING |
| <b>objectType</b>                              | 'gmd:CI_Citation' (static)                     | NC_STRING |

### 12.2.3.68 Group “gmd:date” in “gmd:thesaurusName”

Reference date for the cited resource.

**Attributes in SO<sub>2</sub>\_\_\_/METADATA/ISO\_METADATA/gmd:identificationInfo/gmd:descriptiveKeywords#1/gmd:thesaurusName/gmd:date**

| Group attributes attached to gmd:date |                        |           |
|---------------------------------------|------------------------|-----------|
| Name                                  | Value                  | Type      |
| <b>gmd:date</b>                       | '2008-06-01' (static)  | NC_STRING |
| <b>objectType</b>                     | 'gmd:CI_Date' (static) | NC_STRING |

### 12.2.3.69 Group “gmd:dateType” in “gmd:date”

What date is used for the reference date.

**Attributes in SO<sub>2</sub>\_\_\_/METADATA/ISO\_METADATA/gmd:identificationInfo/gmd:descriptiveKeywords#1/gmd:thesaurusName/gmd:date/gmd:dateType**

| Group attributes attached to gmd:dateType |                                                                                             |           |
|-------------------------------------------|---------------------------------------------------------------------------------------------|-----------|
| Name                                      | Value                                                                                       | Type      |
| <b>codeList</b>                           | 'http://www.isotc211.org/2005/resources/Codelist/gmxCodelists.xml#CI_DateTypeCode' (static) | NC_STRING |
| <b>codeListValue</b>                      | 'publication' (static)                                                                      | NC_STRING |
| <b>objectType</b>                         | 'gmd:CI_DateTypeCode' (static)                                                              | NC_STRING |

### 12.2.3.70 Group “gmd:descriptiveKeywords#2” in “gmd:identificationInfo”

Provides category keywords, their type, and reference source. These keywords are taken from the Climate and Forecast metadata conventions' standard name list [ER5]. The keywords listed below identify the most important parameters in the product.

**L2\_AER\_AI (KNMI)** ultraviolet\_aerosol\_index

**L2\_\_AER\_LH (KNMI)** height\_of\_elevated\_aerosol\_layer

**L2\_\_NO2\_\_ (KNMI)** troposphere\_mole\_content\_of\_nitrogen\_dioxide, stratosphere\_mole\_content\_of\_nitrogen\_dioxide, atmosphere\_mole\_content\_of\_nitrogen\_dioxide

**L2\_\_O3\_\_PR (KNMI)** mole\_fraction\_of\_ozone\_in\_air

**L2\_\_O3\_TPR (KNMI)** mole\_fraction\_of\_ozone\_in\_air

**L2\_\_CH4\_\_ (SRON)** atmosphere\_mole\_fraction\_of\_methane\_in\_dry\_air

**L2\_\_CO\_\_ (SRON)** atmosphere\_mole\_content\_of\_carbon\_monoxide

**L2\_\_FRESCO (KNMI)**

**L2\_\_CLOUD\_ (DLR)**

**L2\_\_HCHO\_\_ (BIRA)** troposphere\_mole\_content\_of\_formaldehyde

**L2\_\_O3\_\_ (DLR/BIRA)** atmosphere\_mole\_content\_of\_ozone

**L2\_\_O3\_TCL (DLR/IUP)** troposphere\_mole\_content\_of\_ozone

**L2\_\_SO2\_\_ (BIRA)** atmosphere\_mole\_content\_of\_sulfur\_dioxide

**L2\_\_NP\_BDx (RAL)**

**Attributes in SO2\_\_/\_METADATA/ISO\_METADATA/gmd:identificationInfo/gmd:descriptiveKeywords#2**

Group attributes attached to gmd:descriptiveKeywords#2

| Name                 | Value                      | Type      |
|----------------------|----------------------------|-----------|
| <b>gmd:keyword#1</b> |                            | NC_STRING |
| <b>objectType</b>    | 'gmd:MD_Keywords' (static) | NC_STRING |

**12.2.3.71 Group “gmd:thesaurusName” in “gmd:descriptiveKeywords#2”**

Name by which the cited resource is known.

**Attributes in SO2\_\_/\_METADATA/ISO\_METADATA/gmd:identificationInfo/gmd:descriptiveKeywords#2/gmd:thesaurusName**

Group attributes attached to gmd:thesaurusName

| Name              | Value                                                    | Type      |
|-------------------|----------------------------------------------------------|-----------|
| <b>gmd:title</b>  | 'CF Standard Name Table v29' (static)                    | NC_STRING |
| <b>xlink:href</b> | 'http://cfconventions.org/standard-names.html' (dynamic) | NC_STRING |
| <b>objectType</b> | 'gmd:CI_Citation' (static)                               | NC_STRING |

**12.2.3.72 Group “gmd:date” in “gmd:thesaurusName”**

Reference date for the cited resource.

**Attributes in SO2\_\_/\_METADATA/ISO\_METADATA/gmd:identificationInfo/gmd:descriptiveKeywords#2/gmd:thesaurusName/gmd:date**

Group attributes attached to gmd:date

| Name              | Value                  | Type      |
|-------------------|------------------------|-----------|
| <b>gmd:date</b>   | '2015-07-08' (static)  | NC_STRING |
| <b>objectType</b> | 'gmd:CI_Date' (static) | NC_STRING |

### 12.2.3.73 Group “gmd:dateType” in “gmd:date”

What date is used for the reference date.

**Attributes in SO2\_\_\_/METADATA/ISO\_METADATA/gmd:identificationInfo/gmd:descriptiveKeywords#2/gmd:thesaurusName/gmd:date/gmd:dateType**

| Group attributes attached to gmd:dateType |                                                                                             |           |
|-------------------------------------------|---------------------------------------------------------------------------------------------|-----------|
| Name                                      | Value                                                                                       | Type      |
| <b>codeList</b>                           | 'http://www.isotc211.org/2005/resources/Codelist/gmxCodeLists.xml#CI_DateTypeCode' (static) | NC_STRING |
| <b>codeListValue</b>                      | 'publication' (static)                                                                      | NC_STRING |
| <b>objectType</b>                         | 'gmd:CI_DateTypeCode' (static)                                                              | NC_STRING |

### 12.2.3.74 Group “gmd:resourceConstraints” in “gmd:identificationInfo”

Provides information about constraints which apply to the resource.

**Attributes in SO2\_\_\_/METADATA/ISO\_METADATA/gmd:identificationInfo/gmd:resourceConstraints**

| Group attributes attached to gmd:resourceConstraints |                                                                                                         |           |
|------------------------------------------------------|---------------------------------------------------------------------------------------------------------|-----------|
| Name                                                 | Value                                                                                                   | Type      |
| <b>gmd:useLimitation</b>                             | 'no conditions apply' (static)<br>Limitation affecting the fitness for use of the resource or metadata. | NC_STRING |
| <b>objectType</b>                                    | 'gmd:MD_LegalConstraints' (static)                                                                      | NC_STRING |

### 12.2.3.75 Group “gmd:accessConstraints” in “gmd:resourceConstraints”

Access constraints applied to assure the protection of privacy or intellectual property, and any special restrictions or limitations on obtaining the resource or metadata.

**Attributes in SO2\_\_\_/METADATA/ISO\_METADATA/gmd:identificationInfo/gmd:resourceConstraints/gmd:accessConstraints**

| Group attributes attached to gmd:accessConstraints |                                                                                                |           |
|----------------------------------------------------|------------------------------------------------------------------------------------------------|-----------|
| Name                                               | Value                                                                                          | Type      |
| <b>codeList</b>                                    | 'http://www.isotc211.org/2005/resources/Codelist/gmxCodeLists.xml#MD_RestrictionCode' (static) | NC_STRING |
| <b>codeListValue</b>                               | 'copyright' (static)                                                                           | NC_STRING |
| <b>objectType</b>                                  | 'gmd:MD_RestrictionCode' (static)                                                              | NC_STRING |

### 12.2.3.76 Group “gmd:spatialRepresentationType” in “gmd:identificationInfo”

Method used to spatially represent geographic information.

**Attributes in SO2\_\_\_/METADATA/ISO\_METADATA/gmd:identificationInfo/gmd:spatialRepresentationType**

| Group attributes attached to gmd:spatialRepresentationType |                                                                                                               |           |
|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|-----------|
| Name                                                       | Value                                                                                                         | Type      |
| <b>codeList</b>                                            | 'http://www.isotc211.org/2005/resources/Codelist/gmxCodeLists.xml#MD_SpatialRepresentation-TypeCode' (static) | NC_STRING |
| <b>codeListValue</b>                                       | 'grid' (static)                                                                                               | NC_STRING |
| <b>objectType</b>                                          | 'gmd:MD_SpatialRepresentationTypeCode' (static)                                                               | NC_STRING |

### 12.2.3.77 Group “gmd:spatialResolution” in “gmd:identificationInfo”

Ground sample distance.

#### Attributes in SO2\_\_\_/METADATA/ISO\_METADATA/gmd:identificationInfo/gmd:spatialResolution

| Group attributes attached to gmd:spatialResolution |                              |           |
|----------------------------------------------------|------------------------------|-----------|
| Name                                               | Value                        | Type      |
| <b>gmd:distance</b>                                | 7.0 (dynamic)                | NC_FLOAT  |
| <b>uom</b>                                         | ‘km’ (static)                | NC_STRING |
| <b>objectType</b>                                  | ‘gmd:MD_Resolution’ (static) | NC_STRING |

### 12.2.3.78 Group “gmd:characterSet” in “gmd:identificationInfo”

#### Attributes in SO2\_\_\_/METADATA/ISO\_METADATA/gmd:identificationInfo/gmd:characterSet

| Group attributes attached to gmd:characterSet |                                                                                                 |           |
|-----------------------------------------------|-------------------------------------------------------------------------------------------------|-----------|
| Name                                          | Value                                                                                           | Type      |
| <b>codeList</b>                               | ‘http://www.isotc211.org/2005/resources/Codelist/gmxCodeLists.xml#MD_CharacterSetCode’ (static) | NC_STRING |
| <b>codeListValue</b>                          | ‘utf8’ (static)                                                                                 | NC_STRING |
| <b>objectType</b>                             | ‘gmd:MD_CharacterSetCode’ (static)                                                              | NC_STRING |

### 12.2.3.79 Group “gmd:extent” in “gmd:identificationInfo”

Extent information including the bounding box, bounding polygon, vertical, and temporal extent of the dataset.

#### Attributes in SO2\_\_\_/METADATA/ISO\_METADATA/gmd:identificationInfo/gmd:extent

| Group attributes attached to gmd:extent |                          |           |
|-----------------------------------------|--------------------------|-----------|
| Name                                    | Value                    | Type      |
| <b>objectType</b>                       | ‘gmd:EX_Extent’ (static) | NC_STRING |

### 12.2.3.80 Group “gmd:geographicElement” in “gmd:extent”

Geographic position of the granule. This is only an approximate reference so specifying the coordinate reference system is unnecessary. The usual limitations apply:  $-180^\circ \leq \vartheta \leq 180^\circ$  and  $-90^\circ \leq \delta \leq 90^\circ$ . Note that for full orbits these values provide little information as at least one pole will be present in the data, ensuring full longitudinal coverage.

#### Attributes in SO2\_\_\_/METADATA/ISO\_METADATA/gmd:identificationInfo/gmd:extent/gmd:geographicElement

| Group attributes attached to gmd:geographicElement                                                                                                                     |                                         |           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|-----------|
| Name                                                                                                                                                                   | Value                                   | Type      |
| <b>gmd:eastBoundLongitude</b>                                                                                                                                          | 180.0 (dynamic)                         | NC_FLOAT  |
| <b>gmd:northBoundLatitude</b>                                                                                                                                          | 90.0 (dynamic)                          | NC_FLOAT  |
| <b>gmd:southBoundLatitude</b>                                                                                                                                          | -90.0 (dynamic)                         | NC_FLOAT  |
| <b>gmd:westBoundLongitude</b>                                                                                                                                          | -180.0 (dynamic)                        | NC_FLOAT  |
| <b>gmd:extentTypeCode</b>                                                                                                                                              | ‘true’ (static)                         | NC_STRING |
| Indication of whether the bounding polygon encompasses an area covered by the data or an area where data is not present. The value “true” indicates <i>inclusion</i> . |                                         |           |
| <b>objectType</b>                                                                                                                                                      | ‘gmd:EX_GeographicBoundingBox’ (static) | NC_STRING |

### 12.2.3.81 Group “gmd:temporalElement” in “gmd:extent”

#### Attributes in SO2\_\_\_/METADATA/ISO\_METADATA/gmd:identificationInfo/gmd:extent/gmd:temporalElement

| Group attributes attached to gmd:temporalElement |                                  |           |
|--------------------------------------------------|----------------------------------|-----------|
| Name                                             | Value                            | Type      |
| <b>objectType</b>                                | ‘gmd:EX_TemporalExtent’ (static) | NC_STRING |

### 12.2.3.82 Group “gmd:extent” in “gmd:temporalElement”

Time period covered by the content of the dataset.

#### Attributes in SO2\_\_\_/METADATA/ISO\_METADATA/gmd:identificationInfo/gmd:extent/gmd:temporalElement/gmd:extent

| Group attributes attached to gmd:extent |                                                                                                                     |           |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------|-----------|
| Name                                    | Value                                                                                                               | Type      |
| <b>gml:beginPosition</b>                | ‘2014-11-14T19:58:00’ (dynamic)<br>Time of the start of the granule, expressed as ISO 8601 [RD35] date-time string. | NC_STRING |
| <b>gml:endPosition</b>                  | ‘2014-11-14T20:08:00’ (dynamic)<br>Time of the end of the granule, expressed as ISO 8601 [RD35] date-time string.   | NC_STRING |
| <b>objectType</b>                       | ‘gml:TimePeriod’ (static)                                                                                           | NC_STRING |

### 12.2.3.83 Group “gmd:dataQualityInfo” in “ISO\_METADATA”

This group contains a general assessment of the quality of the dataset. In addition, the package contains information about the sources and production processes used in producing a dataset, which is of particular importance for imagery and gridded data.

For the TROPOMI 2 products the use of the contained class LI\_Lineage (group “gmd:lineage”, section 12.2.3.91 on page 146) is important for describing the sources which are either used or produced (output) in a series of process steps. The sources refer to the various L1b data products used as inputs (and the L0 products used in producing *those* products) and the auxiliary data (static and especially dynamic) when producing the L2 products.

#### Attributes in SO2\_\_\_/METADATA/ISO\_METADATA/gmd:dataQualityInfo

| Group attributes attached to gmd:dataQualityInfo |                               |           |
|--------------------------------------------------|-------------------------------|-----------|
| Name                                             | Value                         | Type      |
| <b>objectType</b>                                | ‘gmd:DQ_DataQuality’ (static) | NC_STRING |

### 12.2.3.84 Group “gmd:scope” in “gmd:dataQualityInfo”

The specific data to which the data quality information applies.

#### Attributes in SO2\_\_\_/METADATA/ISO\_METADATA/gmd:dataQualityInfo/gmd:scope

| Group attributes attached to gmd:scope |                         |           |
|----------------------------------------|-------------------------|-----------|
| Name                                   | Value                   | Type      |
| <b>objectType</b>                      | ‘gmd:DQ_Scope’ (static) | NC_STRING |

### 12.2.3.85 Group “gmd:level” in “gmd:scope”

Hierarchical level of the data specified by the scope.

#### Attributes in SO2\_\_\_/METADATA/ISO\_METADATA/gmd:dataQualityInfo/gmd:scope/gmd:level

Group attributes attached to gmd:level

| Name                 | Value                                                                                    | Type      |
|----------------------|------------------------------------------------------------------------------------------|-----------|
| <b>codeList</b>      | 'http://www.isotc211.org/2005/resources/Codelist/gmxCodeLists.xml#MD_ScopeCode' (static) | NC_STRING |
| <b>codeListValue</b> | 'dataset' (static)                                                                       | NC_STRING |
| <b>objectType</b>    | 'gmd:MD_ScopeCode' (static)                                                              | NC_STRING |

#### 12.2.3.86 Group “gmd:report” in “gmd:dataQualityInfo”

Value (or set of values) obtained from applying a data quality measure or the outcome of evaluating the obtained value (or set of values) against a specified acceptable conformance quality level.

#### Attributes in SO2\_\_\_/METADATA/ISO\_METADATA/gmd:dataQualityInfo/gmd:report

Group attributes attached to gmd:report

| Name              | Value                               | Type      |
|-------------------|-------------------------------------|-----------|
| <b>objectType</b> | 'gmd:DQ_DomainConsistency' (static) | NC_STRING |

#### 12.2.3.87 Group “gmd:result” in “gmd:report”

Value (or set of values) obtained from applying a data quality measure or the outcome of evaluating the obtained value (or set of values) against a specified acceptable conformance quality level.

#### Attributes in SO2\_\_\_/METADATA/ISO\_METADATA/gmd:dataQualityInfo/gmd:report/gmd:result

Group attributes attached to gmd:result

| Name                                                                 | Value                                                                                                                           | Type      |
|----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>objectType</b>                                                    | 'gmd:DQ_ConformanceResult' (static)                                                                                             | NC_STRING |
| <b>gmd:pass</b>                                                      | 'true' (static)                                                                                                                 | NC_STRING |
| Indication of conformance result. The value “true” indicates “pass”. |                                                                                                                                 |           |
| <b>gmd:explanation</b>                                               | 'INSPIRE Data specification for orthoimagery is not yet officially published so conformity has not yet been evaluated' (static) | NC_STRING |

Explanation of the meaning of conformance for this result. Within the context of INSPIRE conformance can currently not be determined.

#### 12.2.3.88 Group “gmd:specification” in “gmd:result”

Citation of product specification or user requirement against which data is being evaluated.

#### Attributes in SO2\_\_\_/METADATA/ISO\_METADATA/gmd:dataQualityInfo/gmd:report/gmd:result/gmd:specification

Group attributes attached to gmd:specification

| Name              | Value                                                                              | Type      |
|-------------------|------------------------------------------------------------------------------------|-----------|
| <b>objectType</b> | 'gmd:CI_Citation' (static)                                                         | NC_STRING |
| <b>gmd:title</b>  | 'INSPIRE Data Specification on Orthoimagery - Guidelines, version 3.0rc3' (static) | NC_STRING |

#### 12.2.3.89 Group “gmd:date” in “gmd:specification”

Reference date for the cited resource.

#### Attributes in SO2\_\_\_/METADATA/ISO\_METADATA/gmd:dataQualityInfo/gmd:report/gmd:result/gmd:specification/gmd:date

Group attributes attached to gmd:date

| Name              | Value                  | Type      |
|-------------------|------------------------|-----------|
| <b>gmd:date</b>   | '2013-02-04' (static)  | NC_STRING |
| <b>objectType</b> | 'gmd:CI_Date' (static) | NC_STRING |

### 12.2.3.90 Group “gmd:dateType” in “gmd:date”

Meaning of the reference date for the cited resource.

**Attributes in SO2\_\_\_/METADATA/ISO\_METADATA/gmd:dataQualityInfo/gmd:report/gmd:result/gmd:specification/  
gmd:date/gmd:dateType**

Group attributes attached to gmd:dateType

| Name                 | Value                                                                                           | Type      |
|----------------------|-------------------------------------------------------------------------------------------------|-----------|
| <b>codeList</b>      | 'http://www.isotc211.org/2005/resources/Codelist/<br>gmxCodeLists.xml#CI_DateTypeCode' (static) | NC_STRING |
| <b>codeListValue</b> | 'publication' (static)                                                                          | NC_STRING |
| <b>objectType</b>    | 'gmd:CI_DateTypeCode' (static)                                                                  | NC_STRING |

### 12.2.3.91 Group “gmd:lineage” in “gmd:dataQualityInfo”

Non-quantitative quality information about the lineage of the data specified by the scope.

**Attributes in SO2\_\_\_/METADATA/ISO\_METADATA/gmd:dataQualityInfo/gmd:lineage**

Group attributes attached to gmd:lineage

| Name                 | Value                                                                                                | Type      |
|----------------------|------------------------------------------------------------------------------------------------------|-----------|
| <b>objectType</b>    | 'gmd:LI_Lineage' (static)                                                                            | NC_STRING |
| <b>gmd:statement</b> | 'L2 %(product)s dataset produced by %(processingcenter)s from the S5P/TROPOMI L1B product' (dynamic) | NC_STRING |

General explanation of the data producer's knowledge about the lineage of a dataset. Insert short description of the actual Level 2 product in this string (at the %(... )s).

### 12.2.3.92 Group “gmd:processStep” in “gmd:lineage”

Information about an event or transformation in the life of the dataset including details of the algorithm and software used for processing.

**Attributes in SO2\_\_\_/METADATA/ISO\_METADATA/gmd:dataQualityInfo/gmd:lineage/gmd:processStep**

Group attributes attached to gmd:processStep

| Name                   | Value                                                                                                                                | Type      |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>objectType</b>      | 'gmi:LE_ProcessStep' (static)                                                                                                        | NC_STRING |
| <b>gmd:description</b> | 'Processing of L1b to L2 %(product)s data for orbit<br>%(orbit)d using the %(institute)s processor version<br>%(version)s' (dynamic) | NC_STRING |

Description of the event, including related parameters or tolerances. Insert short description of the actual Level 2 product, the orbit number, the name of the institute responsible for the CFI and the software version in this string (at the respective %(... )s and %(... )d).

### 12.2.3.93 Group “gmi:output” in “gmd:processStep”

Description of the output.

**Attributes in SO2\_\_\_/METADATA/ISO\_METADATA/gmd:dataQualityInfo/gmd:lineage/gmd:processStep/  
gmi:output**

Group attributes attached to gmi:output

| Name                                                                     | Value                    | Type      |
|--------------------------------------------------------------------------|--------------------------|-----------|
| <b>gmd:description</b>                                                   |                          | NC_STRING |
| Short description of the output, a copy of the global 'title' attribute. |                          |           |
| <b>objectType</b>                                                        | 'gmi:LE_Source' (static) | NC_STRING |

#### 12.2.3.94 Group “gmd:sourceCitation” in “gmi:output”

Reference to the actual filename of the output data and production date and time.

#### Attributes in SO<sub>2</sub>\_\_\_/METADATA/ISO\_METADATA/gmd:dataQualityInfo/gmd:lineage/gmd:processStep/gmi:output/gmd:sourceCitation

Group attributes attached to gmd:sourceCitation

| Name                                | Value                            | Type      |
|-------------------------------------|----------------------------------|-----------|
| <b>gmd:title</b>                    | '%(logical_filename)s' (dynamic) | NC_STRING |
| Output file name without extension. |                                  |           |
| <b>objectType</b>                   | 'gmd:CI_Citation' (static)       | NC_STRING |

#### 12.2.3.95 Group “gmd:date” in “gmd:sourceCitation”

Production date and time of the output file.

#### Attributes in SO<sub>2</sub>\_\_\_/METADATA/ISO\_METADATA/gmd:dataQualityInfo/gmd:lineage/gmd:processStep/gmi:output/gmd:sourceCitation/gmd:date

Group attributes attached to gmd:date

| Name                                                                                                                                                        | Value                      | Type      |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|-----------|
| <b>gmd:date</b>                                                                                                                                             |                            | NC_STRING |
| Production date and time of the output file. Note that the definition in the XML schema appears to allow the use of a “CI_DateTime” instead of a “CI_Date”. |                            |           |
| <b>objectType</b>                                                                                                                                           | 'gmd:CI_DateTime' (static) | NC_STRING |

#### 12.2.3.96 Group “gmd:dateType” in “gmd:date”

Meaning of the reference date for the cited resource.

#### Attributes in SO<sub>2</sub>\_\_\_/METADATA/ISO\_METADATA/gmd:dataQualityInfo/gmd:lineage/gmd:processStep/gmi:output/gmd:sourceCitation/gmd:date/gmd:dateType

Group attributes attached to gmd:dateType

| Name                 | Value                                                                                       | Type      |
|----------------------|---------------------------------------------------------------------------------------------|-----------|
| <b>codeList</b>      | 'http://www.isotc211.org/2005/resources/Codelist/gmxCodeLists.xml#CI_DateTypeCode' (static) | NC_STRING |
| <b>codeListValue</b> | 'creation' (static)                                                                         | NC_STRING |
| <b>objectType</b>    | 'gmd:CI_DateTypeCode' (static)                                                              | NC_STRING |

#### 12.2.3.97 Group “gmd:identifier” in “gmd:sourceCitation”

Identification of the output product.

#### Attributes in SO<sub>2</sub>\_\_\_/METADATA/ISO\_METADATA/gmd:dataQualityInfo/gmd:lineage/gmd:processStep/gmi:output/gmd:sourceCitation/gmd:identifier

Group attributes attached to gmd:identifier

| Name | Value | Type |
|------|-------|------|
|------|-------|------|

|                                                                                                        |                              |           |
|--------------------------------------------------------------------------------------------------------|------------------------------|-----------|
| <b>gmd:code</b>                                                                                        | '%(shortname)s' (dynamic)    | NC_STRING |
| The product short name, a copy of the 'ProductShortName' attribute in '/METADATA/GRANULE_DESCRIPTION'. |                              |           |
| <b>objectType</b>                                                                                      | 'gmd:MD_Identifier' (static) | NC_STRING |

### 12.2.3.98 Group “gmi:processedLevel” in “gmi:output”

Process level of the output file.

**Attributes in SO2\_\_\_/METADATA/ISO\_METADATA/gmd:dataQualityInfo/gmd:lineage/gmd:processStep/gmi:output/gmi:processedLevel**

| Group attributes attached to gmi:processedLevel |                              |           |
|-------------------------------------------------|------------------------------|-----------|
| Name                                            | Value                        | Type      |
| <b>gmd:code</b>                                 | 'L2' (static)                | NC_STRING |
| <b>objectType</b>                               | 'gmd:MD_Identifier' (static) | NC_STRING |

### 12.2.3.99 Group “gmi:processingInformation” in “gmd:processStep”

Description of the processor in more detail.

**Attributes in SO2\_\_\_/METADATA/ISO\_METADATA/gmd:dataQualityInfo/gmd:lineage/gmd:processStep/gmi:processingInformation**

| Group attributes attached to gmi:processingInformation |                              |           |
|--------------------------------------------------------|------------------------------|-----------|
| Name                                                   | Value                        | Type      |
| <b>objectType</b>                                      | 'gmi:LE_Processing' (static) | NC_STRING |

### 12.2.3.100 Group “gmi:identifier” in “gmi:processingInformation”

Identification of the processor.

**Attributes in SO2\_\_\_/METADATA/ISO\_METADATA/gmd:dataQualityInfo/gmd:lineage/gmd:processStep/gmi:processingInformation/gmi:identifier**

| Group attributes attached to gmi:identifier                                                                                                                 |                                                                         |           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|-----------|
| Name                                                                                                                                                        | Value                                                                   | Type      |
| <b>gmd:code</b>                                                                                                                                             | '%(institute)s L2 %(product)s processor, version %(version)s' (dynamic) | NC_STRING |
| Descriptive name of the processor, with the %(...)s placeholders replaced with the responsible institute's name, product name and software release version. |                                                                         |           |
| <b>objectType</b>                                                                                                                                           | 'gmd:MD_Identifier' (static)                                            | NC_STRING |

### 12.2.3.101 Group “gmi:softwareReference” in “gmi:processingInformation”

Reference to document describing processing software.

**Attributes in SO2\_\_\_/METADATA/ISO\_METADATA/gmd:dataQualityInfo/gmd:lineage/gmd:processStep/gmi:processingInformation/gmi:softwareReference**

| Group attributes attached to gmi:softwareReference |                                           |           |
|----------------------------------------------------|-------------------------------------------|-----------|
| Name                                               | Value                                     | Type      |
| <b>gmd:title</b>                                   | 'UPAS L2 %(product)s processor' (dynamic) | NC_STRING |
| Title of processor description.                    |                                           |           |
| <b>objectType</b>                                  | 'gmd:CI_Citation' (static)                | NC_STRING |

### 12.2.3.102 Group “gmd:date” in “gmi:softwareReference”

Release date (compile date) of the processor.

**Attributes in SO2\_\_\_/METADATA/ISO\_METADATA/gmd:dataQualityInfo/gmd:lineage/gmd:processStep/gmi:processingInformation/gmi:softwareReference/gmd:date**

| Group attributes attached to gmd:date                                      |                            |           |
|----------------------------------------------------------------------------|----------------------------|-----------|
| Name                                                                       | Value                      | Type      |
| <b>gmd:date</b>                                                            |                            | NC_STRING |
| Release date of the processor expressed as an ISO 8601 date string [RD35]. |                            |           |
| <b>objectType</b>                                                          | ‘gmd:CI_DateTime’ (static) | NC_STRING |

### 12.2.3.103 Group “gmd:dateType” in “gmd:date”

Confirm that this is the release date of the processor.

**Attributes in SO2\_\_\_/METADATA/ISO\_METADATA/gmd:dataQualityInfo/gmd:lineage/gmd:processStep/gmi:processingInformation/gmi:softwareReference/gmd:date/gmd:dateType**

| Group attributes attached to gmd:dateType |                                                                                             |           |
|-------------------------------------------|---------------------------------------------------------------------------------------------|-----------|
| Name                                      | Value                                                                                       | Type      |
| <b>codeList</b>                           | ‘http://www.isotc211.org/2005/resources/Codelist/gmxCodelists.xml#CI_DateTypeCode’ (static) | NC_STRING |
| <b>codeListValue</b>                      | ‘creation’ (static)                                                                         | NC_STRING |
| <b>objectType</b>                         | ‘gmd:CI_DateTypeCode’ (static)                                                              | NC_STRING |

### 12.2.3.104 Group “gmi:documentation#1” in “gmi:processingInformation”

Reference to the ATBD of the product.

**Attributes in SO2\_\_\_/METADATA/ISO\_METADATA/gmd:dataQualityInfo/gmd:lineage/gmd:processStep/gmi:processingInformation/gmi:documentation#1**

| Group attributes attached to gmi:documentation#1                        |                            |           |
|-------------------------------------------------------------------------|----------------------------|-----------|
| Name                                                                    | Value                      | Type      |
| <b>objectType</b>                                                       | ‘gmd:CI_Citation’ (static) | NC_STRING |
| <b>gmd:title</b>                                                        | ‘%(title_atbd)s’ (dynamic) | NC_STRING |
| The filename of the current release of the ATBD of the current product. |                            |           |
| <b>doi</b>                                                              | ‘%(atbd_doi)s’ (dynamic)   | NC_STRING |
| DOI for the algorithm theoretical basis document.                       |                            |           |

### 12.2.3.105 Group “gmd:date” in “gmi:documentation#1”

Release date of the ATBD.

**Attributes in SO2\_\_\_/METADATA/ISO\_METADATA/gmd:dataQualityInfo/gmd:lineage/gmd:processStep/gmi:processingInformation/gmi:documentation#1/gmd:date**

| Group attributes attached to gmd:date                                 |                           |           |
|-----------------------------------------------------------------------|---------------------------|-----------|
| Name                                                                  | Value                     | Type      |
| <b>gmd:date</b>                                                       | ‘%(date_atbd)s’ (dynamic) | NC_STRING |
| Release date of the ATBD expressed as an ISO 8601 date string [RD35]. |                           |           |
| <b>objectType</b>                                                     | ‘gmd:CI_Date’ (static)    | NC_STRING |

### 12.2.3.106 Group “gmd:dateType” in “gmd:date”

Confirm that this is the date of publication.

**Attributes in SO<sub>2</sub>\_\_\_/METADATA/ISO\_METADATA/gmd:dataQualityInfo/gmd:lineage/gmd:processStep/gmi:processingInformation/gmi:documentation#1/gmd:date/gmd:dateType**

| Group attributes attached to gmd:dateType |                                                                                             |           |
|-------------------------------------------|---------------------------------------------------------------------------------------------|-----------|
| Name                                      | Value                                                                                       | Type      |
| <b>codeList</b>                           | 'http://www.isotc211.org/2005/resources/Codelist/gmxCodelists.xml#CI_DateTypeCode' (static) | NC_STRING |
| <b>codeListValue</b>                      | 'publication' (static)                                                                      | NC_STRING |
| <b>objectType</b>                         | 'gmd:CI_DateTypeCode' (static)                                                              | NC_STRING |

### 12.2.3.107 Group “gmi:documentation#2” in “gmi:processingInformation”

Reference to the PUM of the product.

**Attributes in SO<sub>2</sub>\_\_\_/METADATA/ISO\_METADATA/gmd:dataQualityInfo/gmd:lineage/gmd:processStep/gmi:processingInformation/gmi:documentation#2**

| Group attributes attached to gmi:documentation#2 |                                                                                                     |           |
|--------------------------------------------------|-----------------------------------------------------------------------------------------------------|-----------|
| Name                                             | Value                                                                                               | Type      |
| <b>objectType</b>                                | 'gmd:CI_Citation' (static)                                                                          | NC_STRING |
| <b>gmd:title</b>                                 | '%(title_pum)s' (dynamic)<br>The filename of the current release of the PUM of the current product. | NC_STRING |
| <b>doi</b>                                       | '%(pum_doi)s' (dynamic)<br>DOI for the product user manual.                                         | NC_STRING |

### 12.2.3.108 Group “gmd:date” in “gmi:documentation#2”

Release date of the PUM.

**Attributes in SO<sub>2</sub>\_\_\_/METADATA/ISO\_METADATA/gmd:dataQualityInfo/gmd:lineage/gmd:processStep/gmi:processingInformation/gmi:documentation#2/gmd:date**

| Group attributes attached to gmd:date |                                                                                                  |           |
|---------------------------------------|--------------------------------------------------------------------------------------------------|-----------|
| Name                                  | Value                                                                                            | Type      |
| <b>gmd:date</b>                       | '%(date_pum)s' (dynamic)<br>Release date of the PUM expressed as an ISO 8601 date string [RD35]. | NC_STRING |
| <b>objectType</b>                     | 'gmd:CI_Date' (static)                                                                           | NC_STRING |

### 12.2.3.109 Group “gmd:dateType” in “gmd:date”

Confirm that this is the date of publication.

**Attributes in SO<sub>2</sub>\_\_\_/METADATA/ISO\_METADATA/gmd:dataQualityInfo/gmd:lineage/gmd:processStep/gmi:processingInformation/gmi:documentation#2/gmd:date/gmd:dateType**

| Group attributes attached to gmd:dateType |                                                                                             |           |
|-------------------------------------------|---------------------------------------------------------------------------------------------|-----------|
| Name                                      | Value                                                                                       | Type      |
| <b>codeList</b>                           | 'http://www.isotc211.org/2005/resources/Codelist/gmxCodelists.xml#CI_DateTypeCode' (static) | NC_STRING |
| <b>codeListValue</b>                      | 'publication' (static)                                                                      | NC_STRING |
| <b>objectType</b>                         | 'gmd:CI_DateTypeCode' (static)                                                              | NC_STRING |

### 12.2.3.110 Group “gmi:report” in “gmd:processStep”

Short report of what occurred during the process step.

**Attributes in SO<sub>2</sub>\_\_\_/METADATA/ISO\_METADATA/gmd:dataQualityInfo/gmd:lineage/gmd:processStep/gmi:report**

Group attributes attached to gmi:report

| Name                                                                                                                      | Value                                                                                                              | Type      |
|---------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|-----------|
| <b>gmi:description</b>                                                                                                    | ‘Sentinel 5-precursor TROPOMI L1b processed to L2 data using the %(institute)s L2 %(product)s processor’ (dynamic) | NC_STRING |
| Textual description of what occurred during the process step. Replace %(. . .)s as indicated.                             |                                                                                                                    |           |
| <b>gmi:fileType</b>                                                                                                       | ‘netCDF-4’ (static)                                                                                                | NC_STRING |
| Type of file that contains the processing report, in our case the processing report is contained in the main output file. |                                                                                                                    |           |
| <b>gmi:name</b>                                                                                                           | ‘%(logical_filename)s.nc’ (dynamic)                                                                                | NC_STRING |
| <b>objectType</b>                                                                                                         | ‘gmi:LE_ProcessStepReport’ (dynamic)                                                                               | NC_STRING |

### 12.2.3.111 Group “gmd:source#1” in “gmd:processStep”

Information about the source data used in creating the data specified by the scope. Repeat group as needed, incrementing the number of the source (after the # mark).

**Attributes in SO<sub>2</sub>\_\_\_/METADATA/ISO\_METADATA/gmd:dataQualityInfo/gmd:lineage/gmd:processStep/gmd:source#1**

Group attributes attached to gmd:source#1

| Name                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Value                    | Type      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|-----------|
| <b>objectType</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 'gmi:LE_Source' (static) | NC_STRING |
| <b>gmd:description</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | NC_STRING                |           |
| Description of the input data, including L1B, L2, dynamic auxiliary input data and semi-static auxiliary input data. Base strings are “TROPOMI L1B %s radiance product”, “TROPOMI L1B %s irradiance product”, “TROPOMI L2 %s product”, “Auxiliary ECMWF %s Meteorological forecast data”, “Processor %s configuration file”, “Auxiliary %s reference data”, “Auxiliary %s algorithm lookup table”, “Auxiliary CTM %s model input data”, “Auxiliary snow and ice input data” and “Auxiliary NPP/VIIRS cloud screening input data”. The %s to be replaced with specific descriptors. |                          |           |

### 12.2.3.112 Group “gmi:processedLevel” in “gmd:source#1”

**Attributes in SO<sub>2</sub>\_\_\_/METADATA/ISO\_METADATA/gmd:dataQualityInfo/gmd:lineage/gmd:processStep/gmd:source#1/gmi:processedLevel**

Group attributes attached to gmi:processedLevel

| Name              | Value                        | Type      |
|-------------------|------------------------------|-----------|
| <b>gmd:code</b>   | Empty!                       | NC_STRING |
| <b>objectType</b> | ‘gmd:MD_Identifier’ (static) | NC_STRING |

### 12.2.3.113 Group “gmd:sourceCitation” in “gmd:source#1”

Reference to the actual filename of the input data.

**Attributes in SO<sub>2</sub>\_\_\_/METADATA/ISO\_METADATA/gmd:dataQualityInfo/gmd:lineage/gmd:processStep/gmd:source#1/gmd:sourceCitation**

---

Group attributes attached to gmd:sourceCitation

---

| Name              | Value                      | Type      |
|-------------------|----------------------------|-----------|
| <b>objectType</b> | 'gmd:CI_Citation' (static) | NC_STRING |

---

#### 12.2.3.114 Group “gmd:date” in “gmd:sourceCitation”

**Attributes in SO<sub>2</sub>\_\_\_/METADATA/ISO\_METADATA/gmd:dataQualityInfo/gmd:lineage/gmd:processStep/gmd:source#1/gmd:sourceCitation/gmd:date**

---

Group attributes attached to gmd:date

---

| Name            | Value | Type      |
|-----------------|-------|-----------|
| <b>gmd:date</b> |       | NC_STRING |

Production date and time of the input file(s) in this group expressed as an ISO 8601 date-time string [RD35]. Note that the definition in the XML schema appears to allow the use of a “CI\_DateTime” instead of a “CI\_Date”.

|                   |                        |           |
|-------------------|------------------------|-----------|
| <b>objectType</b> | 'gmd:CI_Date' (static) | NC_STRING |
|-------------------|------------------------|-----------|

---

#### 12.2.3.115 Group “gmd:dateType” in “gmd:date”

Meaning of the reference date for the cited resource.

**Attributes in SO<sub>2</sub>\_\_\_/METADATA/ISO\_METADATA/gmd:dataQualityInfo/gmd:lineage/gmd:processStep/gmd:source#1/gmd:sourceCitation/gmd:date/gmd:dateType**

---

Group attributes attached to gmd:dateType

---

| Name                 | Value                                                                                       | Type      |
|----------------------|---------------------------------------------------------------------------------------------|-----------|
| <b>codeList</b>      | 'http://www.isotc211.org/2005/resources/Codelist/gmxCodelists.xml#CI_DateTypeCode' (static) | NC_STRING |
| <b>codeListValue</b> | 'creation' (static)                                                                         | NC_STRING |
| <b>objectType</b>    | 'gmd:CI_DateTypeCode' (static)                                                              | NC_STRING |

---

#### 12.2.3.116 Group “gmd:title” in “gmd:sourceCitation”

**Attributes in SO<sub>2</sub>\_\_\_/METADATA/ISO\_METADATA/gmd:dataQualityInfo/gmd:lineage/gmd:processStep/gmd:source#1/gmd:sourceCitation/gmd:title**

---

Group attributes attached to gmd:title

---

| Name                       | Value | Type      |
|----------------------------|-------|-----------|
| <b>gco:characterString</b> |       | NC_STRING |

Textual description of the input file group (same as the “gmd:description” attribute in the “gmi:LE\_Source” object).

#### 12.2.3.117 Group “gmd:alternateTitle#1” in “gmd:sourceCitation”

All filenames in this group, in case more files of a particular file type are delivered, for instance for meteorological or model input. Repeat group as needed, incrementing the number of the input file (after the # mark).

**Attributes in SO<sub>2</sub>\_\_\_/METADATA/ISO\_METADATA/gmd:dataQualityInfo/gmd:lineage/gmd:processStep/gmd:source#1/gmd:sourceCitation/gmd:alternateTitle#1**

---

Group attributes attached to gmd:alternateTitle#1

---

| Name | Value | Type |
|------|-------|------|
|------|-------|------|

---

|                                 |               |           |
|---------------------------------|---------------|-----------|
| <b>gmx:FileName</b>             | <i>Empty!</i> | NC_STRING |
| The basename of the input file. |               |           |

### 12.2.3.118 Group “gmi:acquisitionInformation” in “ISO\_METADATA”

Metadata regarding the acquisition of the original data.

#### Attributes in SO2\_\_\_/METADATA/ISO\_METADATA/gmi:acquisitionInformation

| Group attributes attached to gmi:acquisitionInformation |                                          |           |
|---------------------------------------------------------|------------------------------------------|-----------|
| Name                                                    | Value                                    | Type      |
| <b>objectType</b>                                       | ‘gmi:MI_AcquisitionInformation’ (static) | NC_STRING |

### 12.2.3.119 Group “gmi:platform” in “gmi:acquisitionInformation”

The platform we are on.

#### Attributes in SO2\_\_\_/METADATA/ISO\_METADATA/gmi:acquisitionInformation/gmi:platform

| Group attributes attached to gmi:platform |                                 |           |
|-------------------------------------------|---------------------------------|-----------|
| Name                                      | Value                           | Type      |
| <b>gmi:description</b>                    | ‘Sentinel 5 Precursor’ (static) | NC_STRING |
| <b>objectType</b>                         | ‘gmi:MI_Platform’ (static)      | NC_STRING |

### 12.2.3.120 Group “gmi:identifier” in “gmi:platform”

Short identifier of the platform.

#### Attributes in SO2\_\_\_/METADATA/ISO\_METADATA/gmi:acquisitionInformation/gmi:platform/gmi:identifier

| Group attributes attached to gmi:identifier |                                |           |
|---------------------------------------------|--------------------------------|-----------|
| Name                                        | Value                          | Type      |
| <b>gmd:code</b>                             | ‘S5P’ (static)                 | NC_STRING |
| <b>gmd:codeSpace</b>                        | ‘http://www.esa.int/’ (static) | NC_STRING |
| <b>objectType</b>                           | ‘gmd:RS_Identifier’ (static)   | NC_STRING |

### 12.2.3.121 Group “gmi:instrument” in “gmi:platform”

The instrument used for the observations.

#### Attributes in SO2\_\_\_/METADATA/ISO\_METADATA/gmi:acquisitionInformation/gmi:platform/gmi:instrument

| Group attributes attached to gmi:instrument |                                                 |           |
|---------------------------------------------|-------------------------------------------------|-----------|
| Name                                        | Value                                           | Type      |
| <b>objectType</b>                           | ‘gmi:MI_Instrument’ (static)                    | NC_STRING |
| <b>gmi:type</b>                             | ‘UV-VIS-NIR-SWIR imaging spectrometer’ (static) | NC_STRING |
| Type of the instrument.                     |                                                 |           |

### 12.2.3.122 Group “gmi:identifier” in “gmi:instrument”

Unique identifier for the instrument.

#### Attributes in SO2\_\_\_/METADATA/ISO\_METADATA/gmi:acquisitionInformation/gmi:platform/gmi:instrument/gmi:identifier

| Group attributes attached to gmi:identifier                           |                                |           |
|-----------------------------------------------------------------------|--------------------------------|-----------|
| Name                                                                  | Value                          | Type      |
| <b>gmd:code</b>                                                       | 'TROPOMI' (static)             | NC_STRING |
| The actual identifier.                                                |                                |           |
| <b>gmd:codeSpace</b>                                                  | 'http://www.esa.int/' (static) | NC_STRING |
| Name or identifier of the organization responsible for the namespace. |                                |           |
| <b>objectType</b>                                                     | 'gmd:RS_Identifier' (static)   | NC_STRING |

## A Flag descriptions

The following tables describe the Measurement flags, Processing quality flags (processing failures and filter conditions, errors and warnings) and Surface classifications.

Please be aware that this section is work in progress and the flags are not included in the product yet. The aim of this section is for review only.

**Table 12:** Processing quality flags, errors, processing failures and filter conditions for S5P Level 2. Warnings are listed in table 13. The value in the first column is the result of a bitwise ‘and’ of 255 (0xFF) and the value in the “processing\_quality\_flags” variable.

| #  | Short name                     | Description                                                                                                                                                                                                                                                                                                                                                        | Algorithm                                         |
|----|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| 0  | success                        | No failures, output contains value. Warnings still possible.                                                                                                                                                                                                                                                                                                       | All                                               |
| 1  | radiance_missing               | The number of spectral pixels in the radiance due to flagging is too small to perform the fitting.                                                                                                                                                                                                                                                                 | All                                               |
| 2  | irradiance_missing             | The number of spectral pixels in the irradiance due to flagging is too small to perform the fitting.                                                                                                                                                                                                                                                               | All                                               |
| 3  | input_spectrum_missing         | The reflectance spectrum does not contain enough points to perform the retrieval. This is different from (ir)radiance_missing in that the missing points may not be aligned.                                                                                                                                                                                       | All                                               |
| 4  | reflectance_range_error        | Any of the reflectances is out of bounds ( $R < 0$ or $R > R_{\max}$ ).                                                                                                                                                                                                                                                                                            | FRESCO                                            |
| 5  | ler_range_error                | Lambert-equivalent reflectivity out of range error.                                                                                                                                                                                                                                                                                                                | CO, CH <sub>4</sub>                               |
| 6  | snr_range_error                | Too low signal to noise to perform retrieval.                                                                                                                                                                                                                                                                                                                      | CO                                                |
| 7  | sza_range_error                | Solar zenith angle out of range, maximum value from configuration.                                                                                                                                                                                                                                                                                                 | All                                               |
| 8  | vza_range_error                | Viewing zenith angle out of range, maximum value from configuration.                                                                                                                                                                                                                                                                                               | Development phase only                            |
| 9  | lut_range_error                | Extrapolation in lookup table (airmass factor, cloud radiances).                                                                                                                                                                                                                                                                                                   | NO <sub>2</sub>                                   |
| 10 | ozone_range_error              | Ozone column significantly out of range of profile climatology.                                                                                                                                                                                                                                                                                                    | Total O <sub>3</sub> column                       |
| 11 | wavelength_offset_error        | Wavelength offset exceeds maximum from configuration.                                                                                                                                                                                                                                                                                                              | FRESCO, NO <sub>2</sub>                           |
| 12 | initialization_error           | An error occurred during the processing of the pixel, no output was generated. The following errors raise this flag: Mismatch between irradiance and radiance wavelengths; The on-ground distance between band 1 and band 2 ground pixels exceeds a threshold set in the configuration. Derived a-priori information does not validate, no processing is possible. | All                                               |
| 13 | memory_error                   | Memory allocation or deallocation error.                                                                                                                                                                                                                                                                                                                           | CO, CH <sub>4</sub>                               |
| 14 | assertion_error                | Error in algorithm detected during assertion.                                                                                                                                                                                                                                                                                                                      | CO                                                |
| 15 | io_error                       | Error detected during transfer of data between algorithm and framework.                                                                                                                                                                                                                                                                                            | CO, ALH, CH <sub>4</sub> , O <sub>3</sub> profile |
| 16 | numerical_error                | General fatal numerical error occurred during inversion.                                                                                                                                                                                                                                                                                                           | CO, FRESCO                                        |
| 17 | lut_error                      | Error in accessing the lookup table.                                                                                                                                                                                                                                                                                                                               | CH <sub>4</sub>                                   |
| 18 | ISRF_error                     | Error detected in the input instrument spectral response function input data.                                                                                                                                                                                                                                                                                      | CH <sub>4</sub>                                   |
| 19 | convergence_error              | The main algorithm did not converge.                                                                                                                                                                                                                                                                                                                               | All                                               |
| 20 | cloud_filter_convergence_error | The cloud filter did not converge.                                                                                                                                                                                                                                                                                                                                 | CO                                                |

**Table 12:** Processing quality flags, errors, processing failures and filter conditions for S5P Level 2 (continued).

| #  | Short name                                | Description                                                                                                                                                                                 | Algorithm                          |
|----|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| 21 | max_iteration_convergence_error           | No convergence because retrieval exceeds maximum number of iterations. Maximum value from configuration.                                                                                    | ALH                                |
| 22 | aot_lower_boundary_convergence_error      | No convergence because the aerosol optical thickness crosses lower boundary twice in succession.                                                                                            | ALH                                |
| 23 | other_boundary_convergence_error          | No convergence because a state vector element crosses boundary twice in succession. Note that a separate failure flag is defined for non-convergence due to crossing of lower AOT boundary. | ALH                                |
| 25 | ch4_noscat_zero_error                     | The CH <sub>4</sub> column retrieved by the non-scattering CO algorithm from the weak band or strong band is 0.                                                                             | CH <sub>4</sub>                    |
| 26 | h2o_noscat_zero_error                     | The H <sub>2</sub> O column retrieved by the non-scattering CO algorithm from the weak band or strong band is 0.                                                                            | CH <sub>4</sub>                    |
| 27 | max_optical_thickness_error               | Maximum optical thickness exceeded during iterations.                                                                                                                                       | CH <sub>4</sub>                    |
| 28 | aerosol_boundary_error                    | Boundary hit of aerosol parameters at last iteration.                                                                                                                                       | CH <sub>4</sub>                    |
| 29 | boundary_hit_error                        | Fatal boundary hit during iterations.                                                                                                                                                       | CH <sub>4</sub>                    |
| 30 | chi2_error                                | $\chi^2$ is not-a-number or larger than 10 <sup>10</sup> .                                                                                                                                  | CH <sub>4</sub>                    |
| 31 | svd_error                                 | Singular value decomposition failure.                                                                                                                                                       | CH <sub>4</sub>                    |
| 32 | dfs_error                                 | Degree of freedom is not-a-number.                                                                                                                                                          | CH <sub>4</sub>                    |
| 33 | radiative_transfer_error                  | Errors occurred during the radiative transfer computations, no processing possible.                                                                                                         | O <sub>3</sub> profile             |
| 34 | optimal_estimation_error                  | Errors occurred during the optimal estimation, processing has been terminated.                                                                                                              | O <sub>3</sub> profile             |
| 35 | profile_error                             | Flag that indicates if there were any errors during the computation of the ozone profile.                                                                                                   | O <sub>3</sub> profile             |
| 36 | cloud_error                               | No cloud data.                                                                                                                                                                              | Cloud                              |
| 37 | model_error                               | Forward model failure.                                                                                                                                                                      | Cloud, Total O <sub>3</sub> column |
| 38 | number_of_input_data_points_too_low_error | Not enough input ozone columns to calculate a tropospheric column.                                                                                                                          | Tropospheric O <sub>3</sub> column |
| 39 | cloud_pressure_spread_too_low_error       | Cloud pressure variability too low to estimate a tropospheric column.                                                                                                                       | Tropospheric O <sub>3</sub> column |
| 40 | cloud_too_low_level_error                 | Clouds are too low in the atmosphere to assume sufficient shielding.                                                                                                                        | Tropospheric O <sub>3</sub> column |
| 41 | generic_range_error                       | Generic range error.                                                                                                                                                                        | All                                |
| 42 | generic_exception                         | Catch all generic error.                                                                                                                                                                    | All                                |
| 43 | input_spectrum_alignment_error            | Input radiance and irradiance spectra are not aligned correctly.                                                                                                                            | All                                |
| 44 | abort_error                               | Not processed because processor aborted prematurely (time out or user abort)                                                                                                                | All                                |

**Table 12:** Processing quality flags, errors, processing failures and filter conditions for S5P Level 2 (continued).

| #  | Short name                      | Description                                                                                                                                                                                                 | Algorithm                |
|----|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| 45 | wrong_input_type_error          | Wrong input type error, mismatch between expectation and received data.                                                                                                                                     | All                      |
| 46 | wavelength_calibration_error    | An error occurred in the wavelength calibration of this pixel                                                                                                                                               | All                      |
| 47 | coregistration_error            | No colocated pixels found in a supporting band                                                                                                                                                              | All                      |
| 51 | signal_to_noise_ratio_error     | The signal to noise ratio for this spectrum is too low for processing                                                                                                                                       | All                      |
| 52 | configuration_error             | Error while parsing the configuration                                                                                                                                                                       | All                      |
| 53 | key_error                       | Key does not exist                                                                                                                                                                                          | All                      |
| 54 | saturation_error                | Saturation in input spectrum                                                                                                                                                                                | All                      |
| 55 | max_num_outlier_exceeded_error  | The number of outliers detected in the DOAS fit exceeds a maximum set for healthy spectra.                                                                                                                  | NO <sub>2</sub>          |
| 64 | solar_eclipse_filter            | Solar eclipse.                                                                                                                                                                                              | All                      |
| 65 | cloud_filter                    | The cloud filter triggered causing the pixel to be skipped.                                                                                                                                                 | CO, ALH, CH <sub>4</sub> |
| 66 | altitude_consistency_filter     | Too large difference between ECMWF altitude and DEM altitude value.                                                                                                                                         | CO, CH <sub>4</sub>      |
| 67 | altitude_roughness_filter       | Too large standard deviation of altitude in DEM.                                                                                                                                                            | CO, ALH, CH <sub>4</sub> |
| 68 | sun_glint_filter                | For pixels over water, viewing direction inside sun glint region. Definition of sun glint angle and threshold value from ATBD.                                                                              | ALH                      |
| 69 | mixed_surface_type_filter       | Pixel contains land and water areas (e.g. coastal pixel).                                                                                                                                                   | ALH                      |
| 70 | snow_ice_filter                 | Pixel contains snow/ice: Snow/ice flag according to dynamic input OR climatological surface albedo at VIS wavelength is larger than 0.5.                                                                    | ALH                      |
| 71 | aai_filter                      | AAI smaller than 2.0.                                                                                                                                                                                       | ALH                      |
| 72 | cloud_fraction_fresco_filter    | Pixel contains clouds: The FRESCO effective cloud fraction is larger than threshold. Threshold value from ATBD.                                                                                             | ALH                      |
| 73 | aai_scene_albedo_filter         | Pixel contains clouds: The difference between scene albedo at 380 nm from AAI calculation and the climatological surface albedo exceeds threshold. Threshold value from ATBD. This test filters out clouds. | ALH                      |
| 74 | small_pixel_radiance_std_filter | Pixel contains clouds: Standard deviation of radiances in small-pixel column exceeds threshold. Threshold value from ATBD.                                                                                  | ALH, CH <sub>4</sub>     |
| 75 | cloud_fraction_viirs_filter     | Pixel contains clouds: The cloud fraction from VIIRS / NPP exceeds threshold. Threshold value from ATBD.                                                                                                    | ALH                      |
| 76 | cirrus_reflectance_viirs_filter | Pixel contains clouds: Cirrus reflectance from VIIRS / NPP exceeds threshold. Threshold value from ATBD.                                                                                                    | ALH                      |

**Table 12:** Processing quality flags, errors, processing failures and filter conditions for S5P Level 2 (continued).

| #  | Short name                    | Description                                                                                                                                                                                           | Algorithm       |
|----|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| 77 | cf_viirs_swir_ifov_filter     | Fraction of cloudy VIIRS pixels within S5P SWIR ground pixel exceeds a priori threshold from configuration.                                                                                           | CH <sub>4</sub> |
| 78 | cf_viirs_swir_ofova_filter    | Fraction of cloudy VIIRS pixels within S5P SWIR OFOVa exceeds a priori threshold from configuration.                                                                                                  | CH <sub>4</sub> |
| 79 | cf_viirs_swir_ofovb_filter    | Fraction of cloudy VIIRS pixels within S5P SWIR OFOVb exceeds a priori threshold from configuration.                                                                                                  | CH <sub>4</sub> |
| 80 | cf_viirs_swir_ofovc_filter    | Fraction of cloudy VIIRS pixels within S5P SWIR OFOVc exceeds a priori threshold from configuration.                                                                                                  | CH <sub>4</sub> |
| 81 | cf_viirs_nir_ifov_filter      | Fraction of cloudy VIIRS pixels within S5P NIR ground pixel exceeds a priori threshold from configuration.                                                                                            | CH <sub>4</sub> |
| 82 | cf_viirs_nir_ofova_filter     | Fraction of cloudy VIIRS pixels within S5P NIR OFOVa exceeds a priori threshold from configuration.                                                                                                   | CH <sub>4</sub> |
| 83 | cf_viirs_nir_ofovb_filter     | Fraction of cloudy VIIRS pixels within S5P NIR OFOVb exceeds a priori threshold from configuration.                                                                                                   | CH <sub>4</sub> |
| 84 | cf_viirs_nir_ofovc_filter     | Fraction of cloudy VIIRS pixels within S5P NIR OFOVc exceeds a priori threshold from configuration.                                                                                                   | CH <sub>4</sub> |
| 85 | refl_cirrus_viirs_swir_filter | Average VIIRS cirrus reflectance within SWIR ground pixel exceeds a priori threshold from configuration.                                                                                              | CH <sub>4</sub> |
| 86 | refl_cirrus_viirs_nir_filter  | Average VIIRS cirrus reflectance within NIR ground pixel exceeds a priori threshold from configuration.                                                                                               | CH <sub>4</sub> |
| 87 | diff_refl_cirrus_viirs_filter | Difference in VIIRS average cirrus reflectance between SWIR and NIR ground pixel exceeds a priori threshold from configuration.                                                                       | CH <sub>4</sub> |
| 88 | ch4_noscat_ratio_filter       | The ratio between [CH <sub>4</sub> ] <sub>weak</sub> and [CH <sub>4</sub> ] <sub>strong</sub> is below or exceeds a priori thresholds from configuration.                                             | CH <sub>4</sub> |
| 89 | ch4_noscat_ratio_std_filter   | The standard deviation of [CH <sub>4</sub> ] <sub>weak</sub> /[CH <sub>4</sub> ] <sub>strong</sub> within the SWIR pixel and the 8 neighbouring pixels exceeds a priori threshold from configuration. | CH <sub>4</sub> |
| 90 | h2o_noscat_ratio_filter       | The ratio between [H <sub>2</sub> O] <sub>weak</sub> and [H <sub>2</sub> O] <sub>strong</sub> is below or exceeds a priori thresholds from configuration.                                             | CH <sub>4</sub> |
| 91 | h2o_noscat_ratio_std_filter   | The standard deviation of [H <sub>2</sub> O] <sub>weak</sub> /[H <sub>2</sub> O] <sub>strong</sub> within the SWIR pixel and the 8 neighbouring pixels exceeds a priori threshold from configuration. | CH <sub>4</sub> |

**Table 12:** Processing quality flags, errors, processing failures and filter conditions for S5P Level 2 (continued).

| #  | Short name                     | Description                                                                                                                                                 | Algorithm       |
|----|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| 92 | diff_psurf_fresco_ecmwf_filter | Difference between the FRESCO apparent surface pressure and the ECMWF surface pressure exceeds a priori threshold from configuration.                       | CH <sub>4</sub> |
| 93 | psurf_fresco_stdv_filter       | The standard deviation of the FRESCO apparent surface pressure in the NIR pixel and the 8 surrounding pixels exceeds a priori threshold from configuration. | CH <sub>4</sub> |
| 94 | ocean_filter                   | The ground pixel is over ocean (and ocean glint retrievals are not switched on).                                                                            | CH <sub>4</sub> |
| 95 | time_range_filter              | Time is out of the range that is to be processed.                                                                                                           | All             |
| 96 | pixel_or_scanline_index_filter | Not processed because pixel index does not match general selection criteria.                                                                                | All             |
| 97 | geographic_region_filter       | Pixel falls outside the specified regions of interest.                                                                                                      | All             |

**Table 13:** Processing quality flags, warnings for S5P Level 2. Errors, processing failures and filter conditions are listed in table 12. If a bitwise ‘and’ of the mask value and the value in the “processing\_quality\_flags” variable is not zero, then the warning applies to the specific retrieval.

| Bit # | Mask (hex) | Short name                     | Description                                                                                                                                                                                                                                       | Algorithm                                       |
|-------|------------|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| 0–7   | 0x000000FF | error                          | If non-zero an error has occurred when processing the pixel, see table 12 for details.                                                                                                                                                            | All                                             |
| 8     | 0x00000100 | input_spectrum_warning         | Number of good pixels in radiance, irradiance or calculated reflectance below threshold from configuration.                                                                                                                                       | All                                             |
| 9     | 0x00000200 | wavelength_calibration_warning | Offset from wavelength fit is larger than limit set in configuration.                                                                                                                                                                             | Most                                            |
| 10    | 0x00000400 | extrapolation_warning          | Pressure or temperature outside cross section LUT range, other lookup table extrapolation.                                                                                                                                                        | CO, CH <sub>4</sub>                             |
| 11    | 0x00000800 | sun_glint_warning              | Sun glint possibility warning.                                                                                                                                                                                                                    | All                                             |
| 12    | 0x00001000 | south_atlantic_anomaly_warning | TROPOMI is inside the south Atlantic anomaly while taking these measurements.                                                                                                                                                                     | All                                             |
| 13    | 0x00002000 | sun_glint_correction           | A sun glint correction has been applied.                                                                                                                                                                                                          | Cloud                                           |
| 14    | 0x00004000 | snow_ice_warning               | Snow/ice flag is set, i.e. using scene data from the cloud support product.                                                                                                                                                                       | NO <sub>2</sub> , Cloud                         |
| 15    | 0x00008000 | cloud_warning                  | Cloud filter based on FRESCO apparent surface pressure (VIIRS not available), cloud fraction above threshold or cloud pressure adjusted to force cloud above surface. In case of Cloud product this flag indicates the possibility of ice-clouds. | CH <sub>4</sub> , O <sub>3</sub> profile, Cloud |
| 16    | 0x00010000 | AAI_warning                    | Possible aerosol contamination as either indicated by the AAI (O <sub>3</sub> profile).                                                                                                                                                           | O <sub>3</sub> profile                          |

**Table 13:** Processing quality flags, warnings for S5P Level 2 (continued).

| Bit # | Mask (hex) | Short name                          | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Algorithm                                                             |
|-------|------------|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| 17    | 0x00020000 | pixel_level_input_data_missing      | Dynamic auxiliary input data (e.g.. cloud) is missing for this ground pixel. A fallback option is used.                                                                                                                                                                                                                                                                                                                                                                                                                        | All                                                                   |
| 18    | 0x00040000 | data_range_warning                  | Carbon monoxide column tends to negative values; Water column tends to negative values; Heavy water (HDO) column tends to negative values; others. In case of the O <sub>3</sub> product this flag indicates VCD or effective albedo values outside a valid range. In case of the SO <sub>2</sub> or the HCHO product this flag indicates AMF values outside a valid range. For O <sub>3</sub> profile this warning indicates an out of range cost function, or an out of range RMS difference between retrieval and a priori. | CO, CH <sub>4</sub> , O <sub>3</sub> , SO <sub>2</sub> , HCHO profile |
| 19    | 0x00080000 | low_cloud_fraction_warning          | Low cloud fraction, therefore no cloud pressure retrieved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Cloud                                                                 |
| 20    | 0x00100000 | altitude_consistency_warning        | Difference between ECMWF surface elevation and high-resolution surface elevation exceeds threshold from configuration.                                                                                                                                                                                                                                                                                                                                                                                                         | CH <sub>4</sub>                                                       |
| 21    | 0x00200000 | signal_to_noise_ratio_warning       | Signal to noise ratio in SWIR and/or NIR band below threshold from configuration. For the O <sub>3</sub> and HCHO products this flag indicates an RMS above a certain threshold.                                                                                                                                                                                                                                                                                                                                               | CH <sub>4</sub> , O <sub>3</sub> , HCHO                               |
| 22    | 0x00400000 | deconvolution_warning               | Failed deconvolution irradiance spectrum (not pixel-specific, but row-specific).                                                                                                                                                                                                                                                                                                                                                                                                                                               | CO, CH <sub>4</sub>                                                   |
| 23    | 0x00800000 | so2_volcanic_origin_likely_warning  | Warning for SO <sub>2</sub> BL product, UTLS products: volcanic origin except for heavily polluted sites. For O <sub>3</sub> profile this warning is issued in case of a large SO <sub>2</sub> column which has an impact on the O <sub>3</sub> profile retrieval.                                                                                                                                                                                                                                                             | SO <sub>2</sub> , O <sub>3</sub> profile                              |
| 24    | 0x01000000 | so2_volcanic_origin_certain_warning | Warning for SO <sub>2</sub> BL product, UTLS products: volcanic origin certain.                                                                                                                                                                                                                                                                                                                                                                                                                                                | SO <sub>2</sub>                                                       |
| 25    | 0x02000000 | interpolation_warning               | Warning for interpolation on partially missing data. In this case the valid available data is used, potentially leading to a bias.                                                                                                                                                                                                                                                                                                                                                                                             | All                                                                   |
| 26    | 0x04000000 | saturation_warning                  | Saturation occurred spectrum, possibly causing biases in the retrieval                                                                                                                                                                                                                                                                                                                                                                                                                                                         | All                                                                   |
| 27    | 0x08000000 | high_sza_warning                    | Warning for high solar zenith angle. In this case, the processing can be performed with less final quality.                                                                                                                                                                                                                                                                                                                                                                                                                    | All                                                                   |
| 28    | 0x10000000 | cloud_retrieval_warning             | Warning occurring when the retrieval diagnostic indicates a degraded quality of the cloud retrieval.                                                                                                                                                                                                                                                                                                                                                                                                                           | Cloud                                                                 |
| 29    | 0x20000000 | cloud_inhomogeneity_warning         | The cloud coregistration inhomogeneity parameter is above a given threshold. This warning is also set when the coregistration weight sums are less than 1.                                                                                                                                                                                                                                                                                                                                                                     | Cloud                                                                 |
| 30    | 0x40000000 | thermal_instability_warning         | Input spectra have been labeled with a thermal instability warning flag.                                                                                                                                                                                                                                                                                                                                                                                                                                                       | All                                                                   |

**Table 14:** Surface classification for S5P Level 2. This is a combined land/water mask and surface classification data field. For land the “Global Land Cover Characteristics Data Base Version 2.0” is used [ER11], specifically the “USGS Land Use/Land Cover System (Modified Level 2)” classification. Over water the classification from the NASA SDP toolkit [ER12], which is based on [RD46].

| Bit # | Mask (hex) | Short name                                        | Description                                                                                                                                          |
|-------|------------|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0     | 0x03       | Land                                              | The pixel is over land, for more than 50 %                                                                                                           |
| 1     | 0x03       | Water                                             | The pixel is over water, for more than 50 %                                                                                                          |
| 2     | 0x03       | some_water                                        | Pixel contains water (however small the fraction), i.e. at least one of the 15 × 15 arcsecond subpixels in the SDP dataset is classified as water    |
| 3     | 0x03       | coastline                                         | Pixel is water, but contains land (coastline)                                                                                                        |
| 0     | 0x04       | mixed_surface                                     | Pixel has a mixed surface type. Classification is result of highest bin, not overwhelming majority, i.e. type covers less than 50 % of pixel surface |
| 4     | 0x04       | value_covers_majority_of_pixel                    | Pixel is dominated by surface type, i.e. type covers more than 50 % of pixel surface                                                                 |
| 9     | 0xF9       | Water+Shallow_Ocean                               | Water, shallow ocean                                                                                                                                 |
| 17    | 0xF9       | Water+Shallow_Inland_Water                        | Water, shallow inland water (lake)                                                                                                                   |
| 25    | 0xF9       | Water+Ocean_Coastline-Lake_Shoreline              | Water, mixed with land; coastline                                                                                                                    |
| 33    | 0xF9       | Water+Intermittent_Water                          | Intermittent water, for instance the Wadden Sea                                                                                                      |
| 41    | 0xF9       | Water+Deep_Inland_Water                           | Deep inland water                                                                                                                                    |
| 49    | 0xF9       | Water+Continental_Shelf_Ocean                     | Water, continental shelf ocean                                                                                                                       |
| 57    | 0xF9       | Water+Deep_Ocean                                  | Water, deep ocean                                                                                                                                    |
| 8     | 0xF9       | Land+Urban_And_Built-up_Land                      | Land, urban areas                                                                                                                                    |
| 16    | 0xF9       | Land+Dryland_Cropland_And_Pasture                 | Land, Dryland Cropland and Pasture                                                                                                                   |
| 24    | 0xF9       | Land+Irrigated_Cropland_And_Pasture               | Land, Irrigated Cropland and Pasture                                                                                                                 |
| 32    | 0xF9       | Land+Mixed_Dryland-irrigated_Cropland_And_Pasture | Land, Mixed Dryland/Irrigated Cropland and Pasture                                                                                                   |
| 40    | 0xF9       | Land+Cropland-grassland_Mosaic                    | Land, Cropland/Grassland Mosaic                                                                                                                      |
| 48    | 0xF9       | Land+Cropland-woodland_Mosaic                     | Land, Cropland/Woodland Mosaic                                                                                                                       |
| 56    | 0xF9       | Land+Grassland                                    | Land, Grassland                                                                                                                                      |
| 64    | 0xF9       | Land+Shrubland                                    | Land, Shrubland                                                                                                                                      |
| 72    | 0xF9       | Land+Mixed_Shrubland-grassland                    | Land, Mixed Shrubland/Grassland                                                                                                                      |
| 80    | 0xF9       | Land+Savanna                                      | Land, Savanna                                                                                                                                        |
| 88    | 0xF9       | Land+Deciduous_Broadleaf_Forest                   | Land, Deciduous Broadleaf Forest                                                                                                                     |

**Table 14:** Surface classification for S5P Level 2 (continued).

| Bit # | Mask (hex) | Short name                        | Description                        |
|-------|------------|-----------------------------------|------------------------------------|
| 96    | 0xF9       | Land+Deciduous_Needleleaf_Forest  | Land, Deciduous Needleleaf Forest  |
| 104   | 0xF9       | Land+Evergreen_Broadleaf_Forest   | Land, Evergreen Broadleaf Forest   |
| 112   | 0xF9       | Land+Evergreen_Needleleaf_Forest  | Land, Evergreen Needleleaf Forest  |
| 120   | 0xF9       | Land+Mixed_Forest                 | Land, Mixed Forest                 |
| 128   | 0xF9       | Land+Herbaceous_Wetland           | Land, Herbaceous Wetland           |
| 136   | 0xF9       | Land+Wooded_Wetland               | Land, Wooded Wetland               |
| 144   | 0xF9       | Land+Barren_Or_Sparsely_Vegetated | Land, Barren or Sparsely Vegetated |
| 152   | 0xF9       | Land+Herbaceous_Tundra            | Land, Herbaceous Tundra            |
| 160   | 0xF9       | Land+Wooded_Tundra                | Land, Wooded Tundra                |
| 168   | 0xF9       | Land+Mixed_Tundra                 | Land, Mixed Tundra                 |
| 176   | 0xF9       | Land+Bare_Ground_Tundra           | Land, Bare Ground Tundra           |
| 184   | 0xF9       | Land+Snow_Or_Ice                  | Land, Snow or Ice                  |